PROVISIONAL APPLICATION FOR PATENT

UNDER 35 U.S.C. § 111(b)

PRIVACY-PRESERVING AMBIENT CONSENT SIGNALING GATED ON CONTINUOUSLY RENEWED AFFIRMATION, WITH MUSIC, LAPSE DETECTION AND ESCALATION TO HELP

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Inventor: David Morris Clayman

Residence: Boca Raton, Florida, United States

Applicant: David Morris Clayman (the inventor)

Assignee: Lifesaver Labs Public Benefit Corporation (a Delaware public benefit corporation), under the inventor's employment agreement

Related applications: U.S. Patent No. 12,347,301 B1; U.S. Provisional Patent Application No. 64/016,800; U.S. Patent Application Publication No. 2016/0196735 A1 (each incorporated by reference)

Entity status: Micro entity (certification on Form PTO/SB/15A filed herewith)

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is related to U.S. Patent No. 12,347,301 B1, titled "Consent management system over a communications network," issued to the present inventor (the "'301 patent"), the entire disclosure of which is incorporated herein by reference. The subject matter disclosed herein may be practiced together with, or independently of, the consent scoring module, consent monitoring module, emergency response mechanism, cryptographic storage module, consent customization modes, smart lighting device, and smart button described in the '301 patent. This application is also related to U.S. Provisional Patent Application No. 64/016,800, titled "Networked Emergency Dispatch and Preparedness Simulation System" (the "peer-response application"), which is incorporated herein by reference; the responder alerting, location-based responder selection, and access and entry-management subject matter described below may be practiced with, and in some embodiments is provided by, the platform of that application. This application also incorporates by reference, in its entirety, U.S. Patent Application Publication No. 2016/0196735 A1 of the present inventor (published under the name Adam Clayman), titled "Systems and Methods for Monitoring Health in a Shared Living Environment."

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not applicable.

TECHNICAL FIELD

[0003] The disclosed subject matter relates to consent management systems and, more specifically, to systems and methods that condition the continued output of media content, such as music, on the continuous, real-time detection of affirmative (and, where configured, enthusiastic) consent from one or more persons present, such that the media output itself serves as an ambient, mutually perceptible, fail-closed indicator of consent status.

BACKGROUND

[0004] Consent between persons engaged in an intimate interaction is not a single event but a continuing state that may be affirmed, qualified, or withdrawn at any moment. Existing technical approaches to consent fall into a small number of categories, none of which treats consent as a continuously re-verified precondition for an ongoing, ambient output that both parties can perceive without looking at a screen.

[0005] A first category consists of pre-act consent records. Mobile applications exist in which two parties tap, type, or enter a code on a smartphone before an encounter to record that consent was given, optionally with a time stamp and location. These systems capture a single moment of agreement and provide no mechanism for detecting whether that agreement persists minutes later, and they require

the parties to divert attention to a screen. Such applications have also drawn a substantive criticism, which contributed to the removal of at least one of them from a major application store in 2014 and has shaped the public reception of others since: that by recording a single "yes" they treat consent as a contract rather than as a continuing state, and that a stored record of agreement can later be used against a person who withdrew it. The system described here is designed to answer that criticism. An affirmation in it expires; withdrawal always takes precedence over affirmation; and the record documents a lapse, a stop, a brake, or an unanswered check-in as fully as it documents an affirmation, so that it cannot be used to show consent beyond the window in which consent was actually renewed.

[0006] A second category consists of voice-controlled devices, including systems that transcribe spoken commands and detect vocal stress to generate control signals for a stimulation device. These systems treat the voice as a remote control for an appliance: a command produces an action, and silence produces nothing. They are not designed to treat the absence of affirmation as a safety event, do not require agreement among multiple parties before acting, do not verify that a control action actually took effect, and are not integrated with an emergency response pathway.

[0007] A third category consists of affect-responsive music systems that select or adjust music according to a listener's measured arousal, mood, or spoken emotional tone. Such systems optimize the listening experience of a single user; they neither model consent nor default to silence when a positive signal is absent.

[0008] A fourth category consists of so-called dead-man's switches applied to media, in which a listener must periodically press a button to keep an audiobook playing. These are single-user convenience mechanisms; they take no account of the semantic content of speech, the emotional quality of vocalization, disagreement among multiple parties, or the need to escalate to an emergency response.

[0009] Dating platforms have added safety features of their own (photo verification, a button that summons a safety partner, sharing date details with a friend) but these end at the door: none carries the identity established in the app into the room, none knows whether the person present is the person matched, and none continues to protect either party once the date turns intimate. A fifth category consists of lone-worker and consumer personal-safety applications, in which a user must periodically check in (or hold a button and release it only with a PIN, or let a timer run down) and, failing that, an escalating sequence of reminders, alarms, and calls to contacts or a monitoring center follows. These systems monitor a single person, treat any check-in as sufficient regardless of its content or emotional quality, have no concept of mutual consent among persons present, and provide no pleasant, ambient positive output whose presence signals that all is well; their only output is the alarm.

[0010] Music already accompanies intimacy. The choice of music as the ambient carrier of affirmation follows what people already do. In a 2024 survey of more than one thousand adults, forty-seven percent said that they regularly listen to music during sex (ZipHealth, 2024). In a 2019 survey of 12,010 adults aged twenty-one to fifty in twelve countries, fifty-nine percent said that the right music makes sex better, fifty-six percent that it makes them more adventurous, twenty-two percent that they choose music according to the kind of sex they want to have, and twenty-seven percent that they had turned up music so that others would not hear them; twenty-seven percent of Americans said they kept a playlist specifically for sex (Sonos, 2019). In 2015 Spotify reported that its users had created some two and a half million playlists with "sex" in their names, and in a 2016 study of some thirty thousand people in eight countries, those who most often played music aloud at home with their partners reported sixty-seven percent more sex than those who did so least (Sonos, 2016). These surveys were conducted or commissioned by companies with an interest in music and are self-reported, and the last is correlational; but they agree that music is a common, and for many people a welcome, part of intimacy. A system that gives that music a second meaning, as the audible sign that consent is being renewed, asks participants to adopt no new behavior at the moment of intimacy, only to let something they already do carry information it did not carry before.

[0011] From FRIES to FRIES SAVER. A widely taught framework describes valid consent as FRIES: freely given, reversible, informed, enthusiastic, and specific. The framework makes consent revocable and scoped, but it does not make consent expire. Under it, a yes given at the start of an encounter remains in force until it is withdrawn, so that the burden of reversal falls on the person who must find the words, in the moment, to withdraw it. It says nothing about how long a yes lasts while an encounter rises toward high voltage, or while it continues into plans that neither party consented to at the outset; and it contemplates neither a signal that can be

raised instantly at the first sign of a greyzone, nor any means outside the two people by which consent is checked, prompted, or recorded. The affirmative consent envelope described here supplies those missing properties, and this disclosure names the extended framework FRIES SAVER: consent that is, in addition, staged, alarmable, verified, expiring, and responsive. Expiry matters most. FRIES makes consent reversible, while FRIES SAVER makes it lapse by default unless it is renewed, so that silence, freezing, or fawning ends consent rather than continuing it.

[0012] Existing consent systems record a decision, but the difficulty they leave unaddressed is that the decision itself is often made in a state that distorts it. Research on the hot-cold empathy gap (Loewenstein, 1996 and later work; Ariely and Loewenstein, 2006) shows that people in a "cold," unaroused state systematically underestimate how differently they will judge, want, and act once aroused, and that people in a "hot" state as reliably underestimate how differently they will see the same acts once calm. In the 2006 study, male participants answering while sexually aroused reported markedly greater willingness to engage in acts they had, when calm, rated as unacceptable, including persisting after a partner said no. The gap runs in both directions and applies to both partners: the aroused party misjudges the calm party's experience, the calm party misjudges what the aroused party will do, and a person who is aroused cannot reliably access the preferences he or she held minutes earlier. A boundary negotiated in the moment is therefore negotiated between two people neither of whom is in a position to speak for the person each will be an hour later.

[0013] The consequence shows up in how sexual assault is measured. Survey instruments of the kind introduced by Koss and colleagues (the Sexual Experiences Survey and its revisions) deliberately avoid the words "rape," "assault," and "abuse," and instead ask neutral, behaviorally specific questions, whether particular acts occurred, and whether they occurred through particular means such as continued pressure, incapacitation, or force. Decades of administration of such instruments have found consistently that a large share of persons whose answers meet the legal definition of rape or sexual assault do not, when asked directly, describe what happened to them in those terms, and that a share of persons whose answers describe having obtained sex through such means do not understand themselves to have assaulted anyone. Many such encounters were not experienced, by either party at the time, as a rupture: they were grey-zone encounters in which a hesitation went unasked-about, a "no" was heard as a "not yet," a yes was given under a pressure the giver did not name, or one party's arousal filled in the other's silence. The hot-cold empathy gap is one reason such encounters are so common and so poorly recognized while they are happening: the party pressing forward is not in a state to perceive the other's hesitation, and the party hesitating is often not in a state to voice it as a refusal.

[0014] A second body of research explains why the hesitating party so often does not speak. Under threat, the human stress response takes one of several forms, commonly summarized as fight, flight, freeze, or fawn. Fight is active resistance; flight is escape; freeze is the involuntary stilling of the body and voice (tonic immobility) that is documented in a substantial fraction of sexual-assault survivors, who report being unable to move or call out even when they wished to; and fawn is appeasement, the reflexive effort to placate a perceived threat by complying, agreeing, or soothing, which in a sexual encounter can produce a "yes" that is a survival response rather than a choice. These responses are fast, largely involuntary, and shaped by the person's history, and a given person tends toward one or another. A consent system built around a single expected behavior (say, voicing a clear "no," or pressing a button) serves the person who fights and fails the person who freezes or fawns, and those are the people whom the empathy gap already leaves least able to be heard. The affirmation envelope is designed to serve all four tendencies. Because affirmation must be renewed rather than withdrawal voiced, a person who freezes protects himself or herself by doing nothing: the envelope lapses, the ambient signal fades, and the system checks in. A person who fawns is protected by the system's refusal to treat a bare, reluctant, or tone-discordant "yes" as the enthusiastic renewal the envelope requires, and by the private check-ins that give him or her a way to answer honestly without the partner hearing. A person inclined to flight has the brake and the scoped request as face-saving exits that do not require leaving the room, and a person inclined to fight has the stop word, the panic button, and the help pathways. The '301 patent's emergency-response preferences, which let a participant choose in advance how he or she expects to react under stress, are extended here to the whole envelope: the lapse timing, the check-in channel, and the default response can each be set to the tendency the participant knows in himself or herself. Asserting a grey zone, a yellow flag, a stubbornness, or a brake is deliberately a face-saving gesture and an interim measure: it names no one and accuses no one, it asks only for a pause or a question, and it can be studied later

(by the participants in a cold state, or, in the event of a more serious experience of pressure or harm, by whoever the affected participant chooses to tell) for its true severity. A person who tends to fawn may be unable to say "stop" and yet be entirely willing to say "yellow," or "wait," or "I'm not sure," in one form or another; and for the person who cannot bring himself or herself to say even that aloud, each of these modes of concern (grey zone, yellow flag, brake, and red alert) may be bound at enrollment to a secret, personalized safeword, a touch, a gesture, or a vocal cue of the participant's own choosing (a hummed tune, a whistled phrase, a sung line of a song, a double throat-clear, a tongue click, a tapped rhythm, a whispered or counted word, or another sound that is natural to make and means nothing to anyone who has not been told) so that the concern can be registered and classified without the partner knowing that any particular word, sound, or melody was the one that registered it. A vocal cue of this kind has the further advantage that it is natural to produce during intimacy, carries no lexical content for a partner to object to or a classifier to misread, and can be recognized on-device by melodic contour and rhythm rather than by words, so that it remains private even from the speech recognizer.

[0015] Film and theater have, in the past decade, adopted the intimacy coordinator (a trained professional who choreographs intimate scenes, confirms each performer's boundaries in advance, and remains present to see that those boundaries hold take after take) on the recognition that performers, like anyone, cannot be relied upon to object in the moment to a director or a scene partner. No comparable role exists for private life, where it is most needed and where a human third party is least welcome. The system described here is, in part, an attempt to supply that role's functions (advance agreement, continuous quiet confirmation, and a sanctioned way to pause) through devices the participants already own, including, as described in section 8E, a phone mounted on the headboard as a shared ambient display that both partners can see.

[0016] These two findings, that in-the-moment judgment is predictably distorted and that the resulting harms are predominantly unlabeled rather than overt, define what a technical system for consent must do. It cannot rely on either party to recognize and firmly voice (without prompting or backup support) a need for a break, a grey zone, or even a red-alert problem as a red-alert emergency, immediately, and to extricate himself or herself or press a panic button to call for support; the research shows that recognition is exactly what is impaired. A panic button within reach is helpful, and is part of a complete sexual-assault prevention system. But if the affected party must first think of himself or herself as under assault in that moment, and in need of the panic button, while inside an "all gas, no brakes" hot-cold empathy gap experience, and without a face-saving break or grey-zone exit ramp through which the problem can be triaged, better understood, and mitigated in a cold state, then many affected persons will not come forward and use a blunt, panic-only electronic system to call in an electronic or human bedroom referee or to signal for immediate help. It must instead make affirmation frequent, easy, and expected so that its absence is informative; make hesitation and scoped requests cheap to voice and impossible to overlook; give either party a way to call for a cooling-off that carries no accusation; and keep a record that neither party has to characterize in charged terms, because a neutral, behaviorally specific record (what was said, what was asked, what was answered, when consent was renewed and when it lapsed) is, like the survey instruments, what allows an honest account to emerge afterward. The systems and methods described herein are built to those requirements.

[0017] Preventable guilt. The two findings above together describe a category of harm that this disclosure calls preventable guilt: the culpability of a person who, in the grip of a hot-cold empathy gap he or she could not navigate alone, crossed a line that he or she would not have crossed had a referee been present; the trauma of the person on the other side of that line; and the self-blame that survivors so often carry for not having stopped it, all of which were preventable, not by better character on anyone's part, but by the ordinary presence of a predictable, agreed, and understood third presence in the room. It is the inventor's thesis that a large share of sexual assaults (whether reported immediately, recognized by the survivor only in retrospect, or surfaced through the patient questioning of someone trained in the behaviorally specific items and classification of the Sexual Experiences Survey) occur in encounters that began consensually and went wrong in the gap, and could have been prevented had the technology described in this application, or other prevention-oriented sensor systems, been something close to a universal standard: a referee that helps a person not fully aware of the gap, or struggling in it, to keep to, or return to, his or her best possible bedside manner, and that mitigates the situations in which consent wanes, or that start to turn sour, or that break under stress, before they become traumatic or more traumatic. Nothing in this thesis excuses a person who ignores a stop; the system honors a stop before it does anything else. The thesis

is rather that a large share of the harm is done by people who, with a check-in, a brake, or a visible lapse in front of them, would have stopped. The systems and methods described herein are offered as a step toward making that presence ordinary.

[0018] Preventable victimhood. Every instance of preventable guilt has a counterpart, and the counterpart matters more: preventable victimhood, the harm suffered by a person that a system available at the time would have stopped or cut short. The word preventable here describes what systems, institutions, and the people responsible for them failed to provide; it never describes anything the person harmed failed to do. A survivor bears no responsibility for a fade that never sounded, a stop word that no device was listening for, or an alarm that no one had built, and nothing in this disclosure should be read to suggest otherwise. Preventable guilt and preventable victimhood are the same events seen from the two sides of the same encounter, and this disclosure considers them together for that reason, and wherever it tallies the costs of sexual harm, it counts the harm to survivors first.

[0019] Intelligence in the room, and shared accountability. Research in ubiquitous and ambient computing has long held that the most profound technologies are those that disappear into the fabric of everyday life (Weiser, "The Computer for the 21st Century," Scientific American, 1991), and research on the design of everyday things has shown that much of the knowledge people need to act well is carried not in their heads but in the world around them, in the shape, labeling, and feedback of the objects they use (Norman, The Design of Everyday Things, 1988). The built environment, its furniture, and the devices within it are therefore places where intelligence can be embedded, so that people need not carry all of it themselves. The inventor proposes, as a deliberate play on words, that the devices of the bedroom should be performing de-vicing: the removal, or at least the interruption, of vice. Legal systems, by contrast, continue to operate on the assumption that responsibility for an act of sexual assault lies entirely and exclusively within the body of the perpetrator or the accused. They do not ask whether some part of the responsibility, or at least of the accountability, should also attach to the speechless and clueless room in which the act occurred, and to those who furnished that room with devices able to listen, speak, and call for help that were never set to do so. This disclosure does not propose subtracting any responsibility from a person who causes harm; it proposes copying and doubling accountability, so that the environment carries a share of its own alongside the person's undiminished share. Given the historical record, society cannot rely entirely on men and women to navigate high-voltage, hot-state consent questions adeptly, least of all when partly or fully incapacitated by alcohol, which is involved in roughly half of sexual assaults (Abbey et al., Alcohol Research & Health, 2001), without a security-hardened, intelligent bedroom environment, augmented with a plurality of user devices, that offers assistive guidance, and correctives when things go wrong. Sometimes the corrective can be very simple: a room that, once a person has raised an alarm, repeats only what that person says through a loudspeaker, louder and more clearly than they could, turns a word that might be missed or ignored into an intense command that is impossible to ignore.

[0020] Proportion for the impulsive middle, and severity rating. Because of the hot-cold empathy gap, much misconduct in intimate settings is committed not by people acting with full intent but by people in the impulsive middle: people who drift into a grey-zone situation in a hot state, and who act subintentionally rather than with full intent in the pressure or harm that a partner experienced. For such cases a proportionate response may serve everyone better than either silence or the gravest available sanction. The system therefore lets the affected participant, entirely at his or her election, record privately how serious he or she considers what happened, on a graduated scale from a minor lapse to serious harm, and state the range of response, from a minimum to a maximum, that he or she would want to see given what he or she faced. The range may span no formal action or a repair conversation; nonmonetary measures, such as an apology, completion of training, or a period of separation; recordkeeping, such as a private or institutional note of the incident; an administrative fine; civil remedies; and criminal prosecution. The rating and the stated range express only the affected participant's own view. They inform the responses the affected participant chooses to pursue and, where he or she chooses to share them, the deliberations of a community, institution, court, or other decision-maker, but the system imposes no penalty, and the rating and range do not replace or limit any criminal, civil, or institutional remedy. The rating is never applied to diminish the response to the practices the core treats as non-consentable, or to acts committed with full intent. The rating and the stated range stay in the affected participant's own copy of the log and are disclosed to no one unless he or she chooses.

[0021] The distinction matters for what prevention looks like. Some persons who commit rape or sexual assault act with full criminal intentionality, and some go so far as to intrude into private homes where they have no right to be and attack strangers; those events are the ones most likely to be reported to police, and they are the ones for which heavy punishment and prolonged separation from society is the fitting and necessary answer. But sexual assault is the most under-reported of serious crimes, and one reason is a broad, often unexpressed understanding that hot-cold empathy gaps are partly determinative in moments of passion, that there was a degree of subintentionality, confusion, or confused expectation at play when there was no outside authority or digital bedside referee in the room, and no deterrence from knowing that if something went wrong a neighbor could be called, discreetly or openly, and be knocking at the door in less time than it takes to get dressed. That understanding keeps survivors silent and keeps the people who harmed them from recognizing what they did, and it is exactly the category of harm that a present, predictable, low-threshold system is built to prevent, interrupt, and record. We failed, as a society, to imagine, invent, commercialize, and popularize such systems, and to shift norms around them, and the prevalence of sexual assault we still see has hardly changed for decades, and arguably for centuries and millennia.

[0022] An acknowledged bias, and the communities it underserves. Much of the research and many of the examples in this Background describe sexual assault through a heterosexual lens, and the inventor acknowledges that bias. Sexual violence is not confined to it. In the national survey of the Centers for Disease Control and Prevention, sixty-one percent of bisexual women and forty-four percent of lesbian women reported rape, physical violence, or stalking by an intimate partner in their lifetimes, against thirty-five percent of heterosexual women, and lesbians and gay men reported intimate partner and sexual violence at rates equal to or higher than those of heterosexuals (Walters, Chen, and Breiding, 2013); in the largest survey of transgender people in the United States, nearly half of respondents reported having been sexually assaulted at some point in their lives (James and colleagues, 2016). These communities face, in addition, particular barriers to recognizing and reporting harm: the belief that a woman cannot assault a woman, or a man a man; legal definitions of rape that historically turned on particular anatomy; the fear of outing oneself or a partner by reporting; and, for transgender people, the risk that disclosure of one's history becomes itself the occasion of violence. Nothing in the system described here assumes the gender, sex, anatomy, or orientation of any participant: every role, word, threshold, and default is defined per participant, and the hot-cold empathy gap, the freeze and fawn responses, and the spectrum of sexual harm from impulsive boundary-pushing to rape are human phenomena that the system addresses wherever they occur. Asymmetries of physical size and strength, of age, of experience, and of social power exist in relationships of every composition, and where they exist they raise the stakes of a lapse in the same way.

[0023] Lessons from situational crime prevention. The most reliable findings in criminology concern not the reform of offenders but the design of settings, and they point in the same direction as this disclosure. The interventions with the largest and best-measured effects fall into two families. The first makes the harmful act structurally harder by changing the default state of the thing at risk. The clearest example is the electronic engine immobilizer, which prevents a car from starting unless the correct key is present: a study of more than one hundred and forty thousand vehicle thefts in the United States, comparing cars fitted with immobilizers at manufacture against matched cars without them, concluded that immobilizers reduced vehicle theft by roughly eighty percent over three decades, and described the immobilizer as possibly the most important crime-prevention device of modern times (Farrell, 2024); in England and Wales, cars with central locking and an immobilizer were found to be up to twenty-five times less likely to be stolen than cars with neither. The immobilizer did not persuade anyone not to steal cars. It changed what happened by default when someone tried: without the key, nothing ran. The system described here applies the same design principle to an intimate encounter. Without renewed affirmation, the music does not continue; the fail-closed default does the work that exhortation cannot be relied upon to do in the moment. A second example in this family is lighting: in a randomized trial in New York City public housing, developments assigned new outdoor lighting saw nighttime outdoor index crimes fall by at least thirty-six percent relative to developments that were not (Chalfin and colleagues, 2022). The lights confronted no one; they made the setting one in which conduct was visible, which is the function the shared ambient display and the music perform in a room.

[0024] The second family works through the sense of being observed, and its most striking result comes from an experiment with a picture. In a university coffee room that ran on an honesty box, researchers alternated, week by week, the image at the top of the price list between a pair of eyes and a picture of flowers; in the weeks with eyes, people paid nearly three times as much for their drinks (Bateson, Nettle, and Roberts, 2006). That study was small, and later attempts to show that images of eyes make people more generous have produced mixed results. But the question that matters here is not generosity but misconduct, and on that question the evidence is stronger: a meta-analysis of fifteen experiments found that the presence of watching-eye cues reduced the risk of antisocial behavior by about thirty-five percent, compared with the roughly sixteen percent reduction in crime that systematic reviews attribute to closed-circuit television cameras (Dear, Dutton, and Fox, 2019). The comparison is instructive. A camera records conduct for later review and is quickly forgotten; a cue that feels personal and present acts on conduct as it happens. This is the mechanism of the Hawthorne effect described in section 7A, and it is the mechanism this disclosure relies on: a referee that the participants agreed to, that is audibly and visibly present throughout the encounter, and that acts in the moment rather than replaying it afterward.

[0025] A study of an honor-system bagel business shows why both families are needed, and how they map onto the spectrum described above. For many years a former government economist delivered bagels to offices, left them with a box for payment, and kept careful records of what he was paid (Levitt, 2006). When the payment box was an open basket, the money itself was too often taken; once it was replaced with a closed wooden box with a slot, theft of the box became vanishingly rare, on the order of one box a year out of thousands. Yet roughly one bagel in ten continued to go unpaid for, and the closed box did nothing to change that. Two different behaviors were at work. Taking the cash was a deliberate act that a simple physical barrier all but eliminated; taking a bagel without paying was a softer, opportunistic lapse, committed by people who would never have stolen the box, and it responded to the social character of the office and to the sense of being noticed rather than to any lock. The same division runs through sexual assault. At the deliberate end of the spectrum, barriers, deterrence, and punishment are the fitting answer. In the impulsive, confused, and opportunistic middle, where most of the harm is done, what changes conduct is presence and gentle, timely prompting, which is what the affirmation window, the check-in, and the brake provide. Two further findings reinforce the point. In a field experiment with honor boxes for street newspapers, about two-thirds of buyers paid nothing; a sign reminding them of the moral norm increased how much those who paid chose to pay but did not change whether people paid at all, and a sign reminding them of the legal rule had no effect (Pruckner and Sausgruber, 2013), an indication that exhortation and legal threat, the two instruments institutions have relied on most against sexual assault, are weak on their own. And when welfare benefits in Missouri were converted from paper checks to debit cards, removing cash from the streets, overall crime fell by about nine percent, but rape did not change at all (Wright and colleagues, 2017): the general measures that reduce crime for gain do not reach sexual assault, which therefore requires prevention designed for it.

[0026] The same literature contains a caution that this disclosure takes seriously. Technologies that mainly add a means of reporting after the fact, or a general capacity to summon a guardian (burglar alarms, cameras, and mobile phones) have shown smaller and less consistent effects on crime than barriers and visible presence. A system that merely recorded intimate encounters, or that merely gave a person a way to call for help afterward, would belong in that weaker category. The system described here is designed instead around the two mechanisms that the evidence supports, with reporting as a third function rather than the first: it is fail-closed by default, as an immobilizer is; it is present, attended to, and acting during the encounter, as the watching eyes are; and it makes the state of the encounter visible to everyone in the room, as the lights are. Its record and its help pathways matter when prevention fails, but its design objective is that they be needed as rarely as possible.

[0027] Where the invention sits in criminological theory. Two frameworks help locate the system. Routine activity theory holds that a direct-contact crime occurs when a motivated offender and a suitable target converge in the absence of a capable guardian (Cohen and Felson, 1979). The bedroom is the clearest example of a setting without a guardian: it is private by design, and a human guardian there would be unwelcome. The system supplies a guardian that the participants have themselves agreed to, present for the whole encounter, and acceptable precisely because it is not a person. Situational crime prevention, in the form developed by Ronald Clarke and his colleagues, sorts preventive interventions into five mechanisms and twenty-five techniques (Cornish and Clarke, 2003). The system acts strongly through three of the five. It increases the risks of misconduct: it extends guardianship into the room, reduces anonymity

by attributing each utterance to a participant, enlists place managers such as resident advisers, doormen, and peer responders, and keeps a tamper-evident record. It reduces provocations, chiefly through the brake and cooldown, which correspond almost word for word to the technique of reducing emotional arousal, and through the clarifying exchange, which helps avoid disputes. And it removes excuses, the mechanism it serves most completely: enrollment sets the rules; the session announcement posts the instructions; the check-in alerts conscience at the moment it matters; the whole design assists compliance; and the impairment lockout of section 6A is, exactly, the technique of controlling drugs and alcohol. Removing excuses is also the mechanism least addressed by existing sexual-assault prevention, which has relied on punishment after the fact rather than on removing, before the act, the "I thought she wanted to" on which so much of the harm depends. The system acts only at the margins through a fourth mechanism, increasing the effort, by deflecting some offenders through the vetting of section 8D and by making the room less isolated through the panic button and responder entry; it does not physically prevent an assault as an immobilizer prevents a theft. And it lies largely outside the fifth, reducing the rewards, for the reason the welfare-card study illustrates: sexual assault is not committed for a removable reward, so its prevention must rest on guardianship, de-escalation, and the removal of excuses rather than on locks or on making the crime unprofitable.

[0028] Certainty rather than severity. Research on deterrence has converged on a finding with direct consequences for this field: the certainty of being caught deters far more effectively than the severity of the punishment that follows (Nagin, 2013). Sexual assault is the clearest case of a crime whose penalties are severe and whose certainty of detection is close to zero. Most incidents are never reported, most reports are not prosecuted, and most perpetrators can expect, correctly, that nothing will follow. Raising sentences further therefore does little, because almost no one expects to face them. The dredgable record changes the variable that matters. A contemporaneous, time-stamped, tamper-evident account of when affirmation lapsed, when a stop was given, and when the alarm sounded exists whether or not anyone reports, and every participant knows that it exists, so that the expected certainty of accountability rises for everyone in every session; that is what deters. Heavy punishment and prolonged separation from society remain the fitting answer at the deliberate end of the spectrum, and certainty is what makes even that answer an effective deterrent rather than a rare event.

[0029] Bystanders, human and technological. The best evidence that community presence prevents sexual violence comes from bystander-intervention programs, which teach peers to recognize and interrupt situations that are turning harmful. In a cluster-randomized trial of the Green Dot program in twenty-six Kentucky high schools, schools that received the program saw significant reductions in sexual violence perpetration and in other interpersonal violence relative to control schools (Coker and colleagues, 2017). Bystander programs work where a bystander is present: at the party, in the hallway, on the walk home. They cannot reach the room behind the closed door. The neighbor and peer-responder layers of this system extend the bystander into that room by technological means: a trusted or trained person who can be summoned discreetly and who arrives within minutes, at the moment a situation is turning, without anyone having had to watch.

[0030] Domestic and intimate partner violence. Much sexual violence occurs not between strangers or new acquaintances but within established relationships, and it is often one part of a wider pattern of domestic abuse. In the national survey of the Centers for Disease Control and Prevention covering 2016 and 2017, nearly one woman in five (19.6 percent) and about one man in thirteen (7.6 percent) reported contact sexual violence by an intimate partner in their lifetimes, and nearly half of women and more than two in five men reported contact sexual violence, physical violence, or stalking by an intimate partner. Sexual violence within a relationship differs from violence between new partners in ways that matter for design. It is repeated rather than single. It is embedded in coercive control, so that a survivor may be unable to refuse, or to call for help, without consequence. The abuser often has access to the survivor's devices and accounts, and technology is itself a common instrument of abuse. And the risk of serious harm rises at predictable moments, above all when the survivor attempts to leave. One warning sign deserves particular attention: in a study of women killed or nearly killed by intimate partners, a prior non-fatal strangulation increased the odds of attempted homicide roughly sevenfold, and of homicide roughly sevenfold as well (Glass and colleagues, 2008), while choking during sex has become common among young adults, with 26.5 percent of women in a random sample of 4,168 undergraduates at one large public university reporting that they were

choked during their most recent sexual event (Herbenick et al., 2021), half of the undergraduates in a later campus-representative survey of 4,989 reporting that they had ever been choked during sex, and women overwhelmingly the ones choked in sex between women and men (Herbenick et al., 2024). For these reasons the marital exemption from rape law, which was not removed in all fifty states until 1993, is not merely history: the bedroom of an established couple is where much of the harm this application addresses actually occurs, and where guardianship, the removal of excuses, and discreet pathways to help may matter most. The system accordingly provides, as described in section 8C, a duress affirmation for a person who cannot refuse openly, protections that prevent it from being turned into an instrument of coercive control, a specific response to pressure on the neck, and provisions for the dangerous period of separation.

[0031] Limitations. The system has three limitations. The first is selection: the most deliberate offenders are the least likely to arm a system of this kind, and the encounters most in need of a guardian may be the least likely to have one. That is the rationale for Solo mode, which does not depend on the other party's participation; for the conspicuous bedroom panic button, which does not depend on anyone's participation; and for institutional and community defaults, which make the system present unless someone chooses otherwise. The second is displacement: an offender deterred in one setting might act in another. Reviews of situational crime prevention have found displacement to be the exception rather than the rule; across some one hundred studies it was observed in roughly a quarter of cases, and a diffusion of benefits to neighboring places and times was observed about as often (Guerette and Bowers, 2009). The third is evaluation: no technology for preventing sexual assault in private settings has yet been tested in a rigorous trial. The closest analogues in other fields reduced the specific behavior they targeted by roughly a third (street lighting, watching-eye cues) to four-fifths (engine immobilizers) in the settings where they were deployed; those figures are offered as context and not as a prediction, and the magnitude of this system's effect can be established only by the trials the inventor intends.

[0032] The system described here does not depend on institutional reporting. Its record is made in the room, at the time, and belongs to the participant rather than to the institution, so that it is not limited by geography, by whom the participant chose to tell, by the year in which a report was made, or by the classification an office later applies. Where participants elect it, and as described in section 7A, devices may contribute to a community-level count of greyzone, yellow-flag, brake, and red-alert events through privacy-preserving aggregation, so that the base rate of sexual harm, from greyzone events to rape, in a community becomes visible in near real time without any individual being identified or any incident being reported against anyone's will. And an institution may offer every student, at the end of each term and alongside the release of grades, a private attestation in the behaviorally specific form of the Sexual Experiences Survey, asking whether the student experienced sexual coercion or any other crime during the term, with responses aggregated in the same way. Offered every term rather than every few years, such an attestation would bring the counted incidence toward the levels the climate surveys find, much earlier, and keep it there. A response to the attestation is required of every student, as other end-of-term obligations are, but disclosure is not: every question offers "do not wish to say" as a complete answer, and the first time a student chooses it, a confidential counselor follows up briefly, face to face or by call, only to confirm that the student is well and knows that he, she, or they may share at any time. A universal requirement to respond, joined to a universal freedom not to disclose, reaches every student every term without compelling any student to speak. A community that can see its own rate, term by term, and that has the means to interrupt incidents as they happen, can change the culture that produces them, and with it the base rate, in a way that an annual institutional report cannot.

[0033] The cost of what is proposed, against the cost of what it prevents. What is argued for here is cheap. The hardware is already purchased: a phone, a speaker, a watch, a light bulb, perhaps a hundred-dollar radar module or a thirty-dollar remote for those who want them. The software runs on-device, so its marginal cost per session approaches zero, and an inspectable kernel maintained in the open spreads its development cost across every institution that adopts it. The Flip the Script curriculum costs a university a few hours of a trained facilitator's time per student. A campus-wide trial costs less than a single year of one administrator's salary. Against this stands the cost of a sexual assault that reaches the yellow, grey, or red zone without mitigation or rapid therapeutic intervention, and that cost does not end with the event. In the days and weeks after, it is the emergency department, the forensic examination, the prophylaxis, the pregnancy test, the first missed classes or shifts. In the months after, it is the post-traumatic stress, depression, and

anxiety that follow sexual assault at rates higher than those following almost any other crime; the withdrawal from a course, a team, a job, or a campus; the substance use that so often begins as self-medication; the relationships that end; the sleep that does not return. In the years after, it is the lost degree and the lost earnings, the chronic health conditions that trauma leaves in the body, the elevated risk of suicide attempt that persists for decades, the cost to the next partner and the next relationship, the children raised by a parent carrying it, and, for a survivor who does report, the second injury of an adversarial process that may take years and end in nothing. For the person who caused the harm there is the cost of a conviction and incarceration where it occurs, and, where it does not, a life organized around not knowing or not admitting what happened; for the institution there is litigation, settlement, federal enforcement, attrition, and reputation; for the society there is the aggregate of all of it, which the Centers for Disease Control and Prevention has estimated, for rape alone, at a lifetime cost per victim in the low hundreds of thousands of dollars and a population economic burden in the trillions of dollars across the lifetimes of current victims (Peterson and colleagues, 2017), the great majority of it in lost productivity and medical care, borne mostly by the survivors themselves and by public payers. Prevention of a single assault along the spectrum therefore repays the cost of equipping a great many bedrooms, and prevention at scale repays the cost of standards, trials, and committees many times over; the system described here is cheap in the only sense that matters, which is relative to what it prevents, and the lingering costs enumerated above are exactly the costs that early mitigation (a check-in before the line is crossed, a brake honored, a neighbor at the door, and aftercare and resources offered that night rather than years later) is designed to stop from accruing.

[0034] The cost of sexual assault, in detail. The most widely cited estimate for the United States, from the Centers for Disease Control and Prevention, is that the lifetime cost of rape is $122,461 per victim, and nearly $3.1 trillion across the more than twenty-five million adults in the United States who have been raped, in 2014 dollars (Peterson, DeGue, Florence, and Lokey, American Journal of Preventive Medicine, 2017). Of that total, $1.2 trillion (39 percent) is medical cost, $1.6 trillion (52 percent) is lost work productivity of victims and perpetrators, $234 billion (8 percent) is criminal justice cost, and $36 billion (1 percent) is property loss and other cost; government sources bear about $1 trillion of it, or roughly a third. For scale, that government share alone is about one-fortieth of the entire national debt of the United States, which passed $40 trillion in August 2026; expressed in current dollars, it is closer to one-thirtieth. Per victim, that is roughly $48,000 in medical cost, $64,000 in lost productivity, $9,400 in criminal justice cost, and $1,400 in other cost. Adjusted for inflation to the time of this filing, the per-victim figure is roughly $165,000 to $170,000, and the population figure more than $4 trillion. The cost accrues over a lifetime: the estimate assumes an average age at first rape of eighteen and expresses all later costs as a present value discounted at three percent a year, so that the immediate costs of emergency care, forensic examination, and any criminal process are followed by years or decades of mental-health treatment, chronic physical conditions, and reduced work, and the undiscounted total actually paid over a survivor's life is larger than the headline figure. The estimate is also likely conservative as to earnings: it assumes comparatively little lost work time, whereas a study of Finnish administrative records found that employment and earnings fell by about sixteen percent immediately after a rape and that survivors remained on worse economic and mental-health paths a decade later (Adams, Huttunen, Nix, and Zhang). The burden can be expressed per period, on stated assumptions. If, as national surveys suggest, on the order of one to two million adults in the United States are raped for the first time each year (the National Crime Victimization Survey reports fewer, and the National Intimate Partner and Sexual Violence Survey more), each year's new victims carry lifetime costs of roughly $120 to $245 billion in 2014 dollars, or roughly $170 to $340 billion today; each four-year cohort, the span of a typical undergraduate education, roughly $0.5 to $1 trillion in 2014 dollars; and each decade roughly $1.2 to $2.4 trillion. No comparable estimate exists for the world, but the World Health Organization estimates that about one woman in three worldwide has experienced physical or sexual violence by a partner or sexual violence by a non-partner, and if the costs elsewhere were even a fraction of those in the United States the global burden would be measured in many trillions of dollars.

[0035] Against that, the system described here is inexpensive. Its simplest embodiments run on the phones the participants already own, optionally with a smart speaker and a wireless panic button with several activation modes, for a bill of materials under about one hundred dollars per room, and with the service delivered through a music-streaming subscription that most participants already pay to one vendor or another, into which the safety service could be bundled within existing margins or for a small increment on the order of

ten dollars a year. Spread over a few years of use, the cost per person per year is less than the cost of a single session of psychotherapy, and less still as deterrence and the primary and secondary prevention effects described above reduce the need for the escalation and aftercare functions. The break-even point is correspondingly low. At roughly thirty dollars per person per year and $122,461 per assault, the system pays for itself if it prevents one assault in about four thousand person-years of use. Among undergraduate women, of whom about one in four report nonconsensual sexual contact by force or incapacitation over their years of enrollment, that means preventing on the order of one assault in two hundred; even at a general-population risk on the order of one percent a year, it means preventing about one in forty. And every component was available at consumer prices by 2015, when voice-assistant smart speakers had reached the market and the inventor, then advocating in Chicago for rudimentary parts of this system, filed the provisional application from which U.S. Patent Application Publication No. 2016/0196735 claims priority, describing a room hub with voice-recognized safewords, a bed-mounted panic button, and escalating calls for help.

[0036] The cost of investigation, prosecution, defense, and litigation. The same comparison holds when the cost of prevention is set against the cost of the legal machinery that follows an assault. The Centers for Disease Control and Prevention estimate attributes $234 billion of the lifetime burden of rape in the United States to the criminal justice system, about $9,400 per victim in 2014 dollars; because only a small fraction of assaults are ever reported, investigated, and prosecuted, the cost of each case that does pass through the system (police investigation, forensic examination and laboratory analysis, prosecution, public defense, court time, and, on conviction, years of incarceration at tens of thousands of dollars a year) is many times larger, and one analysis that included those costs put the average cost to society of a single rape at roughly $450,000 in 2008 dollars. To this must be added the private costs of defense counsel for the accused, which commonly run to tens of thousands of dollars in a contested criminal case or campus proceeding, the cost to institutions of their own investigations and hearings, and the cost of the civil litigation that follows institutional failure, which for the largest cases has reached into the hundreds of millions of dollars and beyond: Michigan State University agreed in 2018 to pay $500 million to the survivors of a single physician's abuse, and the University of Southern California agreed in 2021 to an $852 million settlement, bringing its settlements to about $1.1 billion, in another. Against this, the system described here costs on the order of thirty dollars per person per year. The criminal justice cost alone of a single average assault would equip about three hundred person-years of use; the societal cost of a single prosecuted rape, on the analysis just cited, about fifteen thousand; and a single settlement of the size just described, the entire student body of a large university for decades. A contemporaneous, tamper-evident record also lowers the cost of every case that does proceed, because the sequence of events that investigators, prosecutors, defense counsel, and hearing panels would otherwise spend months reconstructing from conflicting accounts is already established, which shortens investigations, narrows disputes, and makes just outcomes, whether conviction or exoneration, both faster and cheaper to reach.

[0037] Probative value at each stage. The records the system produces have evidentiary value that the typical case of sexual assault now lacks, and that value runs in both directions. What is recorded is not an interpretation but a sequence of time-stamped machine events: when a session was armed and by whom; which stages were entered, and on whose verified, staged affirmation; when each affirmation window was renewed and when it lapsed without renewal; when a stop, brake, greyzone, concern cue, or duress affirmation was given; whether a participant's breath-alcohol reading or impairment attestation barred a session or barred the crediting of that participant's affirmations; when the check-in, the alarm, and the call for help occurred, and who arrived; and, where the law and the participants' elections permit, a sealed conflict recording that begins at the triggering event. Because the log is hash-chained, signed, and produced by a release whose build is attested, it can be authenticated by the process of digital identification that modern rules of evidence contemplate for data copied from an electronic device, and because its entries are generated by the machine rather than asserted by a person, they present fewer of the difficulties that hearsay presents. At the investigation stage, such a record establishes in hours the timeline that investigators would otherwise spend months reconstructing from conflicting accounts, and directs attention to the minutes that matter. At the charging stage, it gives a prosecutor evidence of the elements that are hardest to prove in a private encounter (the absence or withdrawal of consent, its timing, and the incapacity of a participant) or, as often, shows that a charge cannot be sustained. At the plea stage, because both sides can see the same record, it narrows the dispute to what the record does not settle and favors earlier and more accurate resolution. At trial, it can establish beyond reasonable doubt that an encounter

continued after a stop, after an unanswered check-in, or into a stage for which no affirmation was ever given, or past a lockout for impairment; and it can equally demolish an accusation, or limit it to the narrower claim that the complainant was rendered unable to signal, by showing that the system was running, that all parties had taken its precautions, that each stage was entered on both participants' verified affirmation, that every window was renewed, and that no stop, brake, concern cue, or duress affirmation was given. That narrower claim, of an induced muteness such as freezing or fawning, can be true, and the record's silence does not prove consent: processing an assault takes time, and a person who reports one should be heard and believed. But with priming, practice, and training, and with signals as easy as a single tap, a hum, or an ordinary-sounding phrase, it should become steadily less common for an assault to occur during a running session without even a cryptic marker of at least greyzone or brake character, and the record will carry the weight that such a marker, or its absence, deserves. The design guards against the misuse that has been charged against earlier consent applications, that a recorded yes might be turned against a person who later withdrew it: affirmations expire by default; withdrawal always takes precedence; the duress affirmation records coercion while appearing to comply; and the record documents every lapse, stop, and unanswered check-in as fully as it documents every affirmation. Admissibility and weight remain, as always, for the court under the applicable rules, and the record shows only what the system observed; but within those limits it substitutes evidence for the contest of credibility on which so many of these cases now turn, at a fraction of the cost and with far less confusion, for the accuser and the accused alike.

[0038] A concern of survivors' advocates, and an answer to it. Many survivors' advocates, and some of the college administrators who rely on them, will see in a record of renewed affirmations a nightmare they have been trained to prevent: a survivor who did not say or mark anything at the time, who later comes to understand and describe what was actually happening, and who is then confronted with a record of her own affirmations used in her partner's defense. From that vantage, any evidence that might favor a partner (typically male) who took precautions, and that might limit a survivor's ability to process traumatic memories and to revise her account as understanding comes, is to be resisted, even at the cost of a safety system whose predominant effect is to make clarity, pauses, brakes, stops, and emergency help far easier to reach. The concern deserves to be stated fairly, and it is answered in the design rather than dismissed: affirmations expire; withdrawal always outranks them; the duress affirmation records coercion while appearing to comply; the record documents every lapse and stop as fully as every yes; and, as described in section 7A, a claim that a safeword was spoken and missed can be tested against the very model that missed it. But the inventor finds the position that rejects any such record outright frustrating and, in the end, mistaken. It treats the absence of neutral evidence as preferable to its presence, when neutral evidence of what happened serves the truth, and the truth is in the interest of survivors. And it underestimates communities of peers, juries, and judges, who can come to understand how people actually use a system of this kind (how the overwhelming majority of women learn, with priming and practice, to trigger it when they meet stubbornness or coercion) and who can appreciate and credit, with the help of psychologists, the reasons a person may give for having affirmed repeatedly in confusion or under duress. A record of this kind does not end that inquiry; it gives it a more resolvable starting point, from which competing accounts can be reconciled into a truer version of what happened.

[0039] Three analogies, and a rule. The objection can be seen most clearly by analogy. First and most nearly: it is as if fire marshals opposed putting smoke detectors in every home because a detector's log might someday show that a homeowner heard the beeping for two minutes before calling for help, and an insurer might try to use that against her, even when the fire was caused by faulty wiring that violated the code. No one would accept that trade. The detectors save lives in countless homes for every case in which a log is misused, and the answer to the misuse is a rule that the log cannot be used that way, not a house with no detector. Second: nearly every new car with an airbag also carries an event data recorder that logs speed and braking in the seconds before a crash, and no one argues against airbags because the recorder might show that an injured driver braked half a second late; we keep the airbag, and we let courts weigh the data with judgment. Third: no one would refuse to put heart monitors in hospital rooms because a monitor's record might one day show that a nurse responded within a minute and so defeat a claim of neglect; to oppose protection in order to keep evidence unavailable is to put the outcome of a hypothetical dispute ahead of the lives the protection saves. In each case the protective device's record is a by-product of its protective function, its misuse is rare and correctable, and the remedy for misuse is a rule rather than a refusal. The same remedy is available here. Rape-shield laws already bar certain kinds of evidence from being used against

survivors, and a comparable rule could provide that the absence of an alarm, a stop, or any other marker in a record of this kind may never, by itself, be treated as evidence of consent, while leaving every other use of the record to the judgment of the court. With such a rule in place, the system's protective benefits can be delivered in full, and the evidentiary concern that might otherwise stand in its way is met directly.

[0040] What halving incidence would save the justice system and the public. The savings available from prevention can be estimated on the same assumptions used above. If on the order of one to two million adults in the United States are raped for the first time each year, and the criminal justice cost per victim is about $9,400 in 2014 dollars, roughly $13,000 today, then each year's new victims commit the justice system (police, forensic laboratories, prosecutors, public defenders, courts, and corrections, nearly all of them state and local, with the federal Department of Justice bearing a smaller share through its own prosecutions and its grants to state and local programs) to roughly $13 to $26 billion in lifetime costs. Halving the incidence of severe sexual assault would therefore spare the justice system on the order of $6.5 to $13 billion for each year's cohort, or some $65 to $130 billion over a decade, before counting the private costs of defense counsel, the costs of campus investigations and hearings, and the institutional settlements described above, which are not included in that estimate. And the justice system's share is the smallest part of the public burden. Government sources bear about a third of the full lifetime cost of rape, through public health insurance, public benefits, lost tax revenue, and criminal justice together, which on the same assumptions is roughly $55 to $110 billion in today's dollars for each year's new victims; halving incidence would spare the public purse some $27 to $55 billion a year in commitments, and the economy as a whole, counting the costs borne privately by survivors, families, employers, and insurers, roughly $85 to $170 billion a year. These figures rest on the stated assumptions and on an estimate that, as noted above, likely understates lost earnings; they are offered as orders of magnitude. But the order of magnitude is the point: a reduction in incidence of the size that the best prevention technologies have achieved for other harms would return to the public each year many times the entire cost of equipping every bedroom in the country.

[0041] Those who would have been spared the guilt. The benefits of prevention do not fall only on the people who are not assaulted. In the stereotypical heterosexual case, the preponderance of harm along the spectrum of sexual harm, from impulsive boundary-pushing to rape, is done by men, and many of the men who cause it fall within the category this disclosure calls preventable guilt: young men stumbling, by trial and error and without a referee, toward a better understanding of consent, with a picture of women's desire and boundaries drawn in part from pornography, in which, as a widely cited content analysis of popular scenes found, physical aggression appears in the great majority of scenes and its targets, nearly always women, respond with apparent pleasure or with indifference (Bridges and colleagues, 2010). The distance between that picture and real women, with natural desires and needs and with comfort zones and boundaries that are sometimes ill-defined and hard to read, is precisely the distance across which preventable harm is done. A generation of men now carries guilty and remorseful memories of harm they caused in their youth that a check-in, a brake, or a fading song would have interrupted; their worst sexual experiences, as well as their partners', could have been avoided. Preventing that harm spares them too, and spares the families, partners, and communities who bear the consequences of what they did. Nothing in this excuses the harm or the person who caused it; it identifies one more class of people whom prevention would have served.

[0042] Relationships that would have survived. Beyond the harm to individuals, countless intimate relationships, many of them promising, end because of misunderstandings, mistakes, and unforced errors in consent that would not occur if affirmative and enthusiastic consent envelopes were broadly available. A partner overwhelmed by sensation, who wants to be the best lover he or she can be and would welcome guidance and feedback, has no one to remind him or her to ask; a partner who drifts into a freeze or fawn response does not recognize it for what it is, and neither does the other; nothing proactively invites the verbosity and communication that would have caught the problem while it was small; and the encounter ends in hurt, confusion, or silence, and the relationship ends soon after. A bedside digital bystander that checkpoints and maintains a chain of enthusiastic consent through the affirmation envelope, that prompts the partners to speak at the moments they are least likely to, and that offers a sanctioned way to pause, serves as a coach, a referee, and a reminder at exactly the moment such support is most useful and least available. It promotes better bonding, because partners who have heard each other affirm, and have seen their requests honored, learn to trust each other faster; and it

prevents the loss of relationships that would otherwise have been sound, which is a benefit to the partners, to the families they might have formed, and to the communities around them. Over time the effect compounds. A culture of active communication, supported by technology that points partners toward the happy path and handles the sad path earlier and more gracefully, may lead to better and earlier committed bonds, and to less wasted and less catastrophic dysfunction in the bedroom. It offers men and women learnable pathways for making the most of every intimate trial and first attempt they are privileged to enjoy, in place of the common alternative: months, years, or even decades of being wrong without knowing it, and of never receiving enough feedback to learn what was carrying them, or their partners, outside the envelope of enthusiastic, affirmative consent. Each check-in, each renewal, each brake, and each conversation in aftercare is a small lesson, delivered when it can be used, and a person who receives such lessons from the beginning of his or her intimate life may reach in a few relationships the understanding that many now reach, if at all, only after many. Every fade-out, in particular, is a teachable moment delivered in or near the hot state, or in the transition out of it, which is where the hot-cold empathy gap of the Background does its damage and therefore where a lesson about it is most likely to stick: learned in the context of a real bond, with a real partner, at the moment it applies, rather than in an abstract lecture delivered in a classroom, in a cold state, far from the bedroom in which it will be needed. The lesson is then reinforced in the cold state as well: in the aftercare conversation, in the participant's own review of the session record, in the satisfaction notes and cadence reflections of section 4F, and in any classroom or peer-education program the participant later attends, which now has a lived example to attach to. The two states teach together, the hot state making the lesson vivid and the cold state making it durable, where classroom instruction alone offers only the second.

[0043] Repeat offenders and the absence of corrective pressure. A finding that bears directly on prevention is that a large share of sexual assaults are committed by a comparatively small number of repeat offenders. In the best-known study, about six percent of college men admitted, on behaviorally specific questions that never used the word rape, to acts meeting the legal definition of rape or attempted rape, and most of those who admitted one admitted several, averaging on the order of half a dozen each (Lisak and Miller, 2002); later work has debated how concentrated the repetition is and found more offenders who desist than the original study suggested (Swartout and colleagues, 2015), but no study disputes that repetition is common, or that the repeat offender is almost never reported after the first assault, or the second. The consequence is that the people who do most of the harm are never subjected to sufficient adaptive and corrective pressure (never confronted at the first incident, never interrupted at the second, never made to adjust before the third) and that each unreported assault teaches the person who committed it that the behavior has no consequence. Prevention technology that acts in the moment changes this at the root: a brake honored, a check-in answered, an alarm sounded, or a neighbor at the door at the first greyzone is corrective pressure applied at the first incident rather than, if ever, after the tenth, and a log that exists whether or not she reports, per community policy or local law, means that the second incident isn't the first on the dredgable record.

[0044] How many offenders act with full intent, and how the system meets each group. How large a share of offenders act with high, premeditated intentionality (planning, choosing a target, and using alcohol or drugs to incapacitate) rather than in the confusion and heightened willingness of the hot state, has not been measured directly, and the two leading studies bracket the answer because they count different things. Counting incidents, Lisak and Miller (2002) found that about six percent of college men admitted acts meeting the legal definition of rape or attempted rape, that about two-thirds of those men had offended repeatedly, averaging about six rapes each, and that this repeat group accounted for roughly nine in ten of the rapes reported; the repeat offenders typically planned their assaults, selected their targets, and used alcohol to incapacitate them. Counting offenders over time, Swartout and colleagues (2015), following 1,642 college men in two samples, found that about 11 percent reported at least one rape between age fourteen and the end of college; that of the men who raped during college, 72.8 percent did so in only one academic year; and that no group followed a consistently high trajectory of offending, the trajectories instead being low or time-limited, decreasing, or increasing. The authors concluded that multi-year offenders are a significant minority of college rapists, not the highest-risk group at entry, and that prevention guided exclusively by a model of serial predation is misguided. The debate between these views continues, but read together the studies suggest that offenders acting with sustained, premeditated intent are a minority of offenders, perhaps a quarter to a third, who may nonetheless account for a disproportionate share of incidents because each offends repeatedly, while most offenders

occupy a situational and opportunistic middle. That middle is not innocent: in an experiment on the effect of arousal on decision-making, the same men reported a far greater willingness to use coercive tactics, including drugging a partner, when aroused than when not (Ariely and Loewenstein, 2006). The hot state does not make that willingness unreal; it reveals a capacity that self-knowledge in the cold state misses, which is the hot-cold empathy gap described above.

[0045] The system described here is designed to meet every group. For offenders in the situational middle (the time-limited, the one-time, and those whose willingness is revealed only in the hot state), it acts before the line is crossed: the stage gates, the affirmation window, the check-in, the brake, and the impairment lockout interrupt the encounter at the moment the gap opens, and each fade-out is a lesson delivered when it can still change conduct. For offenders on an increasing trajectory, it applies corrective pressure early: a greyzone or yellow-flag event is logged, raised in aftercare, and made part of a dredgable record while the behavior is still small, rather than after it has grown. For offenders on a decreasing trajectory, who are learning, it accelerates the learning. And for the deliberate minority, who would never arm the system themselves, it does not depend on their participation: Solo mode, the guard state, the conspicuous bedroom panic button, institutional and community defaults, and the duress affirmation protect the person they target; the tamper-evident record and the community count raise the certainty of detection on which deterrence depends; and the record of one incident means that a second is not the first on the record, so that a repeat offender meets, for the first time, the accumulating evidence that has so far been missing.

[0046] Why survivors do not report, and how this invention answers each reason. The reasons that women, who bear the brunt of sexual assault, and men who are assaulted, typically do not report are well documented in the victimization surveys and the campus climate literature, and they are worth taking one at a time, because the present invention, in combination with the '301 patent, is designed to resolve each of them in the direction of reporting, or, better, of the assault not occurring. (1) The survivor does not label what happened as assault. The largest group of unreported assaults are the unacknowledged ones described earlier, in which an encounter that began consensually went wrong in the gap and the survivor, lacking any outside reference, concludes it must have been her own confusion. The system supplies the outside reference: the log shows that affirmation lapsed, that a stop was given and when, that a brake was called; the greyzone classifier names the greyzone at the time; and the aftercare of section 7 asks, that night, the behaviorally specific questions that the Sexual Experiences Survey asks years later. (2) Shame, embarrassment, and self-blame. The survivor believes she should have stopped it, said no more clearly, not had the drink. The system removes the premise: the check-in, the brake, and the escalation sequence are the system's responsibility and not hers, the record shows what she did say, and the Background's account of the freeze and fawn responses is built into the design so that a survivor who froze is not asked why she did not fight. (3) Fear of not being believed, and lack of evidence. In an encounter with two people in a room, the survivor knows her account will stand against his. The system supplies contemporaneous, tamper-evident, time-stamped evidence held under her own key (the event log, and where armed the conflict recording) so that the question is no longer whose word to take. (4) Fear of retaliation by the perpetrator, his friends, or the community. The system reduces the need for the survivor to be the one who raises the alarm: the escalation sequence raises it, a responder arrives, and the record exists independently of her courage on any given day; and the log rights of section 7 let her hold the evidence privately until she decides what to do with it, including never. (5) Not wanting to get him in trouble, because he is a partner, a friend, or someone she loves. The system offers a path short of prosecution that still applies corrective pressure: the brake script and clarifying exchange in the moment, the aftercare conversation afterward, the satisfaction notes and cadence tools of section 4F for a couple who want to repair, and a record that can be shown to him rather than to the police. (6) Believing it was a private matter or not serious enough to report. The system's graduated zones (greyzone, yellow flag, brake, red alert) give the survivor a vocabulary and a response proportionate to what happened, so that a greyzone need not be either dismissed as nothing or escalated to a crime, and can still be logged, studied, and raised. (7) Having been drinking, and fearing disbelief or punishment for it. The impairment handling of section 6A records the reading, treats intoxication as a reason the session could not be armed rather than as a mark against the survivor, and places the responsibility where the law places it; the record of incapacity is evidence for her, not against her. (8) Distrust of the police, the Title IX office, or the institution. The system does not require any of them: the help pathways of section 4 run first to trusted contacts and peer responders, the log is hers and not the institution's, and the inspectable code of section 7A lets her verify for herself, on her own terms and in private, what the system did and will do with what it

holds. (9) Fear of the process itself: the retelling, the cross-examination, the years. A contemporaneous record shortens and de-personalizes the process: the timeline is established by the log rather than by repeated interviews, and the evidentiary provisions of section 7 are designed so that a party need not relive the night to prove its sequence. (10) Not knowing where or how to report, or that a window has closed. The aftercare of section 7 presents the options (medical, forensic, institutional, criminal, civil, none) that night, with their time limits, and the record remains available if she chooses one later. (11) The passage of time and the fading of memory. The log does not fade. Taken together, these answers describe a system in which reporting becomes easier, safer, better evidenced, and less necessary, because the first design goal is that the assault not occur, and the second is that if it occurs it be interrupted, and only the third is that if it is neither prevented nor interrupted it be provable. A survivor who has all three is in a different position from one who has none, and a repeat offender who faces all three is, for the first time, under pressure to stop.

[0047] Prevention tiers and sexual violence interruption. As used in this disclosure, primary prevention is action taken before any sexual violence begins; sexual violence interruption is detection of, and intervention in, harm as it begins or unfolds, so that it is stopped and its consequences mitigated while they can still be limited; and tertiary prevention is the response after the fact, including aftercare, evidence, reporting, support, and accountability. Interruption is secondary prevention in the general public-health sense of early detection and intervention. Violence-prevention practice often uses "secondary" instead for the immediate response after violence and "tertiary" for the long-term response, and the system serves each of those as well. Most institutional effort today is tertiary, and it depends on reports that most survivors never make, so it acts late and rarely. The system moves the point of action upstream. Before an encounter, explicit decision points, stage gates, impairment lockouts, and the known presence of a responsive system deter harm and make consent a practiced habit. During it, a lapse fades the music, a stop cuts it, a greyzone or brake opens a pause, and silence advances to a check-in, an alarm, and help. Afterward, each participant holds a private, contemporaneous, sealed record and a graded vocabulary (greyzone, yellow flag, brake, red alert) with which to name an experience anywhere on the spectrum from a confusing moment to an unambiguous assault, and to choose a proportionate response, from a private note or a talk-it-out step to a confidential advocate, an anonymous community tally, or a formal report. The inventor expects that this can raise reporting substantially, above all for grey-zone experiences that now go unnamed, while making reports of clear cases better evidenced.

[0048] A third body of research concerns how often couples are intimate and how that frequency relates to their satisfaction. In large population samples, relationship and personal well-being rise with sexual frequency up to about once a week and plateau beyond it (Muise, Schimmack, and Impett, 2016); couples who drift well below their own preferred frequency report lower satisfaction, and a persistent gap between partners' desired frequencies (desire discrepancy) is among the most common complaints brought to couples' therapy (Mark and Lasslo, 2018). The same literature contains a warning that any system in this area must respect: when researchers instructed couples to double their sexual frequency, the couples' mood and enjoyment of sex declined (Loewenstein, Krishnamurti, Kopsic, and McDonald, 2015). Frequency imposed from outside does not produce the benefit that frequency chosen from within does. A system may help a couple reach the cadence they themselves want; it must never prescribe one.

[0049] Two further findings shape what help looks like. First, desire is often responsive rather than spontaneous (Basson, 2000): for many people of every gender and orientation, wanting follows the beginning of intimacy rather than preceding it, so that the barrier is not desire but initiation. Second, initiation and receptivity are distinct behaviors—proceptivity, the active soliciting of intimacy, and receptivity, the willingness to accept it (Beach, 1976)—and a couple is most satisfied when both partners do some of each, rather than one always asking and the other always answering (Muise and Impett, 2015, on communal sexual motivation). A couple who have drifted apart are also, by hypothesis, a couple with less recent practice at reading each other (the hot-cold empathy gap of the first paragraphs reopens after a long interval) so that the resumption of intimacy after a lapse deserves a somewhat shorter leash, as a first encounter does. These observations apply without regard to the couple's composition; nothing in them, and nothing in the system described here, assumes which partner initiates.

[0050] The '301 patent addresses many limitations of the prior art by providing a consent management system with dynamic consent assessment (including consent typing, consent leveling, and, in some embodiments, consent scoring), real-time detection of stop words and go words, a persistent link to an emergency dispatch center, tamper-resistant cryptographic records, and visual feedback through a

connected lighting device. The '301 patent does not, however, disclose using an ongoing media stream (most naturally, music the parties have chosen to play) as the primary, fail-closed carrier of consent status; nor does it disclose an affirmation-lapse architecture in which media continues only while enthusiastic affirmation is periodically renewed and in which the absence of affirmation (and then the absence of any interaction at all) drives a graduated escalation from fade-out, to a spoken check-in, to an alarm, to a call for help; nor an analysis of the emotional quality of vocalization to distinguish enthusiastic affirmation from reluctant or coerced acquiescence; nor the control of third-party streaming services and playback devices, with closed-loop verification, in service of consent management.

[0051] Persistence of the need. The need described in this background has been recognized for decades and has remained unmet. The principle that consent must be affirmative and must continue throughout an encounter was given institutional form at least as early as 1991, when Antioch College adopted a policy requiring verbal agreement before each step of sexual contact, a policy widely ridiculed at the time as impractical, including in a nationally televised comedy sketch. The principle has since been written into law: California's 2014 statute governing its colleges provides that affirmative consent must be ongoing throughout a sexual activity and can be revoked at any time, and New York's 2015 law provides that consent may be initially given but withdrawn at any time and that, when consent is withdrawn or can no longer be given, sexual activity must stop. Throughout that period sexual violence has remained common; the most recent national survey data of the Centers for Disease Control and Prevention indicate that nearly half of women and more than one in six men in the United States have experienced some form of contact sexual violence in their lifetimes. Over the same period the general-purpose capabilities on which a technical answer could draw, such as smartphones, wearables, networked speakers, streaming media, and on-device speech recognition, became nearly universal. Yet the approaches offered, described in the preceding paragraphs, did not meet the need. Consent applications that record a single assent were criticized as treating consent as a contract, and at least one was withdrawn. Personal-safety applications and panic buttons require the person at risk to recognize the danger and act, which the research on freezing and fawning described above shows that many cannot do in the moment. Voice assistants act upon spoken requests only after a wake word; the sounds that some of them detect without one, such as smoke and carbon-monoxide alarms and breaking glass, are not words of distress, and their detection leads to a notification of the account holder rather than to a response in the room; and accessibility features that recognize sounds such as shouting likewise only notify the owner of the device. Media players that pause after a period of inactivity are conveniences for a single listener, with no safety pathway. So far as the inventor is aware, apart from the inventor's own earlier filings, no product or published system has made the legal requirement of ongoing consent operational by causing a shared, pleasant, ambient output to continue only while every participant renews affirmation, fading it upon a lapse, cutting it upon a stop, and escalating an unanswered lapse to help.

[0052] The wake-word problem in an unfamiliar room. Each voice platform answers only to its own wake word, and a person in danger cannot be expected to know which one to use. A person in a bedroom that is not his or her own, such as a new partner's, a host's, or a hotel's, may find there smart speakers, displays, or charging phones of a make, model, or platform different from those he or she uses at home, and often cannot see or identify them, particularly in the dark or in distress. A person who calls out "Alexa, call 911" in a room in which the only listening device is an Apple HomePod, which answers to "Siri" or "Hey Siri," will not reach emergency help, even if he or she subscribes at home to an emergency service on the Alexa platform; the same is true in reverse, and of every other pairing of platforms. The wake words in common use at the time of this filing illustrate the difficulty. Amazon's Alexa answers, as its owner chooses, to "Alexa," "Amazon," "Echo," "Computer," or "Ziggy"; Apple's Siri, on iPhone, HomePod, and other Apple devices, to "Siri" or "Hey Siri"; Google's assistant, including Gemini on Nest speakers and Android phones, to "Hey Google" and, on many phones, "OK Google"; Samsung's Bixby to "Bixby" or "Hi, Bixby," or to a custom wake word recorded by its owner; Meta's assistant on its smart glasses to "Hey Meta"; Sonos speakers with Sonos Voice Control to "Hey Sonos"; Huawei's Celia to "Hey Celia"; and, among other widely sold speakers, Xiaomi's Xiao Ai to "Xiao Ai Tongxue," Baidu's Xiaodu to "Xiaodu Xiaodu," Alibaba's Tmall Genie to "Tmall Genie" in Mandarin, and Yandex's Alice to "Alice." A person entering an unfamiliar room would have to choose among a dozen or more possibilities, some of them chosen privately by the device's owner, and to say the right one in the right language, under stress. Even the correct wake word may not suffice, because some smart speakers cannot place an emergency call directly at all; an Echo device, for example, cannot dial 911 without a separate subscription service or accessory. The wake word

is also the most platform-specific and least instinctive part of any spoken request, and a frightened person calls out instinctive words, such as "help," "stop," or "no," without any preface. Other fields in which emergencies arise among strangers have addressed this problem. At sea and in the air, distress and urgency calls were standardized by international agreement in the 1920s and are understood whatever the make of the radio or the language of the crew: "Mayday," spoken three times, signals grave and imminent danger requiring immediate assistance; "Pan-pan," spoken three times, signals urgency short of immediate danger; and "Sécurité" precedes a safety message. Every vessel and aircraft monitors a common distress channel, such as marine VHF channel 16 and the aeronautical emergency frequency of 121.5 MHz, whatever the make of its equipment, and an aircraft's transponder carries standard emergency codes, among them 7700 for a general emergency and 7500, which a crew may set silently, for unlawful interference. Military and fire-safety practice offers a second model: short commands with strongly conserved meanings, such as "Hit the deck," "Fire in the hole," "Man overboard," and "Stop, drop, and roll," which are drilled until they can be spoken and obeyed under stress without deliberation, the last of them taught to schoolchildren so that it is available in a moment of panic to someone who has never been in a fire. Nothing comparable yet exists for the bedroom, where an intimate encounter can become an emergency without warning: there is no agreed word for "I need help now," none for "something is wrong, slow down," and no common channel on which every listening device in the room is bound to hear either. The levels of concern described in this disclosure correspond to those calls, a greyzone, a yellow flag, or a brake to "Pan-pan," a stop, a red alert, or a call for help to "Mayday," and a secret duress cue to the silent transponder code for unlawful interference; and the registry of universal safe words, together with a requirement that conforming devices listen for it, would supply the common channel. A person in distress therefore needs words that work everywhere: universal safe words, recognized by every conforming device of every maker and platform without any wake word, such as a repeated and distinctive phrase ("mango mango") that is unlikely to be spoken by accident; and, for a request that needs more than a single signal, a platform-independent universal wake phrase, agreed on across makers, that may preface an emergency request on any conforming device ("mango mango, please call 911"). The standardized registry of universal safe words described below is intended to supply both.

[0053] There is therefore a need for a system in which a shared, pleasurable, ambient output (music) continues only while every participant is detectably and enthusiastically consenting, in which the cessation of that output is itself an unmistakable and mutually perceived signal, in which the system defaults to silence on any fault, and in which the whole mechanism operates on the participants' own devices with strong privacy guarantees and a pathway to emergency assistance.

[0054] There is likewise a need for a version of such a system in which no music or other media plays at all, and in which the system nonetheless checks in with the participants and verifies ongoing affirmative consent within an affirmation envelope: a window within which affirmed, enthusiastic consent is understood to be current, after which it is understood to be on a timed, fading expiration that must be renewed. In such a system the affirmation envelope need not be fixed. It may be programmatically extended (lengthened, with gentler and less frequent check-ins) as partners grow more trusting and familiar with each other's wants, needs, practices, and ground rules, and as they share signals and a record of satisfied progress with the systems and methods described herein that confirm as much; and it may be shortened again, automatically or at either partner's election, when those signals are absent, when a new partner, setting, or practice is introduced, or when either partner so chooses. The envelope thus becomes a learned, mutually visible measure of established trust rather than a constant imposed on every encounter, while the check-in, lapse, and escalation safeguards remain in force throughout.

[0055] A short envelope is used at first. Two people who are undressed together for the first time are, by definition, experiencing each other's hot-cold empathy gap for the first time: neither yet knows how the other's wanting, hesitating, and changing of mind actually sound and look, and each is guessing from the inside of a state that distorts the guess. A short envelope keeps that first encounter on a reasonably short leash, not out of suspicion, but because the kindest default is one that asks often and makes answering easy. The system promotes verbosity: it invites small, frequent words of affirmation, scoped requests, and questions, and it rewards them, so that the two people say more to each other than they otherwise might and hear more than they otherwise would. The result is that nobody has to wonder where things are headed; both can feel good about it because both keep saying so. As the two come to know each

other's signals, the leash lengthens on its own, and the system recedes into the background it was built for, a quiet witness that asked often at the start and needs to ask less the better things are going.

BRIEF SUMMARY

[0056] This Summary introduces a selection of concepts in simplified form that are further described in the Detailed Description. It is not intended to identify key or essential features of the claimed subject matter, nor to limit its scope.

[0057] A technology of closeness, not only of safety. The system is meant first to be a positive and rewarding part of intimacy, one that partners choose because it brings them closer, and only second to be a safeguard; its safety functions are a floor beneath the experience, not the experience itself. Music the partners choose together sets the mood and carries their consent without interrupting it. A go word becomes a private language between them, happy words let either partner say how much he or she is enjoying the moment, and an enthusiastic yes, renewed at the one or two moments that matter, is itself an expression of desire that each partner hears from the other. The few planned decision points work like a ripcord or a grounding wire: noticed once, they let both partners relax into the encounter, knowing that a pause or a stop will be honored at once and without drama. The inventor's premise is that partners who feel safe can be more open, more adventurous, and more present, and that a partner who knows a hesitation will be met with care rather than pressure is more likely to say yes, and mean it, the next time. Over a relationship the affirmation envelopes lengthen as trust is earned, the steady intimacy support of section 4F helps a couple reach the frequency and balance of initiation that both partners want, and the aftercare step can invite each partner to note what he or she enjoyed as well as what he or she would change. The system is offered in that spirit, as a way for people to enjoy intimacy more, with each other's yes clearly heard.

[0058] Minimal embodiments. Although this disclosure describes many sensors, devices, and refinements, the system does not depend on them. Three minimal configurations, each assembled from inexpensive parts that most households already own or can readily buy, are expected to deliver most of the protective benefit (in the sense of the Pareto principle, the large majority of the prevention for a small fraction of the complexity) and are the configurations the inventor expects to bring to use first. (A) A bedside smart speaker. A voice-assistant smart speaker in the bedroom, with or without a screen or touch surface, plays the music, hears go words, stop words, close phrases, and vocal cues, speaks the check-ins, sounds the alarm, and calls for help; where it has a screen, the screen also serves as the shared ambient display, and where it has none, the two-envelope configuration of section 4G and spoken announcements take the display's place. (B) A bedside smartphone. A smartphone within easy reach of the bed, running an application with the features described here, performs the same functions and adds a screen and touch interface for private prompts, touch affirmation through its biometric sensor, and discreet operation in Solo mode; it may be mounted on the headboard as a shared display or simply kept face down on the nightstand. (C) Either or both, with a button or remote. A wireless panic button, or a remote control of the familiar television kind, adds a physical input that needs no speech, no screen, and no visible application: distinct press-and-hold patterns map to the system's signals (for example a single press to affirm, a double press to brake, a long hold to stop, a triple press to close, and enrolled secret patterns for a concern cue, a duress affirmation, or a silent call for help) each acknowledged by a distinct haptic pulse on the button or on the participant's wearable, so that every signal is confirmed silently. Configuration (C) is as simple to operate, cryptically or openly, as any device in the house. Beyond devices already owned, the bill of materials for any of these configurations is under about one hundred dollars per room. The rest of this disclosure describes how each function may be refined and extended; none of those refinements is required for a participant to be protected by a minimal configuration.

[0059] A roadmap. This application is intended in part as a roadmap, for the inventor and for any larger organizations with which the inventor may later work, of how the overwhelming majority of greyzone, yellow-zone, and red-zone sexual harm might be prevented: first by the simplest means, in the minimal configurations above, and then, with only the complexity that proves necessary, by the refinements described in the remainder of this disclosure. The many sensors, devices, and techniques described below are alternatives and enhancements, not prerequisites.

[0060] Who builds it. This idea and system were envisioned by a man. Nothing prevents the system from being programmed entirely by women, or from being completed in its implementation by women. The gender of the inventor should not bias adoption one way or

the other; the system should stand on its own merits and virtues. The inventor welcomes the opportunity to work with teams of men and women, of mixed genders, and of a broad range of orientations and of values concerning romance, courtship, marriage, and sexuality, as the invention proceeds from the conception set out in this disclosure through construction, actual reduction to practice, and testing of its variations.

[0061] Accessibility. Persons with disabilities, including deaf and hard-of-hearing persons, blind and low-vision persons, persons with speech, motor, or mobility impairments, and persons with other physical, sensory, or cognitive disabilities, naturally share the interest in finding an intimate partner, and they deserve the same happy-path support and the same trust and safety systems as everyone else, particularly at moments when they may be vulnerable. Versions of the system and software described here are therefore made available with accessible affordances, including alternative input and output interfaces that match the capacities of disabled and other nontypical users, and every function is provided through at least one modality usable by each participant. For a deaf or hard-of-hearing participant, every audible prompt, check-in, alarm, and directive has a visual and haptic counterpart, such as the consent color of the lighting feedback device, on-screen captions on a phone, watch, or display, and distinct vibration patterns; affirmations and withdrawals may be given by enrolled gestures, tap patterns, buttons, typed text, or sign-language signs recognized on-device; and the help stage may use text-to-911, real-time text, or video relay services where available, together with the transports described for alerting. For a blind or low-vision participant, every visual cue has an audible or haptic counterpart and the application works with the platform's screen reader. For a participant with a speech impairment or who does not speak, affirmation and withdrawal may be given through buttons, gestures, tap patterns, typed text, or the participant's own augmentative and alternative communication (AAC) device. For a participant with limited mobility or dexterity, inputs may be given by voice, by large or adaptive switches, by sip-and-puff or eye-gaze control, or by any assistive input the platform supports. For a participant with a cognitive or intellectual disability, the interface may be simplified, with plain-language prompts and fewer choices, consistent with the capacity safeguards of this disclosure and with applicable law, and any involvement of a caregiver or supporter occurs only with the participant's own consent. Accessibility settings are enrolled with the participant's other preferences, are announced to the partner only to the extent the participant chooses, and never weaken any protection: an accessible modality carries the same authority, timing, attribution, and logging as the modality it replaces.

[0062] Relationship forms. Although many examples in this disclosure describe a couple, nothing in this disclosure limits the system to a monogamous or serially monogamous couple or pair. The system applies equally, where permitted by law, to encounters among more than two participants, including multiple partners and group encounters; to ethically non-monogamous and polyamorous relationships; to the consensual production of intimate content among adults; and to other forms of consensual relationship. Where a description refers to both partners, to the other partner, or to a pair, it applies to each participant and to every other participant present, and the multi-party rules of this disclosure govern. Whether, and on what terms, a commercial deployment supports any particular form of relationship is a matter for the assignee and the inventor to decide, consistent with the law and with the purpose of preventing sexual assault, and does not limit the embodiments disclosed. The safety network described here may also, someday, extend to commercial sexual encounters, which are among the most vulnerable sexual experiences and can lead to abuse of women and sometimes of men. Whether and to what degree the system is extended to such encounters, where lawful or as a measure of harm reduction for the persons at risk in them, is likewise a decision for all of the partners involved in the core development of this technology, and does not limit the embodiments disclosed.

[0063] Education alongside technology. The system is meant to be combined with education and training, not to replace them: intensive, empirically supported programs such as Flip the Script resistance training, which in a randomized controlled trial cut the one-year incidence of completed rape among first-year university women approximately in half (Senn and colleagues, 2015), together with bystander training and facilitated discussion groups, that empower everyone, regardless of gender or orientation, to protect innocence and to keep their relationships consensual. The inventor's view is that everyone should go through such training, so as to understand, both in a cold state and in a hot state, how to be on his or her best behavior and how best to support the satisfaction and

happiness of a partner. The system may point participants to such training at enrollment and afterward, and may reinforce what the training teaches, such as the scripts for a brake, a clarifying exchange, or a scoped stop, during the moments in which it applies.

[0064] Mission Zero. Society can continue to nibble at sexual assault, through after-the-fact reporting, adjudication, and cold-state instruction, as it long has, or it can try to strike the problem at its root. The inventor calls the second approach Mission Zero: an aim, borrowed from the safety disciplines that pursue zero harm, of driving the incidence of sexual assault as close to zero as technology, training, and culture together permit. In the inventor's view, success on that aim is likely only if it is organized around assistive reliability technology of the kind described here, much as aviation and anesthesia reached their present safety not through exhortation alone but through checklists, interlocks, alarms, and redundancy. The aspiration is a world in which a graduate of a high school or college who has been trained in the protocol, and who continues to use such safety and reliability technology consistently, can reach, with high and measurable reliability, a lifetime of intimate encounters free of consent violations, and can enter any intimate setting with a partner with justified confidence that, by following the protocol, he or she will neither cause nor suffer preventable harm. That reliability is a design and measurement target for the system as a whole, to be established by trials; consistent with the principle that the system is a signaling aid, it is not a certification of any particular encounter or person.

[0065] The hand-hygiene precedent. Health care offers a close analogy: well-meaning professionals who know a protocol and still skip it. Every clinician is taught to clean his or her hands before and after patient contact, yet a systematic review of 96 studies found median compliance of 40 percent, 32 percent among physicians and 48 percent among nurses, and lowest, 21 percent, before patient contact, the moment that protects the patient (Erasmus and colleagues, 2010). Being watched changes the behavior: in one hospital study, hand-hygiene events nearly tripled in hallways where an auditor was visible, compared with the weeks before (Srigley and colleagues, 2014). Health-care-associated infections, which such lapses help spread, remain common; the Centers for Disease Control and Prevention estimated that on any given day in 2023 about one hospitalized patient in 38 had at least one, some 518,000 infections in U.S. hospitals that year (CDC, 2026). Electronic and video monitoring with real-time feedback has raised measured compliance substantially, in one remote-video program from about 30 percent to about 83 percent, and compliance has fallen back when monitoring was removed (as reviewed by the Joanna Briggs Institute, 2021). The lesson is that reliability in a high-stakes routine comes not from knowing the rule but from an environment that prompts, confirms, and records the step at the moment it is due. The consent conversation is such a step. Well-meaning partners, women and men alike, know that before a first encounter, or before renewing affirmation after the first window, they should talk through consent and boundaries, attend to each other, agree on contraception and on barrier precautions against sexually transmitted infection, share recent test results, and settle the plan for the evening, unless all of this was done responsibly long before; and, like clinicians walking past a sink, they skip these steps when things become hot and heavy. The available figures show how often. In a national sample of 817 U.S. young adults aged 18 to 20, 39 percent said they never talked with partners about sexually transmitted infections before sex, and only about a quarter had recently been tested for chlamydia and gonorrhea (Thompson and colleagues, 2021); a systematic review of 32 studies found that only around half of people with a sexually transmitted infection disclose it to a partner before sex, or believe they should (Journal of Sex Research, 2024); and in the National Survey of Family Growth for 2011 to 2015, only 33.7 percent of men and 23.8 percent of women reported condom use at last sexual intercourse (National Center for Health Statistics, 2017). Most of these studies do not separate heterosexual couples from others, but the pattern across them is the same: the shared test result, the barrier precaution, and the conversation about boundaries, contraception, and the evening's plan are, for many partners, steps known to be wise and routinely left out. The pre-intimacy and pre-high-voltage checklists, the private agent's cross-check of the ground rules, and the stage-entry affirmations described here supply the prompt, the confirmation, and the record that hand-hygiene monitoring supplies in a hospital, at the moment the step matters.

[0066] General applicability. Although this disclosure is directed chiefly to intimate encounters, the mechanisms it describes are general. A perceptible signal, such as media, a haptic pattern, or a light, that continues only while one or more persons renew a check-in within a window; a soft change of that signal upon a lapse, distinct from an abrupt change upon a stop; a check-in that distinguishes interaction from affirmation; and escalation to help upon continued absence of any interaction apply equally to a lone worker; an

elderly person, or a person with a medical condition, living alone; a solo runner, cyclist, swimmer, or hiker; a person on a first date, or in a rideshare or other vehicle; a person recovering from a medical procedure or taking a medication that carries a risk of sedation; a person in a hot tub, sauna, or pool; and any other situation in which a person may become unable to call for help. In each, the affirmation may be any enrolled renewal, the media may be any media the person enjoys, and the escalation sequence, policy engine, privacy kernel, and help pathways operate as described. A product may therefore offer the mechanism as a general safety or check-in feature of which the intimate session is one mode, and nothing in this disclosure limits a claim to intimate encounters unless the claim so states. The same architecture applies to any setting in which a shared ambient output should persist only while all present continue to agree, such as a massage or bodywork session, a medical or dental procedure in which a patient's continuing assent is monitored, a shared virtual-reality experience among persons in the same room, or a physical training session.

[0067] Managing complexity. This application opens many cans of worms. It describes a large number of possible configurations, and each additional sensor, device, platform, mode, and policy dimension multiplies the cases that must be designed, tested, and supported; at the higher levels of complexity, each addition is integrated and tested in the ordinary course of product engineering. The inventor notes that the fuller system calls for careful product management of that complexity, and urges anyone who takes up this disclosure not to be overwhelmed by the many roads it opens. The system can be built out stage by stage, and its complexity managed with care, with the end goal always in view: to reduce sexual harm as much as possible, while not reducing, and if anything gently and autonomously increasing, intimate bonding and the frequency of intimacy that couples themselves want, with better support for long-term, happy-path bonds and for communication about intimacy and desire. Even the simple, simpler, and simplest forms of this system of privacy-preserving, hardware- and software-backed affirmative consent envelopes would do an enormous amount of good in resetting the culture of consent. Much of that good can be achieved in a simpler way, and with less risk of controversy over how sexual choices and boundaries ought to be negotiated, by having the system do no more than technically provide and support the affirmative consent envelope itself (the enthusiastic yes, the pause, the brake, and the greyzone, yellow-zone, and red-zone statuses and their study) leaving the content of the couple's choices entirely to the couple.

[0068] A modular ecology of variants. The system may be built and distributed as modules, split across an ecology of modular choices, rather than as a single product that must settle every question for everyone. At its center is a core: the basic affirmative-consent envelope safety net (the affirmation window, the fade and the abrupt stop, the stop word, the brake, the lapse and its escalation to check-in, alarm, and help, and the privacy kernel) deliverable through a single smart speaker or phone. A person may choose that core alone, or pair it with a software flavor or variant that reflects his or her own values and needs. One variant might be developed, or reviewed and endorsed, by a religious body, for example under the auspices of the Roman Catholic Church and an orthodox Catholic understanding of marital privacy in the bedroom, careful consent, and the prevention and reduction of harm and assault. Another might be a licensed application built by developers of a libertine or hedonist outlook that welcomes and facilitates fully informed, ethical non-monogamy. Others may reflect any other school of thought that is lawful, ethical, and within the range that the licensor and the legislatures are prepared to tolerate. Each variant plugs into the core across a defined technical boundary: a published interface through which a variant may add vocabulary, prompts, music, stages, boundary-map categories, guidance, and rules of its own, and may tighten the core's defaults, but may never weaken the core's invariants. Those invariants are enforced by the core, not by the variant: a stop word, the brake, a concern cue, or a duress affirmation is always honored; silence always lapses the envelope rather than extending it; the escalation to help remains available to every participant; no participant's private settings are disclosed to another; and no data leaves the device except as the privacy kernel's data-flow manifest permits. A variant is signed, declares the interface version and the permissions it uses, and runs in a sandbox that cannot reach the core's audio, sensors, or event log except through the interface. Where the participants in a session have chosen different variants, the core applies, for each invariant, whichever of their settings is more protective, and announces any conflict it cannot reconcile, so that neither participant's variant governs the other. The deeper, more controversial, and more disputed questions of consent, sexual responsibility, and sexual freedom, within marriage and before it, can thus be debated and decided within the variants, in their separate spheres, without embroiling the core in those controversies. Every school of thought within those bounds can feel supported in its own sphere without stepping on the toes of another, and none need obstruct, or seek to dominate, the development or rollout of the core safety envelope for fear that its views will

not be tolerated or honored, at least within reasonable boundaries. Without such a boundary, the core safety envelope would be drawn prematurely into an undecided, and perhaps undecidable, wicked problem and political contest, frustrating investment in it and integration of it by major technology companies and other institutions that hope to avoid the blowback of appearing to pick sides with respect to the sexual revolution and the highly charged fragmentation of dating, romance, and gender ethics in contemporary societies. It is very common for sexual conservatives to want a technology of this kind to bend everyone toward sexual conservatism, and for sexual moderates, liberals, and libertines to want the same for their own views; people typically want their own patterns and practices to be the standard by which everyone else lives. The modular design offers another course. When registering, a person may answer, privately and on his or her own device, a few questions about where he or she stands in a kind of sorting hat of sexual practices and values, and be matched with app experiences, or with intercompatible applications, that support that way of life separately, within reason. No conservative, moderate, or liberal is thereby forced into libertine tolerance or behavior, and no libertine is forced into conservative restraint, while all of them share the same core safety envelope beneath.

[0069] What the core will not compromise on. There are practices that the core treats as non-consentable under every variant. The simplest and most important example is strangulation: every form of strangling, choking, or pressure on the neck is an immediate red card, a red-zone event, and assault, even if ostensibly requested by the person to whom the force is applied. This stance rests on the public health consequences of strangulation. The research of Debby Herbenick and her colleagues has documented how common choking during sex has become, particularly among young adults: in a random sample of 4,168 undergraduates at a large public U.S. university, 26.5 percent of women, 6.6 percent of men, and 22.3 percent of transgender and non-binary students reported having been choked during their most recent sexual event (Herbenick et al., Journal of American College Health, 2021); in a later campus-representative survey of 4,989 undergraduates, half had been choked during sex at some point, and in sex between women and men women were overwhelmingly the ones choked (Herbenick et al., Perspectives on Sexual and Reproductive Health, 2024); undergraduates more often than graduate students reported first choking or being choked in adolescence; and, among students who had been choked, 43 percent reported feeling that they could not breathe, about 15 percent had noticed neck bruising, and about 3 percent had lost consciousness (Herbenick et al., Archives of Sexual Behavior, 2022). Imaging studies of young adult women choked frequently during sex have found differences in brain activation and structure compared with women never choked, though whether choking causes those differences is not yet established (Huibregtse et al., 2022; Hou et al., 2023), and the medical literature on strangulation describes how little pressure and how few seconds can produce loss of consciousness; how hypoxic brain injury, carotid or vertebral artery dissection, and stroke can follow, sometimes hours or days later and sometimes without any visible mark; and how prior non-fatal strangulation is among the strongest predictors of later homicide in abusive relationships (Glass et al., 2008, cited above). In the inventor's judgment, a person who invites strangulation or erotic asphyxiation does not understand those risks well enough, or lacks enough of an instinct for injury avoidance, for the request to carry any grantable validity, so that the practice is permanently unconsentable. The case of a fully informed request is, moreover, the exception. Much more often, in the inventor's reading of that research, those who are strangled during sex were never asked at all, or were pressured or coerced into it, and almost none were given a full understanding of how dangerous it is. In one campus-representative survey of about five thousand students, 21 percent of those who had been choked during sex said that their partners never asked first, and a further 32 percent said that their partners asked only sometimes (Herbenick, presentation to the International Society for the Study of Women's Sexual Health, January 2026); and among men in a campus probability sample, frequent pornography use predicted choking a partner by way of the belief that choking is pleasurable, the belief that it is safe, and the disbelief that it requires the consent of the person choked (Wright, Herbenick, and Tokunaga, Health Communication, 2023). Accordingly, no variant may add neck pressure, choking, or strangulation to the boundary map as a permitted activity; every variant strongly encourages, in its onboarding, its rehearsals, and its guidance, a person subjected to any pressure on the neck to abort the encounter at once and to report it, either openly or cryptically, by stop word, concern cue, button pattern, or cryptic arming input; any such report is mapped in every variant to an immediate red-zone response; and the person, or any other participant, may answer it with the full escalation (the red alert, the call for help through Neighbor 911 or another peer-response or emergency pathway, and the voice relay) so that strangulation practices are stamped out rather than normalized. The

same exclusion may apply to blood play and scat play, each of which carries direct risks of injury and of bloodborne or enteric infection.

[0070] Listening in the bedroom, and how design answers the concern. Some members of the public will find a technology that listens in the bedroom creepy or invasive, to varying degrees, even when it comes with a formal demonstration that everything it hears is processed on the device and nothing leaves it except where an emergency requires. That reaction deserves respect, and the design of the software logic and the hardware can do much to meet it. The system can always be implemented behind a local wake word, or within a local, controlled vocabulary of go words, stop words, and cues, processed entirely on the device, rather than by continuous analysis of everything said and of what the speakers intend. The privacy concern is greatest where audio is processed off the device; it is far smaller where small local models perform only wake-word detection, keyword spotting, transcription, speaker diarization, or the classification of intent in narrowly defined circumstances, and discard the audio as they go. As the computing required to relate speech to intention grows, the configurations described here therefore move that computation onto the device rather than off it, and offer the participant at each level a choice of how much the system listens for, from a single stop word upward. The inventor believes that a balance of privacy and security can be struck that the overwhelming majority of the public will accept as the right trade, in return for driving the share of women who experience sexual assault in their lifetimes from roughly one in five, by the measure of national surveys, toward one percent or less.

[0071] Inconspicuous form factors. For residents who prefer it, the sensing elements of the system may be housed in devices that do not announce themselves as intimate-safety equipment, for example, an upgraded smoke and carbon-monoxide alarm whose housing also carries the microphone array, a non-imaging presence sensor, a speaker, and a status light, so that a single ceiling device serves as fire alarm, carbon-monoxide alarm, and bedroom safety system at once. Such a form factor broadens the options for discretion available to a resident who does not wish the system to be noticed, and it places the system where building codes already require a safety device. It remains subject to every rule of this disclosure: non-imaging sensing only, processing on the device, the data-flow manifest and its review, and, where the jurisdiction table of section 8C requires notice to others present, the notice that the law requires.

[0072] Toward legal protection and building codes. Tenants and guests are generally forbidden by law to disable a smoke detector, whatever their preference, because the safety of everyone in a building depends on it. The inventor hopes that the law will in time recognize a comparable protection for a resident's bedroom safety system: that a resident who maintains a system meeting a published residential standard for such systems, including its privacy safeguards, will be protected from coercive pressure by a visitor or partner to switch it off or remove it, and that visitors will be on notice that pressing a resident to do so is not acceptable. The inventor further hopes that residential building codes will one day include bedroom safety provisions of this kind as default equipment, as they now include smoke and carbon-monoxide alarms, on the condition that the privacy safeguards are guaranteed in hardware or held under continuous, published audit, as described in section 7A. In the same spirit, the inventor hopes that beds will someday be required to be sold with a panic button, either wireless or integrated into the frame or headboard, in a standardized location on every bed, so that a person in any bed, including an unfamiliar one, knows where to reach for help without searching, much as the emergency stop of a machine or the call button of a hospital bed is found in a predictable place. Such a button need carry no microphone or camera; its only function is to signal for help, or to pair with a system of the kind described here.

[0073] In one embodiment, a consent-gated media playback system includes a consent-gated media controller executing on at least one user device having a microphone. The controller includes an expressed-channel analyzer that performs on-device automatic speech recognition and keyword spotting to detect predefined go words, stop words, and safe words; an optional implied-channel analyzer that estimates the affective quality of a participant's speech and non-speech vocalizations (for example valence and arousal, laughter, distress) and that each participant may set for his or her own voice to one of three modes: off, private distress signal, or tone-graded affirmation; a consent state fusion component that assigns each participant a consent state from the tier of go word spoken (for example AFFIRMED or ENTHUSIASTIC, as enrolled by that participant), applies that participant's elected tone mode (withholding refresh, prompting privately, or both, when the estimated affect contradicts the go word), and computes a joint consent state as the

minimum across all participants; a consent keepalive comprising an affirmation timer that signals a lapse when the joint consent state is not re-affirmed within a configurable affirmation window; and a media control adapter layer that causes media content to play only while the joint consent state is affirmed and no lapse has been signalled, and otherwise causes the media content to fade out and stop upon a lapse, and to stop abruptly, without fading, upon a stop word, with the silent condition being the default on any fault, disconnection, or ambiguity.

[0074] In another embodiment, the lapse is followed by an escalation sequence (check-in, alarm, help) driven by the absence of interaction rather than by any affirmative event. When the media has been softened, muted, reduced in volume, or stopped for lack of affirmation and no interaction of any kind (speech, motion, touch, or button press) is detected for a first period, the controller issues a check-in alert through an alert output subsystem to one or more alert channels: a spoken check-in through the room speaker or the user device's speaker (for example, "Is everything OK? Say your go word to continue or your close phrase to end the session"); a haptic or on-screen alert delivered over a wireless interface to a participant's smartwatch, wearable, phone, earpiece, ring, or other electronic accessory; and a change in the color, intensity, or pattern of a connected lighting device in the room; repeating the alert at increasing intensity and adding channels as time passes; when no interaction follows for a second period, the controller sounds an alarm intended to be heard by the participants and by anyone nearby; and when no interaction follows for a third period, the controller calls out for help through a pre-configured sequence of pathways (a trusted contact, a peer emergency-response network, and finally the emergency dispatch link of the '301 patent) reporting an unattended, unclosed consent session with the last known consent state and location. Any interaction pauses the escalation sequence; only a verified affirmation restarts the media; a stop word, safe word, or distress indicator bypasses the escalation sequence and escalates immediately; and only an explicit close (the close phrase or another enrolled closing input) ends the session.

[0075] In another embodiment, the help stage may also grant entry. Where a participant has designated an authorized responder and an access device in advance, the controller, upon reaching the help stage and confirming that the responder has accepted the alert and is at the door, commands the access device to admit that responder for a time-limited window, so that a neighbor, roommate, friend, doorman, security professional, peer responder, EMT, paramedic, firefighter, or emergency service can reach a participant who may be unable to answer. Entry is never granted before the help stage, never to a person who is not both designated (or qualified for entry under local policy and law) and verified at the door, and never without being announced inside the room and recorded; and a participant who is able to respond may cancel the grant by the OK phrase before it takes effect.

[0076] In another embodiment, the media control adapter layer includes a plurality of adapters for controlling media playback through different pathways (including an application-remote interface to a native third-party streaming application, a cloud or "connect"-style remote playback interface, a first-party operating-system music framework, an operating-system audio-session interruption mechanism that pauses whatever audio source is active, a networked playback device, and a local media library) together with an adapter selector that chooses an adapter based on discovered capabilities and measured latency, and a closed-loop verifier that subscribes to player-state notifications, confirms that each directive took effect within a latency bound, and escalates to a fallback adapter when it did not.

[0077] In another embodiment, the playback reference signal (the audio the system itself caused to be played) is supplied to the analyzers to perform acoustic echo cancellation, so that song lyrics and instrumental content are not misrecognized as go words or stop words, and so that the consent analysis remains reliable while music is playing at ordinary listening volume.

[0078] In another embodiment, a response mapper maps the joint consent state and, optionally, the dynamic consent score of the '301 patent onto graded media directives, such that an enthusiastic-tier state may, if the participants so configure, yield access to an additional playlist tier, an ambiguous state yields ducked playback with an audible prompt for re-affirmation, and a withdrawn state yields an immediate pause followed, optionally, by a pre-agreed cooldown or de-escalation track that accompanies the separation protocol described in the '301 patent. The media directive may further select or shape the media itself rather than only its playback: the response mapper may assemble a dynamically generated playlist whose features—genre, lyrical or messaging content, tempo in beats per minute, energy, and the consent-zone quality the music is meant to support (steady, enthusiastic, cooling, cooldown)—are adjusted

to the joint consent state, to the position within the affirmation envelope of the session, and to the longer envelope of the relationship; may select, for a behavioral procedure the participants have pre-agreed, music whose structure cues that procedure, such as a slow, counted track that supports settling into a resting posture like child's pose during a brake or cooldown, or a sharply structured track that cues a drilled sequence like stop, drop, and roll where a participant has enrolled one; may, during the help stage, play a distinct, pre-agreed track or motif that indicates that a neighbor, peer responder, or second responder has acknowledged the alert and is on the way (so that a participant who cannot reach a phone or read a screen knows from the music alone that help is coming, and so that the other party present hears the same thing) with the motif changing again on the responder's arrival at the door, the controller choosing by policy whether that track is reassuring, neutral, or deliberately conspicuous; such a help-is-coming motif may be a short, catchy, and teachable melodic figure that a whole community can learn, serving both as a vocal cue that the system recognizes when a participant whistles or hums it (section 3A) and as the echo the system plays back when the cue has been received and help is on the way (a call-and-response in which the system answers in the participant's own signal), and one candidate the inventor intends to test, if it is legally available for the purpose and proves as catchy and teachable in large communities as it is in its fictional one, is the four-note whistled call associated with the character Rue in The Hunger Games, which is already understood by a generation of readers and viewers as a signal of safety passed between allies; if that figure is not available under fair use or by license, other tones can be found (a public-domain figure, a newly composed one, or one a community or institution adopts for itself) under the same mechanism, and the controller treats whichever figure is enrolled identically; nothing in the mechanism depends on any particular melody; and may, where a generative music model is available on-device or through a permitted service, produce and mix music custom-made for the conditions the sensors report (the participants' breathing and heart rates where a wearable supplies them, the room's sound level, the time remaining in the affirmation window, and the consent state) so that the media is composed for the moment within the consent envelopes of the session and of the relationship rather than chosen from a fixed library, with the same fail-closed, verification, and privacy rules applying to generated media as to any other. The response mapper may further time a fade, crossfade, track change, or resumption to a boundary of the music (an internal boundary such as the end of a bar, phrase, chorus, or verse, or an external boundary such as the end of one song and the start of the next) within a bounded deferral, while stops, brakes, and distress indicators take effect immediately regardless of any boundary.

[0079] In another embodiment, the system operates on a single shared device present in the room, attributing speech to individual participants by speaker diarization; in other embodiments, each participant's device runs the controller and the devices exchange consent-state heartbeats over a local link, the joint state being computed as the minimum of the individually reported states.

[0080] In another embodiment, all audio analysis is performed on-device, raw audio is never persisted or transmitted, and only consent-state transitions, media directives, verification acknowledgements, and their latencies are recorded in a hash-chained event log compatible with the cryptographic storage module of the '301 patent.

[0081] In another embodiment, the privacy- and reporting-relevant logic of the system is isolated into a bounded privacy kernel accompanied by a machine-readable data-flow manifest enumerating every egress from the device by privacy mode; the kernel, where commercial and contractual arrangements permit, is published as source code or executable pseudocode; releases are built reproducibly and their hashes published to a transparency log against which the running application attests; automated reviews, artificial-intelligence reviewers operating under the direction of a privacy and security balance and extreme privacy guarantee committee, and independent third-party audits are run and published on a frequent cadence; and the controller performs all processing locally wherever technically feasible, calling a server only where a security need outweighs the privacy cost, and never in Extreme Privacy mode.

[0082] In another embodiment, and only for a participant who has enabled it, the controller issues an optional private renewal reminder to that participant's wearable, phone, or accessory a lead time before the affirmation window expires (a haptic tap that says, in effect, "time to say yes again") and a private receipt cue when an affirmation is credited, so that a participant who wishes to maintain an unbroken chain of affirmed consent is not required to keep time or guess the pacing: the wearable keeps time for them, tells them when a renewal is due, confirms that it landed, and does so without any audible change in the room and without informing

the other participant. The reminder lead time, pattern, channel, and repetition are configurable per participant, and the controller may adapt the reminder to the participant's observed rhythm so that it arrives shortly before that participant would otherwise have lapsed.

[0083] In another embodiment, detection is not limited to enrolled words. An on-device semantic classifier (a compact language model or an equivalent intent model executing on the user device) classifies the meaning of any transcribed utterance and of consent-relevant non-lexical vocalizations into at least go, greyzone, and stop categories, each with a risk weight, using thresholds that are asymmetric so that stop-like meanings are detected more readily than go-like ones; enrolled go words and stop words remain guaranteed overrides in both directions so that a classifier failure can never suppress a stop or manufacture an affirmation. The policy engine maps meaning categories and risk weights, not only states, to response patterns. When an utterance falls in the greyzone, an attending device or the participants' own devices may ask a short, supportive clarifying question (aloud, audiovisually on a display, privately on the speaker's own device, or by directing one or both participants to check their phones or other devices, as the participant elects) to establish the speaker's intent as to scope, duration, intensity, and the response desired from the partner, including a pre-agreed reaction script such as a pause in a neutral or resting pose or a full stop-and-separate sequence; the answer resolves the greyzone to a state and, where a script was requested, the system announces the script to the partner and tracks compliance by interaction and presence sensing.

[0084] In another embodiment, the parties agree in advance to a cooldown command layer: when either says a brake phrase, or when a clarifying exchange calls for it, the system enters a BRAKE state in which both are to stop, move a short distance apart, adopt a pre-agreed calming posture such as a child's pose, and remain there (with the media faded, optionally the lighting changed, and optionally private breathing-paced haptics on each wrist) until each has had time to become de-aroused and calm, each has said and the other has heard that he or she is calm, and, in some embodiments, the system has sensed separation and a settled heart rate. Only then does the system open a structured talk-it-out step, and only after that may the session resume on both parties' go words. The purpose is to take decisions about what happens next out of the hot state in which the hot-cold empathy gap distorts judgment, and to give both parties a scripted, face-saving way to call for that without it being a withdrawal.

[0085] In another embodiment, for participants who prefer not to listen to music or other media during the session, the system carries no continuous signal at all. Music has value as an ongoing ambient signal because it is varied and pleasant; a constant drumbeat of heartbeat-like haptic pulses, a steady light, or a latent tone would not. It would become a sea of signal without deep meaning, which participants would soon tune out, so that the result would be signal exhaustion and a loss of signal in noise. Instead, the envelope runs silently and the reminder load is minimized. Nothing is presented while the envelope is comfortably within its window. A check-in on schedule, or a more insistent haptic pattern or beat on each participant's wearable, phone, or accessory (or, where elected, a change of lighting), arises only when the envelope is near its expiry or past it. If no affirmation follows, the controller runs the warning notifications, then the check-in, and ultimately the alarm and help stages, in the same sequence as with music, so that the consent phase-out is the same whether or not music is the carrier while the signals themselves stay rare enough to keep their meaning.

[0086] In another embodiment, for an established couple who elect it, the system supports a steady intimacy cadence of the couple's own choosing: each partner privately sets a preferred frequency range and an initiation preference; the controller computes the overlap and, where the partners' ranges overlap, treats the overlap as the couple's target cadence; it tracks recency of sessions and the balance of initiation between partners on-device; it offers, never imposes, private low-stakes bids and reminders to whichever partner has elected to receive them; it couples the affirmation window to recency, so that the window is somewhat shorter after a long interval and lengthens with regular practice; and it never discloses one partner's private settings or declines to the other.

[0087] In another embodiment, the controller includes a policy engine through which each participant independently configures, and a mode preset supplies defaults for, the behavior of the system when affirmative consent is not continuously detected. Configurable dimensions include what counts as affirmation and from whom; every timing parameter; the media response on lapse (fade to silence, fade to a floor volume, duck and hold, crossfade to a prompt or "yellow" track, step down a playlist tier, stop, change lighting only, or make no audible change); the form, channel, addressee, language, and wording of prompts; which escalation stages are enabled and in what order, and what sensor gates must be satisfied before advancing; which help pathways are used, in what order, with what data,

audibly or silently; who has authority to cancel an escalation or resume the media, including a requirement that cancellation be spoken in a specified participant's verified voice; lockout and re-checklist rules after an escalation; and what is logged and where. Where participants' settings differ, the engine merges them protectively: the shorter window governs, any participant's stop governs, all participants' affirmation is required to resume, help pathways are the union, and cancel authority is honored per participant.

[0088] In another embodiment, affirmations are attributed to individual participants by voiceprint. Where a participant has given separate, informed, revocable consent to biometric processing, and where the platform is determined to support on-device speaker-embedding extraction and secure template storage, the participant enrolls a voiceprint template that is held in the device's secure element, is never transmitted in the clear, and may be leased in encrypted, session-scoped or relationship-scoped form to a partner's or shared device solely for verification during a session or, under a relationship-scoped lease, during the relationship while consent continues. Each detected go word or stop word is embedded on-device, matched against enrolled templates, and checked for liveness, yielding an attribution level of VERIFIED, PROBABLE, or UNATTRIBUTED, or a CHALLENGE in which the system asks for a random word and re-verifies. Policy determines which attribution level is required for an affirmation to be credited to a participant, for a cancellation to be honored, or for media to resume; stop words are honored at any attribution level. Where biometric consent is withheld or the platform is not capable, the controller degrades to diarization, device-proximity, or joint-only attribution and records the mode in use. Each affirmation is logged with a pseudonymous participant identifier, attribution level, score band, liveness result, and time stamp.

[0089] In another embodiment, a participant may renew affirmation physically rather than vocally, by a momentary biometric reading on a device the participant already owns and has enrolled with, for example: a fingerprint sensor on a phone, tablet, computer, or panic button; a wearable's heart-rhythm, pulse-waveform, or vein-pattern sensor; or the secure biometric unlock of the device itself, which releases a signed affirmation token to the controller without the biometric ever leaving the device's secure hardware; or any other biometric or touch sensor, now known or later developed, the foregoing being examples only, in a list that is not exhaustive and may continue. Such a touch affirmation is attributed to the participant at a physically-verified level, requires no camera and no spoken word, is silent in the room, and may be used alone, as the renewal in response to a haptic reminder, or as a second factor alongside a go word where the policy requires one.

[0090] In another embodiment, the controller maintains a binding between each participant and the wearable devices, phones, and accessories that participant owns, established at enrollment by pairing each device from within that participant's own account and optionally confirmed at session start by a check that the device is on that participant's body or in that participant's hand. An affirmation given on a bound device is attributed to its owner and not to the other participant; a bound device that is not on its owner, or that has been removed and re-worn, loses its binding for the session until re-confirmed; and reminders, receipt cues, and private prompts are delivered only to the devices bound to the participant they concern.

[0091] In another embodiment, the system is deployed with a voice-assistant smart speaker in one or more of three topologies: (A) the speaker serves as the playback, prompt, alarm, and help stage endpoint while a phone or wearable runs the continuously listening controller, the speaker being driven through its platform's skill, cast, remote-streaming, or wireless-audio interface and its media-status events being consumed by the closed-loop verifier; (B) the speaker serves as a wake-word-gated affirmation channel in which a participant addresses the platform's assistant to affirm, the platform's own native speaker recognition attests the speaker's identity to the controller as a platform-attested attribution level, the affirmation window is adapted for explicit check-ins, the platform's native stop command is treated as a withdrawal signal, and a phone continues to listen for red words (stop words), brake words, grey words, yellow words, duress affirmations, secret concern cues, megaphone words, and every other interruption word or cue in the cue vocabulary defined in section 1; and (C) where the platform or device maker permits, the controller executes natively on the speaker using its far-field microphone array. The controller discovers which topology each speaker supports, composes them, and degrades safely when a platform forbids continuous listening or third-party code.

[0092] In another embodiment, a session is started hands-free by a spoken invocation to a smart speaker's assistant (for example, "<wake word>, start Safer Yes"), which launches the system's voice application, triggers a platform routine, or invokes an application

intent, and causes the controller to arm the session, announce the composed configuration and protection level aloud, identify the invoking speaker where the platform attests identity, bind any phones or wearables present, conduct mode selection and any pre-interaction checklist by voice, and then begin media only when every participant has spoken a go word within a mutual window. The invoking voice arms the session but cannot consent for another participant; the same voice interface accepts "<wake word>, end Safer Yes" as the close phrase and treats the platform's native stop command as a withdrawal signal.

[0093] In another embodiment, and without altering the standalone operation of the system, the system may optionally integrate with a dating platform so that two people who have matched can move, by mutual opt-in, from the platform's chat into successive layers of protection: pre-date setup of words, policy, and trusted contacts on each person's own device; a date check-in timer whose help pathways may include the platform's safety partner; a meeting handshake by device proximity or a scanned code that binds the platform's pseudonymous match token to the enrolled participant and to the device in the room, so that the person present is verifiably the person matched; a voice-started intimate session as described elsewhere; and a close after which the platform receives at most a minimal outcome signal. The platform never receives audio, words, consent states, or voiceprints; either person may unlink at any time; and declining the integration has no effect on matching. The optional layers may further include sharing of sexually transmitted infection (STI) test results, exchanged directly between the two matched people's devices at each holder's election, optionally bearing a testing provider's signed attestation and an expiry date, and never retained by the platform; and the establishment, within the dating interface, of ground rules for first intimacies, which are passed into the consent-gated system described herein as each participant's boundary map and policy. Those ground rules are a natural place to make clear, whenever two people who have matched on a dating platform meet, that strangulation and any other pressure on the neck are disallowed: such practices are extremely widespread among young adults, are medically irresponsible, and are, in the inventor's understanding, typically criminal under state law and, in many jurisdictions, not a matter to which consent is a defense. In a further embodiment, the system is built into a dating application with a dating front-end interface, so that people enter through the ordinary funnel of making themselves available as a potential date. Every person whom a user meets through such a platform has already completed the safety provisions and precautions, and has indicated an interest in, and support for, guardrails during early intimate encounters, so that both people can trust from the outset that safety precautions are supported, protected, and fiercely advocated for, and mutual trust is raised before they meet.

[0094] Methods corresponding to each of the foregoing embodiments are also disclosed, as are non-transitory computer-readable media storing instructions for performing such methods.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0095] The accompanying drawings illustrate embodiments of the disclosed subject matter and, together with the description, explain its principles. The embodiments shown are presently preferred, it being understood that the disclosed subject matter is not limited to the precise arrangements shown.

[0096] FIG. 1 is a block diagram of the architecture of a consent-gated media playback system, including a consent-gated media controller executing on a user device and a plurality of media playback targets, according to one embodiment.

[0097] FIG. 2 is a block diagram of the consent signal fusion pipeline, including echo cancellation against the playback reference signal, speaker attribution, the expressed channel, the optional implied channel with its per-participant tone modes, and the fusion rules, according to one embodiment.

[0098] FIG. 3 is a state diagram of the affirmation window and lapse escalation (playing, lapse, check-in, alarm, and help) together with the withdrawal path and the closed state, according to one embodiment.

[0099] FIG. 4 is a block diagram of the response mapper, adapter selector, media control adapters, and closed-loop verification component, according to one embodiment.

[0100] FIG. 5 is a block diagram of the non-affirmation policy engine, including per-participant settings, mode presets, protective merge, and the configurable policy dimensions, according to one embodiment.

[0101] FIG. 6 is a block diagram of voiceprint-attributed affirmation, including biometric consent and feasibility checking, enrollment, session-scoped template leasing, embedding, matching, liveness checking, and attribution-level assignment, according to one embodiment.

[0102] FIG. 7 is a block diagram of three deployment topologies for use with voice-assistant smart speakers (speaker as endpoint, speaker as wake-word-gated affirmation channel, and controller native on the speaker) according to embodiments.

[0103] FIG. 8 is a sequence diagram of a voice-initiated session start on a smart speaker, from the spoken invocation through announcement, mode selection, mutual go words, and verified playback start, together with the guards applied at start, according to one embodiment.

[0104] FIG. 9 is a hardware block diagram of a computing device (a user device, wearable, smart speaker, dedicated bedside device, or robotic system) on which the controller or a part of it executes, showing the processor, memory, secure element, wireless interfaces, microphone, speaker, haptic actuator, display and status light, sensors, physical button, audio playback path with reference tap, power, and clock, according to one embodiment.

[0105] FIG. 10 is a flow diagram of integration with a dating platform, from match and mutual opt-in through pre-date setup, date timer, meeting handshake, session, and close, with the data-minimization boundary between the platform and the participants' devices, according to one embodiment.

[0106] FIG. 11 is a sequence diagram of the exchange of signed renewal-token messages among the participants' devices and a playback target, including verification, a lapse, and a stop, together with the fields of a renewal-token message, according to one embodiment.

[0107] FIG. 12 is a state diagram of consent-gated physical contact by a robot, including the invariants that hold in every state, according to one embodiment.

DETAILED DESCRIPTION

[0108] The following detailed description refers to the accompanying drawings. Wherever possible, the same reference numerals are used in the drawings and the description to refer to the same or similar elements. Reference numerals in the 100 series that coincide with reference numerals of the '301 patent (for example server 102, network 106, users 111 and 113, devices 112 and 114, smart light 115, smart button 116, communications link 150, emergency services 160) denote the same or corresponding elements. While specific embodiments are described, modifications, adaptations, and other implementations are possible, and the description does not limit the disclosed subject matter.

1. Definitions

[0109] "Affirmative consent" means a positive, expressed indication of willingness to start or continue an interaction, detected from a predefined go word or go phrase, a predefined gesture or button input, a momentary biometric touch as described in section 4D, or another positive input chosen by the participants during enrollment, including a vocal cue as defined below: a hummed, whistled, or sung melodic cue, a non-lexical vocalization, a rhythmic cue, or a speech-form cue recognized by signature, contour, or rhythm rather than by words.

[0110] "Cue vocabulary" means the full set of categories of words, phrases, and other cues to which the system listens and responds, each of which a participant may enroll, in any language, or accept by default, and each of which changes the action of the system in a defined way. The categories are: (a) go words, the affirmations that start or renew the affirmation envelope, including enthusiastic go words, as defined below, and happy words, by which a participant may say how happy he or she is; (b) stage-entry words, go words that open a higher stage of intimacy as described in section 4G; (c) grey words, which signal uncertainty, discomfort, or a wish to slow down or check in without stopping, and which assign the GREYZONE consent state; (d) yellow words, which signal caution or a request to change course, for example to slow down, to change an activity or position, or to step back to a lower stage, without ending the encounter; (e) brake words or brake phrases, which call the cooldown BRAKE state, in which both participants stop, separate, and

calm themselves before a structured talk-it-out step; (f) red words, which are stop words: words of withdrawal, including any enrolled safe word of the '301 patent, that stop the media abruptly within the stop latency, assign the WITHDRAWN consent state, and, according to the policy, invoke the emergency response mechanism and any elected voice relay; (g) duress affirmations, which sound like go words to anyone listening but are privately bound to a request for help; (h) secret concern cues, privately bound to the grey, yellow, brake, or red mode of concern; (i) megaphone words, which enlist the device, through the voice relay, to repeat the speaker's words louder; and (j) session words, namely invocation, close, OK, and cancel phrases. Each category may take the form of a single word, a phrase, another linguistic form, or a vocalization, such as a hum, a whistle, a sung or chanted figure, or another sound, whose meaning is carried by pitch, tone, rhythm, or musicality, as described for vocal cues in section 3A. The categories listed are not exhaustive: the system is open to further categories and forms of cue not enumerated here, each defined by the change in the system's action that it calls for. Not every participant will use or learn every category. Many will use only a small, stable core subset, at minimum a go word and a red word and preferably a brake phrase, that is taught to every user and emphasized above all the rest, so that it is drilled in and available under stress. A participant who wants more may learn, practice, and rehearse the full library of cues, commands, and supports in the practice or rehearsal mode described in section 4, as often as he or she likes, and so draw on the full, supple support of the system in any direction he or she chooses, to the extent that those options are supported by the modules and variants available to that participant within his or her network of devices. Throughout this disclosure and its claims, every reference to a go word, a stop word, a red word, a brake phrase, or any other category of cue refers to the categories defined here, and participants, and any responder, are to listen for and respond to each category as described.

[0111] "Enthusiastic consent" means affirmative consent expressed in words the participant has enrolled as enthusiastic (for example, a participant may enroll "okay" as an affirmation and "yes," "more," or "keep going" as enthusiastic affirmations) so that the degree of enthusiasm is carried by the words the participant chose and spoke, not by a judgment of how they were spoken. Go words may be enrolled at two or more tiers (for example AFFIRMED and ENTHUSIASTIC), and the participant decides which tier the policy requires. An implied-channel estimate of vocal affect ("tone grading") is available as an option, never required and never excluded: a participant may elect, for his or her own voice, (a) no tone analysis; (b) a private distress signal that prompts only that participant; or (c) tone-graded affirmation, in which a go word counts at its enrolled tier only when the manner of its utterance is concordant with it, and a discordant go word yields an AMBIGUOUS state that does not refresh the timer. Each participant makes this election for himself or herself; a participant's tone is never graded under option (c) unless that participant chose it.

[0112] "Consent state" means a discrete value assigned to a participant, drawn from a set that includes at least ENTHUSIASTIC, AFFIRMED, AMBIGUOUS, GREYZONE, YELLOW, BRAKE, EMERGENCY, HAZARDOUS, and WITHDRAWN, and optionally UNKNOWN, and other states as configured. States are ordered, WITHDRAWN being lowest among the ordinary states, EMERGENCY and HAZARDOUS being treated as at least as low as WITHDRAWN. YELLOW is the state assigned upon a yellow word, in which the encounter continues only after the requested change of course. EMERGENCY is the state assigned, where the policy so directs and alongside WITHDRAWN, upon a red word, a distress indicator, a duress affirmation, or a cue or button pattern bound to emergency, and it invokes the abrupt stop and the escalation to help. HAZARDOUS is the state assigned when a participant reports, or a sensor indicates, a hazard such as pressure on the neck, a fall, smoke, or an intruder, and it is handled as the participant's policy for that hazard directs, up to and including the escalation to help. GREYZONE is the state assigned when a participant's utterance carries consent-relevant meaning that is neither an affirmation nor a withdrawal, short of a stop but sharp enough that review and support are urgently necessary (hesitation, a condition, a request to slow or change, discomfort short of a stop) as determined by the semantic classifier of section 3A; it does not refresh the affirmation timer, it triggers a clarifying exchange, and under the lowest-state rule it governs the joint state until resolved. BRAKE is a session state, entered by any participant's brake phrase or by the clarifying exchange, in which the parties have agreed in advance to apply the brakes: to cease activity, separate by a short distance, adopt a pre-agreed calming posture, and wait until both are de-aroused and calm before working anything out; it is distinct from WITHDRAWN in that it is a cooperative pause with a defined exit, and from LAPSE in that it is requested rather than timed out. "Brake phrase" means an enrolled word or phrase (for example "brake," "cooldown," "time out," or a word the participants choose) that invokes the BRAKE state; the enrolled stop word remains available and always takes precedence. The words "brake" and "break"

are used interchangeably in this disclosure for this state, in which a participant applies the brakes by asking for a break in the encounter. The two homophones support each other in that request, and a participant who says either is understood to have invoked the BRAKE state. Where a single fused form is wanted, the state and the request may be written BRÆKE, signifying at once a brake and a break, as when a person faces a partner who is all gas and no brakes.

[0113] "Joint consent state" means the minimum of the consent states of all participants in a session. The lowest state governs.

[0114] "Affirmation window" (T_a) means a configurable period, for example between 1 and 20 minutes, within which the joint consent state must be re-affirmed for media playback to continue without interruption. The affirmation window may be adjusted manually or algorithmically according to the level of established trust between the parties: lovers may graduate from short windows to long ones, and the window may shrink again according to how the more vulnerable party, or both parties, represent their experience during and after their encounters and in periodic check-ups, as described for envelopes that track the relationship in section 4G. The partners may also set the affirmation window so that they are reciprocally responsible, in round-robin fashion, for the enthusiastic affirmation that continues the session. For example, the first renewal might be due from the more vulnerable party, who in a heterosexual encounter may be the woman, and the next window from her male partner, who is then prompted to say something enthusiastic. Each participant's affirmation remains required at least once in every round, and any participant's red, brake, grey, or yellow word acts at once in every window. The rotation keeps the consent going and prompts feedback and conversation about how each person is feeling in the midst of the encounter: even if not a deep conversation, at least the start of one, and an opportunity for it at the moment when one or the other may be drifting in how he or she feels, or may want to emphasize how happy he or she is with how things are going.

[0115] "Lapse" means the condition in which the affirmation window expires without the required affirmation. "Fade period" (T_f) means a configurable period, for example between 10 and 30 seconds, over which the media is faded toward silence following a lapse, during which a soft cue invites re-affirmation.

[0116] "Renewal reminder" means a private cue (by default a haptic pattern on the participant's own haptic-capable wearable of any kind, or, absent one, on the participant's phone) issued a configurable lead time T_n before the affirmation window would expire, so that the participant can renew affirmation before any lapse occurs. "Receipt cue" means a private cue confirming that a participant's affirmation was recognized and credited. Both are optional conveniences that a participant may enable or leave off; neither is required for the system to operate, neither is audible in the room, and neither is delivered to the other participant.

[0117] "Interaction" means any detected sign that a participant is present and responsive (speech of any content, laughter, movement detected by a device's motion sensors or a room presence sensor, a touch on a device, a wearable's motion or heart-rate change, or a button press) as distinguished from "affirmation," which requires an enrolled go word or gesture. Interaction may be any mode of action (touch, motion, gesture, speech of any content, a press of a button, or another detectable act) and may be detected at any device present: the participant's phone, a smart speaker, a smart display, a multi-use panic button (such as button 116 or a platform-native button accessory), a wearable device, a tablet, a computer, a smart bed or bedside sensor (a load-cell or pressure-sensing mattress or pad, or a bedside radar, reporting motion, presence, and respiration or heart rate without any device being worn), a camera or room presence sensor (a smart display or security camera, or an mmWave or passive-infrared occupancy sensor, used for gesture and presence detection with processing on-device and only where the participants have enabled it), a vehicle's cabin system (its infotainment head unit, cabin microphone, seat occupancy sensors, and controls), an intimate device having a button or motion sensor, smart glasses or another head-worn display, earbuds or hearing aids, or any attending device as defined below, including a home robot present as a chaperone. Where an attending device or head-worn display has a camera, camera frames are processed on-device for gesture and presence only, are never stored or transmitted, and are used only where the participants have enabled them. Interaction halts escalation; only affirmation resumes media.

[0118] "Check-in period" (T_c) means a configurable period of no interaction, following the stop of media, after which the controller issues a spoken check-in. "Repeat interval" (T_r) means the interval at which the check-in is repeated with increasing intensity. "Alarm period" (T_l) means the period of no interaction after which an alarm is sounded. "Help period" (T_h) means the period of no interaction after which the controller calls out for help. Example defaults are T_a of 3 minutes, T_f of 20 seconds, T_c of 90 seconds, T_r of 30 seconds, T_l of 5 minutes, and T_h of 10 minutes measured from the last interaction; all are configurable at enrollment and may be scaled by the consent mode and by the participants' emergency preferences.

[0119] "Close phrase" means an enrolled word, phrase, or gesture by which participants explicitly end a session. A session that is not closed is presumed to be ongoing; a session that is ongoing but shows no interaction is presumed to warrant attention.

[0120] "Gesture" means, wherever used in this disclosure, any deliberate movement or touch that a device can distinguish, including a hand or body movement sensed by a wearable or other sensor and a touch gesture on a touchscreen or touch-sensitive surface. Any gesture that is distinctive, unique, and reserved, so that it is rarely used for any other purpose, may serve for any signal the system receives, including affirmation, a go word's equivalent, a stop, a brake, a mode of concern, a close, or a cancellation. For example, on a tablet, a four-finger spread gesture, which users rarely make for any other purpose, may be reserved and, where the tablet's platform permits it to be reassigned, may signify that a consent envelope has been initiated with an affirmative consent sequence started, so that a couple who prefer a tablet can begin with that one small gesture. The same applies on a smartphone or other device whose touch sensor reliably detects multi-finger touch.

[0121] "Critical participant" means a participant whose affirmation or acceptance the session requires under its mode and active policy: in Duo mode, every enrolled participant; in Solo mode, the enrolled participant who armed the session.

[0122] "Affirmation input" and "stop input" mean, respectively, any input by which a participant affirms, or withdraws or halts, the activity of a session, whether enrolled or universal, including a spoken word or phrase, a vocal cue, a gesture, a touch, a biometric touch, a button press or press pattern, a selection on a screen, and a renewal-token message received from a device bound to the participant. Wherever this disclosure describes a go word, it describes equally any affirmation input, and wherever it describes a stop word, it describes equally any stop input.

[0123] "Attribution level" means the degree of confidence with which an utterance is assigned to a particular participant: VERIFIED (matched to that participant's enrolled voiceprint template above a verification threshold and passing liveness), PROBABLE (assigned by diarization cluster, device proximity, or wearable correlation without a voiceprint match), UNATTRIBUTED (assigned to no particular participant), CHALLENGE (a borderline match pending re-verification), PHYSICALLY VERIFIED (the affirmation was given by a momentary, non-camera biometric reading on a device the participant touches or wears, such as a fingerprint sensor, a wearable's heart-rhythm or vein-pattern sensor, or a device's secure biometric unlock, that identifies the participant physically rather than by voice), and PLATFORM-ATTESTED (identity asserted to the controller by a voice-assistant platform's own native speaker recognition, such as a per-skill person identifier, when the participant has enabled that feature on the platform). "Voiceprint template" means a speaker embedding, or set of embeddings, derived from a participant's enrolled speech and stored for later comparison; it is a biometric identifier and is handled as such. "Device binding" means the controller's record, made at enrollment and optionally confirmed at session start, of which wearable device, phone, or accessory belongs to which participant, so that an affirmation, stop, cancellation, or interaction received from a bound device is attributed to that participant and not to another. "Attending device" means any device present in the room or vehicle, mobile or fixed, that has at least one sensor and a wireless interface and is paired to the controller, in any of three roles: a sensor role, in which it reports interaction and presence; a chaperone role, in which it also renders prompts, alarms, and help-stage output; and a witness role, in which it also contributes signed, time-stamped entries to the event log. Attending devices include a home robot, a robotic pet, a robotic vacuum whether parked or roaming, smart furniture (a bed frame, nightstand, or chair with sensors), a fixture (a lamp, ceiling fan, thermostat, smoke or occupancy detector, or room-control panel), an appliance, a smart display or television, a ranging beacon or tracker tag, and a hotel or venue room-control panel; a home robot

present as a chaperone is one example of an attending device in the chaperone role. "Imaging sensor" means any sensor whose raw output can be rendered, with or without processing, into a visible or near-visible image from which a person or a part of a person's body is recognizable, including visible-light, near-infrared, and full-resolution depth cameras; "non-imaging sensor" means a sensor that cannot, including millimeter-wave radar, a low-resolution thermal array, a coarse-zone time-of-flight sensor, an optically blurred sensor, ultrasonic or acoustic ranging, passive infrared, and load, pressure, motion, and physiological sensors. "Authorized responder" means a person or service designated in a participant's policy, in advance, as permitted to be alerted at a given escalation stage and, where so designated, to be granted entry to the premises for a wellness check or emergency response, including a security professional, a monitoring or dispatch service, a neighbor, roommate, flatmate, friend, or family member, a building doorman, concierge, receptionist, or manager, a peer responder from a response network, an emergency medical technician, paramedic, firefighter, or police officer, and emergency services, as well as a robot acting in any of the foregoing roles, for example as a robotic paramedic, firefighter, emergency medical technician, receptionist, concierge, or building doorman, which may respond to an alert under this framework and act as an active bystander, separating the parties or mediating between them, or serve as an impromptu sexual-crisis counselor or intimacy coordinator, in each case according to the consent and agreement of the parties, local policy or law, or the contracted terms of the system. "Access device" means a lock, door, gate, or building-entry system with a wireless or network interface that the controller can command to grant entry, including a smart lock, a building intercom or access-control system, a garage or gate controller, and a hotel or venue door system. In an interventional embodiment of the chaperone role, a chaperone robot operating with privacy at the maximum (non-imaging sensing where available, any camera confined as described in section 7, and no recording beyond what the conflict-recording settings permit) may respond to a stop, an EMERGENCY or HAZARDOUS state, or an unresolved discomfort by escalating its physical presence in steps: first, turning to face the participants and asking aloud that they separate and move apart; then, if they do not, moving its body between them; and, only where the participant it protects has enabled it in advance, applying a light touch to the shoulder of one participant to indicate that he or she should move to a different part of the room, or dress and leave the room, as the case may be. A robot has a unique capability to use its mechanical capacities to help separate people who are too close to each other while aroused and in conflict, so that everyone feels safe. The intervention is bounded: contact force and speed are limited to values safe for incidental human contact, for example within the limits recognized for collaborative robots; the robot never grips, restrains, strikes, or confines anyone, never touches the head, neck, or genitals, and never blocks the exit of either participant; it withdraws on resistance; and each step is announced aloud, logged, and accompanied by whatever escalation to help the participants' policy specifies.

[0124] "Non-affirmation policy" means the set of parameters, selected per participant and merged as described below, that governs what the system does when affirmative consent is not continuously detected.

[0125] "Invocation phrase" means a spoken phrase, following a voice-assistant platform's wake word, that the platform recognizes and routes to the system (by launching the system's voice application, by triggering a routine the participant has configured on the platform, or by invoking an application intent on a companion device) and that the controller treats as a request to arm a session. Example invocation phrases are "start Safer Yes," "open Safer Yes," and a participant-chosen alias, or a request that is more covert or combined, such as the device's wake word followed by a request to play a particular playlist that the participant has already tied to the start and launch of Safer Yes. Other, more cryptic or covert ways of activating Safer Yes may also be used, such as asking a smart speaker or phone for the weather in a particular construction of words that the participant has pre-registered as a Safer Yes activation cue. Such an activation may begin the vigilance of the system, arming the guard state described in section 8B, or, where the participant has set it up that way, may also open an affirmative consent envelope. In particular, the mere selection of a specific playlist, track, artist, or station that a participant has enrolled as an activation cue, whether by voice, by touch in a music application, or through any other media-control interface, and without any command that names the system, may by itself activate Safer Yes. Such covert, natural activation arms the safety system, including stop-word listening and the escalation to help, without an audible announcement where the policy and applicable law permit, and it may optionally open an enthusiasm check-in in which playback continues only while each participant renews affirmation within the affirmation window.

[0126] "Fail-closed" means that, in the absence of a positive, verified signal, the system transitions to and remains in a paused or silent condition. Silence is the safe state. "Vocal cue" means any enrolled sound that a participant produces with the voice, breath, or body (other than the lexical content of speech) that the controller recognizes by its acoustic signature, pitch contour, or rhythm and treats as an enrolled input; it includes a melodic cue (a hummed tune, a whistled phrase, or a sung line of a song, recognized by pitch contour), a non-lexical vocalization (for example a double throat-clear, a tongue click, a sigh, a sharp inhale, a laugh, or a pitched vowel sound), a rhythmic or percussive cue (a tapped, snapped, clapped, or knocked rhythm), a speech-form cue whose content is not the signal (a word spoken in a particular cadence or register, a counted sequence, a repeated word, or an invented word), and a breathing cue. A vocal cue may be enrolled as any input (a go word at either tier, a stop, a brake, a close, an OK phrase, or a concern cue) and may be enrolled as a secret personalized cue bound to a mode of concern and not disclosed to the partner. "Target cadence" means the overlap, computed on-device, of the frequency ranges that two partners have each privately entered under section 4F, and is the only frequency the system ever treats as a goal; the system has no default cadence of its own. "Recency" means the interval since the couple's last session relative to the target cadence. "Bid" means a private, low-stakes invitation to intimacy sent by one partner to the other through the partner's private channel and answerable by accept, later, or not now, no response being reported to the sender as a tally. "Duo mode" means a session in which every party present is aware of the system, is enrolled permanently or as a guest with accounts mutually attached for the session, and has given reasonably crystal-clear consent to participate, as recorded by the session-start announcement. "Solo mode" means a session in which one enrolled participant runs the system as a discreet protective safety layer for himself or herself, without the other party's enrollment and, where local law permits and the participant so elects, without the other party's awareness, the controller processing only the enrolled participant's own side of the interaction as described in section 8C. "Dredgable record" means a sealed, hash-chained record (the event log of section 7 and, where armed, a conflict recording) that is not surfaced to anyone on its own, is held under the participant's own key, and can be retrieved later, by the participant, by both participants jointly, or by lawful process, under the log rights of section 7 and the community policy or local law in force, so that an incident is on the record whether or not it is ever reported. "FRIES SAVER" means the framework of consent described in this disclosure, which extends the FRIES framework of the '301 patent (consent that is freely given, reversible, informed, enthusiastic, and specific) with five further elements: staged, in that consent is scoped to each stage of intimacy and a separate affirmation is required to ascend; alarmable, in that a greyzone, yellow-flag, or red-alert condition can be raised instantly and easily, openly or cryptically, by word, cue, touch, or button; verified, in that affirmations and alarms are attributed, recorded, and supported by signals; expiring, in that an affirmation lapses by default at the end of its window unless it is explicitly renewed, so that silence reverses it automatically; and responsive, in that a digital sounding board prompts, checks in, and answers throughout the encounter. "Intelligent bedroom environment" means a room in which people sleep or are intimate, together with the speakers, microphones, displays, lights, locks, buttons, wearables, phones, and other connected devices placed or carried within it, configured under the policies described herein to sense the enrolled signals, render media and alerts, and summon help; an ape- and rape-hostile, security-hardened intelligent bedroom environment is one so configured and augmented with a plurality of user devices that it interrupts escalation without affirmation, amplifies a participant's refusal, and calls for help when warranted. "Voice relay" means the function, described below, by which, upon a selected emergency condition, the system repeats at an elevated volume, verbatim, in paraphrase, or in summary, what is said by the participant who triggered the condition.

[0127] "Media content" means audible or audiovisual content perceptible by the participants, including music streamed from a third-party service, music stored locally, ambient soundscapes, spoken-word audio, video, and, in some embodiments, coordinated lighting effects. Music is the preferred embodiment because it is pleasurable, commonly present during intimate encounters, ambient, and mutually perceived without either participant looking at a device. "Ambient signal" means whatever ongoing, perceptible output the participants have chosen to carry consent status; it is media content in the preferred embodiment, but it may instead be a silent signal (a periodic haptic pulse, a lighting state, or a low ambient tone) for participants who prefer no music or media during the session.

2. System architecture (FIG. 1)

[0128] Referring to FIG. 1, a first user 111 and a second user 113 are present in a shared physical space. At least one user device 112 (a smartphone, tablet, or similar computing device having the hardware described with respect to FIG. 9, including at least a microphone 910, a processor 902, memory 904, a wireless interface 908, and at least one of a speaker 912, a haptic actuator 914, and a display 916) executes a consent-gated media controller 120. In a two-device embodiment a second user device 114 of like construction executes a second instance of the controller and the devices are joined by a local link 118 carried over their wireless interfaces, which may be Bluetooth, Bluetooth Low Energy, peer-to-peer Wi-Fi, ultra-wideband, an audio watermark exchange, or a server-mediated channel through network 106. In a single-device embodiment device 114 is absent and device 112 serves both users, attributing utterances to individual speakers as described below. Each participant may additionally wear a wearable device 117a, 117b—any device worn on the body that has a haptic actuator 914 and a wireless interface, including a smartwatch on any platform, a fitness band, a smart ring, earbuds, a hearing aid or earpiece (with tap, head-gesture, and in-ear pulse sensing where present), smart glasses or another head-worn display including an augmented- or virtual-reality headset (with microphone, inertial sensing, hand tracking, and, where present, eye tracking, so that a glance, gaze dwell, or nod may serve as an enrolled affirmation or interaction), a pendant, or a haptic strap or garment, whether it runs a companion application of the system or exposes only a generic notification or vibration channel—having the haptic actuator 914, optionally a microphone 910, an inertial measurement unit and photoplethysmographic or skin-conductance sensors 920, and a Bluetooth Low Energy or ultra-wideband interface 908 by which it is paired to a user device or directly to the controller, and which is bound to its owning participant as described in section 4E so that inputs from it are attributed to that participant alone. Other electronic accessories, such as a bedside device with a status light 918 and button 922, may be paired in the same way. The user devices, wearables, and accessories may also be smart speakers, smart displays, or head-worn devices, and the controller may be partly or wholly embodied in a networked playback device 144 or smart speaker 706 as described with respect to FIG. 7. A wearable may further include a surface electromyography (surface EMG) band, for example on the forearm, that reads the electrical activity of the muscles. A participant may train, in the rehearsal mode, an EMG pattern as a private emergency cue, for example the pattern of a planned and intended slap, so that the neuromuscular signal associated with beginning to slap a partner raises a red alert at once, with no actual slap required; another clear sign of anger, such as making a fist and beginning to punch, may likewise be encoded through EMG sensors as a further entry, so that it raises an alert before the participant has to follow through on the action. More generally, a participant may use any peripheral neural interface that can pick up micro-gestures, or gestures intended before they occur, including non-invasive head-mounted EEG sensor caps (sometimes called thinking caps), as further sources of enrolled cues, subject to the neural-data rules of section 8E.

[0129] Controller 120 is a set of program modules stored in memory 904 and executed by processor 902 (including, where present, a neural or digital-signal co-processor for the speech, embedding, and affect models) and comprises an expressed-channel analyzer 122, an optional implied-channel analyzer 124, a consent state fusion component 126, an affirmation timer 128 that signals a lapse, a media control adapter layer 130, an event logger 132, an alert output subsystem 134, a policy engine 136, and an escalation sequencer 138. The analyzers operate on frames delivered by microphone 910 through the device's audio front end; the fusion component and timer maintain state in memory 904; the adapter layer and alert subsystem issue commands over wireless interfaces 908 and the device's audio output path 924; and the event logger writes to non-volatile memory under keys held in secure element 906.

[0130] Controller 120 is operatively coupled, through media control adapter layer 130, to one or more media playback targets: a third-party streaming service 140 reached through an application-remote interface to the service's native application on the same device or through a cloud playback interface that addresses any of the user's registered playback devices; a local media player or operating-system audio session 142 on the user device itself, rendering through speaker 912 or audio output path 924; and a networked playback device 144 such as a wireless speaker, smart speaker, television, or multi-room audio group reached over Wi-Fi or a casting or wireless-audio protocol. Server 102 of the '301 patent, hosting the consent scoring module and the persistent communications link 150 to emergency services 160, is optionally coupled over network 106.

[0131] Controller 120 is further coupled, through alert output subsystem 134, to a set of alert channels, each of which is a hardware output reached over a specific path: (i) audible alerts rendered through speaker 912 of a user device or through networked playback device 144 or smart speaker 706 by the same adapters used for music, so that the check-in is heard through whichever speaker the music was using; (ii) haptic alerts rendered by haptic actuator 914 of the user device or of a wearable 117 over local link 118; (iii) visual alerts rendered on display 916 of a user device or wearable, which the subsystem wakes and drives with large touch targets for the check-in question; (iv) lighting alerts rendered by a connected lighting device 115 (a color-capable bulb, strip, or lamp) reached over Wi-Fi, Zigbee, Thread, Matter, or a smart-home bridge, by commands that set color, intensity, and flash pattern; and (v) accessory alerts rendered by the status light 918, speaker, or actuator of any other paired electronic accessory. A physical button 116 (a device button 922, a wearable crown or side button, a platform-native button accessory, or a smart-home button) is coupled as an input to fusion component 126 over the same links, so that a press is received as a stop or, if so enrolled, an affirmation, regardless of the state of the audio channels. The alert output subsystem selects channels and intensities according to the escalation stage and the active policy, as described in sections 4 and 4A: before any lapse, a private renewal reminder and receipt cue are delivered only to the participant's own haptic actuator 914 over local link 118; a lapse produces at most a soft cue; a check-in produces a spoken prompt plus a haptic tap and a lighting color change; an alarm produces the alarm tone, continuous haptics, and lighting flash; and the help stage additionally drives every speaker in the dwelling.

[0132] A playback reference signal 146—the audio stream, or a sufficiently precise description of it (track identifier, playback position, volume), that controller 120 caused to be played—is tapped from audio output path 924 or reported by the playback target and fed back to analyzers 122 and 124 to permit acoustic echo cancellation and lyric masking, as described with respect to FIG. 2.

[0133] In a preferred embodiment, all components of controller 120 execute on the user device and no raw audio leaves the device. The automatic speech recognition (ASR) and/or speaker diarization engine of analyzer 122 may be a streaming, on-device engine (for example an engine built on the sherpa-onnx runtime executing a compact transducer or CTC model on processor 902) operating continuously on microphone frames with a total end-to-end latency, from utterance to media directive, that is preferably below 500 milliseconds and more preferably below 250 milliseconds.

3. Consent signal fusion (FIG. 2)

[0134] Referring to FIG. 2, microphone audio 202 from one or more devices is passed, together with playback reference signal 146, to an acoustic echo cancellation and voice activity detection stage 204. Because controller 120 itself selected and initiated the media playing in the room, it possesses a near-exact reference for the interfering signal. Stage 204 subtracts the reference (aligned by cross-correlation and compensated for room impulse response and playback latency) from the microphone signal, so that lyrics, vocals within the music, and instrumental sounds are not presented to the downstream analyzers as participant speech. In an alternative or additional embodiment, stage 204 consults the lyric or transcript metadata of the currently playing track and masks any recognizer output that coincides in time and content with the known lyrics; a further embodiment simply excludes, during enrollment, any go word or stop word that appears in the lyrics of tracks in the participants' chosen playlists, or ducks playback for the duration of a detected utterance. Conversely, rather than excluding a word from enrollment, the controller may screen the music: as tracks are chosen for the participants' playlists, and again before each track plays, the controller checks the track's lyric or transcript metadata, or its own on-device transcription of the track, against the batch of conserved safety words and phrases in force for the session, including the participants' enrolled go words, stop words, brake words, and other cues and any universal safe words, and excludes any track that contains one, skips or mutes the segment in which it occurs, or substitutes an edited version of the track. Playlist screening may be used as an alternative to maintaining a signal analysis of the audio channel, where echo cancellation is unavailable or unreliable, or in addition to it, so that the music does not trigger speech recognition or keyword spotting in the verbal channel analyzed for the participants in the room.

[0135] A speaker attribution stage 206 assigns each voiced segment to a participant at one of the attribution levels defined above. In the preferred embodiment, where the participant has consented and the platform is capable, attribution is by voiceprint verification as described with respect to FIG. 6, yielding VERIFIED attributions. Where voiceprint verification is unavailable, attribution is by

speaker diarization using speaker embeddings clustered within the session, by the relative signal energy at each device's microphone, by ultra-wideband or Bluetooth ranging, or by correlation with a wearable's detection of laryngeal vibration or breathing, yielding PROBABLE attributions. Where none of these is available, segments are UNATTRIBUTED and the controller operates in joint-only mode, in which any go word counts toward a single joint affirmation and the policy engine restricts what may be credited. Attribution enables per-participant consent states, prevents one participant's affirmation from being credited to another, and (critically for the cancel-authority policies described below) prevents one participant from cancelling an escalation that another participant has configured to require his or her own verified voice.

[0136] The expressed-channel analyzer 122 performs on-device ASR and keyword spotting (and, as described in section 3A, melodic-cue recognition by pitch contour) on attributed speech to detect: (a) go words and go phrases (for example "yes," "keep going," "more," "green," or any phrase chosen at enrollment); (b) stop words and safe words (for example "stop," "red," "safe word," or the participants' chosen safe word), which are detected with a lower detection threshold and shorter latency than go words because a missed stop word is more harmful than a false stop; (c) qualifying phrases that adjust boundaries without withdrawing consent (for example "slower," "yellow"); and (d) optionally, a coached separation word, such as "Mayday". Keyword spotting may be implemented with a small acoustic keyword model running in parallel with the general ASR so that stop words are detected even if the general recognizer lags. Detections carry a confidence value, a time stamp, and an attribution.

[0137] The implied-channel analyzer 124 is optional. Where present, each participant elects for his or her own voice one of three modes: OFF, in which no affect estimate is produced for that participant; PRIVATE, in which the estimate is used only as a distress signal that prompts that participant; or GRADED, in which the estimate also conditions whether that participant's go word counts at its enrolled tier. The analyzer estimates the affective quality of speech and non-speech vocalizations attributed to that participant. Features may include pitch contour, pitch variability, energy, speaking rate, spectral tilt, jitter and shimmer, harmonic-to-noise ratio, presence of laughter, presence of crying or sobbing, presence of strained or choked phonation, and breathing pattern. A compact neural classifier trained on labeled vocal-affect corpora outputs at least an estimated valence (negative to positive) and arousal (calm to excited), and preferably discrete indicators for laughter and distress. The analyzer is calibrated per participant during enrollment by recording a short baseline of neutral speech and a short baseline of the participant's chosen go word spoken as that participant would say it when meaning it, so that the distress and concordance tests of rule R3 are calibrated to that participant's own voice rather than to a population average. Its output is attributed to, and governed by the election of, the participant whose voice it analyzes; in PRIVATE mode it is never shown to or logged against the other participant, and in GRADED mode only the resulting consent state, not the affect estimate, is shared or logged. Where a wearable is available, biometric signals such as heart rate variability, skin conductance, and motion are fused into the implied estimate. Distress indicators from the implied channel are routed both to fusion 126 and, except for a participant in PRIVATE mode, directly to the emergency response mechanism of the '301 patent. The implied-channel analyzer, like the expressed-channel analyzer, processes audio from a ring buffer and does not record it. A conflict recording, which is a sealed audio recording that begins at a triggering event, is a separate function; whether it may be armed, and whether an audible announcement is required, are governed by the conflict-recording settings and by the recording-consent laws of the participants' jurisdiction, including the one-party and all-party consent rules discussed under Duo mode and Solo mode and under Jurisdictions and Solo mode in section 8C.

[0138] The consent state fusion component 126 applies rules 210 to produce a per-participant state S_i and a joint state S. In the illustrated embodiment: (R1) a stop word or safe word from any participant, or a press of button 116, sets that participant's state (and therefore the joint state) to WITHDRAWN immediately, overriding all other evidence; (R2) a go word whose confidence exceeds a threshold sets the participant's state to the tier at which that word was enrolled (AFFIRMED or ENTHUSIASTIC) and refreshes the affirmation timer if the tier meets the policy's requirement; (R3) according to the participant's elected tone mode: in PRIVATE mode, a go word accompanied by a distress indicator in that participant's own voice (crying, strained phonation, or a wearable distress signal) causes a private haptic prompt to that participant and, at the participant's option, withholds refresh; in GRADED mode, a go word whose accompanying affect estimate is concordant (positive valence, arousal within the participant's calibrated band, no distress

indicator) counts at its enrolled tier and refreshes the timer, while a discordant go word (flat, negative, tearful, or strained) sets the state to AMBIGUOUS, does not refresh the timer, and causes a prompt for re-affirmation through the channel the participant chose (private haptic, or audible to all); in OFF mode the rule is not applied; the rationale being that words spoken under coercion may be lexically affirmative while acoustically distressed, and that the participant whose voice is analyzed should decide whether and how that check applies; (R4) positive affect without a go word, such as laughter or an enthusiastic non-lexical vocalization, may refresh the timer only if the participants enabled "implied refresh" at enrollment and only for a limited number N of consecutive refreshes before an expressed go word is again required; (R5) a distress indicator from the implied channel (except for a participant in PRIVATE mode, to whom rule R3 applies) or a biometric distress signal sets the state to WITHDRAWN and escalates; (R6) the joint state is the minimum over all participants, so that the least-consenting participant governs playback; and (R7) a grey word sets that participant's state to GREYZONE. For example, a participant may enroll the word "foul," as in a party foul or a foul in sport, something that calls for a referee, as a grey word: it signals a grey-zone situation that halts the action, calls for rapid review, and opens a discussion before anything resumes.

[0139] Enthusiasm in this system is lexical by default: it is carried by which enrolled word the participant chose to say, at the tier the participant assigned to it. Tone grading is an option layered on that default, and the choice belongs to the participant whose voice is analyzed. The reasons for making it optional rather than mandatory are that a system that always graded the manner of an affirmation would penalize quiet, flat, or dissociated speech and would place a machine's verdict on a participant's performance into the room; the reasons for offering it rather than excluding it are that some participants want the protection against their own coerced or distressed "yes," some couples want a shared signal that responds to how things sound and not only to what is said, and prior voice-controlled systems act on a keyword regardless of how it is spoken. The three modes of R3 let each participant place himself or herself anywhere on that range; the policy engine records the election, and the log records which mode was in force for each affirmation.

[0140] Fusion 126 optionally exchanges its outputs with the consent scoring module of the '301 patent, contributing per-participant states and affect estimates as inputs to the Main Event Score, and receiving the dynamic consent score as an additional input to the response mapper of FIG. 4. In Private Practice and Extreme Privacy modes of the '301 patent, this exchange is suppressed and fusion operates entirely on-device.

3A. Semantic classification, greyzone, and clarifying exchange

[0141] Vocal and melodic cues. In addition to words, the expressed channel recognizes enrolled melodic cues (a hummed tune, a whistled phrase, or a sung line of a song) by pitch contour and rhythm rather than by lexical content, using on-device query-by-humming techniques in which the enrolled cue is stored as a contour template and incoming vocalization is matched to it after the playback reference signal 146 has been cancelled, so that music the system itself is playing cannot be mistaken for a participant's cue. A melodic cue may be enrolled as any input (a go word at either tier, a stop, a brake, a close, an OK phrase, or a concern cue) and is attributed, logged, and acted upon exactly as the corresponding word would be. Melodic cues have three properties that make them valuable here: they are natural to produce during intimacy, they carry no lexical content for a partner to object to or for the semantic classifier to misread, and, where enrolled as a secret personalized cue bound to a mode of concern (grey zone, yellow flag, brake, or red alert), they allow a participant who cannot bring himself or herself to say even a soft word aloud to register and classify a concern without the partner knowing that any particular sound was the one that registered it. The controller never announces which cue triggered a response; it announces only the response. Melodic cues are one subtype of the broader class of vocal cues defined in section 1, all of which the expressed channel recognizes by the same enrolled-template matching and treats identically. Non-lexical vocal cues include a cough or throat-clear (a deliberate double throat-clear being a classic discreet signal), a sigh, a sharp inhale or exhale, a sniff, a tongue click, a kiss sound, a laugh, a gasp, a pitched vowel sound, or a growl or purr, each recognized by its acoustic signature; rhythmic and percussive cues include a tapped rhythm on a headboard or a wrist, a snapped finger, a clap pattern, or a knock, each recognized by inter-onset timing with the same matcher used for the audibly verified knock of section 4; speech-form cues that are not content include a pet name spoken in a particular cadence, a counted sequence, a word in another language, a nonsense or invented word, a repeated word (so that "yes, yes, yes" may be enrolled as a signal distinct from a single "yes"), and a

change of register such as a whisper where speech was normal or a sung rather than spoken word; and breathing cues include a held breath or a deliberately slowed breathing rhythm, which a wearable may corroborate. Any vocal cue may be enrolled as any input or as a secret personalized concern cue, and the rules of this paragraph (on-device matching after echo cancellation, identical attribution and logging, and announcing only the response and never the trigger) apply to all of them. Keyword spotting is exact and fast but narrow. People do not withdraw consent only with enrolled words; they say "wait," "I'm not sure," "not like that," "that hurts," "maybe later," "can we slow down," or nothing at all after a question, and they affirm, withdraw, and signal a grey zone with words they never enrolled. The expressed-channel analyzer therefore includes, in addition to the keyword path, a semantic classifier 123 that operates on the transcribed text of each utterance, on the few seconds of transcript preceding it, and on consent-relevant non-lexical vocalizations (a sharp inhale, a whimper, a laugh, silence following a prompt), and that assigns the utterance to a meaning category with a risk weight. The classifier is a compact language model (a small instruction-following or intent-classification model quantized to run on the user device's processor or neural co-processor within the latency budget of section 2) or an equivalent intent model, and it runs entirely on-device; no transcript leaves the device.

[0142] Meaning categories and risk weights. The classifier outputs at least the categories GO (affirmation in substance), GREYZONE (consent-relevant meaning that is neither affirmation nor withdrawal, including hesitation, a condition, a request to slow, change, or pause, discomfort short of a stop, a question about what is happening, or a deferral), and STOP (withdrawal in substance), and preferably sub-categories within GREYZONE (slow, pause, change, uncertain, discomfort, question, defer) and within STOP (stop, hurt, fear, help). Each output carries a risk weight on a scale, representing the classifier's estimate of how strongly the utterance indicates that the interaction should not continue unchanged, together with a confidence. Participants may supply their own example phrases for each category at enrollment, in their own words and languages, and the classifier is adapted to those examples on-device, so that a couple's private vocabulary is understood without ever being transmitted.

[0143] Scoped stops and bare requests. The most important ambiguity the classifier resolves is that the same word carries different meanings in intimate speech. "Stop" spoken alone, "stop, that hurts," and "stop, you're using too much lube" share a word and differ entirely in meaning: the first is a withdrawal, or a withdrawal pending clarification; the second is a withdrawal with a reason that should also reach the partner; the third is a scoped request (stop one specific thing, continue otherwise) that is not a withdrawal of consent at all and is consistent with enthusiastic, ongoing consent. A keyword system treats all three as the same stop and pauses the session three times; a human partner hears three different requests. The classifier is designed to make these distinctions, and the policy governs what follows: a bare "stop" is honored immediately as a stop (the enrolled keyword path guarantees this) and is then clarified ("Stop everything, stop, drop, and roll, or stop one specific thing?"), so that the speaker can scope it after the fact, with the session resuming only on the speaker's go word; a stop with an expressed reason of pain, fear, or distress is honored as a withdrawal and the reason is relayed to the partner and, where the policy directs, to the emergency mechanism; and a stop whose scope is already explicit in the utterance ("stop with the lube," "stop talking," "not there") is classified as a scoped request in the GREYZONE change sub-category, with the media ducked rather than paused, the request announced to the partner in the speaker's own words, and the affirmation timer neither refreshed nor expired until the speaker's next affirmation. Participants who want every utterance of the word to pause the session regardless of scope may set the policy so; the default is that the enrolled stop word always pauses, while the classifier decides how an unenrolled stop-like utterance is scoped. In some embodiments, the classifier is joined by a private AI agent that clarifies intent in the green zone (affirmation), the grey zone, and the red zone (withdrawal) to the extent each participant has enabled it, tolerates it, and welcomes it, serving in effect as a just-in-time sex therapist and intimacy coordinator present in the room. The agent runs on-device or under end-to-end encryption, retains nothing beyond the event log that the participants have authorized, and protects the participants' privacy to the maximum extent permitted by local law, disclosing only as mandatory reporting requirements or other applicable law compels. The agent may support, or itself engage in, the disclosure of hard truths and other sensitive facts that bear on comfort or safety (for example, an anatomical variation that may cause pain or call for added sensitivity, such as a penis that is larger or smaller in length or girth than a partner may anticipate) by prompting, scripting, or relaying such a disclosure at a time and in words the disclosing participant chooses. The purpose of such a disclosure is not to make any body seem abnormal or to unsettle a partner, but only to inform: to see that every participant has the information he or she needs, without anyone

being made to feel that his or her body is monstrous or unwanted, while recognizing that a given body or act may not be for everyone. The same holds for anatomy on the female side and for persons of any gender, including transgender and intersex persons. Disclosures of this kind are highly sensitive, and many participants may welcome having support and backup to make them kindly and gently, as a systematic guarantee that the information is delivered and reaches the partner before the encounter rises into higher and higher voltage. Some private facts about a person are hard to bring out at all (out of embarrassment, fear of rejection, or simple lack of words) unless the person has the support of an intimacy coordinator, or of a large language model (LLM) agent working sensitively on that person's behalf. The agent serves each participant on his or her own behalf and, in balance, on the partner's behalf as well, favoring neither, so that each participant can make a fully informed decision under the FRIES SAVER framework of this disclosure, in particular, so that consent is informed as well as freely given, specific, and enthusiastic, and the encounter proceeds by a safer yes. The agent may also cross-check the session against the participants' stated ground rules and records and flag an apparent gap before ascent to a higher voltage, for example: "Hold on—have you shared sexually transmitted infection test results with each other? I don't see that you have yet. Just confirm that you have, and I'll get out of your way." A participant's confirmation is logged and the session proceeds. The agent may, optionally, recite critical ground rules at the start of a session or before ascent, and it treats an early relationship as probationary: a pair new to each other begins with short consent envelopes and short affirmation windows that expire automatically, and possibly before reaching high voltage, and earns longer affirmative consent envelopes only as trust is established through sessions completed without lapse, withdrawal, or flagged concern. Like the classifier, the agent can add caution, a check-in, a greyzone, or a withdrawal, but can never supply an affirmation on a participant's behalf, and each participant may silence or disable it at any time.

[0144] The same treatment applies to the bare request, the common pattern in intimate speech in which a preference is voiced only partly, or only by its first word, or only by a sound: "softer," "not yet," "wait," "hmm," "can you—," a wince, a hand moved away, a change of breathing. Each is consent-relevant and under-specified. The classifier places it in the appropriate GREYZONE sub-category with a risk weight, and the clarifying exchange of this section supplies, with a question or two, the dimensions the speaker did not voice (what it applies to, for how long, how much, and what the partner should do) so that the speaker's intent and the most considerate response to it are made explicit without the speaker having to compose a full sentence in the moment. The exchange is designed around the observation that in intimate communication the burden of expressing a preference fully falls on the person least able to bear it at that moment; the system carries that burden instead, by asking the small questions, relaying the answer to the partner in neutral words, and recording only the category and the resolved intent, never the words. Where a speaker has enrolled example phrases for his or her own bare requests (a sound that means slower, a word that means not there), the classifier learns them, and over sessions the clarifying questions become fewer because the speaker's shorthand resolves directly to a scoped request and a reaction.

[0145] Asymmetric thresholds and guaranteed overrides. The thresholds at which the classifier's output is acted upon are asymmetric: a STOP meaning is acted upon at a low confidence and a low risk weight, because the cost of ignoring a withdrawal is far higher than the cost of pausing unnecessarily; a GO meaning is credited as an affirmation only at a high confidence, and, under a policy that requires enrolled words for refresh, is not credited at all; a GREYZONE meaning is acted upon at a moderate threshold and never by refreshing the timer. Enrolled stop words, safe words, and button presses are detected by the keyword path independently of the classifier and are honored regardless of anything the classifier says; enrolled go words are likewise credited by the keyword path; the classifier is an additional channel that can only add a withdrawal or a greyzone, or add an affirmation where the policy permits, and can never suppress an enrolled stop or override an enrolled go word's tier. A classifier fault, timeout, or unavailable model leaves the keyword path operating and is announced and logged.

[0146] Fusion and response patterns. The fusion component 126 receives the classifier's category, sub-category, risk weight, and confidence alongside the keyword detections and attribution. A STOP meaning above its threshold is treated as rule R1; a GO meaning above its threshold is treated as rule R2 at the AFFIRMED tier unless the policy maps particular meanings to the ENTHUSIASTIC tier; a GREYZONE meaning sets the participant's state to GREYZONE, which does not refresh the affirmation timer and, under rule R6, governs the joint state. The policy engine, in Dimension C and Dimension D, maps meaning categories and

risk-weight bands (not only states) to response patterns, so that, for example, a low-weight GREYZONE (uncertain) produces a private haptic prompt to the speaker only; a moderate-weight GREYZONE (slow, pause) ducks the media and issues a clarifying question; a high-weight GREYZONE (discomfort) pauses the media and issues the clarifying question aloud; and a STOP sub-category of hurt or fear pauses immediately and advances to the emergency mechanism without waiting for the help period. Response patterns are participant-configurable per category and band, and the mapping in force is recorded in the log with each classified utterance, as are the category, sub-category, weight band, and confidence band, never the transcript.

[0147] Clarifying exchange. When a participant's state becomes GREYZONE, or any other state when called for, the controller may conduct a brief, supportive clarifying exchange to learn what the speaker meant and what the speaker wants, rather than guessing. The exchange may be delivered through any channel the speaker's policy elects: audibly, through an attending device in the chaperone role, a smart speaker, or the room's speaker; audiovisually, through a smart display, television, or head-worn display showing the question with large touch targets while speaking it; privately, through the speaker's own phone, wearable, or earpiece; or by a short audible or haptic direction to one or both participants to check their phones, wearables, or other devices, on which the question is then presented so that the content of the exchange is not spoken aloud in the room. The same choice applies to the announcement of a reaction script to the partner. The exchange consists of at most a few short questions in the speaker's language, generated from a template or by the on-device language model and constrained to the following dimensions: scope (what, specifically, the utterance applies to—everything, a particular act, a particular place or touch), duration (a pause for a moment, for a set number of minutes, for the rest of the session, or an end), intensity (slower, gentler, less, or a change rather than a reduction), and desired reaction from the partner (continue with the change, pause and wait, move to a pre-agreed resting posture such as a neutral pose or a child's pose, separate, or the full stop sequence). Questions are phrased to be answerable by a single word, a go or stop word, a touch affirmation, or a button, and they are phrased non-judgmentally and without pressure toward continuation, for example, "Do you want a pause, or to stop?" rather than "Are you OK to keep going?", because the purpose of the exchange is to make it easy to say less, not to elicit a yes. The exchange is time-limited; no answer within the limit is treated as a lapse at minimum and, for a high-weight greyzone, as a withdrawal.

[0148] Reaction scripts. Participants may pre-agree, at enrollment, one or more reaction scripts that the clarifying exchange can invoke, each a short named sequence the partner is expected to follow and the system announces and tracks: a neutral pose script (both participants stop and rest in a pre-agreed neutral or resting posture, such as lying side by side or a child's pose, until the speaker gives a go word); a pause-and-wait script (the partner withdraws contact and waits, with media ducked, for a set time or until a go word); a change script (the partner adjusts as the speaker specified, with the media continuing); and a stop-drop-roll script, being the full stop sequence of the '301 patent's separation protocol (stop all activity, drop contact and move apart, and roll through the coached separation and de-escalation steps) with the cooldown track and, where configured, the emergency mechanism. When the speaker selects a script, the controller announces it to the partner through the room's speaker, a chaperone device, or the partner's own wearable as policy directs, in neutral language and with the speaker's chosen wording, and then tracks whether the script is being followed by interaction and presence sensing (separation by ranging between the participants' devices, cessation of motion by the bed sensor or radar, posture where a non-imaging sensor can infer it) and by the speaker's subsequent utterances; a script that is not followed within a short period escalates as a withdrawal would, and the announcement, the script, and compliance are logged. A reaction script may also be invoked directly by an enrolled phrase without a clarifying exchange.

[0149] Language, culture, and accessibility. Because the classifier operates on meaning, it may be run in any language the on-device model supports, and a couple who speak different languages may each enroll examples in their own; the clarifying exchange is conducted in the speaker's language. For a participant who cannot speak, the exchange may be conducted on a wearable or phone by touch, and the classifier may be applied to typed text or to an enrolled gesture vocabulary. Nothing in this section is required: a participant may disable the semantic classifier entirely, in which case the system operates on enrolled words alone, and may disable the clarifying exchange, in which case a greyzone resolves by lapse.

3B. Cooldown command layer (brake state)

[0150] Judgment during intimacy is made in a hot state. People reliably mispredict, when aroused, what they would want when calm, and mispredict, when calm, how they will feel when aroused (the hot-cold empathy gap) so that a boundary negotiated in the moment, or a disagreement worked out while both parties are still aroused, is settled under conditions neither would choose. The system therefore provides a cooldown command layer: a pre-agreed protocol, invoked by a single word, under which both parties apply the brakes and become calm before anything is discussed or decided. The layer sits between the scoped requests and clarifying exchange of section 3A, which handle adjustments that can be made without stopping, and the withdrawal and separation protocol, which ends the session. It is cooperative: calling for a brake is not a withdrawal, carries no implication that anything was wrong, and is something either party can do without cost, which is what makes it usable.

[0151] Invocation and the brake script. Each participant enrolls a brake phrase, and the participants agree at enrollment to a brake script with the following elements, each of which the controller announces in turn and tracks where it can. (1) Stop: all activity ceases; the media fades to silence or to the silent carrier's slowed pulse, and the lighting device changes to the pre-agreed brake color. (2) Separate: the parties move a short distance apart (to opposite sides of the bed, or one to a chair) so that neither is in contact with the other; the controller instructs this and, where the parties' devices or wearables support ranging or the bed sensor or radar resolves two positions, confirms it, prompting again privately if the parties remain in contact. (3) Settle: each party adopts the pre-agreed calming posture (by default a child's pose, which is private, physically self-contained, and lowers the head below the heart, or alternatively lying on the back with hands at the sides, sitting with feet on the floor, or any posture the parties chose) and the controller delivers, on each party's own wearable or phone, a slow breathing-paced haptic rhythm (for example, a four-second inhale and four-second exhale pattern) for a minimum settling period the parties set, typically two to five minutes, during which nothing is asked of either party and the room is quiet: the controller interrupts any premature attempt to speak, refereeing attempts to communicate until every party has reached a calm, de-aroused ("cold") state. (4) Confirm: at the end of the settling period, the controller asks each party privately (on his or her own device, so that neither answers under the other's gaze) whether he or she is calm and de-aroused, and continues the settling period in further increments until each has answered yes; where a wearable reports heart rate or heart-rate variability, the controller may require, as an additional and participant-elected condition, that the party's heart rate have fallen to within a band of that party's enrolled resting rate and remained there for a set interval, and it may show each party his or her own trend privately as feedback. (5) Declare: when both have confirmed, the controller announces, through the room's speaker or chaperone device, that both parties have said they are calm, so that each hears the other's confirmation through a neutral voice rather than having to ask or to claim it. (6) Talk: the controller then opens a talk-it-out step, structured as the parties chose at enrollment (a timed turn for each to speak uninterrupted, a set of prompts the parties wrote in advance for themselves, a reading of the pre-interaction checklist, or nothing more than an announcement that they may now talk) with the media still off and the lighting still in the brake color, and with the enrolled stop word and the help pathways available throughout. (7) Resume or close: the session resumes only on a go word from each party given after the talk step, as at session start, or ends by the close phrase or by either party's silence, which the system treats as a lapse that proceeds to check-in.

[0152] Sensing and trust. The brake script does not depend on sensing: a session with no wearables and no room sensor runs it on announcements and private confirmations alone. Where sensing is available it is used to help the parties trust the state rather than to police them: separation confirmed by ranging or by a two-position bed sensor or radar reading means neither party has to wonder whether the other moved away; a settled heart rate shown to its owner means a party can see his or her own calm rather than guess at it; and the declared confirmation from a neutral voice means neither party has to take the other's word under pressure. Sensed values are used only for the brake script's own conditions, are shown only to their owner, and are logged only as pass or fail against the condition, never as values. A party may also enroll a private brake (a brake phrase or touch that invokes the script while attributing the invocation to no one, so that the announcement says only that a brake was called) for the case in which naming the caller would itself be a cost.

[0153] Relationship to the other states. A brake may be invoked by either party's brake phrase, by a touch affirmation enrolled for it, by the clarifying exchange of section 3A when the speaker selects it as the desired reaction, by the semantic classifier when an utterance is classified as a request for a break (for example "I need a minute," "let's take a breath"), or by the policy engine automatically when both parties' consent states have been GREYZONE for longer than a set period or when a disagreement is detected by the classifier in the exchange. A brake is never a withdrawal: the event log records it as a cooperative pause, and the consent scoring module of the '301 patent, where used, does not score it against either party. The enrolled stop word, safe word, button, and distress indicator take precedence over the brake script at any point and move the session to WITHDRAWN; the help pathways and the escalation sequence remain armed throughout, so that a brake in which one party becomes unresponsive proceeds to check-in, alarm, and help as any lapse would. The brake script is also the default reaction script offered by the clarifying exchange when the speaker's answer indicates that the parties need to work something out rather than merely adjust, and it may be configured as the mandatory first step before any renegotiation of enrolled boundaries during a session, so that boundaries are only ever changed by parties who have first become calm.

4. Affirmation window, lapse, and escalation (FIG. 3)

[0154] The controller is organized as a consent keepalive, especially in the early phases of a new intimate relationship or when intimacy is going through a troubled period: affirmations are heartbeats, media plays only while they keep arriving, and the lapse of that signal (rather than any affirmative command) is what changes the system's behavior. Because a lapse ends playback without anyone having to voice a withdrawal, reversibility—the R in the FRIES model of consent (Freely given, Reversible, Informed, Enthusiastic, Specific)—becomes automatic rather than something a participant must speak. This is the inverse of a voice-controlled appliance, in which silence means "no change." Here silence means "something may be wrong," and the longer the silence, the more insistent the system becomes.

[0155] Referring to FIG. 3, the controller begins a session in an ARMED/SILENT state 302 in which no media plays. The session is initiated as in the '301 patent (mode selection, optional pre-interaction checklist, optional biometric authentication, device linking) (by touch on a phone or by a spoken invocation phrase to a smart speaker as described in section 8B) during which the participants select or confirm a playlist, their go words, safe words, and close phrase, and confirm the timer parameters. Upon receipt of a go word from every participant within a short mutual window (at the tier the active policy requires) the controller transitions to PLAYING 304 and issues a play directive through adapter layer 130. The start of music is itself the perceptible confirmation that mutual consent (or, in Solo mode, adequate, enthusiastic individual consent) has been registered.

[0156] In PLAYING 304 the affirmation timer counts down from the affirmation window T_a, typically a few minutes. Each affirmation satisfying rule R2 (or R4, if implied refresh is enabled) resets the timer. The loop is deliberately undemanding when things are going well: participants who are enjoying themselves audibly will refresh it without effort, and the affirmation window may be lengthened by the response mapper while the dynamic consent score remains high. To remove the burden of timekeeping for a participant who wants it, the controller may issue, through alert output subsystem 134 and only for a participant who has enabled the feature, a private renewal reminder to each participant whose affirmation has not yet been received in the current window, at a lead time T_n before T_a expires (for example 30 seconds before) as a haptic pattern on that participant's wearable 117 or phone over local link 118 (or on the participant's earpiece, ring, or other accessory), optionally repeated more firmly at a shorter lead time (for example 10 seconds before). The reminder is silent in the room, is not delivered to the other participant, and does not itself change the media. When the participant's affirmation is credited, the same device delivers a brief receipt cue (a single soft tap) so that the participant knows the affirmation was heard and counted and need not repeat it. A participant may configure the reminder to be omitted, to be delivered only when the window is more than half elapsed, or to be delivered on every window; and may configure the receipt cue independently. The controller may adapt T_n per participant from that participant's observed renewal rhythm, moving the reminder earlier if the participant tends to renew late and later if the participant renews early, within bounds set at enrollment. The intent is that a participant who wants to maintain a continuous, never-lapsing chain of affirmation can do so by responding to the wrist rather than

by watching a clock, and that the absence of a reminder is itself reassurance that the current window is already satisfied. The reminder and receipt cue are optional in every respect: no wearable or haptic device is required, a participant may enable one, both, or neither, and the affirmation window, lapse, and escalation sequence operate identically whether or not any participant has enabled them or wears any device. In some embodiments the media itself carries these signals: the controller mixes the media in real time, as a disc jockey would, using crossfades, beat-matched transitions, and generative media splices (for example, synthesized bridges, risers, drops, or filter sweeps) to mark affirmation-envelope transitions, such as the opening of a new window, ascent to a higher envelope, or the approach of expiry, and to deliver prodromal cues that announce, within the music, that a reaffirmation will soon be needed. A participant may pre-register a particular musical artist, track, or transitional motif as his or her own signal of a need for reaffirmation; for example, during a session of electronic music, a mix-in of a popular track by an artist such as Fatboy Slim, or of another artist or motif the participant has chosen, signals that an affirmation check-in is due. Such a cue is heard by everyone in the room, but its meaning is known only to the participants who registered or were told of it, and it supplements, rather than replaces, the private reminder and the lapse fade. If T_a expires without an affirmation from every participant (a lapse), the controller enters LAPSE 306: according to the active policy the media is faded toward silence, faded to a floor volume, ducked, or muted over the fade period T_f; or the lapse may be addressed to particular participants (targeted to one participant, directed first to the more vulnerable party, or rotated round-robin among the participants) or signaled by a music mix-in known to one or both of the parties; and a soft, nonalarming cue (a change in the music's envelope, a gentle chime, or a haptic tap on a wearable) invites re-affirmation. A fade rather than an abrupt stop is used at this stage because the most common cause of a missed affirmation is simply that the participants were quiet, and the fade gives them an unmistakable but unhurried signal to say their go word. Affirmation from all participants during the fade restores full playback and returns to PLAYING.

[0157] If the fade completes without affirmation, the controller enters CHECK-IN 308. The media is now softened, muted, or silent according to the policy, and the controller watches for interaction rather than for affirmation. If no interaction is detected for the check-in period T_c, the controller issues a spoken check-in in the participants' language, for example: "Is everything OK? Say your go word to keep going, or say your close phrase to end the session." The check-in is issued by alert output subsystem 134 and is repeated every T_r at increasing intensity and progressively through additional channels: the spoken prompt through speaker 912 or the room's smart speaker; a haptic pattern on each participant's smartwatch, wearable 117, or phone over local link 118; a change in the color of lighting device 115 (for example from the session color to amber) and then a pulse; and, on a phone or watch, the display 916 wakes and shows the same question with large touch targets. A participant may configure that the check-in reach only his or her own wearable first, privately, before any room-audible prompt. Any interaction (speech of any content, movement, a touch) halts the escalation and resets the check-in period, but does not restart the media. Only a verified affirmation returns the controller to PLAYING, and only the close phrase (or a touch confirmation) transitions to CLOSED 330. This distinction between interaction and affirmation is important in this embodiment: the system does not restart intimate-session music merely because someone moved, but it also does not alarm merely because the participants have gone quiet while plainly present. Lighting device 115 may be an overhead smart bulb, a dedicated color LED panel, or a light strip, with or without feedback for clarity and with or without touch sensitivity, and it changes the ambient light in a notable, easily visible way whose hue, saturation, or color represents the state of consent at that moment as assessed or inferred from the signals in the room. An affirmative, enthusiastic consent signal from one or both participants may shift that color at once, for example from a color trending toward orange or yellow, or deeper into red, back to green or dark green. An inexpensive, older smartphone of any make, mounted at the headboard or on a wall in a central place that both participants can see, serves the same purpose, so that participants in flagrante, while they are intimate, can glance at it, or glance back at it, and see the color of their consent at that moment. While a phone serves in this display role, its cameras are not used: each camera may be covered with opaque tape or another privacy shield, closed by a mechanical shutter, or disabled electronically by the controller, by the application running the display, or by the operating system, so that the device can show the room its color without being able to see the room. The display, whether a light bar, a square or rectangular panel, a bulb, an older phone, or any other compatibly connected

device, need not run the controller itself: it may be activated and driven remotely by a smart speaker, by the current smartphone of one of the participants, or by whichever device is serving as the main driver of the consent safety system that the participants have jointly agreed to or that one participant has activated on his or her own initiative in Solo mode.

[0158] If no interaction is detected for the alarm period T_l, the controller enters ALARM 310. Alert output subsystem 134 plays an alarm tone through speaker 912 and every networked speaker at a volume intended to be heard by the participants and by anyone in adjacent rooms; drives lighting feedback device 115 to flash red; drives haptic actuators 914 of wearables 117 and phones continuously; and the check-in message is replaced by an explicit warning: "No response for [n] minutes. I will call for help in [m] minutes unless someone, or [participant name], says [close phrase] or [OK phrase]." Any interaction returns the controller to CHECK-IN 308 and resets the alarm timer. The alarm stage serves two purposes: to rouse a participant who may be asleep, intoxicated, or otherwise unresponsive, and to attract the attention of others nearby before any external call is made.

[0159] If no interaction is detected for the help period T_h, the controller enters HELP 312 and calls for help through a pre-configured escalation sequence of pathways, each tried in turn and each logged: (a) a message and automated voice call to one or more trusted contacts designated at enrollment, reporting an unattended, unclosed consent session with the last known consent state, the time of last interaction, and the location; (b) a request to a peer emergency-response network, such as a neighbor-response platform, asking a nearby vetted responder to check on the location; (c) a loud, repeated spoken help stage from the device itself, such as "Help is needed at [apartment or room], please check on the occupants," intended for neighbors or building staff; and (d) invocation of the emergency dispatch communications link 150 of the '301 patent, transmitting geolocation, the disambiguated room or floor information collected under the '301 patent, and an incident summary classified as "no-interaction / unattended session" so that dispatchers can distinguish it from an active stop-word emergency. The order, inclusion, and timing of these pathways are set by the participants' emergency preferences under the '301 patent, including the silent-alarm and flight/fight/freeze/fawn preferences described there. An "OK phrase" spoken during HELP cancels pending pathways and returns to CHECK-IN; the close phrase cancels and transitions to CLOSED; and every cancellation is logged with the pathways already invoked so that trusted contacts or responders can be stood down.

[0160] Help-is-coming motif. Where the policy enables it, the help stage is accompanied by an audible acknowledgement loop. When a responder accepts the call for help through the peer-response platform, or a trusted contact acknowledges it, the controller fades or stops any media content and renders through the speaker a short pre-enrolled melodic motif (by default four to eight notes, distinct from any track in the participants' library) repeated at an interval until the responder's arrival is verified at the door, at which point the controller renders a changed form of the motif (for example, the same figure resolved to its tonic or played in a higher register) and announces who is being admitted. The same motif is enrolled as a vocal cue under section 3A: the expressed-channel analyzer extracts a pitch track from the echo-cancelled microphone signal, converts it to a sequence of pitch intervals and inter-onset intervals so that the match is independent of key and tempo, and compares the sequence with the enrolled template by dynamic time warping, using a threshold set at enrollment from the participant's own two or three renditions. A match is treated as a request for help that advances the escalation sequence to the help stage regardless of the stage then in force, and the controller immediately renders the motif back through the speaker as an acknowledgement, so that a participant who cannot speak a word or reach a device knows that the request was received. A community or institution may adopt a common motif for all its members, so that the call and the answer are recognized by anyone within earshot; the controller treats any enrolled motif identically, and the event log records that the motif was recognized and echoed, and when, without recording the audio.

[0161] Responder entry. Where the participants' policy designates one or more authorized responders and an access device, the help stage may also open the door. The controller first alerts the designated responders in the order configured, and each alert carries a one-time acceptance link or code; when a responder accepts, the controller tracks the responder's approach by the responder's own device location (shared only for the response), by the building's intercom or doorbell camera, or by a code the responder enters at the door, and verifies at the door that the person present is the accepted responder, by the one-time code, by a credential the responder's device presents over near-field or Bluetooth to the access device, by the building's access-control system confirming the responder's own badge, or by a live confirmation from a monitoring operator. Only then does the controller command the access device to unlock, for a

time-limited window (for example, 60 seconds) and, for a building, along a path (street door, elevator floor, apartment door) that the building system supports. At the same time the controller announces inside the room, through every chaperone device and speaker, who is being admitted and why, so that a participant who is able to respond can say the OK phrase to cancel the grant before it takes effect, or the close phrase to end the session, and so that the arrival is never a surprise and permission to enter is clear. This matters especially here, where one or both participants may need about 30 seconds to dress, yet speed remains essential: the responder's prompt arrival serves the mutual security of the participants, provides conversational refereeing and support, aids memory while events are fresh, and serves a secondary witness purpose regarding the aftermath, including whether everyone, from the moment the active bystander arrives, conducts and acquits himself or herself honorably and compassionately. A concerned neighbor arriving as a responder works to keep everyone calm and to de-escalate and, if each participant is willing to share it, hears a brief version of each person's account and experience soon afterward, with the participants separated or together as best practice indicates. The grant, the verification method, the responder's pseudonymous identifier, the time of unlock, and any cancellation are written to the event log, and the access device's own log is referenced where it has one.

[0162] The entry grant is governed by the policy engine with these constraints, which cannot be relaxed by the other participant under the protective merge: entry is available only at the help stage, never at the check-in or alarm stage; it is available only to responders designated in advance by the participant whose policy it is, or who meet a qualifying emergency-responder level for entry under local policy and law, as lawfully permitted or as an emergency requires for that entry, each with the stage at which that responder may be alerted and whether that responder may be admitted; a responder who is designated for alert only is never admitted; a responder is admitted only after verification at the door; the window is time-limited and the lock re-secures after it; and a participant may designate a tiered order—for example a flatmate first, the building's front desk second, a peer responder third (for example, through the Neighbor 911 peer-response platform, which has patent-pending technical systems, including those of U.S. Provisional Patent Application No. 64/016,800, that support some of these dynamics), and EMS, fire, or police fourth—so that the nearest trusted person who is near enough is tried, according to the participant's preferences, before, while, or after a more or less distant qualified professional or Samaritan arrives as a stranger. For a participant who lives alone, the designated responder may be a neighbor who holds no key, which is precisely the case the entry grant serves: the responder need not possess a key, the controller supplies entry only when the system has decided help is needed and the responder has been verified. The participant may also require that entry be granted only if a second signal corroborates the no-interaction condition (no motion from the bed sensor or radar, no wearable heart-rate change) so that a working door is not opened on a silent but well participant.

[0163] Audibly verified knock. As a further verification at the door, available alone or in combination with the others, the policy may require that the responder knock, and that the knock be verified acoustically before the access device is commanded. The responder is told, in the acceptance message, a knock pattern to use (a pattern chosen for the session, for example three knocks, a pause, and two knocks, or a pattern the participant enrolled for a specific responder) and the controller listens for it through the microphone of a device on the inside of the door or in the room: a user device, a smart speaker, an attending device in the chaperone role, a doorbell or intercom microphone, or a dedicated sensor on the door itself. The controller verifies the knock by three tests: that the pattern of impacts matches the expected pattern within a timing tolerance; that the impacts are consistent with a knock on that door rather than a recording or a sound from elsewhere, by comparing the impulse signature (its spectral shape and decay) to a signature the controller learned for that door at setup or during the session's first prompts, and, where an accelerometer or contact sensor is mounted on the door, by requiring that the vibration channel agree with the microphone; and, where a camera-free presence sensor covers the door, that a person is present at the door at the moment of the knock. The expected pattern is never spoken aloud inside the room, so that a person inside cannot learn it and an aggressor cannot stage it. A knock that fails verification is logged, the responder is told to try again or to use another verification method, and after a configured number of failures the controller falls back to the other door-verification methods or, if none is available, declines entry and tells the responder so. The knock may be required in addition to the code, credential, or operator confirmation, which binds the responder's identity to the responder's physical presence at the door at that moment, and the controller may require the knock to be repeated when a building path has more than one door. After a verified knock, the controller may hold the door for a wait of a duration specified by the policy before unlocking it. The wait is a courtesy: like a

moment in a changing room, it gives each participant a pause in which to re-dress and to compose his or her face and attitude before the responder enters. The wait is announced inside the room and to the responder, and it may be shortened or waived by a participant, or by the policy, when the emergency requires it.

[0164] Location proximity. Where the responder has shared the location of his or her own device for the duration of the response, the controller requires, as a further condition of entry, that the responder's reported location be within a configured distance of the dwelling unit (for example, 25 meters of the building entrance, or on the correct floor where the device reports altitude or the building's access system reports the elevator stop) at the moment the door-verification succeeds, and that the location track show an approach consistent with travel from the responder's location when the alert was accepted. A location that is absent, stale, implausibly distant, or inconsistent with the knock or credential event may cause the grant to be withheld pending another verification. Location is used only for the response, is shared only with the controller and, where the policy permits, with the participant's trusted contacts, is coarsened to the precision needed, and is not retained after the response is logged.

[0165] Precise bedroom location, elevation, and wayfinding. Because help that cannot find the room arrives too late, the system registers, and keeps current, a precise and dispatchable location for each bedroom in which it is used. At registration, the participant scans a room tag affixed in the bedroom, such as a near-field communication (NFC) tag, a printed QR code or other optically readable code, or an ultra-wideband or Bluetooth beacon, and the controller records, with the participant's consent, a dispatchable bedroom location comprising the civic address, building and unit identifiers, floor, room identifier, and wayfinding instructions from the building entrance to the room (for example, the entrance to use, the elevator or stair, the turn from the landing, and the position of the door), in a form consistent with local standards for uniquely identifying a location and directing emergency responders to it, such as the dispatchable-location and civic-address conventions used for emergency calling where they apply. The controller confirms the location on later sessions by a fresh scan of the room tag or by position estimates: elevation from the device's barometric pressure sensor, calibrated against a reference where one is available, so that the correct floor can be identified; and horizontal position from satellite positioning fused with nearby wireless signals, such as overlapping Wi-Fi networks, Bluetooth and ultra-wideband anchors, and low-power community networks such as Amazon Sidewalk where available. Significant movement of a device away from the registered location prompts the participant to confirm or update the location before a session is armed. The floor may also be determined by barometer-based calculation alone, from the pressure difference between the device and the ground-floor or street level, and narrowed further by any other beacon signals available in the building, such as floor-specific Bluetooth, ultra-wideband, or Wi-Fi access points, or the elevator stop reported by the building's systems. The controller also recognizes when a fixed device, such as a smart speaker or bedside device, has been moved from one bedroom to another or from one dwelling to another, by changes in the set of available network and beacon signals, and prompts the participant to update the registered address and room before the device is used again. At periodic intervals, for example semi-annually or annually, the controller reminds the participant to confirm that the registered address and wayfinding instructions are current, and it monitors the batteries of any battery-powered sensors, buttons, tags, and other devices in the bedroom environment, reminding the participant to replace a battery that is running low before the device fails. Where a community so provides, and only with the consent of the residents or as the law requires, the registration may be verified by periodic inspection, for example annually, by a private entity, an advocacy organization, or a strictly privacy-focused government inspector associated with the sexual violence prevention function of local government, who confirms that the room tag and the registered wayfinding instructions match the dwelling, without collecting anything about the residents' private lives. The call for help carries the dispatchable bedroom location and wayfinding instructions to each responder permitted under the policy, whether neighbor, peer, friend, family member, or official emergency responder, as permitted, consented to, or required. The inventor hopes for a world in which every door in every community that leads into a bedroom carries a small NFC tag. Such tags are extremely inexpensive, typically costing less than 25 cents each, and each one encodes the location of that bedroom and minimal wayfinding information for reaching it in an emergency, and nothing about its occupants. Whenever a person walks into or toward a bedroom, whether with a new potential lover, with a person of a matching sexual orientation, or in any situation in which interest may be mutual or one-sided, that person can scan the tag with a phone or another mobile device, such as a watch, and become fully aware of where he or she is. In an emergency, that person's device can then send his or her precise location, and the way to reach it, to anyone whom a

peer-response service such as Neighbor 911, or a next-generation emergency-calling (NG911) system, may make available to help in a moment of need.

[0166] Scope of the grant and alternatives to a commanded unlock. The grant may apply to an exterior access device (a street or lobby door, a gate, or a garage), to one or more interior access devices (an apartment door, a bedroom door, or an interior gate), or to a path through several, each commanded in turn as the responder's verified presence advances. Where a door has no access device the controller can command, the grant instead carries instructions: the acceptance message, or a spoken message at the door through a chaperone device or intercom, tells the responder how to enter: the location of a lockbox and a one-time code the controller sets or reveals, the location of a spare key the participant has disclosed to that responder, a building entry code, or the name and extension of the front desk. Where a key is held by a trusted neighbor, the controller may alert that keyholder as a responder in his or her own right, or may alert the keyholder to meet and admit the primary responder, so that a professional responder who has never visited and a neighbor who holds the key arrive together, and the keyholder's presence at the door is itself logged as corroboration. The participant designates, per door, which of these applies, and a building operator may pre-configure the exterior path for all units.

[0167] Relationship to a peer emergency-response platform. The responder alerting, acceptance, location sharing, approach tracking, and keyholder coordination described here may be provided by the peer-response platform of the applicant's related application, in which the help stage of the escalation sequence raises an event on that platform, the platform selects and alerts nearby responders according to its own rules, and the platform returns to the controller the identity, acceptance, and location of the accepting responder for the door verification and entry grant described here. The present disclosure adds, to any such platform, the consent-session context in which the event arises, the verification of the responder at the door of an intimate space, the announcement inside the room, the participant's ability to cancel, and the constraints of the protective merge.

[0168] From any state, a qualifying stop word, safe word, qualifying semantic phrase, button press, or distress indicator bypasses the escalation sequence and transitions the controller to WITHDRAWN/EMERGENCY 320. The controller issues an immediate pause directive (through the fastest available adapter, in parallel with a fallback adapter), suppresses any pending play directive, and invokes the emergency response mechanism of the '301 patent with the current joint state, the triggering event, and its time stamp. Cessation of music, or the introduction of emergency-coded, safety-themed music or of calming, commanding, or de-escalatory signals or music, at this moment is abrupt rather than faded, so that it cannot be mistaken for an ordinary fade; the two acoustic behaviors (fade for absence, cut for withdrawal) are themselves a signal that both participants learn to read, alongside audio and visual directives on a plurality of present devices.

[0169] From WITHDRAWN 320 the controller may, according to the participants' enrollment preferences, play a pre-agreed de-escalation track at low volume (distinct from the session media and never indicating consent) to accompany the separation protocol of the '301 patent. Resumption of session media from WITHDRAWN, CHECK-IN, ALARM, or HELP is never automatic and always requires a fresh mutual affirmation as at session start. From WITHDRAWN, from a brake, or from any other state in which the policy calls for it, the controller may also issue stage instructions to either or both participants, aloud or privately, directing each to go to his or her own phone and interact with a compatible application or with the system's own first-party application. There each participant is shown his or her role from that point and how he or she can best serve the partner, the partnership, and the community in which they live, by working to resolve the situation in the gentlest and simplest way possible; is given the opportunity to record his or her own understanding of what happened, privately and in his or her own words, in the event log or the evidence locker; and is offered the choice, depending on the situation and subject to the other participant's choice and to any safety requirement, either to separate for the night or to spend the rest of the night together.

[0170] False-alarm mitigation. Because an unwarranted call for help during an intimate encounter is itself harmful, the escalation sequence is designed so that a help stage reached through a lapse ordinarily requires a conjunction of conditions: the session was opened and never closed; the media stopped for lack of affirmation; every check-in went unanswered; the alarm went unanswered; and no interaction of any kind (including motion sensed by the device or wearable) occurred throughout. Participants may additionally require that the controller confirm continued presence by short-range ranging (for example, ultra-wideband or Bluetooth proximity

between participants' devices) before escalating past CHECK-IN, so that a session left open after both participants have simply left the room resolves to an auto-close with a notification rather than an alarm. The controller also recognizes ordinary session endings that omit the close phrase (both participants' devices leaving the room, a phone being plugged in and left idle for a configured period, or an explicit sleep mode) and treats them as a soft close that issues a single notification rather than an escalation. Conversely, participants in a high-risk configuration may shorten every period and skip the alarm stage so that help is summoned silently.

[0171] Wind-down and the happy-path close. Participants who have finished an encounter happily, perhaps after one or both have reached orgasm, often forget to close the session, and the next lapse of the affirmation window would then begin a fade, a check-in, and in time an alarm, at the very moment they have turned to pillow talk. The controller therefore recognizes a wind-down state. After a session has reached a high stage, wind-down cues (a declared finish such as "that was amazing" or an enrolled close word, relaxed conversation and laughter in place of intimate cues, a heart rate that peaks and then declines steadily on a wearable, and stillness of both participants reported by a non-imaging sensor) move the session into wind-down, in which the affirmation window is suspended, the media shifts to an aftercare treatment, and the controller asks once, softly, whether to close the session; an affirmative word, a tap, or simply continued calm conversation closes it or holds it in wind-down until it closes. The session may also be closed, cryptically or openly, by an enrolled ordinary phrase ("goodnight", "want some water?"), a triple press on the remote or the panic button, switching off a connected light, placing a phone on its charger, or the detection by a wearable or a non-imaging sensor that both participants are asleep.

[0172] Afterglow grace and brief absence. A participant may also leave the room after intimacy, for example to urinate to reduce the risk of a urinary tract infection, without having closed the session, and in Solo mode may be the only person who knows the session is running. If the participant carries a wearable or the phone, walking, a normal heart rate, and device ranging all count as interaction, so that escalation halts; the check-in is delivered privately to the participant's wrist or phone, and a single tap answers it or closes the session. If the participant has left every device in the bedroom, the controller recognizes the departure of the enrolled participant after a high stage, in the absence of any risk indicator, and grants an afterglow grace period, by default ten to fifteen minutes and configurable, during which escalation advances no further than a private check-in on the participant's own devices and nothing audible is made in the room. If the participant returns and interacts, the session passes into wind-down. If the grace period expires without a response and without any risk indicator, the controller closes the session softly rather than sounding an alarm, announces on the participant's own device that the session was closed and may be reopened, and records the reason; a participant may elect that such a closure also send a brief, neutral message to a trusted contact. A participant may enroll a post-intimacy routine ("I usually get up afterward") that sets the grace period in advance, and the controller may learn the routine from the participant's own history of closures, within bounds that never shorten any period below its safety minimum.

[0173] Guardrails on wind-down. Because leaving the room is also what a person fleeing does, departure alone is never treated as safe. Any risk indicator ends the wind-down state or the grace period at once and restores the ordinary escalation sequence: a stop or brake earlier in the session, crying or a distress vocalization, an impairment flag under section 6A, a secret concern cue or duress affirmation, an elevated heart rate on the departing participant's wearable, or a stop from any participant. Wind-down cues are credited only from the participants whom the session protects: in Solo mode only the enrolled participant's own cues, at the verified attribution level where available, and in Duo mode the cues of both participants, so that one person cannot talk the safety layer into standing down. The event log records whether each session was closed by phrase, by wind-down, by expiry of a grace period, or by detected sleep, and the participant's private record keeps a history of false alarms and near-misses from which, within the bounds stated, timings may be tuned to that participant's habits.

[0174] The in-room alarm as the participant's informed choice. Whether the alarm stage sounds in the room is, in Solo mode and wherever a participant elects it in Duo mode, the enrolled participant's own choice, made at enrollment and changeable at any time, and the controller informs that choice with the evidence available about likely outcomes. On one side, research on self-protective action by victims of attempted rape has found that most forms of resistance, including making noise and calling for help, are associated with a lower likelihood that the rape is completed, and that injury, where it occurs, more often precedes the resistance than

follows it (Tark and Kleck, 2014). On the other, many women fear, and the literature on domestic violence and witness intimidation recognizes, that an alarm may provoke an outlier violent person who fears exposure to retaliate on the spot, adding assault or worse to what has already been done in an attempt to intimidate the person who raised it. The controller therefore offers several options: an audible alarm in the room; a silent alarm that invokes the help pathways only; an alarm audible only outside the room, through a hallway or neighboring device; and a help-adjacent alarm, whose audible output in the room is deferred until a responder has accepted the call for help and is reported to be within a configured distance, so that the alarm sounds when help is moments away and the window for retaliation is shortest. As the community data of section 7A accumulate, the controller may present aggregated outcomes for each option. The inventor's own recommendation is the audible alarm, but the choice belongs to the person whose safety is at stake.

[0175] Voice relay (the megaphone). Among the emergency conditions that a participant may select in the non-affirmation policy (for example, a stop word, the brake, an emergency-mode concern cue, a duress affirmation, or a button pattern), the participant may attach a voice-relay property. When a condition carrying that property is triggered, the bedside device or phone, in addition to stopping the media abruptly and taking any other response the condition defines, opens a relay in which it repeats only what is said by the participant who triggered the condition, through the loudspeaker, at an elevated and unmistakable volume, to the other person in the room. The triggering participant is identified by the input that raised the condition: by the voiceprint attribution component at the verified or probable level where the condition was spoken, or by the enrolled device, wearable, or button that raised it. Speech attributed to any other voice is never relayed, so that the relay cannot be turned against the person it is meant to serve. Whether that speech is also transcribed or retained as evidence is not decided by the relay; it is governed, neutrally and like all other speech in the room, by the conflict-recording settings described in section 8C, including the default-on conflict recording of Duo mode and the jurisdiction table that governs Solo mode. The participant elects in advance one of three relay forms: verbatim, in which the recognized words are repeated as spoken; paraphrase, in which they are restated in plain, direct language; and summary, in which a sustained utterance is condensed to its operative request (for example, "Stop has been said. Stop now and step back."). By default the relay is rendered in a synthesized voice rather than as a replay of the captured audio, so that the participant's own voice is not itself broadcast, and each relayed utterance is preceded by a short distinctive tone so that it cannot be mistaken for ordinary speech or for media content. Where the recognizer's confidence is low or a phrase is ambiguous, the relay offers optional support for clarification: it asks the triggering participant a short confirming question ("Did you say stop?" or "Do you want help to come?"), privately through an earpiece, wearable, or phone where one is available and otherwise aloud, relays only what is confirmed, and treats a confirmed request for help as advancing the escalation sequence to the help stage. The effect is that of having a sister or brother, or a supportive, loyal, and protective friend, in the room in a pinch: someone who hears the participant, repeats the participant's preferences, needs, and commands loud and clear, and helps make them understood. Its activation is not dispositive, in any way, of who was right or who was wrong: the relay is support for the person who called on it, not a finding about the encounter, and nothing in its activation or its output is to be treated as establishing fault. What the relay does is answer the immediate need of a person who believes the encounter has just entered the grey, yellow, or red zone to be heard, and to know that he or she is being heard. It can also help the counterpart realize, faster, that this is not a moment that calls for more attempted physical stimulation, or for another, different attempt at arousing the person toward the sexual act in question, and that it may make more sense to give her or him space, get dressed, apologize profusely, and go home, or somewhere else, for the night. The lesson, as with clothing on fire, is to stop, drop, and roll: to stop, to step back, and to pay heed. A counterpart who does not may find that the person who invoked the megaphone only grows louder, and that the hammer of punishment may come, unless the counterpart takes special care to cause no further harm and does not attempt to debate his or her way back toward the act to which he or she was pressuring that person. For the inventor, the idea goes back to Frank Herbert's Dune novels, in which the Bene Gesserit train in the Voice, a controlled use of tone and timbre that commands attention and compliance, alongside the weirding way, their discipline of combat (rendered in the 1984 film adaptation as "weirding modules" that amplify the voice). The relay compels no one and is not meant to; but there is no reason why women, and men who are assaulted, should not have access to a version of the Voice when they need it, within reasonable limits, as a mode of emphasis and self-protection in a confusing or uncontrolled situation. Informally, the voice relay serves as an anger interpreter, a role made familiar by

the "Obama's Anger Translator" sketches of the comedy series Key & Peele, in which a companion restates a restrained speaker's measured words with the force and clarity they deserve. Here the role is applied, without the comedy, to a sexual assault context, where amplifying a person's voice can be decisive in getting the message across fast and clear. The relay remains open, repeating only the words of the participant it was first activated to protect, until the participants separate (as indicated, for example, by a door sensor, by presence sensing, or by the departure of either participant's device from the room) or that participant deactivates the feature. The event log records that the relay was opened, in which form, how many utterances were relayed, and when it closed; whether the audio, a transcript, or the words relayed are retained is governed by the participant's election and by the conflict-recording settings, not by the relay.

[0176] Guided configuration and rehearsal. The software and hardware of the bedroom tool suite described herein (the stop words, brake, concern cues, duress affirmation, button patterns, alarm election, voice relay, help pathways, motif, stages, and boundary map) may be configured to a user's or a couple's preferences through a deterministic artificially intelligent or superintelligent chat or interactive session, before anything has happened and well before any crisis. The session is deterministic in that it runs a fixed, versioned model with fixed decoding (for example, greedy decoding or a recorded seed), or a fixed question tree whose wording alone is generated, so that the same answers always produce the same configuration and the configuration can be reproduced and audited later. The session asks each participant, in plain language, what each emergency condition should do; asks privately, on that participant's own device, about anything the other participant is not to see; compiles the answers into the non-affirmation policy; and reads the result back in plain language for each participant's approval before it takes effect. The session then offers practice rehearsals in a labeled drill mode, in which the participants speak their stop words softly, whispered, and in a distressed voice, press their button patterns, trigger the voice relay, and hear the alarm and the help-is-coming motif at a reduced volume, while the help pathways are simulated rather than dispatched and any trusted contact who has agreed to take part is told that the message is a drill. The rehearsal measures how reliably each cue is recognized for each participant, suggests adjustments to thresholds, cues, or relay forms where recognition is weak, and may be repeated at intervals, so that what is to happen in an emergency has been chosen, heard, and practiced in calm conditions rather than first encountered in a crisis. The session may run on the user device, or in a private or incognito mode under the privacy kernel, and retains nothing beyond the approved configuration and the rehearsal results the participants elect to keep.

[0177] Timer parameters may be adjusted by the response mapper according to the dynamic consent score: a sustained high score may lengthen T_a within a bounded range to reduce prompting; a falling score shortens it. Participants may also configure a quiet interval during which R4 implied refreshes are accepted so that the system does not demand speech at moments when speech is unwelcome, subject to the limit N.

[0178] Reference implementation of the affirmation envelope. The controller may be implemented as an event-driven state machine scheduled on a monotonic clock, so that its timing is unaffected by changes to the wall clock and its behavior can be reproduced from the event log. In one embodiment it executes the following handlers on processor 902, in which every timer is a deadline on the monotonic clock, every directive to the media playback target is confirmed against the player state returned by the media control adapter, and a watchdog restarts the controller, with the media paused, if any handler fails to complete within a bound:

on session start: state := ARMED; media := paused; log(start).

on affirmation from every required participant within the mutual window: state := PLAYING; deadline_a := now + T_a; play(); verify(playing).

on affirmation(p) in PLAYING or LAPSE: credited[p] := now; if every required participant has been credited since the last reset: deadline_a := now + T_a; if state = LAPSE: restore volume and state := PLAYING.

on now >= deadline_a in PLAYING: state := LAPSE; fade(T_f, policy media response); soft cue; deadline_f := now + T_f.

on now >= deadline_f in LAPSE: state := CHECK-IN; hold media at the policy floor; prompt; deadline_c := now + T_c.

on interaction in CHECK-IN: deadline_c := now + T_c (hold); on interaction in ALARM: silence alarm; state := CHECK-IN;

deadline_c := now + T_c. Interaction never resumes the media; only affirmation from every required participant does.

on now >= deadline_c in CHECK-IN: state := ALARM; alarm(per the participant's alarm election); deadline_l := now + T_l.

on now >= deadline_l in ALARM: state := HELP; call for help through the configured pathways in order; motif where enabled.

on brake, in any state: state := BRAKE; fade(Tf); run the brake script of section 3B; log(brake, attribution level).

on stop word, concern cue, duress affirmation, distress indicator, or button pattern, in any state: state :=

WITHDRAWN/EMERGENCY; issue pause through the fastest adapter in parallel with a fallback adapter, completing within T_{stop} <

T_f; suppress any pending play directive; apply the response bound to the condition (including voice relay where elected); log(event, attribution level, model identifiers).

on adapter error, loss of player state, failed verification, or watchdog expiry, in any state: fail closed by pausing or muting through any available adapter; log(fault).

on close phrase from every participant: state := CLOSED; log(close).

[0179] Technical character of the envelope. The affirmation envelope is not a rule of conduct applied on generic hardware but a specific arrangement of signal processing and real-time control that solves technical problems that an ordinary media player or voice assistant does not. (i) Self-triggering: because the media being gated may itself contain spoken or sung go words, the expressed-channel analyzer cancels the playback reference signal 146 from the microphone input before recognition, so that the system cannot renew its own envelope from its own output. (ii) Inverted keepalive semantics: an ordinary voice-controlled player treats silence as "no change," whereas here the absence of a verified signal by a deadline drives the change of state, which requires deadline scheduling on a monotonic clock, crediting per participant, and fail-closed handling of crashes, adapter errors, and lost player state. (iii) Bounded actuation latency: a withdrawal must silence the media within a stop latency T_{stop}, preferably below 500 milliseconds and more preferably below 250 milliseconds from the end of the utterance, and in every case shorter than the fade period T_f (by default about twenty seconds), so that the abrupt stop and the fade are acoustically distinct; the controller meets that bound by on-device streaming recognition, parallel directives to a primary and a fallback adapter, and suppression of pending play directives. (iv) Closed-loop actuation: each directive is confirmed against the state reported by the playback target and is retried, rerouted, or escalated when the target does not comply. (v) Privacy by construction: recognition, attribution, and timing execute on the device, so that no audio need be transmitted for the envelope to operate. Together these features reduce false renewals and missed withdrawals, bound the time from a refusal to silence, and keep the system in its safe state when its own components fail.

4A. Non-affirmation policy engine (FIG. 5)

[0180] Referring to FIG. 5, the behavior of the controller when affirmative consent is not continuously detected is not fixed but is determined by an active policy 510 produced by a policy engine from each participant's own settings 502, 504 and from a mode preset 506. The engine exists because the correct response to silence differs radically between participants and situations: a couple in a long relationship at home may want a gentle fade and nothing more; a first encounter in an unfamiliar place may warrant a short window, an audible escalation sequence, and a trusted contact on call; a participant who fears that an audible alarm could provoke an aggressor may want no audible change at all and a silent call for help. The policy dimensions 512 are enumerated below; each is independently settable per participant, and presets supply coherent defaults based on evolving public-safety and health research in this domain, including research into the user experiences that such technology systems, and their fused sensors and devices, allow persons to have.

[0181] Dimension A: what counts as affirmation, and from whom. Options include: an enrolled go word alone; a go word enrolled at the enthusiastic tier (enthusiastic consent); a go word graded by the speaker's own elected tone mode (OFF, PRIVATE, or GRADED); implied refresh by laughter or positive non-lexical vocalization, with a limit N; an enrolled gesture detected by a camera or wearable; a press of a physical or on-screen button; a double-tap on a wearable; a momentary biometric touch affirmation (section 4D) on a fingerprint sensor, wearable physiological sensor, or secure device unlock; a touch pattern on a shared device; and, for participants who cannot or prefer not to speak, any combination of the foregoing. Separately, the policy specifies whether the affirmation timer is reset by affirmation from all participants, from any participant, or from a named participant (for example, the participant who identified himself or herself at enrollment as the one whose consent is at issue), and the length of the mutual window within which the required affirmations must arrive.

[0182] Dimension B: timing. Each of the affirmation window, fade period, check-in period, repeat interval, alarm period, and help period is settable; each may be made adaptive to the dynamic consent score within bounds; a quiet interval may be defined during which only implied refresh is expected; and time-of-day or location rules may select among stored timing sets.

[0183] Dimension C: media response on lapse. Options include: fade to silence; fade to a floor volume and hold; duck by a fixed amount and hold; crossfade to a designated prompt track whose meaning both participants have learned; switch to the designated "yellow" track; step down one playlist tier; stop immediately without fade; stop and clear the queue; change only the lighting device 115 while music continues; deliver only a haptic cue to wearables with no audible or visible change (a discreet policy); and no change, in which the lapse is logged and the escalation sequence proceeds without any media effect. Options may be assigned per stage, so that, for example, the lapse produces a fade, the check-in produces a switch to the prompt track, and the alarm stage produces a stop. Dimension C also selects the carrier of the ambient signal: media content (the default); a silent carrier as described in section 4C; or both, with the silent carrier continuing when media is paused by the participants for reasons unrelated to consent. Dimension C further maps the semantic classifier's meaning categories and risk-weight bands (section 3A) to media responses, and Dimension D maps them to prompts and to whether a clarifying exchange is conducted, privately or aloud, and which reaction scripts it may offer. Media selection may also serve as an input in the other direction, as a cue for lockdown. A participant may enroll one or more tracks, artists, or playlists as belonging to a safe-word track category, either openly or cryptically (an ordinary-seeming song whose significance only that participant knows), so that selecting such a track, by voice, by touch, or by skipping to it, is treated as a stop or a red alert. According to the policy, the resulting lockdown and emergency sequence may be visible to both persons present; or it may proceed invisibly, with no change in the room, so that the first sign that the call for help was sent out to the community of responding active bystanders or professional responders is a knock at the door some minutes later, preferably in fewer than 3 minutes and typically within 3 to 10 minutes.

[0184] Musical boundaries. Optionally, and only for the softer media responses, the media control adapter layer respects the boundaries of the music being played, so that a fade, a duck, a crossfade, a change of track, or a resumption lands where a listener would expect music to change rather than in the middle of a phrase. Boundaries are of two kinds. Internal boundaries are positions within a track (the end of a bar or of a four- or eight-bar phrase, the start or end of a chorus, verse, bridge, or drop, and the end of a lyric line) obtained from the streaming service's audio-analysis or section metadata where the service provides it, from time-synchronized lyrics, or from on-device analysis of the playback reference signal by beat tracking and structural segmentation. External boundaries are positions between tracks (the end of the current song, the gap or crossfade before the next one, and the start of the next item in the queue) obtained from the player's queue and track-duration state. From these the controller maintains a short list of upcoming boundary times. When the affirmation timer is about to expire, the controller schedules the lapse response to begin at a boundary rather than at the instant of expiry: it may begin the fade at the last suitable boundary before expiry, so that, for example, the music thins out over the final bars of a chorus; or it may defer the start of the fade to the first suitable boundary after expiry, but by no more than a maximum deferral set in the active policy (by default a few seconds, and never more than a small fraction of the affirmation window) after which the fade begins wherever the music is. An affirmation received during a deferral cancels the pending fade as if it had arrived before expiry, and a deferral is never counted toward any escalation period. The same scheduling may align a

crossfade to the prompt track or the yellow track to a beat, matching tempo and beat phase so that the transition is mixed rather than cut; may hold back the start of a new track after a lapse so that the next song does not begin into a lapsed state; may place the renewal reminder of section 4 a set number of bars before a chorus ends, so that renewal and the return of the chorus coincide; and may hold a resumption after renewed affirmation until the next downbeat, phrase start, or track start, again within a bounded wait. Boundary alignment never applies to a stop word, a safe word, a brake, a distress indicator, a button press, a secret concern cue, or any stage of the escalation sequence beyond the lapse: those take effect immediately wherever the music is, and the abrupt, unaligned cut is itself part of what tells everyone in the room that something other than a lapse has occurred. The deferral applied, the boundary type, and its source are recorded with each media directive in the event log. The music may also be chosen to fit the envelope, not merely trimmed to it. The response mapper may select and sequence songs by their duration, or by the positions of their internal boundaries, so that a song, or a chorus or other section within it, ends at or shortly before the expiry of the current affirmation window, making the turn from one song or section to the next the natural moment for renewal; because each affirmation resets the timer, the controller re-plans the upcoming queue from the time remaining whenever the window is renewed. As the window lengthens or shortens over a session or a relationship (with trust under section 4G, with recency under section 4F, or with the score), the selection follows, so that a short first-session window is matched with short songs or with sections that turn over every minute or two, and a longer earned window with longer tracks. For any period of the encounter the participants designate (a slow opening, the main session, a cooldown after a brake, aftercare), the selection may be constrained to the durations, tempi, and boundary spacings the participants associate with that period. Where a generative or editing model is available, a track may be composed, extended, or shortened, for example by repeating or omitting a chorus or adding a bridge, so that its boundaries fall where the window requires. Song selection is an aid to renewal and never a substitute for it: it does not lengthen a window, and it does not delay a lapse response beyond the deferral bound stated above.

[0185] Dimension D: prompting and renewal reminders. Options for the renewal reminder include: none; a single private haptic at lead time T_n; a two-stage private haptic at T_n and a shorter lead; a receipt cue on each credited affirmation; adaptive T_n learned from the participant's rhythm; and delivery to a chosen device (including, without limitation, a watch, phone, earpiece, ring, bedside accessory, other wearable, robot attendant, or other attending device). Options for check-in and later prompts include: no prompt; haptic only; visual only (screen or lighting); spoken; spoken and escalating in volume; prompts addressed to all participants through the room speaker or privately to one participant through a wearable or earpiece; the language, voice, and wording of prompts, including participant-authored wording; the number of repetitions; and whether the prompt names the time remaining before the next stage.

[0186] Dimension E: escalation sequence shape. Each stage of FIG. 3 (lapse, check-in, alarm, help) may be enabled or skipped; stages may be reordered within safety constraints (a help stage may never precede a check-in unless the policy is expressly a silent-alarm policy); each advancement may be gated on a presence check (ranging between devices, motion, or wearable signal) so that an empty room resolves to a soft close rather than an alarm; a second-sensor gate may require corroboration from a wearable (no heart-rate change, no motion) before the help stage; and a silent-mode flag suppresses every audible stage while leaving the help stage active.

[0187] Dimension F: help pathways. The policy lists the pathways to be used and their order—trusted contacts by name and channel, a peer emergency-response network, the device's own spoken help stage, an emergency SMS option (including text-to-911 where available), integrations with 911 and Next Generation 911 (NG911) systems, with their counterparts in other nations (such as the 112 emergency number used across the European Union and much of the rest of the world, and next-generation 112 services), and with peer-response platforms such as Neighbor 911, and the emergency dispatch link of the '301 patent—together with, for each, whether it is audible or silent, what data is shared (location only; location and last consent state; location, state, and time of last interaction; or a fuller incident summary), whether an automated voice call or a message is used, and whether a pathway auto-cancels on interaction or requires an explicit OK phrase. Each integration applies a privacy model that discloses to each recipient only what that recipient needs for emergency handling, such as the dispatchable location, the nature and urgency of the call, and a callback or relay channel, and withholds sensitive private information unnecessary for that purpose, such as the nature of the encounter, the participants' identities beyond what the responder requires, consent-state history, audio, and the cadence data of section 4F. Where a phone or wearable

offers an emergency-message or SOS capability over a satellite link, that capability is an available help pathway when no terrestrial network is reachable, and the controller composes the incident summary to fit the capability's message limits. Dimension F further designates, per responder, whether that responder may be admitted by the access device at the help stage, the verification required at the door (including whether an audibly verified knock is required, alone or with another method, and the knock pattern or that a per-session pattern be generated), the unlock window, the path through a building's access system, and any second-signal corroboration required before entry. Dimension F also designates whether a location-proximity condition applies and its distance, which exterior and interior access devices are in the grant and in what order, and, per door without an access device, the entry instructions or the keyholder neighbor to be alerted.

[0188] Dimension G: cancel and resume authority. The policy specifies who may halt or cancel an escalation and who may resume media: any participant; all participants; or a named participant speaking in his or her VERIFIED voice. The last option is the mechanism by which a participant who fears coercion ensures that an aggressor cannot silence the escalation sequence by speaking a cancel phrase: the controller honors a cancellation only when voiceprint verification attributes it to the participant who set the policy, and in any other case continues the escalation sequence while logging the attempted cancellation. The policy further specifies a lockout (after the escalation sequence reaches a chosen stage, media may not resume until the session is closed and reopened) and whether resumption after a lapse requires re-affirmation only, re-affirmation at the enthusiastic tier, a repeat of the pre-interaction checklist of the '301 patent, or a cooling-off minimum before reopening. The policy further specifies, per participant, the attribution level a device-based affirmation from that participant's bound wearable must carry (section 4E) and whether an UNBOUND device's inputs are ignored or treated as UNATTRIBUTED. Dimension G further holds the brake script of section 3B: the brake phrase, the calming posture, the settling period and its increments, whether heart-rate settling is required and its band, whether separation is sensed, the form of the talk-it-out step, whether a brake is mandatory before any boundary is renegotiated, and whether a private (unattributed) brake is enabled.

[0189] Dimension H: logging and sharing. The policy specifies whether the log records escalation stages only, or also attribution levels and score bands, or also full records of interactions, safe words, and safe phrases; whether it is held on-device only, mirrored encrypted to server 102, or discarded at session close; and which pathways, if invoked, may receive which fields.

[0190] Policy merge 508. Where the participants' settings differ and two or more participants have relevant settings, the engine merges them protectively rather than by either participant's preference alone: for any timing window the shorter value governs; a stop from any participant is honored regardless of the other's settings; resumption requires whatever each participant requires, so that the union of resume conditions applies; help pathways are the union of each participant's chosen pathways, each carrying only the data its owner authorized; and cancel authority is honored per participant, so that a participant's requirement of his or her own verified voice cannot be relaxed by the other. A participant may also mark certain settings as private, in which case the other participant is informed only that a protective setting is in effect and not of its content. Presets 506 correspond to the consent modes of the '301 patent and add a Discreet preset (no audible change, haptic prompts, silent help stage), a High-vigilance preset (short windows, full escalation sequence, dispatch enabled, verified-voice cancel authority), and a Practice preset (full behavior with all external pathways stubbed).

4B. Voiceprint-attributed affirmation (FIG. 6)

[0191] Referring to FIG. 6, affirmations and other consent-relevant utterances are, where the participant consents and the platform permits, attributed to the individual who spoke them by comparison with that individual's enrolled voiceprint template. The purpose is evidentiary and protective: the record should show not merely that a go word was heard, but that a particular person said it; and policy decisions such as cancel authority depend on knowing who spoke.

[0192] Biometric consent and feasibility check 602. Voiceprint processing is opt-in and is presented separately from general enrollment, with a plain-language explanation that a voiceprint is a biometric identifier, how it is stored, that it never leaves the device in usable form, how it may be deleted, and what the system will do without it. Consent is recorded in the event log with its own time stamp and may be withdrawn at any time, whereupon the template is deleted and the controller falls back to non-biometric attribution.

Independently, the controller determines whether the platform is technically capable: whether an on-device speaker-embedding model of adequate accuracy is available for the device's processor, whether a secure element or equivalent protected storage is available for the template, and whether the microphone and audio path deliver sufficient quality. The controller may run a short self-test at enrollment, measuring the separation between the participant's embeddings and a set of reference embeddings, and refuses to enable voiceprint attribution (reporting the reason) if the measured equal-error rate on that device exceeds a threshold. The feasibility determination is repeated when the device, operating system, or model changes.

[0193] Enrollment 604. The participant speaks his or her go words, safe word, close phrase, OK phrase, and a short passage of free speech, in the environment and at the distance at which the device will be used, and preferably with representative background music playing so that the template reflects real conditions after echo cancellation. Embeddings are extracted on-device, aggregated into a template, and stored encrypted under a key held in the secure element. The template is bound to a pseudonymous participant identifier rather than to a name. It may be synchronized between the participant's own devices only by end-to-end encryption under keys the participant controls.

[0194] Session-scoped template lease 606. In the single-device embodiment, or when one participant's device must verify the other's voice, a template must be present on a device the participant does not own. The controller supports a lease: the template is transmitted to the partner or shared device in encrypted form, over the local link or over a transient or sustained encrypted server link, usable for verification only by the controller's attribution component, marked with the session identifier, and automatically destroyed at session close, on link loss, or on the owner's revocation, whichever occurs first. The lease is itself logged. A participant may decline to lease, in which case the partner's device attributes that participant's utterances at PROBABLE at most, and the policy engine applies whatever restriction the participant has chosen for that case. The lease may instead be relationship-scoped. Where the participant whose template it is consents, a template leased to a partner's device, or to a shared device in a common bedroom, may persist across sessions for the duration of the relationship, so that the partners need not lease their templates afresh each time; it remains encrypted, usable only for verification by the attribution component, and bound to a relationship identifier rather than a session identifier. A relationship-scoped lease continues only so long as that consent continues: it is destroyed when either participant terminates the relationship in the system, when the participant whose template it is revokes the lease or deletes the template, or when it expires. The lease carries an expiry that is renewed only while consent continues, and the controller may ask the participant at configured intervals to reconfirm it, so that a device that can no longer be reached loses the template when the expiry passes. Termination is unilateral and requires no reason, and every grant, renewal, reconfirmation, and termination is logged.

[0195] Verification 608–612. For each detected go word, stop word, close phrase, OK phrase, or cancel phrase, the controller extracts an embedding from the utterance (after echo cancellation), compares it with each enrolled or leased template to produce a match score s, and applies a liveness check. The liveness check draws on several independent signals: the playback-reference cancellation already excludes any recording replayed through the system's own speaker; spectral and channel cues distinguish speech radiated by a human vocal tract in the room from speech emitted by a loudspeaker; consistency between the utterance and a wearable's contemporaneous laryngeal-vibration or breathing signal, when available, confirms the utterance came from that participant's body; and, when the score is borderline or at a policy-defined random interval, the controller issues a challenge (asking the participant to repeat a randomly chosen word that was not enrolled) and verifies the response, defeating replay of a previously recorded go word.

[0196] Attribution level assignment 614. A score at or above the verification threshold with liveness passed yields VERIFIED; a score between the probable and verification thresholds yields CHALLENGE and triggers the challenge procedure; a score below the probable threshold, or the absence of any template, yields PROBABLE if diarization or proximity supports an assignment and UNATTRIBUTED otherwise. Thresholds are set per device from the enrollment self-test. The policy engine then decides what each level is good for: by default, VERIFIED or PROBABLE affirmations reset the affirmation timer, only VERIFIED affirmations satisfy a named-participant requirement or a verified-voice cancel authority, and UNATTRIBUTED affirmations count only in joint-only mode. Stop words, safe words, and distress indicators are honored at any attribution level (attribution is recorded for them but is never a precondition) because the cost of ignoring a genuine stop is far higher than the cost of honoring a spurious one.

[0197] Template maintenance and deletion. Templates may be updated with verified in-session utterances to track changes in voice over time, subject to a drift limit. A participant may view the existence (not the content) of his or her template, delete it, and see every lease that has been granted. Deletion is executed on all of the participant's devices and all leases are revoked; the deletion is logged. No raw enrollment audio is retained after the template is computed.

[0198] Emergency evidence preservation. When a session reaches an emergency, such as a red alert, a withdrawal that escalates, or the help stage, the controller may freeze the critically involved data that the participants consented to use during that session, including the voiceprint templates and leases, biometric matches, attribution records, device bindings, and the event log, so that they are not destroyed at session close as they otherwise would be. The frozen data is sealed in encrypted form in an evidence locker on the device, or exchanged with server 102 in encrypted form where law, contract, and the participants' preferences allow, in order to serve the needs of evidence production in response to lawful process, and the personal review of the data by either or both participants who are entitled to see and review it. Each access to the locker is authenticated and logged; the locker is retained for a period set by the policy and by applicable law and is then destroyed, unless lawful process requires that it be held longer; and freezing does not enlarge what was collected, but only preserves what the participants had already consented to use during the session.

[0199] Logging. Each consent-relevant utterance is logged as a tuple comprising the pseudonymous participant identifier, the utterance class (go word, stop word, close phrase, OK phrase, cancel phrase, challenge response), the attribution level, a coarse score band rather than the raw score, the liveness result, the attribution mode in force (voiceprint, diarization, proximity, joint-only), and the time stamp, appended to the hash-chained log 132. The raw embedding and the audio are not logged. This record allows a later reader to see that a specific person, verified by voice, affirmed at a specific time, or that the system could only say that someone did.

4C. Silent ambient signal for sessions without media

[0200] Some participants do not want music or any other media playing during intimacy. For them the controller offers a silent ambient signal in place of media content, chosen in Dimension C of the policy engine. In the preferred embodiment, described above, the silent signal is not continuous: nothing is rendered while the envelope is comfortably within its window, and a scheduled check-in, a more insistent haptic pattern or beat, or an elected change of lighting arises only when the envelope is near its expiry or past it, so that the signal keeps its meaning and participants are spared signal exhaustion. Where participants nonetheless elect a continuous carrier, it is rendered by the alert output subsystem 134 through one or more of: a periodic haptic pulse on each participant's wearable 117, phone, or accessory over local link 118 (for example a soft, heart-pulse-like double tap at an interval the participants choose, adjusted for growing or collapsing trust and for the assertiveness, verbosity, and confidence of the participants' unprompted affirmations, for example every 20 to 60 or 120 seconds) whose steady continuation signifies that affirmation is current; a lighting state on lighting device 115, for example a warm steady color; or a low ambient tone or soundscape through speaker 912 or a networked speaker at a volume the participants set, which may be near-inaudible. The participants choose which of these carriers to use, and each participant may choose a different carrier for himself or herself (one participant a wrist pulse, the other a light), since the signal is a reminder for each person rather than a shared performance.

[0201] The silent signal follows the identical sequence as media. In PLAYING, the pulse, light, or tone continues steadily and each credited affirmation produces the receipt cue of section 4. At the lapse, instead of a fade, the carrier changes character over the fade period: the pulse slows or weakens and then stops, the light shifts toward amber and then dims, the tone fades. In the default haptic embodiment, however, no pulse is felt while affirmation is current: the start of the pulse is the lapse-window warning, and the pulse intensifies as the session enters and passes through the lapse, marking the move into the disambiguation of an emergency from mere absence and the management of whichever it proves to be. Where the participants instead elect a continuous carrier, the cessation of the pulse is the lapse signal; it is as perceptible on the wrist as the stopping of music is in the room, and because both participants' carriers stop together it is equally mutual. CHECK-IN, ALARM, and HELP then proceed exactly as in section 4, with the check-in alert, alarm, and help-stage output delivered through whatever channels the policy selects: a haptic check-in pattern distinct from the pulse, a spoken prompt through the phone or a speaker at the participants' chosen volume, and the lighting and help pathways

unchanged. A stop word produces an abrupt cessation of the pulse and an immediate distinct haptic pattern, preserving the fade-for-absence, cut-for-withdrawal grammar in the haptic domain.

[0202] The renewal reminder of section 4 is available with the silent carrier and is, for many participants, the natural way to use it: no pulse is felt while affirmation is current, a distinct pre-lapse pattern at lead time T_n invites renewal, the receipt cue confirms it, and the sequence of consent phase-out (silence, reminder, lapse, check-in, alarm, help) is learned by touch. The pulses are signals rather than a background: an ever-present pulse would lead to signal exhaustion, and is not the preferred embodiment, although a continuous, reassuring vibration, signifying that the system is present and attentive to the participant, may be offered as an option and evaluated in testing. Participants may also combine carriers, so that media plays and the wrist channel serves as a private backup if the media is paused for an unrelated reason (a phone call, a track change, a speaker disconnection): in that case the wrist, rather than the music, carries the state, delivering the renewal reminder and the lapse warning as described above, and the controller does not treat the media pause as a lapse. All timers, the policy engine, voiceprint attribution, the event log, and the smart speaker and dating integrations operate identically with a silent carrier; only the media control adapter layer is idle. The silent-carrier mode is also the appropriate mode where media control cannot be verified (no playback target is reachable, or the adapter layer reports a fault) so that the system degrades to a silent signal rather than to no signal, and the policy engine may select it automatically in that case and announce the change.

4D. Momentary biometric touch affirmation

[0203] A participant may renew affirmation by touch rather than by voice. A touch affirmation is a momentary biometric reading on a device the participant already owns and has enrolled with the controller, taken at the moment of renewal and discarded immediately, that identifies the participant physically. Three sources are supported. First, a fingerprint sensor on the participant's phone, tablet, computer, wearable, or a multi-use panic button equipped with one: the participant rests a finger on the sensor and the device confirms the match. Second, a wearable's physiological sensor: a photoplethysmographic pulse-waveform or heart-rhythm signature, a vein-pattern reader, or a bioimpedance measurement, matched against a template enrolled on that wearable, so that a deliberate press-and-hold on the watch face or crown is confirmed to have come from the enrolled wearer. Third, the secure biometric unlock of the device itself, meaning the operating system's fingerprint or face unlock, used as a signing oracle: the controller's companion application requests a device-attested signature over a session-scoped nonce, the operating system performs the biometric match inside its secure hardware and returns the signature only on success, and the controller verifies the signature against a key enrolled at setup. In the third source the biometric data is never exposed to the application at all; the controller learns only that the enrolled person (where the device's biometric unlock is enrolled to that person alone) authorized this affirmation at this moment.

[0204] A touch affirmation carries the attribution level PHYSICALLY VERIFIED, which the policy engine treats by default as equivalent to VERIFIED, and which a participant may designate as required for his or her own renewals, as required for cancel authority (section 4A, Dimension G), or as a second factor to be combined with a go word within a short window. Because the reading is momentary and local, no camera is used by the controller; face unlock, where it is the device's only biometric, is performed by the operating system inside its secure hardware and the image is never available to the controller. Nothing is recorded except the attribution level, the source (fingerprint, wearable physiological, or device unlock), the device identifier, and the time stamp in the hash-chained log. Where a participant has withheld biometric consent, or no enrolled sensor is present, touch affirmation is unavailable for that participant and a plain button press remains an UNATTRIBUTED affirmation subject to the policy. Such a press may also come from a Bluetooth remote control interface, as described for the compatible bedside remote among the device configurations of this disclosure, whose buttons are assigned to the system's signals according to their natural analogs: play as affirmation, pause as a brake, stop as a stop, directional buttons as directional guidance (up, down, left, right, here, not there), and volume or skip buttons as harder and softer, or as more and less. The same assignment applies to other widely available, inexpensive peripherals whose controls have natural analogs, such as media remotes, camera-shutter remotes, presentation clickers, game controllers, and smart buttons, so that a couple needs only to assign the buttons of a device they can buy anywhere. A custom-made

hardware remote may later be produced for Safer Yes or similar systems, designed to be kept bedside and dedicated to that purpose, rather than borrowed from, or identical to, the television remote that the couple uses elsewhere in the home.

[0205] Touch affirmation is designed to pair with the renewal reminder of section 4 and the silent carrier of section 4C: the wrist pulses to invite renewal, the participant presses the watch face or rests a finger on the phone's sensor, and the receipt cue confirms, making a complete renewal with no sound in the room and no camera, attributable to that participant alone. It also serves participants who cannot or prefer not to speak, situations in which speech is unwelcome, and the asymmetric and guest sessions of section 8C, in which a guest may enroll a touch affirmation on his or her own device for the session only. A stop may likewise be registered by a biometric touch on a device enrolled for that purpose, but a stop is honored at any attribution level and never requires one.

4E. Device-to-participant binding

[0206] Every affirmation, stop, cancellation, interaction, touch affirmation, renewal reminder, and receipt cue that passes through a wearable, phone, or accessory is attributed or delivered by reference to a device-to-participant binding that the controller maintains. The binding answers the question whose watch is this, so that a tap on one wrist can never be credited to the other participant. It is established in three stages. At enrollment, a participant pairs each of his or her devices to the controller from within that participant's own account and on that participant's own phone: the device's hardware identifier (its Bluetooth address, a platform device identifier, or a key generated on the device at pairing) is recorded against the participant's pseudonymous identifier, and the controller refuses to bind a device that is already bound to another enrolled participant until the first binding is released. In the simplest case, binding is merely presumed: the owner of a wristwatch or other wearable is presumed to continue to own it and to wear it exclusively on his or her own body, rather than having moved it to the body of an intimate partner, and inputs through it are attributed accordingly at the presumed-bound level described with respect to attribution from a bound device. At session start, optionally, the controller confirms that each bound device is present and on its owner by at least one of: the wearable's own on-wrist or on-body detection (optical or capacitive), with the device reporting that it has been continuously worn since its last unlock; a liveness correlation, in which a pulse-waveform or motion signature read by the wearable is matched to the owner's enrolled template or to the owner's phone's motion at the same instant; a momentary biometric touch on the device as in section 4D; or, for a phone, the device's secure unlock performed by its owner. The result is announced in the session-start announcement of section 8B (for example, Watch A is on Participant A and Watch B is on Participant B) and recorded. During the session, the binding is maintained by continuity: a wearable that reports removal, loses its radio link, or is re-worn after removal drops to UNBOUND and its inputs are not credited to anyone until the session-start confirmation is repeated for that device, which the controller requests privately on the device itself; and the controller may, where the two participants' wearables both report a pulse, check that the two pulse-waveforms are distinct from each other and each consistent with its owner's enrolled template, so that two watches worn by one person are detected.

[0207] Attribution from a bound device is assigned the attribution level appropriate to the strength of the confirmation: PHYSICALLY VERIFIED where the confirmation was a biometric touch or a pulse-waveform match; PROBABLE where it was on-wrist continuity or motion correlation alone; UNATTRIBUTED where the device is UNBOUND. The controller thereby distinguishes a presumed-bound device from a proven-bound device. Most devices are presumed bound: a smartwatch, phone, ring, or other device owned by or registered to a participant is presumed to be worn or held by that participant, but nothing at the moment confirms that it is. A device is proven bound when a biometric signature taken as the device is put on or picked up, or while it is worn or held, such as a fingerprint, a pulse-waveform, vein-pattern, or bioimpedance match, or the device's own secure biometric unlock, establishes, to the highest confidence the device's sensors support, that the participant is the person who put on the watch or other device or who is holding the phone; a proven binding lapses when on-body or in-hand continuity is broken. The controller records which kind of binding stands behind each input and respects the distinction to the degree its sensors permit: an input through a presumed-bound device carries at most the PROBABLE level, an input through a proven-bound device may carry the PHYSICALLY VERIFIED level, and the policy may require a proven binding for renewals at higher voltage, for cancel authority, or for any other input it designates. The policy engine decides, per participant, which level a renewal from that participant's device must carry, and whether a device-based affirmation may stand alone or must accompany a go word. Stop inputs from any device, bound or not, are honored. Device binding

also governs delivery: renewal reminders, receipt cues, private distress prompts, and private check-ins are sent only to devices bound to the participant they concern, so that a reminder meant for one participant is never felt by the other. In a guest session (section 8C) the guest's device is bound for the session only, by the guest pairing it from his or her own phone or, where the guest has no phone, by a biometric touch on the device at the moment of binding; the binding and any template are deleted at session close. The binding table, with each confirmation and each change of binding state, is written to the event log with time stamps and attribution levels; device identifiers are stored as salted hashes.

4F. Steady intimacy support

[0208] Purpose and limits. This section describes optional functions for an established couple (of any composition) who wish to be intimate at a mutually satisfying frequency, with sufficient and regular initiation by both partners, and who elect the system's help in getting there. Every function in this section is off by default, is enabled only by both partners separately, may be disabled by either partner alone at any time without the other being told why, and is bounded by the finding of the Background: the system helps the couple reach the cadence they have chosen for themselves and never proposes a cadence of its own, scores the couple against a population norm, or presents frequency as an obligation of either partner.

[0209] Cadence target. At enrollment, or at any later time, each partner privately enters a preferred frequency range (for example, "one to three times per week"), an acceptable minimum interval, and times or days that are good or bad, together with an initiation preference: whether he or she would like to initiate more, be invited more, or keep the present balance. The controller stores each partner's entries in that partner's own encrypted record and computes, on-device, the overlap of the two ranges. Where the ranges overlap, the overlap is the couple's target cadence and is shown to both. Where they do not, the controller shows each partner only that the ranges do not overlap and by roughly how much, never the other's numbers, and offers a structured conversation (a set of prompts the couple may work through together, or a referral to the aftercare and resource pathways of section 7) rather than a computed compromise, because desire discrepancy is a matter for the couple and not for the arithmetic. Either partner may revise his or her range at any time; the other sees only the change in the overlap.

[0210] Recency and balance tracking. The controller records, on-device and under the log rights of section 7, the time of each session, which partner started it (by the invocation of section 8B, by touch, by a bid accepted, or by the first affirmation), and the outcome of each bid. From these it derives two private quantities: recency (the interval since the last session relative to the target cadence) and initiation balance, the share of recent sessions and bids begun by each partner, compared to each partner's stated initiation preference. Neither quantity is computed from anything other than session metadata; no content, duration, or quality of a session enters them unless a partner elects to add a private satisfaction note as described below. Because much of a couple's intimate life takes place without the system running, either partner may also mark an initiation manually after the fact, recording who initiated, approximately when, and whether the bid was accepted, declined, or deferred (for example, "not tonight, maybe tomorrow"); the other partner may optionally confirm the entry, so that the record remains mutual rather than one-sided. Manual marking need not be done event by event: the partners may instead reconstruct and confirm their initiations roughly once a month, in a periodic review paced by a regular calendar period, by a menstrual cycle, or by a virtual cycle of comparable length that the couple chooses. This counting is not meant as bean-counting. It is a baseline measure of mutual satisfaction within the couple's window of desired frequency, and it is designed never to become a source of pressure or stress: it lets each partner become aware, ambiently and privately, that he or she may be overdue, or may be pressing against the comfort zone or the outer bounds of the other partner's stated preferences, and nothing more. For a mature couple, the count may be kept by something as simple as a single short press on a physical button, such as button 116 or a wearable's button, which records a successful initiation for the partner who presses it (or for the partner to whom the button is enrolled), while a long press of the same button remains the panic signal that triggers the escalation of section 4, so that one inexpensive device serves as both the couple's initiation counter and a safeguard. Manually marked initiations count toward recency and initiation balance in the same manner as sessions and bids the system recorded, so that each partner can see whether the couple is finding its way into the zone of overlap between their frequency goals and toward the balance of initiation each desires. A well-

matched rhythm and balance of initiation, kept in view in this way, may strengthen the partners' feelings of closeness and intimacy and help each partner receive the frequency of intimacy he or she wants or needs to remain closely bound to the other.

[0211] Bids. A bid is a private, low-stakes invitation from one partner to the other, sent through the partner's private channel (a distinct haptic pattern on a wearable, a glow on the shared ambient display of section 8E visible to both but meaningless to a visitor, or a short message on the partner's phone) and answered by one of three responses: accept, which arms a session; later, with an optional proposed time that the controller may place on both calendars; or not now, which requires no reason. The design of the bid follows the responsive-desire finding of the Background: the bid is easy to send, because sending it is the step that responsive desire needs; it is easy to decline, because a decline that costs nothing can be given honestly; and it is easy to defer, because "later" is the answer that keeps the door open. The controller never reports to either partner a tally of the other's declines, never escalates a declined bid, and never sends a bid on a partner's behalf. A bid may be scheduled in advance ("Friday evening?"), and an accepted scheduled bid may run the date timer and handshake of section 8D as a warm start.

[0212] Role reciprocity. For couples who elect it, the steady intimacy support of this section may also support reciprocity of roles where the couple's practices involve roles that can be exchanged: which partner initiates, which leads, which gives or receives a particular act, or, in the terms used by many gay and bisexual men, who tops and who bottoms. Each partner privately enters whether he, she, or they wish to keep fixed roles, to reciprocate within every encounter, or to alternate, and if alternating, by encounter, by day, by week, or by month. Where both partners' entries permit alternation, the controller may suggest, at random or in rotation, and afresh with each new relationship registered, which partner takes which role and for how long, and may track the balance over time privately for each partner against that partner's own stated wish, in the same manner as the initiation balance described above. A suggestion is only a suggestion: either partner may decline it without giving a reason, a decline is not reported to the other partner as a tally, the feature is off by default, and a couple whose stated preference is fixed roles receives no suggestions. The feature exists because, where roles come to track physical size, strength, or social power rather than desire, a partner may occupy a role by default rather than by choice, and a private, low-stakes prompt to reconsider, offered only to couples who have asked for it, is one way of keeping role a matter of mutual choice. For couples who would enjoy it, the suggestion may be presented as a game: a randomizer, shown on a shared or private screen as a spinning wheel, a coin, or a deck of cards, that the partners actuate together and whose outcome neither controls. The randomizer may be fair, or may be weighted, by the couple's election, toward whichever assignment moves the balance back toward what both partners have said they want, so that chance does the asking and neither partner has to. Over time the controller maintains a balance meter for the couple's roles. It is informed not only by the recorded history of which partner took which role but by the partners' own estimation: in the aftercare period of section 7, each partner may be asked privately, in a single tap, whether the balance of roles feels right, has drifted too far toward the partner, or has drifted too far toward the participant himself or herself. The meter combines the history with those answers, and where the answers and the history diverge it trusts the answers, because the measure of balance is whether the partners experience it as balanced. Each partner sees the meter privately; both see it together only where both have elected sharing; and, where the couple has asked for it, the meter may be set against a reference range drawn from a published guideline of a community the couple belongs to or chooses, such as the recommendations of a sexual-health organization or a community's own norms of reciprocity, the comparison being offered as information for the couple's own reflection and never as a requirement, a score, or a judgment of either partner. A sustained imbalance, or a divergence between the two partners' estimations of the balance, is surfaced to each partner privately, with the conversation prompts and resources of section 7, in the same manner as the other quantities of this section. The inventor hopes that, over time, role in every community comes to reflect mutual choice rather than physical size or strength.

[0213] Receptive-load sharing. Role reciprocity also has a health dimension, and for couples whose practices include receptive anal intercourse the system may support sharing the receptive role as a means of reducing each partner's risk of injury. Published research on men who have sex with men indicates a dose-response relationship between the frequency of receptive anal intercourse and certain harms: in a French online survey of 21,762 men who have sex with men, eight percent reported fecal incontinence in the preceding month, and the prevalence was 12.7 percent among men reporting receptive anal intercourse at least once a week against 5.7 percent

among men reporting none; in multivariate analysis, high-frequency receptive intercourse (odds ratio 1.64), chemsex (1.67), and fisting (1.61) were each independently associated with incontinence, as were HIV seropositivity and lower socioeconomic status, while what the authors termed simple receptive intercourse, about once a week and without chemsex or fisting, was not associated with a significantly increased risk (Garros and colleagues, Journal of Sexual Medicine, 2021); and in a 2024 survey of 1,100 participants conducted at the University of California, Los Angeles, lifetime receptive exposure was not significantly associated with any of the colorectal or urologic diagnoses examined except anal fissures, which increased linearly with additional exposure, the authors suggesting that fissures may be directly related to trauma of the anal canal from penetration (Gaither and colleagues, LGBT Health, 2024). Both studies are self-reported and cross-sectional, and neither establishes causation, but together they indicate that the risk of injury tracks frequency. A couple who share the receptive role can therefore sustain a higher combined frequency of intimacy while keeping each partner's individual receptive frequency nearer to the range not associated with excess harm, so that neither partner carries the whole of the physiological burden; this is harm-sharing equity, and it is a reason for reciprocity independent of, and in addition to, the fairness reasons given above. Where both partners elect it, the controller maintains privately for each partner a receptive load, the number of sessions in a rolling period in which that partner took the receptive role; compares it with a reference frequency that the partner selects from a published guideline, or sets for himself or herself or with a clinician; and, when one partner's load approaches or exceeds the reference while the other's is below it, weights the randomizer and the role suggestion toward exchange and gives the partner whose load is high a private, non-judgmental notice. A soft balance bound may be elected, under which the weighting grows as the ratio of the two partners' receptive loads departs from equality beyond a configured ratio, for example sixty to forty, so that the interface leans increasingly toward balance without ever locking a choice. Aftercare responses reporting pain, bleeding, or other symptoms increase the weighting and present to the reporting partner information on preparation, lubrication, rest, pelvic-floor care, and when to seek clinical care. Partners differ in anatomy, in sphincter and pelvic-floor tone, in prior injury, and in conditions such as anodyspareunia, and a partner for whom the receptive role is painful or medically inadvisable may exempt himself or herself from the weighting for a reason that is not disclosed to the other partner; the feature never prevents a session or a role the partners choose, and it is offered as health information and a gentle default rather than as medical advice or a rule. The same mechanism applies to any couple, of any composition, whose practices include a receptive act that carries a dose-dependent risk.

[0214] Nudges. A partner who has elected to be nudged receives private, infrequent prompts from the controller when recency falls outside the target cadence or when initiation balance drifts from that partner's own stated preference, for example, a wrist tap on a good evening with the text "it has been a while; want to send a bid?", or "you said you'd like to initiate more; tonight might be good." A nudge goes only to the partner who asked for it, is phrased to that partner's own stated preference and never to the other's, is capped at a frequency the partner sets, and is silent during any period either partner has marked off. A partner who has not elected nudges receives none, and the other partner cannot enable them. The controller may also, where both partners have enabled it, propose a time on the couple's shared calendar that fits both partners' good times and the target cadence, labeled as the couple chooses.

[0215] Coupling the affirmation window to recency. The affirmation window T_a of section 4, and the envelope lengthening of section 4A Dimension B, are coupled to recency when the couple has enabled this section. The rationale is the Background's: a couple returning to intimacy after an interval well beyond their own cadence are, for that session, somewhat closer to a first encounter than to their usual practice, and the hot-cold empathy gap is correspondingly wider. The policy therefore shortens the window somewhat (toward, though not necessarily to, the first-session default) in proportion to how far recency has fallen beyond the target cadence, and lengthens it again, up to the couple's earned cap, as regular practice resumes. In one embodiment the window for a session is computed as $T_a = T_{usual} - k \cdot \min(r - 1, 1) \cdot (T_{usual} - T_{first})$ when r exceeds one, and as T_{usual} otherwise, where r is the interval since the couple's preceding session divided by the longest interval consistent with the target cadence, T_{usual} is the window that would otherwise apply, T_{first} is the first-session default, and k is a gain between zero and one, by default one-half, so that even a very long

interval moves the window at most part of the way toward the first-session default; T_{usual} is then restored in equal steps over a configured number of sessions, by default three, held within the target cadence. The shortening is announced at session start in neutral terms ("it's been a little while; I'll check in a bit more often tonight"), is never framed as a consequence of either partner's conduct, and may be declined for the session by either partner, in which case the previous window applies and the decline is logged as the partner's own election. The same coupling may be applied, at the couple's option, to the renewal-reminder lead time of Dimension D and to the default display configuration of section 8E, so that a long interval brings back the coarse, reassuring display.

[0216] Satisfaction notes and tuning. After a session, in the aftercare period of section 7, each partner may privately record a one-touch satisfaction note and, optionally, a short free-text note that is encrypted to that partner alone. Each partner sees only his or her own notes. Where both partners have elected sharing, the controller shows each a smoothed trend of the other's satisfaction (never individual sessions) and uses the two trends, together with recency and balance, to suggest adjustments to the target cadence that the couple may accept or ignore. A sustained decline in either trend, or a sustained imbalance in initiation against a partner's stated preference, is surfaced to that partner alone with the resource pathways of section 7, never to the other partner and never as a judgment of the couple.

[0217] Privacy. Everything in this section is the most sensitive data the system holds, and it is treated accordingly: computed and stored on-device only, in each partner's own encrypted record, never synchronized to any server, never shared with a dating platform under section 8D or with any third party, excluded from any export unless the exporting partner elects to include it, subject to each partner's log rights under section 7, and erased in full by either partner's election. No quantity in this section is ever used by the escalation sequence, the policy engine's vigilance level, or the attribution system except the recency coupling of the affirmation window, which only ever shortens the window and never relaxes any safety behavior.

[0218] Incentives decoupled from intimacy. Participation may be encouraged by rewards that are computed on the participant's own device and that are independent of whether any intimate activity occurred, of how often it occurs, and of any participant's affirmation. Credit is earned for protective acts: enrollment; configuration of safe words, the universal safe phrase, a bedroom mode, and emergency contacts; completion of consent education; completion of a check-in or an aftercare step; and an honored stop, brake, or concern cue, which earns credit equal to, and never less than, a session that continued. Each participant earns credit individually; no reward requires another participant's enrollment or affirmation, and no reward is reduced by a stop or a declined invitation. Rewards are redeemed only for goods, services, or recognition that are not sexual, such as music-subscription credit, safety hardware, a donation to an organization supporting survivors, or a nonsexual experience the participants choose together, and never for sexual conduct. Partners may also arrange private incentives with each other for taking early, strong, and continuous precautions, such as a chore taken on, a favorite meal, extra snuggle time, or a nonsexual massage. Such a gift is offered freely by the giver, recorded only on the couple's own devices, and redeemable only when the giver confirms at that time that he or she still wishes to give it, so that it is a standing offer and never a debt; either partner may withdraw an offer without giving a reason, and no gift is a sexual act or is conditioned on one. A gift involving touch is itself subject to the consent of both partners when it is given, and may be run as a session of its own, with its own go and stop words. The device discloses to a reward issuer only an anonymous credential, such as a blind-signed token or a zero-knowledge proof, attesting that a qualifying protective act occurred, without the time, partner, content, or occurrence of any session. An institution, such as a college, an employer, an insurer, a dating platform, or a government, may likewise grant a benefit (for example, a fee or tuition credit, a premium discount, a sales-tax exemption for certified devices, or a procurement preference) upon a privacy-preserving attestation that a person or device has the preventive configuration in place, preferably where an alternative path that does not require the device, such as completion of training, earns the same benefit, so that declining the technology carries no penalty. As used in this disclosure, "clinician" means any professional or trained helper who counsels people on health, relationships, or sexuality, including a physician, nurse, nurse practitioner, physician assistant, pharmacist, psychologist, psychiatrist, therapist, mental health counselor, sex therapist, marriage and family therapist, social worker, sex educator, health educator, peer educator, victim advocate, or community health worker. Clinicians who counsel and enroll communities may be

supported through salaried or grant-funded programs and through billing for counseling time, neutral as to vendor and not paid per enrollment.

4G. Stage-aware affirmation envelopes

[0219] Stages of intimacy. An encounter is rarely a single act, and consent to one act is not consent to the next. Where the participants elect it, the controller tracks the stage of the encounter and applies a distinct affirmation envelope to each stage. In one embodiment there are four stages, described by the voltage of the acts they include: (1) low voltage—affection, cuddling, and kissing; (2) mid voltage—making out, touching over clothing, and non-genital caressing; (3) upper-mid voltage—undressing, touching under clothing, fondling, and manual genital touching; and (4) high voltage—oral sex, digital or penile penetration of any orifice, and intercourse, with any act that a participant has entered in the boundary map of section 6 as conditional, such as anal intercourse, treated as a sub-stage requiring its own affirmation. The boundary between upper-mid and high voltage is fixed by default at the line drawn by applicable law between sexual contact and a sexual act (in the United States, for example, at the definition of rape used for federal crime statistics, which includes oral penetration) as determined from the jurisdiction table of section 8C and updated when the law changes, because the category the law treats as gravest is the one in which the cost of a misunderstanding is greatest. Couples differ in whether they regard oral sex as foreplay or as the main event; the default deliberately does not depend on that label.

[0220] Ascending and descending. Ascent to a higher stage requires a stage-entry affirmation: an enrolled stage word or phrase, a request recognized by the semantic classifier of section 3A ("can I take this off?", "do you want me to go down on you?") answered affirmatively, or an enrolled gesture or touch for that stage. In Duo mode a stage-entry affirmation is required from every participant; in Solo mode it can be obtained only from the enrolled participant, and the controller records that only that participant's affirmation was obtained and, before ascent to high voltage, privately prompts the enrolled participant to ask the partner aloud. Consent to a stage never implies consent to the next. An affirmation at a stage renews that stage's window; the windows grow shorter as voltage rises; and entry to a higher stage starts that stage's shorter window. At the couple's election, a lapse at a high stage may first step the media and the shared display down to a lower stage's treatment before the full fade. Descent is always free: any participant may step down a stage by an enrolled downshift word, a gesture, or a scoped stop naming the act ("not that"), and a downshift is not a withdrawal, does not advance the escalation sequence, and is logged as a downshift rather than a stop. A stop word retains precedence over every stage rule.

[0221] Attribution at the top. Entry to high voltage requires, wherever the platform supports it, an affirmation attributed at the verified level (by voiceprint with liveness, by physically verified touch, or by a proven-bound device) from each participant whose affirmation is required. This is the default for the first sessions of a relationship and under the dating preset of section 8D. Where verified attribution is unavailable, the controller announces the lower level, may require a touch affirmation in its place, and records the level obtained.

[0222] Burden and simplification. Four explicit confirmations on the way from low to high voltage may be experienced as burdensome, and the system offers several ways to lighten them without removing the gate that matters most. Stage affirmations may be a single word, a single gesture, or a tap on a remote. The participants may agree at session start a stage plan, an itinerary of the stages they expect to reach, after which each transition within the plan needs only a brief confirmation. They may elect a simplified configuration of three stages, in which low and mid voltage are merged, or of two, in which everything below the legal line forms one stage and everything above it the other; each simplified configuration preserves a gate at the legal line. Relaxation of the gate itself, for example a standing joint authorization to move through every stage within a session, is available only by the express joint election of both participants, reconfirmed at each session start, unavailable in a first session and under the dating preset, and revocable by either participant at any time. Which configuration best balances protection against burden is expected to be determined by testing with users.

[0223] Two-envelope configuration for screenless devices. Where the session is conducted through a voice-assistant smart speaker without a screen, on which several fine-grained stage confirmations are least convenient, the stages may be grouped into two

envelopes. The first envelope covers low and mid voltage and is in force from the start of the session, with its own affirmation window. To pass into the second envelope, which covers upper-mid and high voltage, each participant whose affirmation is required (both participants in Duo mode, the enrolled participant in Solo mode) must register an affirmation at the enthusiastic tier, which the speaker acknowledges aloud ("you're both a yes for more; I'll check in every few minutes"). That registration grants a lease of the second envelope for the remainder of the session: the participants need not register again to remain in it, but the lease is kept alive by renewal within an affirmation window whose duration may be the same as the first envelope's or different, typically shorter. The lease ends upon any stop or withdrawal, or upon a descent back into the first envelope, after which re-entry requires a new registration; a lapse within the second envelope is handled by the escalation sequence in the ordinary way. Because this configuration places the gate below upper-mid voltage rather than at the legal line, it is at least as protective as the default at the point that matters most, while asking for one deliberate, enthusiastic affirmation rather than several; participants who prefer may additionally retain a spoken confirmation at the legal line within the second envelope.

[0224] Few envelopes, and a single decision point. The affirmation envelopes of a session are intended to be few. In a typical configuration, a session is divided into only two phases, a lower-voltage phase and a higher-voltage phase, with the boundary between them at the line drawn by applicable law between sexual contact and a sexual act, so that a single reaffirmation, given at the one planned fade between the phases, carries the participants from the first phase into deeper and deepest intimacy. The participants may instead configure two reaffirmations, or more, up to the four stages described above. Within each phase, the affirmation window may be set long enough to cover the phase, and ordinary go words and, where enabled, implied refresh keep the envelope open without prompting, so that the only planned fade of the session is the one at the decision point; stop words, the brake, concern cues, and the escalation that follows an unanswered lapse remain in force throughout. The system is not intended to be burdensome or intrusive, or to ask for reaffirmation so often that it feels unnatural or annoyingly recurrent. Even a single planned fade, dividing a session into two parts with one decision point on whether to continue deeper, does a great deal: it builds a conversation about consent into the encounter, helps ensure that the conversation takes place at least once, at the point where it matters most, and primes a vulnerable party, or both parties, to stay aware and ready to use the system if needed, much as a skydiver who has been briefed on the ripcord is ready to pull it, or an electrician working on a high-voltage system follows grounding and lockout practices as a matter of routine, just in case.

[0225] Stage awareness without surveillance. The controller learns the stage principally from declarations: stage words, the classifier's recognition of requests and answers, and the vocabulary of each participant's boundary map. Non-imaging sensors may optionally corroborate changes of proximity or position, but recognizing acts from sensor data is neither reliable nor consistent with the privacy design of this disclosure, and the declared stage is the stage of record. The media and the shared ambient display may follow the stage, the energy of the playlist and the color of the display changing with it, so that both participants know where they are without asking. The event log records each stage transition, the attribution level of each stage-entry affirmation, and each downshift; because the stages correspond to the legal categories, that record speaks directly to the questions a later proceeding would ask.

[0226] Envelopes that track the relationship. The envelope timing need not be fixed for the life of a relationship. Where both partners elect it, the controller maintains a relationship profile on their devices, bound to the relationship identifier described above for relationship-scoped template leases, and lengthens the affirmation window, the check-in period, and the renewal reminders, and may reduce the number of stage-entry confirmations, as the couple passes milestones that the partners themselves define and certify. Such milestones may include a number of sessions without a lapse that reached check-in, without a stop, concern cue, or duress affirmation, and without a negative aftercare report; an elapsed time together; or a step such as engagement, cohabitation, or marriage that both partners confirm. In this way the check-in becomes less frequent, less intrusive, and less laborious once the couple is past the red zone of the early part of the relationship. The lengthening is bounded: it never removes the stop word, the brake, the abrupt stop, or the escalation to help, and it never extends a window beyond a ceiling the partners have set. The process runs in both directions. A trust breach, a confusion, or a dissatisfaction, signalled by a stop, a concern cue, a lapse that reaches check-in or alarm, a negative or "do not wish to say" aftercare report, or a private request from either partner that need not be explained, shortens the envelope again, back

toward the first-session default, until the couple has re-earned the longer setting. Either partner may shorten the envelope at any time unilaterally, while lengthening it requires both. Separately, the stages described above may be divided as the couple wishes into any number of divisions: two (for example, a combined low-and-mid-voltage phase and a combined upper-mid-and-high-voltage phase), the four described above, or finer subdivisions, each with its own envelope and stage-entry affirmation, provided that a division boundary always falls at or below the line drawn by applicable law between sexual contact and a sexual act, as in the two-envelope configuration described above.

5. Media control adapter layer and closed-loop verification (FIG. 4)

[0227] Referring to FIG. 4, a response mapper 404 receives the joint consent state 402 (and optionally the dynamic consent score) and emits a media directive. Directives include play, pause, resume, duck (reduce volume by a set amount), restore volume, set volume, skip to a specified track, switch to a specified playlist or playlist tier, and set a tempo or energy target, or a mix of tracks carrying purposeful signal meaning, for automatic or prompted track selection. In a graded embodiment, AFFIRMED maps to normal playback; ENTHUSIASTIC may, where the participants have configured it, additionally unlock a second playlist tier containing tracks the participants designated in advance (optional, and expected to be disabled by default); AMBIGUOUS maps to duck plus prompt; expiry of the affirmation window maps to fade-then-stop; BRAKE maps to fade, brake lighting color, and breathing-paced haptics on each party's own device; WITHDRAWN maps to immediate abrupt pause and, optionally, the cooldown track; and the check-in, local-alarm, and help stages map to spoken prompts, alarm tones, and help-stage audio emitted through the same adapters so that the escalation is heard through whatever speaker the music was using. In a further embodiment specific tracks are themselves consent signals: the participants pre-agree that a request to play a designated "yellow" track signals a boundary adjustment, and that the appearance of the cooldown track signals that the session has ended, so that the musical program communicates status to both participants and to a trusted third party listening remotely if one has been designated.

[0228] An adapter selector 406 discovers which playback pathways are available on the device and in the room, consults user preferences, measures the round-trip latency of each, and orders them for use. Adapters include: an application-remote adapter 410 that connects to a third-party streaming application installed on the same device and issues play(uri), pause, resume, and queue commands to it, subscribing to its player-state stream; a cloud or connect adapter 412 that issues playback-transfer, play, and pause commands through the streaming service's remote-playback interface to any of the user's registered devices, including a speaker or television that is not the user device; a first-party music framework adapter 414 that plays subscription content within the controller's own process using the operating system's music framework; an operating-system audio-session interruption adapter 416 that, by activating a non-mixable audio session or otherwise asserting audio focus, causes whatever audio source is currently active on the device (regardless of which application is producing it) to be interrupted, thereby providing a source-agnostic pause that does not depend on any partner interface; a local library adapter 418 that plays bundled or downloaded audio when no streaming pathway is available; a voice-assistant platform adapter that drives a smart speaker through its skill, casting, wireless-audio, announcement, or device-control interface and consumes its media-status events; and, in further embodiments, adapters for a television or a lighting controller so that lighting device 115 of the '301 patent is driven in synchrony with the music or the consent state. Smart speaker deployments are described further with respect to FIG. 7.

[0229] A closed-loop verifier 420 supports the fail-closed guarantee. Each directive is issued with a deadline L (for example 300 milliseconds for pause, 1000 milliseconds for play, and the fade period plus 500 milliseconds for a fade). The verifier subscribes to player-state notifications from the selected adapter and confirms that the directive took effect (that is, that the reported playback state changed as commanded) within L. If confirmation is not received, the verifier escalates to the next adapter in the selector's order (for example, an app-remote pause that is not acknowledged is followed by an audio-session interruption, then by a system volume mute, then by a command to a networked speaker), continuing until a pathway confirms or all pathways are exhausted, in which case the controller transitions to PAUSED/fault and notifies the participants by haptic and visual means. Pause directives are therefore verified positively, not assumed. The directive, each acknowledgement or timeout, and the measured latency are written to event logger 132.

[0230] Because third-party interfaces may be rate-limited, may disconnect when the native application is terminated, and may be subject to platform policy, the adapter layer treats loss of an adapter connection as a fault that transitions the controller to PAUSED through whichever adapter remains, and never treats an unacknowledged directive as successful. The controller also handles operating-system audio interruptions (an incoming call, another application seizing audio focus) by re-evaluating the joint state before resuming, so that resumption after an interruption is never automatic.

6. Enrollment and calibration

[0231] During enrollment, which may be performed once per participant and refreshed per session, each participant: selects or records go words, qualifying words, a safe word, a close phrase, and an OK phrase; enrolls go words at one or more tiers (AFFIRMED, ENTHUSIASTIC); optionally provides any enrolled input in the form of a vocal cue (a melodic cue such as a hummed tune, a whistled phrase, or a sung line, or a non-lexical vocal cue such as a double throat-clear, a tongue click, a tapped or snapped rhythm, a whispered or sung rendering of a word, or a counted sequence) recorded two or three times for signature, contour, or rhythm matching; optionally enrolls, for each mode of concern (grey zone, yellow flag, brake, red alert), a secret personalized cue (a word, touch, gesture, melodic cue, or other vocal cue) that is not disclosed to the partner and that the system classifies and responds to without revealing which cue registered; elects a tone mode for his or her own voice (OFF, PRIVATE, or GRADED) and, if PRIVATE or GRADED, records a short neutral speech baseline and a short go-word baseline for calibration; separately, and only after biometric consent and a feasibility check, enrolls a voiceprint template as described in section 4B; selects a non-affirmation policy or preset as described in section 4A; selects the session playlist, an optional second playlist tier, an optional yellow track, and a cooldown track; sets the affirmation window, fade period, check-in period, repeat interval, alarm period, and help period, the ordered help pathways and trusted contacts, whether implied refresh is permitted and its limit N, and whether a wearable's biometric signals participate in fusion; and confirms the emergency preferences of the '301 patent. The controller optionally screens the selected playlist for lyrics containing any enrolled keyword, phrase, or vocalization and warns the participants or automatically excludes the track. Enrollment data is stored on-device, encrypted at rest under a key held in the device's secure enclave or equivalent, and, where the '301 patent's server is used, only a cryptographic commitment to the enrollment data is transmitted.

[0232] Boundary map and language. At enrollment each participant may record, privately or for sharing with a partner, a boundary map: the words he, she, or they wish used for his, her, or their body and its parts, and words that must not be used; areas that may not be touched, or may be touched only after a specific affirmation; acts that are welcome, conditional, or excluded; and, where relevant, considerations particular to the participant's body, such as post-surgical sensitivity, dysphoria, a medical device, or pain. The boundary map is presented to the partner, at the participant's election, before a session and in the participant's own words. During a session the semantic classifier treats a reference to a mapped area or act as scoped by the map: a scoped stop naming a mapped area is resolved against it without the participant having to explain, a greyzone arising near an excluded or conditional item is weighted accordingly, and the clarifying exchange of section 3A uses the participant's own words. The boundary map is held in the participant's own encrypted record, is shared only by the participant's election and only for the session or relationship the participant chooses, and is never transmitted to a dating platform or any third party. It is useful to any participant, and is described here with particular regard to transgender and gender-nonconforming participants, for whom unspoken assumptions about anatomy and language are a frequent source of harm.

6A. Impairment handling and breath-alcohol gating

[0233] The '301 patent's Healthy Consent mode asks each participant about intoxication at the pre-interaction checklist. The present system extends that question into measurement and into time. A breath-alcohol sensor 934—a fuel-cell or semiconductor breath-alcohol concentration (BrAC) sensor in a phone accessory, a keychain or wearable device, a dedicated bedside device, an attending device, a vehicle's ignition-interlock or cabin sensor, or a venue's fixed station—is an optional input to the policy engine and to the consent state fusion. A reading may be taken on demand at the pre-interaction checklist, at session start, or at any clarifying exchange

or brake; and, in an embodiment that gives the mechanism particular value, a reading is taken when a participant leaves a drinking occasion, before any session is contemplated, as in leaving a bar or, among youth, a fraternity or sorority party.

[0234] Leaving-the-party reading and the consent lockout. A participant may enroll a rule that, upon leaving a venue or event at which the participant has indicated alcohol was consumed (detected by the participant's own check-out, by a venue's exit station, by geofence departure, or by a scheduled prompt) the participant is asked to provide a breath sample on an enrolled sensor. The result is associated with the participant's account as a time-stamped reading, held on-device and in the participant's own encrypted record, never transmitted to a partner or platform, and used for one purpose: if the reading exceeds a threshold the participant or an institution has set, the controller applies a consent lockout to that participant's account for a period computed from the reading, during which the system will not open a session in which that participant is a consenting party, will refuse to credit that participant's affirmations, and will say so plainly and privately to that participant and, at session start, to any partner ("a session is not available tonight") with or without disclosing the reading or the reason, as the participant elects; neither is disclosed until the participant elects to disclose it, except that community norms, policy, law, or the system's own teaching preferences may in some cases call for disclosure of the reason to the partner, so that a second party understands why he or she cannot try further or push his or her luck any further that night. The lockout does not judge capacity, which the system never adjudicates. It applies a rule the participant adopted in advance, while sober, about what he or she wants the system to do that night.

[0235] Computing the lockout period. The period is computed from the reading by a clearance model and a margin policy, both configurable. The default clearance model assumes a conservative population elimination rate (for example 0.010 to 0.015 grams per deciliter per hour, selectable toward the slow end) and computes the time at which the reading is expected to fall below a clearance threshold; the margin policy then adds padding for the three uncertainties inherent in a single exit reading: that absorption may not be complete at the moment of the reading, so that the true peak is later and higher; that further drinking may occur after leaving, unobserved; and that the sensor itself has error. Margin policies include: a fixed clearance threshold with a fixed elimination rate (for example, lock until the modeled concentration is below 0.02 g/dL); a bottle-to-throttle rule of the kind used in aviation, in which the participant is locked out until both a modeled concentration below a stated level (for example 0.04 g/dL) and a minimum elapsed time (for example 8, 12, or 24 hours from the last drink) are satisfied, whichever is later; a padded rule that adds a fixed number of hours beyond the modeled clearance; and a re-test rule, under which the lockout may be lifted early only by a fresh reading below threshold on an enrolled sensor, with a minimum interval between tests. An institution (a university, a residence, a military unit, a program) may publish a margin policy for its members, setting a drier or wetter standard according to its own judgment of the risks, and a participant may adopt the institution's policy, a stricter personal one, or none. The policy in force is announced to the participant at the time of the reading and recorded with it.

[0236] Personalized clearance calibration. For a participant who drinks in a controlled setting and wishes to know his or her own clearance time rather than a population estimate, the system supports calibration: with the participant's informed consent, a sequence of readings on the same enrolled sensor over the hours following a known quantity of alcohol is recorded on-device, the participant's own elimination curve is fitted from the descending readings, and the fitted rate and its confidence interval are stored in the participant's record and used by the clearance model in place of the population rate, with the margin policy applied on top. The system tells the participant how much time he or she must wait from the time of drinking, personalized to that curve and to the quantity consumed, and may deliver that as a private countdown on the participant's wearable. Calibration data never leaves the device; the fitted rate is a number, not a record of drinking; and a participant may delete it at any time, reverting to the population model.

[0237] Impairment flags without a sensor. Where no breath sensor is enrolled, the semantic classifier of section 3A and the optional implied channel may raise an impairment flag for a participant's own voice from slurred or fragmented speech, inconsistent answers to the clarifying exchange, or an affirmation stream that is internally contradictory. The flag never adjudicates capacity; it shortens the affirmation window, disables implied refresh, requires the enrolled go word rather than a classified GO, requires the enthusiastic tier if enrolled, and may require a brake before any boundary is renegotiated. A participant may set the flag to lock out a session entirely,

and may set it to apply only to his or her own affirmations. Every setting in this section is the participant's own, adopted in advance; the system's role is to carry out, that night, a decision the participant made when clear.

7. Privacy and evidentiary properties

[0238] The controller is designed so that the most sensitive data (the audio of an intimate encounter) never leaves the device and, apart from any conflict recording armed under section 8C, is never persisted. Microphone frames are processed in a ring buffer of no more than a few seconds, are consumed by the on-device ASR and affect classifier, and are discarded. What is retained is an event log 132 consisting of consent-state transitions with time stamps, the triggering event class (go word, stop word, timer expiry, check-in issued, alarm sounded, pathway invoked, cancellation, distress indicator, link loss, adapter fault), confidence values, the media directive issued, the acknowledgement or timeout, conflict recordings, consent agreements, and their metadata, if any, as well as the measured latency. Each entry includes a cryptographic hash of the preceding entry, so that the log is append-only and tamper-evident in the manner of the cryptographic storage module of the '301 patent. Participants may elect that the log be held only on-device (Extreme Privacy mode), mirrored to server 102 in encrypted form (Balanced Privacy mode), or generated only in memory and discarded at session end (Private Practice mode).

[0239] Non-imaging sensing. Room sensing for interaction, presence, and gesture is performed, in the preferred embodiment, by non-imaging sensors, so that the worst privacy failure (an intimate image leaving the room) is prevented by the physics of the sensor rather than only by software. The preferred room sensor is a millimeter-wave radar operating at about 60 GHz, which resolves presence, position, gross motion, gesture, breathing, and heart rate of one or more persons through bedding and in darkness, and whose raw output is range-Doppler data from which no image of a body can be formed. Alternatives are a low-resolution thermal array of roughly 8-by-8 to 32-by-32 thermopile elements, which locates warm bodies and their motion without resolving a face or anatomy; a coarse-zone time-of-flight sensor of a few dozen zones, which reports distance per zone sufficient for gesture and for a participant rising, but not an imageable depth map; an event-based sensor that reports only pixel brightness changes, used with the local-only rule below because a sustained event stream can be partly reconstructed; an optically blurred sensor, in which a defocused lens or diffuser at the sensor makes the optics physically incapable of forming a sharp image, so that the privacy guarantee is in the glass; ultrasonic or acoustic ranging using a device's own speaker and microphone; and passive infrared. Such sensors are available as consumer presence sensors and as components, and may be built into a dedicated bedside device, an attending device, or a smart speaker.

[0240] Imaging sensors are treated as follows. A near-infrared camera, including a night-vision camera and the flood-illuminated camera of a face-unlock system, is an imaging sensor: it produces a recognizable image and, in some bands, renders thin fabric partially transparent, and it is not made safe by being invisible to the eye. A full-resolution depth camera is an imaging sensor: a dense depth map of two people is an image of their bodies. The inventor strongly prefers not to use imaging sensors, for reasons of privacy. Any imaging sensor is used by the controller only where the participants have expressly enabled it at session start, with the session announcement naming it; its frames are processed on-device, in a frame buffer that is overwritten, for gesture and presence inference only; no frame, thumbnail, embedding, or derived image is persisted, transmitted, or exposed to any application; and the controller refuses to enable it on a device that cannot attest, by its operating system or secure hardware, that the frames are confined to on-device inference. An attending device may hold the witness role (section 1) only if its sensors are non-imaging, or if any imaging sensor it has is optically or physically incapable of forming a recognizable image, so that a witness device can never become a source of intimate imagery. The policy engine exposes, per participant, a setting that forbids any imaging sensor in the room from participating, and under a High-vigilance or Discreet preset that setting is on by default.

[0241] Log ownership, disclosure, and portability. Each participant holds his or her own copy of the event log, written to that participant's own device (or, in the single-device embodiment, to a per-participant encrypted partition keyed to that participant). No participant can obtain another's copy through the system; disclosure of a copy is by its holder's own export, which produces a human-readable file independent of the application together with a signed attestation of the log's hash chain, so that a recipient can verify the record was not altered without needing the system. The other participant cannot compel disclosure through the system, the platform of section 8D cannot request it, and the applicant does not hold a copy unless the participant elected mirroring. A participant may export,

delete, or carry the log to another device at any time; deletion of one participant's copy does not affect the other's. These rules apply equally to the brake, clarifying-exchange, responder-entry, and breath-reading records.

[0242] Age assurance. The system is for adults. Enrollment requires age assurance by a method the platform or the participant's jurisdiction accepts (a platform account's existing age attribute, a credential check, or an identity document verified on-device with only the pass result retained) and the controller refuses to enroll, to bind a device to, or to open a session for a participant who has not passed it; a guest enrollment (section 8C) requires the same before the guest's affirmations are credited. The system is not a consent record for, and does not purport to establish consent by, any person under the age of majority, and the event log carries the age-assurance method and pass result for each participant so that a record cannot be presented as something it is not. The personal-safety functions of the system (the lapse-to-check-in-to-help escalation, the stop-word emergency path, responder alerting and verified entry, the renewal reminder as a check-in timer, and the breath-alcohol lockout) are not intrinsically about intimacy, and where law permits they may be provisioned for a minor by a parent or guardian as the account holder, with the guardian as a designated responder and the guardian's consent recorded. In that configuration the consent-session functions, affirmation crediting, voiceprint attribution of affirmations, and the consent record are disabled and cannot be enabled by any setting, so that the system operates for a minor only as a personal-safety check-in and emergency-escalation service and never as a consent system.

[0243] Aftercare and session-end check-in. After the close phrase, the controller offers each participant privately, on his or her own device, a brief aftercare step: a single question (for example, "How was that?") answerable by a word or a touch; an opportunity to attach a private note, voice memo transcribed on-device, reaction, or rating to that participant's own copy of the log, visible to no one else; an optional scheduled check-in the next day on the participant's own device, with the help pathways available from it; and, where both participants elect it, a shared note that both must approve before it is written to both copies. The aftercare step is where a participant who did not raise something during the session can record it while it is fresh, and where a participant who wants to say nothing can decline with one touch. As a relationship matures, these check-ins become less frequent, and the periodic check-ins eventually become linked to a menstrual cycle or to a virtual menstrual cycle, depending on the composition of the couple; two men, for example, would keep a regular monthly period of their own. The periodic check-in helps the partners stay in clear communication about how their relationship and their connection are going, what they might want to reinforce and embrace even further, and what they might want to change, gently, over time.

[0244] The controller is designed to operate without any network connection. Speech recognition, attribution, fusion, the affirmation timer, the escalation sequence through the alarm stage, the alert output subsystem, and the event log all execute on the user device; media may be played from the local library adapter or from content the streaming application has already downloaded; wearables and lighting devices are reached over local radios; and the local link between participants' devices is peer-to-peer. Only the help stage's external pathways and the optional mirroring of the log to server 102 require a network, and the controller states at session start, as part of the announced configuration, whether those pathways are reachable. A session that begins online and loses connectivity continues unchanged except that the help stage falls back to whatever local pathways remain (the device's own spoken call-out, a cellular or satellite interface on a dedicated device, and any reachable neighbor device) and the loss and restoration of connectivity are logged.

[0245] The log has a specific evidentiary character: it records that music was playing at a given time, which under a policy in which the media carries the ambient signal and responds to a lapse implies that the joint consent state was affirmed and verified at that time; and it records the moment music ceased and why. Because the state of the music is contemporaneously perceptible to both participants, as long as both are aware of the ambient consent signal, the log corresponds to a shared experience rather than to a hidden system judgment, which reduces the risk that either participant is surprised by the record afterward. The session may run in Solo mode, with only one person fully aware of the system's operation, or in Duo mode, with both persons fully aware of it; the shared-experience character of the log described here holds fully in Duo mode, and in Solo mode only for the participant who is aware of the signal. In Solo mode, in a one-party-consent jurisdiction, it is possible for only one party to be fully aware that the music envelope is driven by renewed affirmations of affirmative consent and that the safety system has been activated. It may also be possible to operate

Solo mode in an all-party-consent jurisdiction if the system listens to and records only the party who has clearly consented, and neither listens to nor records the party who has not; whether that is permitted in a given jurisdiction is subject to legal interpretation and advice.

7A. Verifiable implementation: inspectable code, attested builds, and the privacy–security balance

[0246] Trust as a design requirement. A system that listens in a bedroom will be trusted only if the public can see how it works, and a promise in a privacy policy is not the same as code that can be read. The inventor's preference, in every embodiment in which commercial and contractual arrangements allow it, is that as much as possible of the logic that touches sensitive data be exposed for ongoing public inspection, as source code where possible, as executable pseudocode where source cannot be released, and as a machine-readable data-flow manifest in every case. Nothing in this section requires any particular code to be published in any embodiment; it describes the mechanisms by which an embodiment that elects to be inspectable makes that inspection meaningful.

[0247] The privacy kernel. The privacy- and reporting-relevant pathways are isolated into a bounded set of modules, the privacy kernel, with a defined interface to the rest of the system: the microphone ring buffer and its discard rule; the on-device speech, attribution, affect, and semantic models' input and output contracts; consent state fusion; the affirmation timer and escalation sequence; the event logger and its hash chain; every export, report, mirroring, and sharing path; the help-stage pathways; the sensor classification and confinement rules of section 7; the neural-data rules of section 8E; the cadence data of section 4F; and the device binding of section 4E. The kernel is the unit of publication and the unit of review. Because every byte that leaves the device must pass through it, reviewing the kernel is sufficient to verify the system's privacy claims without reviewing the media adapters, user interface, or platform integrations around it.

[0248] Data-flow manifest. Each release carries a manifest, in a machine-readable form, enumerating every egress from the device: the pathway, the privacy mode or modes in which it is active, the triggering condition, the data fields carried, their retention at the destination, and the destination's identity. The manifest is published alongside the kernel, is the basis of the session-start announcement's statement of what is reachable and what will be sent, and is the specification against which the automated reviews of this section test. A participant may view, for any past session, the ledger of what actually left the device, derived from the event log and reconciled against the manifest; any egress not in the manifest is a defect that the kernel reports and the ledger shows.

[0249] Reproducible and attested builds. Where the kernel is published, each release is built reproducibly, so that an independent party compiling the published source obtains a binary identical to the one distributed, and the hash of each release is published in a public append-only transparency log. The running application reports its own build hash, and where the platform supports it obtains an attestation from the operating system or secure hardware that the hash it reports is the hash of the code executing; the session-start announcement may include it, a participant's device may display it, and an attending device, responder device, or dating platform under section 8D may require a hash present in the transparency log before it will participate. A participant can thereby verify that the code reviewed is the code running, which is the link that distinguishes inspectable software from merely published software.

[0250] Inspectability in controversy. Where a session becomes the subject of a case or controversy (a complaint, an investigation, a civil or criminal proceeding, or a dispute between the participants themselves) inspectable code makes it straightforward for any party to examine and understand the system's properties: what the controller would have done on a given input, what it did do according to the event log of section 7, and whether the two agree. The log of section 7 records what happened; the kernel and manifest establish what the system does; and the attested build hash recorded in the log ties the one to the other, so that a party need not take the operator's word for how the system behaves and an expert for either side can reproduce the controller's decision from the published logic and the logged inputs. This complements the evidentiary character of the log described above and removes from the proceeding a class of dispute (what the software actually does) that would otherwise be resolved by the party with the better access to the vendor.

[0251] Model versioning and the reproducibility of a missed signal. Every recognition system errs, and a person may truthfully say that she spoke her safeword and the system did not hear it, because of her phrasing, her accent, the acoustics of the room, or the strain in her voice. The system is designed so that such a claim can be tested rather than merely asserted or dismissed. Each entry in the

event log records the identifier and cryptographic hash of every recognition and classification model whose output produced it; every released version of every such model is archived, together with its published error rates measured across accents, speech conditions including distressed and whispered speech, and acoustic environments; and, upon such a claim, the archived model named in the log can be run on the claimant's own renditions of the cue, or on a conflict recording where one was lawfully made, under the conditions the log records. If the model is shown to miss the claimant's cue at a material rate, a record that appears to show a sustained, renewed consent can rightly be understood as one that should have been interrupted earlier, faster, and harder by a stop that a technical flaw failed to catch, and the claim is corroborated by the system itself rather than undermined by it.

[0252] Inspectability in community. Inspectable code also lets the community of users question, rebalance, and adjust the system over time, through issues, suggestions, feedback, targeted technical feedback, and detailed pull requests against the published kernel or pseudocode. In the present day that participation is dramatically more feasible than it once was: a careful privacy or safety enthusiast, activist, or advocate who is not trained in computer science can, with the assistance of an artificial-intelligence model, read the published logic, ask what it does, and propose a change in the system's own terms. A nontechnical college student with the published code or verified pseudocode can ask, in plain language, "What if the affirmation envelope falters?", "What if I get greyzoned?", or "What happens if I or my partner call for a brake?", and receive (at the level of a high-school or college reader in the student's own language, or as a more technical breakdown on request) the reaction pathway the system would follow, with the ability to interrogate further in any direction ("and then what?", "what if my partner disagrees?", "what does the log show afterward?") and to adjust his or her own enrollment and policy settings accordingly. To make that participation available to a nontechnical user without any setup, an embodiment provides a one- or two-click interaction (from the application's settings, from the session-start announcement, or from the published repository) that attaches the user's own preferred artificial-intelligence or superintelligence service, in that service's own interface and on the user's own account and token budget, to the available codebase or verified pseudocode: the interaction packages the kernel, the manifest, and, if the user elects, the user's own policy settings into a form the chosen service can read, opens the service with that material loaded, and supplies a starter set of example prompts—"explain in plain language what happens when my affirmation lapses," "trace what leaves my phone in Balanced Privacy mode," "what would you have done differently in last night's session, given this log?" (from which the user can begin to inspect, question, and propose changes to the code in the interface he or she already trusts), with the option, on by default under the High-vigilance and Discreet presets, to launch the selected service in its private or incognito mode, or in a temporary or ephemeral session where the service offers one, so that the material and the user's questions are not retained in the service's history, are not used for the service's training where the service honors that setting, and leave no trace on the device's browser or in the service's account after the session is closed, with the interaction stating plainly which of these guarantees the selected service actually provides, with no data passing through the system's operator and the choice of reviewer entirely the user's. The effect is that every and any user (the merely curious, a participant who feels supported by the system and wants to understand why, or a party who believes the system failed or injured him or her) can play the part of auditor and privacy or security advocate, at any time of his or her choosing, with very little in the way of technical hurdles: no development environment, no build, no reading of source without assistance, and no permission from the operator. Auditing the system is thereby not a role reserved for the operator's own reviewers, the committee, or retained experts, but a standing capacity of the user base, and an injured party in particular can begin to interrogate the system on the night of the event, in private, before deciding whether or to whom to speak. An embodiment may provide this question-and-answer capability within the application itself, as an on-device or server-side model loaded with the kernel, the manifest, and the participant's own policy, so that the question "what will you do if…" can be put to the system before the session rather than discovered during it. A system that is questioned in this way, and that accepts contributions from those who question it, can get smarter every day, and can do so in the open.

[0253] Questions the public may want to audit and debate. Certain functions of the system are sensitive questions in their own right, and not merely implementation details, and inspectable code exists in part so that the public can audit and debate them. Among them: the circular audio buffer of section 7, which forgets at the speed of a goldfish (a few seconds and no more) and whose length, discard rule, and resistance to being lengthened by any update are exactly the kind of parameter a user should be able to read for himself or herself rather than be told; the conditions under which any recording is retained at all, including the conflict-recording permissions of

section 8C and the '301 patent, which depend on local law (one-party and all-party consent rules differ by jurisdiction and change), on each participant's stated preferences, and on each participant's actual awareness at the time, so that what the system may lawfully and properly retain in one bedroom is unlawful or unwelcome in another; the defaults under each preset, which decide what happens for the many participants who never change a setting; the attribution thresholds and the greyzone boundaries of section 3A; the escalation timings and the help pathways invoked without a participant's affirmative act; and the balance between what the device computes alone and what it sends to a server. On every one of these the design chooses a line between ease of use, privacy, safety, and the permission defaults that a community and a couple will tolerate, and the inventor does not claim to have found the right line, only a defensible one, stated in the open. These are relatively new questions: the agreements and expectations that will govern an attending device in a bedroom have not yet been normalized or negotiated by society, as they have been, imperfectly, for cameras in public and microphones on a phone, and the lines drawn here are an opening position in that negotiation rather than its conclusion. The inventor welcomes that debate, and the mechanisms of this section (the published kernel, the manifest, the committee's published criteria, and the one-click path by which any user can put the question to a reviewer of his or her own choosing) are intended to make it a debate that anyone can join on the merits, with the actual code in front of them, and whose outcome the system can adopt through the same contribution path as any other change. The specific characteristics of these technical features (how many seconds the buffer holds, what the default preset does, which jurisdictions' recording rules the system follows and how it learns which applies, what leaves the device and when) matter enormously to whether widespread consensus approval, widespread buy-in, and enthusiastic, normal, commonly and socially expected acceptance of a system like this (in Duo mode, in which every party is aware, enrolled, and clearly consenting to the system, or in Solo mode, in which one participant runs it discreetly as a protective layer for himself or herself) will be attainable over time in complex local and institutional communities, as a new best-practice social standard for romantic and sexual relationships at every stage: early, mid-stage, and mature; from the first dating exchanges of critical preferences, safety, and health information and choices under section 8D, through the first encounters for which the short envelope of the Background was designed, all the way to the mature patterns of intimacy of married couples who use the steady intimacy support of section 4F to strengthen their happy path and to keep their bedroom consent vibrant, alive, and evidently mutually satisfying, rather than facing the surprise, stated several monthly periods too late, that intimacy has been unsatisfying for one or both partners. Each community will decide on the specifics: a campus, a religious community, a municipality, a hotel chain, a dating platform, or a public-health authority will each ask these questions in its own terms, and each will approve or forbid the system's use on the answers. A parameter that is published, legible, and adjustable can be negotiated with such a community and set to its standard; a parameter that is hidden can only be trusted or refused. Inspectable code therefore serves adoption as well as individual privacy: it is the means by which a community can satisfy itself, on the specifics, that the system meets its norms, and the means by which the system can be configured, by preset or by policy, to meet different communities' norms at once.

[0254] Predictability and the Hawthorne effect. Knowing in advance how the system works is itself a safety property. Participants who know exactly what reaction the system will have (neither mysterious nor unpredictable, neither understated nor overblown) can take part without fear of the technical reaction, and can keep themselves well balanced within the enthusiastic affirmation envelope because they know where its edges are. The underlying mechanism is the Hawthorne effect: the observation, first made in studies of workers at the Western Electric Hawthorne Works in the 1920s and 1930s, that people change their behavior when they know they are being observed, and that the change persisted regardless of the particular condition the experimenters varied. Participants in a session know that the controller is listening for affirmation and for stops, know that a lapse will be checked in on and a stop will be honored, and know that the log will record what happened; the predictable presence of such an observer shifts behavior toward the conduct the participants themselves agreed to (affirming clearly, honoring a brake, asking rather than assuming) before any reaction of the system is needed, and does so more reliably the better the participants understand the observer. Inspectable code is what makes the observer understood rather than merely present. The same effect means that the system's reactions should be conservative and legible by design: a reaction that is surprising, or disproportionate to what the participants expected, weakens the effect by making the observer something to be managed rather than something to be trusted, which is why the reaction pathways of sections 3 and 4 are deliberately graduated, announced, and the same every time.

[0255] Continuous review and certification. The kernel and manifest are subjected to automated and semi-automated privacy and security review on a frequent, published cadence: static analysis of the kernel for undeclared egress; an egress test harness that runs scripted sessions in each privacy mode on instrumented devices and asserts, for Extreme Privacy mode, that no bytes leave the device other than a help-stage call or message sent over the device's own radios and, for Balanced Privacy mode, that every byte leaving matches a manifest entry; fuzzing of the parsers that handle any inbound data; and review of the on-device models' input and output contracts. The guarantees of this section are fulfilled in part by regular processing of the kernel, the manifest, and each release's diff by artificial-intelligence reviewers (large language models and, as they become available, superintelligent systems) that read the code as a human reviewer would, trace every sensitive data flow from its source to its sink, check each against the manifest and against the rules of this disclosure, and produce a written finding for each release; such review is repeatable on every commit at a cost that permits it to be continuous rather than periodic. The AI reviewers operate under the direction of a privacy and security balance and extreme privacy guarantee committee, which sets the review criteria, the interpretation of the privacy–security balance below, and the invariants that Extreme Privacy mode must satisfy; adjudicates findings; and publishes both the criteria and the findings, so that the public can see not only that the code was reviewed but what it was reviewed against. Human reviewers and independent third-party auditors remain in the loop: AI findings are confirmed or overruled by the committee, independent audits and certifications against recognized privacy and security standards are obtained and published at intervals, with the scope and date of each certification shown in the application, and a finding that fails is disclosed with its fix, not withheld.

[0256] The privacy–security balance. The system's governing rule is that processing is done locally, on the participants' own devices, wherever technically feasible, and a server is called only where a security need outweighs the privacy cost of the call. In Extreme Privacy mode no server is called in any pathway: speech, attribution, fusion, logging, and escalation through the alarm stage run on-device, and the help stage uses only the device's own radios: a direct call or message from the device, a cellular or satellite interface, and reachable neighbor devices. In Balanced Privacy mode the controller may make server calls, each declared in the manifest and announced at session start, for purposes that security requires and the device cannot supply alone: delivering a help-stage alert through a monitored service; verifying an authorized responder's credentials and attesting the responder's device under section 4; revoking a compromised device key or voiceprint lease; obtaining signed safety-model and software updates; and mirroring the encrypted event log for a participant who has elected it. Each such call carries the minimum data the purpose requires, never audio, never sensor frames, and never the cadence data of section 4F. The balance is chosen per participant, and when two participants have chosen different modes the stricter governs the session, by the same minimum rule that governs the joint consent state. Private Practice mode, in which nothing is retained, is available in either.

[0257] On-demand, transport-independent alerting. No embodiment requires a persistent or continuous connection between the controller and any server, dispatch center, or other remote system. In the preferred embodiment the controller and its devices operate offline and connect outward only as needed: when a hazardous or emergency condition arises, when a grey zone or other state calls for immediate or near-immediate review beyond the room, or when another non-local activation of this section is required. The connection is opened for that purpose, carries the minimum sealed payload, and is closed. Consent assessment, scoring, typing, and leveling are computed on-device wherever feasible, so that what leaves the device is a result, such as an alert class, a state, a location where the policy directs, and a signed reference to the event log, rather than the inputs from which it was derived. Any one or more of the following transports may carry such a message, directly to an authority or to a server system that relays it to one: a single HTTPS request, including a POST or a webhook call to a monitoring or alert-forwarding endpoint, over TLS 1.3 with mutual authentication or certificate pinning; a session opened on demand and closed after delivery, such as a WebSocket, gRPC, or QUIC connection, or a publish-subscribe message (for example, MQTT over TLS); a platform push-notification channel carrying only an opaque wake token, the content being fetched afterward over an authenticated encrypted channel; an end-to-end encrypted message using a double-ratchet protocol or group messaging under Messaging Layer Security (MLS) to responders, contacts, or a monitoring service; a request routed through an oblivious relay (for example, Oblivious HTTP) or an anonymizing proxy so that the receiving server learns the message but not the sender's network address, authorized by an anonymous token rather than an account identifier; SMS, MMS, or RCS messaging to emergency services (including text-to-911 where available), to responders, to emergency contacts, or to a monitoring

service, sent either automatically where the phone's operating system permits an application to send a message without user intervention, or, where the operating system requires user intervention, by preparing the message (recipient, text, and location) in the native messaging composer so that a single tap or confirmation sends it; a voice call carrying a synthesized spoken message; IP-based next-generation emergency-services interfaces where available; satellite messaging, including emergency satellite-messaging functions of phones and wearables; e-mail encrypted under S/MIME or OpenPGP; a vehicle telematics or platform emergency service; and peer-to-peer or mesh relay through reachable neighbor devices (Bluetooth Low Energy, Wi-Fi Direct, Thread, or long-range low-power radio) in a store-and-forward manner until a device with wider connectivity delivers it. In each case the payload is minimized, encrypted to the intended recipient by public-key or hybrid public-key encryption, and signed by the originating device; and where a remote party must compute a state or score, it may do so inside an attested trusted execution environment, over encrypted data by homomorphic or secure multiparty computation, or by verifying a zero-knowledge proof supplied by the device, so that the remote party learns no more than the result. Where more than one transport is available, the controller may send by several in parallel or in fallback order, and logs which succeeded. These transports and encodings are interchangeable equivalents for the purposes of this disclosure: none is required, the selection among them is an implementation choice, and the mechanisms described here do not depend on any particular one.

[0258] Listening ladder. Separately from the server modes above, which govern what may leave the device, each participant selects a rung on a privacy ladder that governs how much of his or her speech and intent is analyzed at all, even on-device. In ascending order: (L0) no listening, in which the microphone is off and every input is by touch, button, gesture, wearable, or haptic pattern; (L1) wake-word only, in which audio is examined solely for a platform or system wake word, and a safe word, stop word, or go word is recognized only when spoken after it; (L2) continuous safe-word sensing, in which on-device keyword spotting listens without a wake word for the participant's own enrolled safe words, stop words, brake words, and go words, without transcription; (L3) universal safe words, adding recognition, without enrollment, of standardized universal safe words and phrases recognized by everyone (for example, "stop," a platform's native stop command, or a standardized phrase such as "stop, drop, and roll"), so that a guest or a participant who never enrolled is still protected; (L4) non-lexical and tonal cues, adding recognition of distress vocalizations and, where elected under the OFF / PRIVATE / GRADED options, tone; and (L5) full semantic analysis during intimacy, in which, while the session is in an intimacy-active state, utterances are transcribed on-device and assigned meaning categories by the semantic classifier and, where elected, the private AI agent described above, with transcripts never retained beyond the utterance window unless a participant has elected otherwise. Each rung includes the protections of the rungs below it that use the microphone. The analysis of each participant's speech never exceeds the rung that participant has chosen; stepping down takes effect at once at a participant's word or touch, while stepping up requires the affirmation of each participant whose speech would be newly analyzed. Higher rungs are scoped to the session, and the controller falls back to the participant's resting rung (by default L1 or L2) when the session ends. The current rung of each participant is shown by the indicator and announced at session start, and each change of rung is recorded in the event log.

[0259] Standardized universal safe words and renewal tokens. The universal safe words of level L3, and the messages by which devices of a session exchange its state, may be specified as an open, versioned standard so that devices, platforms, and services from different makers interoperate. In one embodiment the standard comprises two parts. The first is a registry of universal safe words: for each language and locale, a versioned list of entries, each comprising the phrase, a language tag, phonetic and dialect variants, and a category selected from at least stop, brake, and help; a minimum recognition performance, such as a maximum false-rejection rate measured while media plays; and a maximum response latency for each category, such as ceasing media playback within 500 milliseconds of a stop. A conforming device advertises the registry version it supports to the other devices of a session, recognizes every entry for its locale on-device without enrollment and without a wake word whenever a consent session or guard state is active at a listening level that permits it, and executes the response of the entry's category within the specified latency. The second part is a renewal-token message format: each message comprises a protocol version, a session identifier generated randomly for the session, a participant pseudonym scoped to the session, an event type selected from at least renewal, stop, brake, concern, interaction, close, and help, the attribution level of the event, a monotonic counter, and a time stamp; it contains no audio and no transcript; it is signed by a

key of the originating device and, where it leaves the local link, encrypted to its recipient under hybrid public-key encryption. A recipient, such as a playback target, a partner device, an attending device, or a server, verifies the signature, rejects any message whose counter or time stamp is not fresh, and, failing to receive a valid renewal for each participant within the affirmation window, enters a lapse state, so that the fail-closed behavior of this disclosure holds across devices of different makers. FIG. 11 illustrates this exchange: the participants' devices 1102 and 1104 establish a session identifier and per-session pseudonyms 1110 and send signed renewal messages 1112 and 1114 to a playback target 1106, which verifies them 1116, fades the media and issues a check-in upon a lapse 1118, and stops the media at once upon a stop message 1120; the fields of the message are shown at 1130. The registry may further define a universal wake phrase, such as "mango mango," that every conforming device of every maker and platform recognizes on-device without its own wake word, both as a standalone entry of the help category and as a preface to a spoken emergency request that follows it ("mango mango, please call 911"). A conforming device executes such a request itself or, where it cannot place an emergency call, hands it to a pathway that can, such as a paired phone, a peer-response service, or an emergency-calling service, and it does so even where its own platform would not otherwise accept a request from a person it does not recognize, so that a person in an unfamiliar room need know neither the make of the devices around him or her nor their wake words. Because a visitor cannot arm a host's device, this protection depends on conforming devices listening for the registry's entries whenever they are in a bedroom mode, in a guard state, or otherwise so set by their owners, or wherever the law of the locale requires it.

[0260] Session interface for third-party applications. The system may also be provided as a software development kit that a third-party application, such as a music, dating, smart-home, or personal-safety application, embeds in order to offer the features of this disclosure without implementing them. The kit exposes to the application operations to open a consent session identifying its participants and a policy; to register renewal, stop, brake, concern, and close inputs; to subscribe to lapse, check-in, alarm, help, and state-change events; and to request control of a media playback target. The speech and sensor processing, the affirmation timer, the escalation sequence, and the event log execute within the privacy kernel of the kit on the user device, not within the application, and the application receives only session states and events, never raw audio, sensor frames, or transcripts. Each data flow the kit performs is declared in the application's data-flow manifest, and the kit exchanges session state with other devices by the renewal-token format described above.

[0261] Consent-gated intimate devices. A connected intimate device, such as a vibrator, stimulation device, or other motorized device, whether operated by its user or by a partner nearby or remotely, may itself be the gated output. Its actuation continues only while affirmation is current from each participant whose affirmation the session requires, including the person on whom the device acts; as expiry approaches, the device may deliver a distinct pulse pattern as the renewal reminder; upon a lapse its intensity ramps down over the fade period to a floor or to rest; upon a stop from the person on whom it acts, or from any participant, it stops at once, within the stop latency; and resumption requires a fresh affirmation. Commands from a remote controller operated by another person are accepted only while the person on whom the device acts has a current affirmation, so that a remote partner cannot continue actuation through a lapse or a stop. The device may also serve as a touch-affirmation surface and as a private haptic output.

[0262] Venue-owned deployment. A hotel, short-term rental, dormitory, cruise ship, or other venue may install the system in its rooms as venue-owned equipment, such as a room-control panel, a smart speaker, a lighting device, a bedside button, and a door system, so that a guest who owns no compatible device can still use it. A guest opts in for a stay from the room panel, a key card, the venue's application, or his or her own phone. The opt-in creates a stay-scoped profile holding the guest's enrolled words and policy, which is erased at checkout unless the guest elects to carry it to his or her own device. The venue's staff and systems receive no session content; where the guest's policy permits, they receive only a help-stage alert, routed to the venue's security staff or front desk as one help pathway together with the room's dispatchable location, and the venue's door system may serve as the access device for verified responder entry.

[0263] Privacy-preserving proof of deployment. An insurer, institution, or regulator that offers a benefit for adopting the system, such as a premium discount, may require evidence that the system was installed and active without learning anything about any session. The controller may provide such evidence as a periodic attestation, signed by a key held in secure hardware of the device and

accompanied by the attested build hash of section 7A, stating that a conforming release was installed, that its self-tests passed, and that it was armed for at least a stated share of a reporting period, the share being computed without times of day; or as a zero-knowledge proof or anonymous credential establishing the same facts without identifying the device or its user beyond the policyholder. No attestation discloses whether, when, or with whom any session occurred.

[0264] Account-attributed shared playback. Where a streaming service offers a shared session in which several accounts join and control a single playback, the system may use that session for attribution. Each participant's phone joins the session under his or her own account, and the service's server gates the one stream playing in the room on renewal tokens attributed to each joined account: playback continues only while a renewal from every account whose affirmation the session requires arrives within the affirmation window, a lapse of any such account fades the stream, and a stop from any joined account stops it at once. Attribution then rests on the account and its device binding, without voiceprints and without one device performing all of the listening.

[0265] Wearable gestures and paired-wearable affirmation. An earbud with an inertial sensor may detect deliberate head gestures, a nod enrolled as a renewal and a shake as a brake, and the earbud also serves as the participant's private channel for renewal reminders and receipt cues. A watch or ring may detect a deliberate gesture, such as a finger double-tap or a squeeze. Where both participants wear such devices, each device signs its wearer's renewals as renewal-token messages, and the two devices exchange them only while ultra-wideband or other short-range ranging confirms that they are within a short distance of each other, so that each renewal is attributed to the wearer of a bound device and shown to have been made in the other's presence, without speech, a phone, or a screen.

[0266] Clasp affirmation, distinguished from hand-holding. A clasp of hands may serve as an affirmation, but only as a deliberate act, because hand-holding is common in intimate settings and does not by itself signify consent to anything. A clasp is credited only when four conditions hold: the session has privately prompted the participants for an affirmation at a decision point, so that touch at other times counts for nothing; each participant's watch, ring, or band senses a deliberate pattern that the participant has enrolled and that ordinary hand-holding does not produce, such as two firm squeezes or a squeeze held for about two seconds, by its inertial sensors and, where the wearable has one, a wrist electromyography or grip-force sensor; both wearables sense the same clasp within a short matching interval; and the relative orientation of the two wearables, together with ultra-wideband ranging and, where available, direction finding, indicates that the participants face each other with their hands joined, rather than walking or lying side by side. Each participant may confirm the clasp privately with a tap, and either may decline it in words at any time. Where the wearables support human body communication, by capacitive or electro-quasistatic coupling through the skin, each wearable may also transmit through the clasped hands a short signed token, comprising the session identifier, the wearer's session pseudonym, a counter, and a time stamp, which the other wearable receives only through the contact, so that the clasp is shown to have joined the two enrolled wearers at that moment; where human body communication is not available, a short vibration pattern of one wearable's haptic actuator, carried through the clasp and sensed by the other wearable's inertial sensor, may serve the same purpose with lower reliability. Because the area and pressure of skin contact between clasped hands set the strength of the capacitive or electro-quasistatic coupling between the two bodies, a squeeze raises the received amplitude of the body-coupled signal and its release lowers it; a wearable may therefore detect the enrolled squeeze pattern in the envelope of the body-channel signal itself, and may require that this envelope match the squeeze sensed by its inertial or grip-force sensors, so that the squeeze and the contact are confirmed through two independent channels. The body channel carries only that token; any larger record, such as an attestation of recent test results or a confirmation of the participants' ground rules, is exchanged only upon the election of both participants, over an encrypted short-range radio link bound to the token. A wearable may also confirm, by electrocardiographic matching at the moment its wearer touches its electrode, that it is worn by its enrolled participant.

[0267] Pairing initiated by a single body-coupled wearable. Human body communication need not be available to both participants. Where only one participant, the initiator, wears a watch, ring, or band having a body-coupling electrode, the two phones nearest the clasp may be joined to a session as follows. The initiator starts the pairing on the wearable by an enrolled gesture or a tap, and the wearable then watches for the onset of skin contact with another person. When the initiator's hand closes on another person's hand, the second body adds its capacitance and its return path to those of the initiator, so that the load and the return signal seen at the wearable's

electrode change in a characteristic way, and the wearable's inertial sensors register the enrolled clasp pattern; the two together mark the onset. The wearable then signals the initiator's phone, to which it is bound, and that phone advertises over Bluetooth Low Energy an ephemeral pairing invitation that carries no identity of either participant. Of the phones that answer, which are those running a compatible application, the initiator's phone selects the nearest other phone by ultra-wideband ranging from the wearable or from the initiator's phone or, where ultra-wideband is not available, by received signal strength, and proceeds only if that phone lies within a threshold distance, such as about one and a half meters, and is nearer by a clear margin than any other phone that answers; where no phone qualifies, the pairing is not completed automatically. The two phones agree a session key by an authenticated key exchange, and each shows a short confirmation code derived from that key. The pairing completes only when each participant confirms on his or her own phone; neither can confirm for the other, and either may decline. Where the second participant confirms by touching the screen of his or her own phone while the clasp continues, and that phone's touch controller supports it, the wearable may also inject a short body-coupled code, which passes through the initiator's body, the clasped hands, and the second participant's body to the touch-sensing electrodes of that screen, where the touch controller receives it, at a low rate over a second or two, as a modulation of the sensed touch signal. Because only a screen touched by the person whose hand is clasped can receive that code, it shows that the phone confirming is held by that person and is not merely the nearest phone in the room. Release of the clasp, or expiry of a short pairing interval, before both confirmations cancels the pairing. A wearable without a body-coupling electrode may start the same pairing from the clasp pattern alone, with lower assurance. The pairing joins the two phones to a consent session and nothing more: it is not itself an affirmation, which the session still requires on its own terms, and no part of it is sent to a server or retained after the session beyond what each participant elects to keep.

[0268] Squeeze-count invitation (the Firm Yes) and a pause before foreplay. A clasp may carry a short, learnable exchange that people can use with or without wearables, which the inventor calls the Firm Yes. The initiator invites with one squeeze. The other person answers with zero to three squeezes: none, no answer within a few seconds, or a hand withdrawn means not now and needs no reason; one means a tentative or slow yes; two, a clear yes; and three, an enthusiastic yes. Each participant may enroll a different count or pattern, and the count describes how that person feels at that moment, never a commitment to any act. Each then makes one further agreed gesture, such as a press of the thumb on the back of the other's hand, by which he or she attests to not being intoxicated; that attestation is the person's own word and never overrides an impairment lockout or a breath-alcohol reading under section 6A. Where wearables are worn, they record the initiating squeeze, the answering count, and the attestations by the inertial, grip-force, or body-channel sensing described above, and a count of zero signs nothing. A positive answer then opens a brief break that the participants take together, before any physical foreplay, in which they start the consent session on their devices, register their affirmations at the tier their answers indicated, and run through any pre-session checklist they keep, such as ground rules, contraception and protection, or recent test results, before raising their voltage to any stage above the first.

[0269] Intimacy focus mode. An operating system may offer a focus or do-not-disturb mode designated for intimacy. Activating it may open the renewal gate described above for the device's audio output; suppress every notification and call except the session's own prompts, emergency calls, and contacts the participant designates; set the device's sensing to the participant's chosen listening level; and close automatically upon the close phrase, upon a soft close, or at a time the participant sets.

[0270] Freeze-response detection. The freeze response described in the Background has a physiological signature that a wearable can observe: an abrupt cessation of movement, often accompanied by a rise in heart rate, a fall in heart-rate variability, a change in breathing, or a change in skin conductance. During an active session, where the participant has enabled it, the controller may compare these signals with the participant's own enrolled baseline and, when a pattern consistent with freezing persists beyond a short interval, deliver to that participant's own device a private check-in requiring the least possible effort to answer: a single tap or squeeze to continue, and no response at all to have the system pause the session and escalate. The detection is never shown to the other participant, and it is never credited as an affirmation.

[0271] Consent provenance for intimate content. Where adults have elected to record intimate content, for example for private use or for publication on a creator platform, a recording device or application may record only while each person being recorded has a

current affirmation, pausing the recording upon a lapse and stopping it upon a stop. Each segment of the output is signed with a content credential, for example in the manner of the C2PA content-provenance standard, asserting that each person recorded had affirmed consent to recording and, where elected, to publication, and identifying each person only by a revocable pseudonymous credential. A person who later revokes consent to publication may issue a signed revocation referencing the credential, which a participating platform honors by removing or restricting the content, so that the credential travels with the content and supports the removal of non-consensual intimate imagery.

[0272] Smart-home hub orchestration and smart beds. A smart-home hub acting as a controller of a smart-home fabric, such as Matter over Thread or Wi-Fi, may receive the session's stage events, such as lapse, check-in, alarm, help, and close, from the consent controller and execute for each stage a scene across the devices of the fabric, for example dimming or coloring lights, setting speaker volume and announcements, and, at the help stage, granting verified responder entry through a lock. A smart bed with occupancy, movement, and respiration sensors may serve as an attending device, reporting presence, interaction, and interruption of breathing to the controller, and may carry a panic button in a standardized location, such as the center of the headboard or the side rail nearest each sleeping position, which signals for help when pressed regardless of the state of any session.

[0273] Legal character of this section. This section describes technical mechanisms and the inventor's design preference. It does not constitute, and no embodiment is required to make, any undertaking to publish source code, pseudocode, or any other artifact, and the licensing terms under which any artifact is published are outside the scope of this disclosure. An embodiment may publish any subset of the artifacts described, and the mechanisms of this section are each independently useful: a data-flow manifest and ledger are valuable even where no code is published, and attested builds are valuable even where only a hash is published.

[0274] Local rules rundown. A participant may optionally request, for a location he or she specifies, a supportive rundown of the policies, procedures, and laws that bear on the use of the system in that location and jurisdiction at the time of the request, such as recording-consent rules, mandatory reporting requirements, emergency-calling and responder-entry rules, age requirements, and biometric-privacy rules, together with which of the system's settings and procedures are active there and why. The rundown draws on the jurisdiction table of section 8C and is presented as the operator's best-effort assembly, with a plain statement that it may not be complete or current at all times, that the operator's understanding of local law may drift, and that it is not legal advice. The rundown invites the participant, and members of the public, to suggest laws, regulations, or local policies that it omits or states out of date; suggestions are reviewed and, where confirmed, incorporated proactively into the jurisdiction table and into the default configuration of the system in the affected locales.

[0275] Signaling aid, not a determination; provider holds no content. The system is a signaling aid. It makes the participants' own continuing signals perceptible to them and summons help when those signals stop; it does not determine, certify, or warrant that consent was given, that capacity existed, or that any person is safe, and no output of the system is presented as such a determination. Consistently with that principle, an embodiment may be implemented so that the provider of the platform, the media, or the device holds no content of any session: consent state, attribution, and escalation are computed on the participants' devices; the event log and any evidence locker are encrypted under keys held only by the participants' devices; and a server operated by the provider receives, if anything, only the minimized help-stage payload of the transports described above, and never audio, transcripts, consent-state history, or cadence data. A provider so configured has no session content to disclose or to lose. The feature may also be presented under neutral names, such as a check-in mode, a keep-playing check, or a safety session, rather than under names that describe it as the verification of consent.

[0276] Community incidence reporting. Where participants elect it, and a community, institution, or public-health authority offers it, the controller may contribute to a community-level count of incidents without disclosing any individual's data. Each device, at a reporting interval, contributes for each event class (greyzone, yellow flag, brake, red alert, help stage invoked, duress) a count perturbed by a local differential-privacy mechanism such as randomized response, or submitted through a secure-aggregation protocol in which the aggregator learns only the sum across devices; no time stamp finer than the reporting interval, no location finer than the community, no identity, and no audio or transcript is contributed. The aggregator publishes the resulting rates with their confidence

intervals, so that a community can see its own base rate across the spectrum of section 1, and its trend, term by term. An institution may further offer, at the end of each academic term and alongside the release of grades, a private attestation on the participant's own device, phrased in the behaviorally specific items of the Sexual Experiences Survey or a successor instrument, asking whether during the term the participant experienced any of the enumerated forms of sexual coercion or assault, or any other crime. A response to the attestation may be required of every enrolled student, in the same way that other end-of-term obligations are, but no student is required to disclose anything: every question offers the response "do not wish to say," which is accepted as a complete answer. The first time a student selects "do not wish to say," the student's device arranges a brief follow-up, in person or by call, with a confidential counselor, whose purpose is only to confirm that the student is well and knows that he, she, or they may share at any time; the counselor is told that a follow-up was requested and is not told any response, and the counselor is designated as a confidential resource so that the conversation does not itself trigger an institutional report. Substantive responses are aggregated by the same privacy-preserving mechanism, and no individual response is visible to the institution. A participant who, in answering, wishes to make an individual report is offered the pathways of section 7, and nothing is reported individually otherwise. The mechanisms of this paragraph lie within the privacy kernel and the data-flow manifest and are subject to the review provisions above.

[0277] Built-in evaluation. The reliability that this disclosure sets as a target is to be established by measurement, and the system can measure it without exposing any person. A community, institution, or research team that elects to evaluate a feature, such as a length of affirmation window, a style of check-in, a renewal reminder, or a brake script, may have participating devices assign themselves, at random and with the participants' consent, to one of two or more variants of that feature. Each device then reports, at a reporting interval, counts of greyzone states, brakes, lapses, withdrawals, and help-stage invocations, labelled only with the variant and the interval and perturbed by local differential privacy or submitted through secure aggregation, as described above for community incidence reporting. The evaluator estimates from the aggregated counts the difference between variants in the rate of each event class, together with its uncertainty; may stop a variant early whenever its rate of help-stage invocations or withdrawals exceeds that of the comparison by a preset margin; and may adopt the better variant as a default. No count identifies a participant, a session, a time of day, or a location finer than a community, so that the system becomes its own instrument for the trials on which Mission Zero depends.

8. Alternative embodiments

[0278] Single shared device. In one embodiment a single device, such as a smart speaker or a phone placed on a nightstand, executes the controller for all participants present, using speaker diarization for attribution. This embodiment requires no pairing and no second device and is the simplest to deploy.

[0279] Wearable-assisted. In one embodiment each participant wears a device with a microphone close to the mouth or a sensor of laryngeal vibration, improving attribution and permitting private haptic prompts (a tap on the wrist meaning "please re-affirm") that are not audible to the other participant.

[0280] Remote trusted listener. In one embodiment, with the express consent of all participants, a designated trusted third party receives only the consent-state stream (not room audio) and may hear the same music through a synchronized stream if affirmation dies off, so that significant fading or cessation of music alerts the third party in real time. This embodiment integrates with the emergency contact preferences of the '301 patent and operates after a predefined interval. It is operated in compliance with local privacy and recording laws.

[0281] Non-musical media. The media content may be an ambient soundscape, a video or video projection, a game, a lighting program, or a combination; the fail-closed principle and verification loop apply identically. Coordinated lighting may be driven so that light and music cease together. The fading or cessation of consent may also act on furniture: for instance, it may cause an electrically adjustable bed to change its conformation from flat to a bent, upright position more conducive to gently separated repose and cooling down. Any such movement is slow, stops at once upon an obstruction or an occupant's resistance, and is not made when the controller suspects a medical emergency or that an occupant is asleep or incapacitated.

[0282] Hardware embodiment. The controller may be embodied in a dedicated bedside device having a microphone array, a speaker, a physical button 116, a light 115, a wireless interface to streaming services and to participants' phones, and a secure element for the event log. Such a device may be sold, rented, or provided by a venue.

[0283] Vehicle embodiment. The controller may execute on, or be coupled to, a vehicle's cabin system: the infotainment head unit plays the media content and renders spoken check-ins and alarms through the cabin speakers; the cabin microphone array, with the head unit's own playback as the reference signal 146, captures go words and stop words; seat occupancy and weight sensors, door and window state, and cabin motion sensing supply interaction and presence; the vehicle's horn, hazard lights, and interior lighting serve as the alarm stage; and the vehicle's telematics unit, with its own cellular connection and location, serves the help stage, including a vehicle-platform emergency service as one help pathway. A participant's phone or wearable remains the private channel for renewal reminders and receipt cues. The vehicle embodiment is otherwise identical to the room embodiments, and the policy engine may select a vehicle preset when the controller detects that it is paired to a vehicle. Reconfigurations of the cabin's furniture, such as seat position, recline, and partition, may be driven by the consent state, and the vehicle's navigation may change in the event of a consent crisis, issue, or question, for example by proposing or, in an autonomous vehicle, driving a route toward a safe, lit, public place, a participant's home, or a responder.

[0284] Attending-device embodiment. Any attending device may take part in the sensor, chaperone, or witness role, and the room need not contain a device dedicated to the system: a docked robotic vacuum has a microphone or obstacle camera, motion and bump sensors, a speaker, and Wi-Fi; a robotic pet has a microphone, touch sensors, an accelerometer, a speaker, and a display; a bed frame has load cells, and some beds are motorized and can switch positions; a lamp, fan, thermostat, smoke or occupancy detector has presence or motion sensing; a hotel room-control panel has touch, a display, and a speaker. Each is paired over wireless interface 908, most simply as an endpoint on a smart-home fabric such as Matter or Thread with the controller or a bridge as the fabric's controller, so that one pairing method works across vendors. In the sensor role the device reports interaction and presence; in the chaperone role the alert output subsystem also uses its speaker, display, light, or motion to deliver check-ins, alarms, and help-stage audio, and a mobile device may position itself to be heard or travel to a door; in the witness role the device signs, with a key provisioned at pairing, the events it observed, and those signed entries are appended to the hash-chained log as corroboration from a device neither participant controls after pairing. A ranging beacon or tracker tag is an attending device in the sensor role only. In a hotel or venue, the operator may pre-pair the room's attending devices for guests, scoped to the stay. The session-start announcement lists the attending devices present and their roles.

8A. Smart speaker embodiments (FIG. 7)

[0285] Voice-assistant smart speakers are commonly present in bedrooms and are, in many households, the device already playing music during an intimate encounter. They are therefore a natural home for the disclosed system, but the major platforms impose constraints that a naive design would violate: third-party code generally cannot listen continuously on the speaker, cannot receive raw microphone audio, may be invoked only after the platform's wake word, may not run on the speaker at all (only on a companion phone), and may control playback only through platform-specific interfaces. The controller accommodates these constraints through three composable topologies, discovered at setup for each speaker and selected per the policy engine.

[0286] Topology A 702: speaker as endpoint. The controller 120 runs on a user device 112 (phone, tablet, or wearable) that performs the continuous, on-device listening, ASR, affect analysis, attribution, fusion, and escalation sequence logic described above. The smart speaker 706 is used only as the room's audio output: for the session music, for the spoken check-in, for the alarm, and for the help-stage audio. The controller drives the speaker through a speaker platform interface 704 appropriate to the platform, which may be: a third-party voice application (a skill or action) whose backend issues long-form audio play and stop directives and announcement or text-to-speech output; a casting protocol by which the phone streams or directs media to the speaker and can issue text-to-speech announcements; a wireless multi-room audio protocol by which the phone renders audio and the speaker plays it; a streaming service's remote-playback interface that targets the speaker as a registered device; or a smart-home device-control interface that at least can stop playback and set volume. Media-status or player-state events returned by the interface (playback started, stopped, paused, track

changed, volume changed) are consumed by the closed-loop verifier 420 so that a fade or stop on the speaker is confirmed, not assumed, and so that the track identifier and position reported by the speaker are used as the playback reference signal 146 for echo cancellation at the phone's microphone. Where an interface cannot report state, the verifier falls back to acoustic confirmation: the phone's microphone confirms that the speaker's output has ceased by measuring the residual after cancellation. Because platform announcement channels can be slow, the adapter selector measures each pathway's latency at setup and routes stop directives through the fastest pathway available (for example the streaming service's remote-playback pause, which is typically faster than a skill round-trip), while prompts and alarms are routed through whichever pathway can synthesize speech.

[0287] Topology B 708: speaker as wake-word-gated affirmation channel. The wake-word-gated affirmation channel is a first-class option. It does not depend on a participant's preferring the speaker to a phone, or on no phone being available for continuous listening: it may be used alone, together with a phone, or as the preferred interface, which, in the inventor's belief, many people will prefer, because a smart speaker is always on, is already positioned correctly in the room, and requires no one to launch an application in the middle of a romantic encounter. In this topology the controller accepts affirmations through the platform's voice application: the participant addresses the assistant with the platform's wake word and an invocation phrase enrolled as a go phrase (for example, "<wake word>, tell <skill name> yes"), or with a custom universal safe-word start, ideally one that works across speaker brands; or by a request to play a specific playlist tied to Safer Yes; or by an instruction such as "volume up by one" given within the first 10 or 20 seconds of music play, which may be pre-coded and pre-registered to start Safer Yes discreetly, possibly without the awareness of the other party where it is initiated cryptically in Solo mode and the law permits. The voice application's backend 710 receives the recognized intent—never audio—together with, where the platform supports it and the participant has enabled it, a platform-generated per-application speaker identifier produced by the platform's own native voice recognition. The controller treats such an identifier as a PLATFORM-ATTESTED attribution: it is not the controller's own voiceprint verification, but it is the platform's attestation that the recognized voice matched a profile the participant enrolled with the platform, and the policy engine may accept it as equivalent to VERIFIED, as equivalent to PROBABLE, or as insufficient, at each participant's election. Because affirmation in this topology requires a deliberate invocation rather than natural speech, the controller adapts the affirmation window: the fade-and-check-in stages become explicit spoken check-ins delivered through the speaker ("Say '<wake word>, tell <skill> yes' to keep the music going") at an interval the participants choose, and implied refresh is disabled for the speaker channel. Stop handling in this topology has two layers. First, the platform's own native stop command ("<wake word>, stop"), which every major platform honors without any third-party application, halts the music natively; the voice application receives the resulting playback-stopped event and the controller treats it as a withdrawal signal, entering WITHDRAWN and invoking the escalation sequence and emergency mechanism according to policy. Second, unless the speaker's operating system natively recognizes, without the wake word, universal safe words adopted across the industry, by that device or operating-system maker, by a community, or under a state, national, international, or linguistic policy agreement or mandate applicable to that locale or that maker, a stop word spoken without the wake word would not reach the speaker platform; in that case the controller requires in this topology that at least one user device 112 continue to listen for stop words continuously, or that a physical button 116, a platform-native button accessory, or a smart-home button be present; if no such backstop is available, the policy engine may refuse to start a session or may start it only in a mode that the participant or participants have expressly accepted as reduced protection.

[0288] Topology C 712: controller native on the speaker. Where the platform or device maker permits third-party code to execute on the speaker with access to its far-field microphone array (through a device-maker software development kit, an open smart-speaker platform, a smart-home standard that exposes microphone and playback capabilities, or a dedicated bedside device 714 manufactured for the purpose), the entire controller 120 executes on the speaker. This is the preferred topology where available: the speaker's microphone array and its exact knowledge of its own playback provide the best echo cancellation; diarization benefits from beamforming toward each participant; the voiceprint template is held in the speaker's secure element; and the escalation sequence's prompts, alarms, and help-stage audio are emitted by the same device that is listening. Phones 112, 114 and wearables 117, if worn during an intimate encounter, remain optional companions for private haptic prompts, ranging-based presence checks, and biometric

signals over local link 118. A dedicated device may additionally provide a physical stop button, a status light corresponding to lighting device 115, and a cellular or satellite fallback for the help stage when the home network is unavailable.

[0289] Composition and discovery. At setup the controller enumerates the speakers in the room and, for each, determines which of topologies A, B, and C it supports and which interfaces are available, measuring latency for each. A single session may compose topologies (for example, Topology A for playback and prompts on a speaker that permits no third-party listening, Topology B for platform-attested affirmations from a participant who prefers the speaker, and the phone for continuous stop-word detection) with the fusion component receiving affirmation events from every channel and applying the same rules R1–R7 regardless of source. Each event is tagged with its channel and attribution level. The policy engine exposes the composed capabilities to the participants in plain language before the session begins ("This speaker will play music and speak prompts; your phone will listen for your stop word; your partner's affirmations will be verified by the speaker's voice profile") and records the composition in the event log.

[0290] Platform-specific notes. Which topology applies to a given speaker platform depends on whether that platform offers third-party voice applications with a per-application speaker identifier (Topology B and platform-attested attribution), only casting and device-control interfaces from a companion phone (Topology A with the phone as listener), or no third-party code at all but wireless audio and application intents (Topology A with the assistant's intents as an additional affirmation channel); a streaming service's remote-playback interface is a platform-independent adapter wherever the speaker is a registered device. The controller is designed so that platform changes alter the adapter and attribution tables without altering the loop, the escalation sequence, the fusion rules, or the policy engine.

[0291] Multi-room and multi-speaker. Where several speakers are grouped, the controller addresses the group for music and the nearest speaker (by the phone's ranging or by the platform's room assignment) for prompts. Alarms and help-stage alerts are, by preference, targeted rather than broadcast: they follow a sequence of escalation in the form dictated by a peer-response system such as Neighbor 911, which alerts, in order, the persons best placed to help, whether inside the dwelling, in a neighboring unit, or nearby, and gives each of them directions to the room or locale where help is needed, so as to provide a new form of umbrella coverage consistent with the signals being received from the room where there is a concern. As an optional strategy, the policy may instead sound the alarm on all speakers in the dwelling and, in the help stage, use every speaker in the dwelling for the spoken help stage so that it is heard beyond the room.

8B. Voice-initiated session start on smart speakers (FIG. 8)

[0292] A session may be started entirely by voice, without touching a phone, by speaking an invocation phrase to a smart speaker's assistant. Referring to FIG. 8, a participant says, for example, "<wake word>, start Safer Yes" 802. The platform recognizes the phrase and routes it 804 to the system in whichever way the platform supports: by launching the system's third-party voice application with a launch intent; by triggering a routine the participant configured on the platform whose action calls the system's backend or casts to the speaker; or by invoking an application intent registered by the companion application on the participant's phone, which then drives the speaker as in Topology A. Where the platform recognizes the speaker's voice against a profile the participant enrolled with the platform and the participant has enabled sharing of a per-application identifier, that identifier accompanies the launch and yields a PLATFORM-ATTESTED attribution for the invoker. The backend 710 forwards a session-start request 806 to the controller 120, which may be running on the speaker (Topology C), on the backend's behalf on a phone bound to the household (Topology A or B), or, if no phone is reachable, in a speaker-only mode described below.

[0293] Arming and announcement 808. The controller enters ARMED/SILENT, discovers the topology and adapters available for this speaker and room, binds any phones or wearables that are present and enrolled (waking the companion application by push notification if necessary), and then optionally announces through the speaker, in plain language, what is about to happen and at what protection level: which device will play the music, which device or button will catch a stop word, whether affirmations will be verified by voice, and whether a phone is bound. This announcement is an option in the preferred embodiment, and it may be legally necessary in some jurisdictions whose all-party-consent rules on recording allow no exception for a system that listens only to one party's side of

the utterances, or for one that presents on-screen prompts and questions in place of an announcement. Where it is given, a participant who did not initiate the session can hear what the system will do before deciding whether to say a go word. The controller then asks for a mode 810 (Quick Consent or Healthy Consent of the '301 patent, or a named preset of the policy engine) and, for Healthy Consent, conducts the pre-interaction checklist by voice, asking each question aloud and accepting each participant's spoken answer, attributed as described in sections 3 and 4B, with the option to defer any question to the phone for privacy.

[0294] Mutual go words 812–820. The controller prompts each participant by voice to say his or her go word, and begins media only when a go word attributed to every participant has been received within the mutual window (rule R6). The invocation phrase itself is never treated as a go word for anyone: the invoking voice arms the session but cannot consent for another participant, and the controller states this in the announcement ("I'll start the music when I hear a go word from each of you"). Participants who have not enrolled a go word may be led through a brief spoken enrollment at this point, subject to the biometric consent flow of section 4B if a voiceprint is to be created. On receipt of the required go words the controller issues the play directive 818 through the selected best or fastest verified adapter, confirms playback from the speaker's media-status events, and the music begins 820, which is, as elsewhere in this disclosure, the perceptible signal that mutual consent has been registered.

[0295] Guards at start 822. Several guards apply specifically to voice-initiated starts. First, the start is logged with the invoker's attribution level, the channel, the topology, the mode, and the announcement text that was spoken. Second, the wake-word-triggered Instant Arm and Instant Alarm modes of the '301 patent are honored as aliases: an invocation phrase enrolled as an Instant Alarm phrase bypasses arming and goes directly to WITHDRAWN/EMERGENCY. Third, a session started by voice may be ended by voice with "<wake word>, end Safer Yes" or an enrolled close phrase, which is attributed and honored per the policy engine's cancel-and-resume authority; and the platform's native stop command, which the platform honors regardless of the system, is treated as a withdrawal signal as described in Topology B. Fourth, where the speaker's platform provides the assistant's own per-household user recognition but not a per-application identifier, the controller still records that the platform recognized a household member, at PROBABLE, so that a stranger's invocation is distinguishable from a household member's in the log.

[0296] Speaker-only mode. If no phone, wearable, screen, or other device is reachable and the platform permits neither continuous third-party listening nor on-speaker execution, the controller may run a more limited session entirely through the platform's voice application, without screen support—affirmations only by wake-word invocation (Topology B), check-ins spoken at the chosen interval, the native stop command as the only voice stop, and a platform or smart-home button as the physical stop—as a distinct, named protection level that is optionally announced, requires explicit acceptance by each participant whose affirmation the session requires, and is logged.

[0297] Routine-based start. On platforms with user-configured routines and custom spoken starters, the companion application offers to create routines whose starters are the invocation, close, and OK phrases and whose actions arm, close, or cancel escalation of a session, set lighting device 115 and speaker volume, and optionally set do-not-disturb on the participant's phone, so the invocation works on a platform with no third-party voice application and with phrases that do not name the product.

[0298] Cryptic arming and the guard state. A participant may wish to raise the system's readiness before anything intimate has been proposed or agreed: for example, when a man steps into a woman's bedroom, or a woman into a man's, a space that is naturally charged, and the participant wants protection in place in case the other person makes an unwanted move. For this purpose the controller provides a guard state, distinct from a consent session. In the guard state no affirmation is awaited and no media is gated, so that nothing about the room changes; but stop words, the brake, concern cues, duress affirmations, button patterns, and distress indicators are all honored, the escalation and help pathways are armed, and the event log records what occurs. The guard state may be armed openly or, more often, cryptically, by an enrolled input that the other person would have no reason to notice: a distinctive turn of phrase addressed to the voice assistant in the form of an ordinary request ("Hey, what time is it, exactly?", "remind me to water the plants tomorrow", "what's the weather like in Lisbon?"), to which the assistant gives its ordinary answer while the controller arms the guard state; a particular press pattern on a button or remote; a touch or twist on a watch; placing the phone face down at a particular angle; or an ordinary command to a connected device, such as dimming a light to a particular level. The arming produces only the

ordinary response to the request and a private haptic acknowledgement on the participant's own device. A guard state may also be offered privately when the participant's devices detect that a second person has entered the participant's bedroom, and may be set to arm automatically in that circumstance. A consent session can begin from the guard state only by the affirmations a session requires, so that the guard state never itself implies consent to anything; it continues, silently, until the participant who armed it closes it.

[0299] Touch messages as arming inputs. Watches and other wearables offer touch messages, such as a pattern of taps, a drawn sketch, or a transmitted heartbeat, sent from one wearable to another, as the Digital Touch feature of Apple Watch has done. A touch message matching an enrolled pattern may serve as a cryptic arming input: sent by a participant from his or her own wearable to his or her own devices, or to a trusted contact, it arms the guard state or the bedroom mode, so that the participant's devices become safe-word sensitive, while appearing to anyone who sees it as an ordinary message. A touch message may likewise answer a private check-in or, bound to a mode of concern, signal a greyzone, a brake, or an emergency to a trusted contact. A person whom the participant has designated, such as a friend who knows where the participant is spending the evening, may also send a touch message that privately asks the participant's devices to offer the guard state, which arms only upon the participant's own confirmation.

[0300] Recognizing a participant's bedroom from behavior. A participant may mark his or her bedroom explicitly, by the room tag described above or by naming the room of a device, but the controller may also recognize it, with the participant's permission, from where the participant's devices spend the night. Over a learning period of several nights, the controller notes, on the participant's own devices, the room in which the participant's phone, watch, or other wearable rests during the participant's sleep period, as inferred from the wake-sleep cycle (the times the phone is last used at night and first used in the morning, alarms set and dismissed, a sleep or focus mode, and sleep detected by a worn wearable) and from the charging cycle (a phone or watch placed on a charger at night and removed in the morning); and it associates that room with the smart speaker, bedside device, or other fixed device nearest to the participant's device during the sleep period, as determined by Bluetooth, ultra-wideband, or Wi-Fi proximity or by an acoustic ranging signal inaudible to people. Other behavior of the participant in the room reinforces the inference, such as a bedtime or wake-up routine run on that device, sleep sounds or a sleep timer played through it, and lights turned off or an alarm set by a request made to it. Where the participant charges a phone outside the bedroom, for example in a kitchen or a home office, the controller disambiguates by giving greater weight to the signals that follow the person rather than the charger: the room in which a worn wearable detects sleep, the room in which the alarm is dismissed, the room whose device last detects the participant's voice before the sleep period and first detects it after, the room in which snoring or other sounds of sleep are detected as described below, and the room through which the phone is carried between being unplugged and its first use of the morning. Where the signals still conflict, the controller asks the participant once which room is the bedroom and makes no designation until the participant answers. The controller learns each participant's bedroom separately, from that participant's own devices, so that the bedroom of partner 1 and the bedroom of partner 2, whether in one dwelling or in two, are each recognized. A room so recognized is stored only as a designation on the participant's own devices and on the device in that room, never as a location history and never on a server; the designation is proposed to the participant for confirmation before it takes effect, may be reviewed or removed at any time, and is relearned when a fixed device is moved, as described above.

[0301] Further signals of the sleeping site. Several other signals indicate where a participant sleeps, and the controller uses whichever of them are available, weighting each by its reliability. A wake-up alarm is a particularly clear one. An alarm set at night and turned off the next morning, after an interval consistent with a full sleep period, whether the participant's sleep-wake timing is regular or irregular, must have sounded within earshot of the sleeping person, so that the device on which it sounded was in or near the room in which that person slept. The controller notes the room in which the alarm sounds and is turned off; the interval between the alarm sounding and its being turned off, a short interval indicating that the device was close to the sleeper and a long interval or repeated snoozing indicating that it was farther away; and, where a fixed device in the room is listening as permitted under this disclosure, whether that fixed device heard the alarm sound and stop, which places the alarm, and the sleeper, within earshot of that fixed device. Physiological markers of sleep sensed by a worn wearable, such as a sustained fall in heart rate, a change in heart-rate variability, slower and more regular respiration, and a change in skin temperature, together with stillness, establish when the participant is asleep;

short-range ranging between the wearable and a fixed device, by Bluetooth signal strength accumulated over several nights or by ultra-wideband ranging where both devices support it, establishes the room; and the satellite position of the participant's phone or wearable, or the identity of the Wi-Fi network to which it is connected, establishes the dwelling. The controller can thereby recognize not only the participant's own bedroom but also a sleeping site away from home, such as a partner's bedroom or a hotel room, at which the participant regularly or occasionally sleeps. A phone that lies for hours without use, still and in darkness as reported by its motion and ambient-light sensors, or that rests on a mattress and senses the motion of breathing, indicates sleep near the phone; and sensors of the room or the bed, such as lights turned off at night, a thermostat entering a sleep schedule, a millimeter-wave or other radar presence sensor, Wi-Fi motion sensing, or the occupancy and breathing sensors of a smart bed or mattress pad, indicate a still, sleeping occupant without any microphone. These signals, like the others described in this section, are processed on the participant's devices; no sleep data, position history, or physiological measurement leaves those devices or is retained, and only the resulting designation of a room as a sleeping site of the participant is kept.

[0302] Safe-word sensitivity in a recognized bedroom. Whenever the controller detects that a person is present in a space matching the bedroom environment of partner 1 or of partner 2, as recognized above, whether by occupancy detected by the devices of that room, by the presence of a participant's phone or wearable, or by a match between the signature of wireless signals and room acoustics of the space and that of the recognized bedroom, the device in that space becomes safe-word sensitive. It enters the guard state described above, or a lighter listening state in which it detects on the device, without a wake word and with any media content being played cancelled from the microphone audio, only the universal safe words of the registry for its locale, the resident's own enrolled safe words, and, where the resident's partner has shared them with the resident's devices, that partner's enrolled safe words, retaining no audio and generating no transcript of anything else. The resident chooses the level of sensitivity, universal safe words only or universal and specific safe words. A device in a room not recognized as a bedroom, such as a kitchen, a living room, or another shared space, does not become safe-word sensitive by inference, though a participant may make it so by an explicit choice. The device shows the resident, by its status light or on the resident's own devices, when it is safe-word sensitive; the sensitivity ends when the space is unoccupied; and it never itself opens a consent session or implies consent to anything. The device that becomes safe-word sensitive may be the participant's own phone, which for many people is the only device with a microphone in the bedroom. During the sleep periods by which the bedroom is learned, the phone records a fingerprint of the place where it rests, comprising the signal strengths of the wireless networks it can detect, including those of neighbors, the barometric pressure, which identifies the floor, and the local magnetic field; and whenever its current surroundings match that fingerprint, the phone itself becomes safe-word sensitive, whether or not any fixed device is present, the fingerprint and the designation being kept only on the phone. The error such a match most often makes is to confuse the bedroom with an adjacent room of the same dwelling, and because the safe-word sensitive state listens only for safe words, on the device, and retains nothing, such an error costs little.

[0303] Bedroom mode launched by conventional bedtime signals. A device may also become safe-word sensitive without any learned designation, by entering a bedroom mode upon a conventional signal that its user is going to bed or is in bed: the phone or other device being connected to electrical power after a configured time of night, as phones are traditionally plugged in at bedtime; a sleep, bedtime, wind-down, or do-not-disturb mode of the operating system being turned on, on its schedule or by the user; a wake-up alarm being set for the following morning; a bedtime routine being run on a voice assistant; a bedtime scene of a smart-home system being activated; or an explicit command to enter the bedroom mode. In the bedroom mode the device is safe-word sensitive in the manner described above. It leaves the bedroom mode upon a wake signal, such as a configured morning time, disconnection from power in the morning, the end of the sleep, bedtime, or do-not-disturb mode, or the turning off of the wake-up alarm. Where a learned designation of the bedroom is also available, the controller may require both, entering the bedroom mode only where the bedtime signal occurs at the recognized bedroom, so that a phone charged overnight in a kitchen does not make the kitchen safe-word sensitive. The bedroom mode is subject to the same minimum age as recognition of a bedroom, described below, and, like every safe-word sensitive state, it never itself opens a consent session or implies consent to anything. In a stronger form, the device enters the bedroom mode only where the bedtime signal occurs while the device is determined to be in the user's bedroom, by a learned designation, by a room tag read by the device, or by the device's assignment to a bedroom in a smart-home application; while in the bedroom mode, it cancels from the

microphone audio any sleep sounds, music, or other content it plays through the night, using that content as a reference; and, upon a safe word, it stops that content and advances an escalation of a private check-in to the user on the user's own device, an alarm, and a call for help, the escalation advancing when the check-in goes unanswered, halting upon the user's response, and advancing directly to the call for help upon a safe word of the help category. Where, during the bedroom mode, the device detects by short-range wireless detection that a device of a second person is present in the bedroom, it privately offers the user, or enters automatically where the user has so set, the guard state described above. However the bedroom mode is entered, whether by a bedtime signal, by recognition of the bedroom, or by the user's explicit command, entering it may by itself make the device safe-word sensitive, automatically and without any further step. That sensitivity may be minimal: the device may listen only for a single universal emergency phrase, for example a repeated word such as "mango mango" adopted by law, by an industry, or by a device maker, whose detection raises at least a greyzone concern and may escalate to an emergency and a call for help; or it may listen for the full set of the user's safe words, safe phrases, and greyzone words together with the universal ones, each bound to the response of its category. A fixed device that its owner has assigned, at registration or later in a smart-home application, to a room or a device group labeled as a bedroom, including a guest bedroom, may enter the bedroom mode, and so become safe-word sensitive, by reason of that assignment alone, without any learned designation or bedtime signal, either continuously or only while the room is detected to be occupied, as the owner sets. A guest bedroom so labeled thereby protects a visitor who could not have armed the device and does not know its wake word.

[0304] Recognizing a bedroom by snoring and other sounds of sleep, and the age floor. A smart speaker or other device with a microphone may also recognize that it is in a bedroom environment from snoring that it senses ambiently. An on-device acoustic event classifier, of the kind used for sleep sensing, detects the periodic sound of snoring, and, as supporting signals, other sounds characteristic of sleep, such as slow regular breathing where it is audible, sleep talking, or teeth grinding, together with breathing sensed by a radar or other motion sensor where the device has one; it distinguishes these from the same sounds produced by a television, a sound machine, or a pet. The recurrence of such sounds in the same room on several nights, during the night hours and in coincidence with the sleep periods of a household member, marks the room as one in which a person sleeps, from which the controller proposes that the device in that room be safe-word sensitive. The classifier retains only a count or flag of the nights on which such sounds occurred and never the sound itself, and such sounds heard on a single occasion, as when a person dozes on a couch, do not suffice. Because many people do not snore audibly, these sounds are a supporting signal and not a requirement; and because they do not by themselves identify whose bedroom the room is, they are combined with the device and behavioral signals described above to identify the resident. A device becomes safe-word sensitive by recognition of a bedroom only where the resident of that bedroom has been established to be at or above a required minimum age, which is at least the age of majority and may be set higher by law or by the rules of the resident's community. The resident's age may be established by attestation, such as the age assurance described in this disclosure, an age-range signal provided by the operating system or app store of the resident's device, or the designation of the resident as an adult by an adult account holder of the household; or by inference from patterns of use, such as the age of the resident's account, the resident's role in the household account, and the resident's patterns of activity. Where these signals are absent, conflict, or suggest that the resident may be below the minimum age, the controller treats the resident as below it, and the device does not become safe-word sensitive by recognition of the bedroom. That limit applies to the consent functions of this disclosure; it does not restrict the personal-safety configuration that a parent or guardian may provision for a minor, as described with respect to age assurance, or any general emergency feature that a device offers in every room.

8C. Asymmetric, guest, and single-participant sessions

[0305] The description above assumes that every participant is enrolled. In practice the person who installs the system will often be with a partner who has not. The controller supports three configurations for this case, chosen by the enrolled participant's policy and announced at session start. In an asymmetric session, affirmation is required only from the enrolled participant (or from participants named in the policy); the unenrolled partner is present but is not a party whose go word is awaited, and the joint consent state is computed over the enrolled participants only. The unenrolled partner's stop words and the platform's native stop command are nonetheless honored, because a stop from anyone in the room is honored under rule R1 whether or not the speaker is enrolled. In a

guest session, the partner is enrolled for the session only, by a brief spoken enrollment at the start (choosing or accepting a go word, a stop word, and, if the guest consents to biometric processing and the device is capable, a voiceprint template that is marked as a guest template and deleted at session close), after which the guest is a full participant under rule R6 for that session and no data about the guest persists beyond the session except the hash-chained event entries, which identify the guest by a session-scoped pseudonym. In a single-participant session, one person uses the system alone, for example, for self-monitoring during solo use of an intimate device, during a medical or bodywork procedure, or during any activity in which the person wants a lapse of responsiveness to escalate to help: the affirmation window, lapse, and escalation sequence operate exactly as described with a single consent state, and the help stage is typically configured to reach a trusted contact or a peer responder.

[0306] Duo mode and Solo mode. The configurations above are grouped, for the participant and for the announcement, into two modes. In Duo mode, every party present is aware of the system, each is enrolled (permanently or as a guest) with accounts mutually attached for the session, and each has given a reasonably crystal-clear consent to participate, which the session-start announcement records; the joint consent state, the shared ambient display of section 8E, and the steady intimacy support of section 4F are Duo-mode functions. In Duo mode, crisis-triggered, privacy-preserving conflict recording (recording that begins only on a stop, brake, concern cue, greyzone, or lapse reaching check-in, captures only the window the applicable rule allows, and is sealed under the participants' keys with the hash chain of section 7) is always permitted and is armed by default for every session, subject to local law, and is disarmed only in one of two ways: both parties jointly opt out; or one party opts out after being fully informed, by the enrollment flow and the announcement, of what the recording captures, when it is triggered, how it is sealed and who can open it, and of the reduced safety profile that forgoing conflict recording brings, and the other party (the party who would otherwise have it armed) allows that opt-out and is satisfied to accept it for the session. Either route is announced and logged, and the party who wanted the recording may decline to proceed. The default is set this way because a recording that exists only when something has gone wrong, and that neither party can open alone, protects both of them (the one who may need evidence and the one who may need exoneration) and because the absence of such a record reduces the safety profile not only of the couple but of the community and society that must adjudicate what happened in rooms with no other witness. In Solo mode, one enrolled participant runs the system as a discreet, and where lawful an unannounced, protective safety layer for himself or herself, without the other party's enrollment and, if the participant so elects, without the other party's awareness. Solo mode exists because a participant may be embarrassed to raise the system with a partner, or may fear that the partner would reject it or withhold consent to it, and a person in that position is often the person who most needs a safety layer; the Background's findings on freeze and fawn responses apply with particular force to someone who could not bring himself or herself to mention the system beforehand. Solo mode is an asymmetric session in which the announcement is private to the enrolled participant (delivered on the participant's own wearable, earbud, or phone rather than aloud) and in which the controller listens, in principle and in implementation, only for the enrolled participant's side of the interaction: the expressed channel processes only speech attributed to the enrolled participant's voiceprint at the VERIFIED or PROBABLE level, together with that participant's touch, vocal-cue, remote, and wearable inputs; speech not so attributed is discarded from the ring buffer without transcription, classification, or logging, except that a stop word from any voice is honored under rule R1 and logged as an UNATTRIBUTED stop without retaining the audio or any derived feature of the speaker; the implied channel, where enabled, runs only on the enrolled participant's own voice; no imaging sensor is available; and log mirroring, sharing paths, and conflict recording are, by default, unavailable. In some jurisdictions and under some community policies, however, conflict recording and certain of the related functions of section 4 and the '301 patent are permitted in Solo mode, as an evidence base in the event of a greyzone, yellow-flag, or red-alert conflict: where the jurisdiction table and the community policy in force for the participant (a campus, employer, venue, or platform policy that the participant has accepted) both permit it, the enrolled participant may elect at enrollment that conflict recording be armed in Solo mode, in which case recording begins only upon the triggering event (the participant's own stop, brake, concern cue, or greyzone, or a lapse that reaches the check-in stage), captures the window that the applicable rule allows, is sealed under the participant's own key with the hash chain of section 7 and a record of the rule under which it was made, and is never transmitted or mirrored except by the participant's later, separate election; where either the table or the policy forbids it, the election is not offered, and where the law permits it only with notice, the controller gives whatever notice the law requires at the moment of arming; and the

event log records only the enrolled participant's own inputs and the controller's own actions. Whether Solo mode may lawfully be operated without the other party's knowledge depends on local law: on whether the jurisdiction permits one party to a conversation to record or process it, on whether any bedroom, intimate-setting, or device-specific rule requires all parties to opt in, and on the principle, which the design is built to honor, that a device processing only the enrolled participant's own vocalizations and interactions is not intercepting the other party's. The controller therefore maintains a jurisdiction table, determined from the device's region setting and, with the participant's permission, its coarse location; where the table shows that discreet operation is permitted, Solo mode is available as described; where it shows that all-party consent is required, Solo mode requires the audible announcement of section 8B and so becomes an announced asymmetric session, or is unavailable for the audio channel and falls back to the participant's own touch, wearable, remote, and non-imaging-sensor inputs only; and where the table is uncertain the controller says so and takes the more restrictive course. The participant attests at enrollment that he or she understands the mode's limits, and the participant's own election and attestation are logged. Solo mode collects nothing about the other party beyond the fact that an unattributed stop, if any, was honored (except where the evidentiary functions described above are lawfully enabled, in which case it collects only what those functions, as configured for the jurisdiction and the community, permit, and only from the moment of the triggering event), is never a means of monitoring a partner, and is converted to Duo mode at any time by the other party's guest enrollment, after which the announcement is made and the session continues as a Duo session.

[0307] Jurisdictions and Solo mode. The availability of Solo mode, and of conflict recording within it, depends heavily on where the session takes place. In the United States, federal law and the laws of most states follow a one-party consent rule, under which a participant in a conversation may record or capture it without the knowledge of the other parties; a minority of states require the consent of all parties to a private conversation. As of the filing of this application, the states generally classified as requiring all-party consent are California, Connecticut, Delaware, Florida, Illinois (for private conversations, under the statute enacted after its former law was held unconstitutional in 2014), Maryland, Massachusetts (which forbids secret recording), Montana, Nevada, New Hampshire, Oregon (for in-person conversations, though not for telephone calls), Pennsylvania, and Washington. Michigan's rule for a participant who records is unsettled; Hawaii follows a one-party rule but requires the consent of all parties for a recording device installed in a private place, which a bedroom device may be; Vermont has no general statute; and the remaining states, and the District of Columbia, follow a one-party rule. Puerto Rico, Guam, the United States Virgin Islands, the Northern Mariana Islands, and American Samoa each have their own rules, which the jurisdiction table records individually. Solo mode is simplest in the one-party jurisdictions, where an enrolled participant's own device may lawfully process the participant's side of an encounter, and may arm a conflict recording, without notice to the other party; in the all-party jurisdictions the controller requires notice or limits Solo mode as described above. Whether a system that processes audio on the device and retains nothing is recording or intercepting a conversation under any of these laws is itself unsettled, and the jurisdiction table is to be maintained by counsel and updated as statutes and decisions change. Comparable regimes apply elsewhere. In Brazil, for example, the Supreme Federal Court has held that an environmental recording made by one of the interlocutors without the knowledge of the other is lawful evidence in the absence of a specific legal duty of secrecy (Tema 237, RE 583.937), and the anticrime legislation of 2019 provides that no crime is committed when one of the interlocutors makes such a capture (Law No. 9,296 of 1996, art. 10-A, § 1), while other countries require the knowledge or consent of all parties. The local and national laws of each community can therefore dramatically slow or accelerate the adoption of this technology, in its minimal or its maximal form, in a given nation, state, or locality, and the controller is designed to adapt to them rather than to require any one of them.

[0308] Domestic and intimate partner violence. Where the participants are in an established relationship in which one may be subject to coercion or abuse, several provisions apply, and a Domestic Safety preset enables them together. (a) Duress affirmation. A participant may enroll, privately, a duress variant of an affirmation (a go word accompanied by a particular enrolled additional word or cadence, a touch pattern, a remote-button sequence, or a vocal cue) that is indistinguishable to the other participant from an ordinary affirmation. On recognizing it the controller does not credit an affirmation; it leaves the media and the room unchanged, so as not to alert the other participant; it silently invokes the help pathways the participant has designated for duress, such as a trusted contact, a domestic-violence advocate or hotline, or a peer responder instructed to arrive on an ordinary pretext; and it records the duress only in

the participant's own sealed record. The principle is that of the duress code long used in alarm systems, applied to the one setting in which a person may be unable to refuse openly. (b) Protection against coercive control. No participant's device receives another participant's location, event log, settings, satisfaction notes, or cadence data except by that participant's own election, made on his, her, or their own device and revocable at any time without notice to the other; enrollment, pairing, and every sharing election require the participant's own biometric or passcode; the participant may hide the application, disguise its icon and notifications, and exit it with a single gesture; and the system never provides one partner a means of monitoring the other. A system built to protect against abuse must not become one of its instruments, and technology is among the most common instruments of coercive control. (c) Strangulation and pressure on the neck. Because non-fatal strangulation is among the strongest known predictors of later homicide, and because choking during sex has become so common that half of the undergraduates in one campus-representative survey had experienced it (Herbenick et al., 2024), choking and any other pressure on the neck are excluded in every variant as permanently unconsentable, as described in the summary above; where a wearable or non-imaging room sensor supplies respiration or blood-oxygen signals and the participant has enabled it, an interruption of breathing or a fall in oxygen saturation beyond an enrolled threshold is treated as a distress indicator and bypasses the escalation sequence; a participant may also wear an optional neck-worn sensor in the form of a choker, necklace, or collar, whose pressure or strain sensor detects compressive force on the neck beyond an enrolled threshold and reports it to the controller over the local link, with the same effect; and the aftercare of section 7 asks specifically, and privately, whether any pressure on the neck occurred, and presents information on its delayed medical risks and on when to seek care. (d) Separation. A participant may privately mark a period, such as one in which he, she, or they are preparing to leave the relationship, during which help pathways are widened, the alarm stage is silent rather than audible, and the participant's sealed record is copied to a location designated in advance, because the period of separation is the period of greatest danger. (e) Resources. The help pathways may include a domestic-violence hotline or a local advocate, and the aftercare presents safety-planning resources privately on the participant's own device and never on a shared display.

[0309] The choice among these configurations is a policy dimension of the enrolled participant, is announced aloud at session start so that the partner hears whether his or her affirmation is being awaited, and is recorded in the event log. An unenrolled partner may at any time become a guest by speaking the enrollment phrase, and a guest may at any time withdraw from the session by the close phrase, which ends the session for everyone.

8D. Dating platform integration (FIG. 10)

[0310] The integration described in this section is optional in every respect: the system operates fully standalone, every other embodiment and claim in this disclosure is complete without any dating platform, and a participant who never links to a platform has access to all layers of protection if they are set up separately, except the match token and the platform's safety partner as a help pathway. Referring to FIG. 10, the system may be linked to a dating platform 170 (a server-based matching and messaging service with its own applications) so that a protected date can be arranged from within the platform and the protection can follow the two people from the chat, to the meeting, into the room, and out again. The link is established by an authorization flow between the platform and the system (or by the platform embedding the system's components as a software layer within its own application), and is used only to pass a pseudonymous match token and to receive the minimal signals and data package described below. The participants' devices 112 and 114 are ordinary phones with the hardware of FIG. 9; the platform is an additional network endpoint reached over wireless interface 908 and network 106.

[0311] Layer 1: match and mutual opt-in 1002. From the platform's chat or social features, either party may propose a protected physical dating relationship; the proposal is a platform-rendered message describing in plain language what the system will and will not do, and the other party accepts or declines within the platform. Only upon acceptance by both does the platform pass a pseudonymous match token to each party's device. A participant may, at election, display a willingness indication on his or her profile and filter matches by it. The platform may treat participation as it chooses: it may make the system optional and provide that declining a proposal or omitting the indication has no effect on visibility or ranking; it may favor, in ranking or in filters, users who have set up the system; or it may require that the system be set up as a condition of using the platform. In each case, features that rest on a separate

biometric or recording consent, such as voiceprint attribution and conflict recording, remain separately optional where the law so requires, and no proposal within a match may be pressed on a person who has declined it.

[0312] Layer 2: pre-date setup 1004. Each participant, on his or her own device and before meeting, completes the enrollment of section 6 with a dating preset applied by default (a short affirmation window, the full escalation sequence, help pathways beginning with a trusted contact, verified-voice cancel authority for the proposer or both) and, subject to section 4B, a voiceprint template. None of this is sent to the platform. Where both have enrolled voiceprints, each device prepares a session-scoped lease keyed to the match token, released only upon the meeting handshake and mutual consent to begin a physical encounter.

[0313] Layer 3: date timer 1006. Optionally, before the meeting, the system runs a check-in timer of the ordinary personal-safety kind, using the same escalation sequence and alert output subsystem: the participant confirms arrival and periodic safety on his or her own device or wearable; a missed check-in escalates to a trusted contact and, where the platform provides one, to the platform's safety partner as one of the help pathways of Dimension F. The meeting time and place, if the participant chooses to share them, go to the trusted contact and not to the platform. Where the date is at a participant's own home, the participant may designate the responders and access device of section 4 in advance, so that a trusted neighbor or the building's front desk can be admitted at the help stage; the platform receives none of this unless a participant elects otherwise.

[0314] Layer 4: meeting handshake 1008. When the two people meet, their devices perform a handshake over local link 118 (Bluetooth Low Energy or ultra-wideband ranging confirming that the two devices holding the same match token are within a short distance, or one device scanning a code displayed by the other), which binds the match token to each enrolled participant and to the device now in the room. This provides identity continuity from the platform's own verification (for example, photo verification) to the enrolled participant to the voiceprint that will attribute affirmations, so that the person present is verifiably the person matched and an impostor holding a stolen phone cannot pass the voiceprint layer. The handshake releases any pending template leases and starts the session in ARMED/SILENT. If the handshake fails or a device is absent, the system falls back to guest enrollment (section 8C) and notifies the enrolled participant of the reduced attribution level.

[0315] Layer 5: session 1010. The intimate session is started and run exactly as described elsewhere; the platform has no role during it and receives nothing from it.

[0316] Layer 6: close and after 1012. On the close phrase the session ends and leases are destroyed, unless the parties remain in a relationship, in which case, with the consent of the participant whose template it is, the lease is extended unless and until the relationship ends or that consent is withdrawn. The system then sends to the platform, if and only if the participant's policy permits, at most one of two signals: that the session closed normally, or that the help stage was invoked. A participant may elect to send nothing. The hash-chained event log remains on the participants' devices. If a participant later makes a report to the platform's trust-and-safety function or to law enforcement, the participant may choose to disclose his or her copy of the log, and the match token allows the platform to associate that log with the match without the system ever having sent the log to the platform unbidden.

[0317] Data minimization 1014 and unlinking. Audio, enrolled words, consent states, escalation stages, locations, voiceprints, and conflict recordings never reach the platform unless a participant elects otherwise; the platform sends the match token and, optionally, its safety partner as a help pathway; the system sends at most the two outcome signals, unless a participant elects to send more, or unless the system is more fully integrated with the dating platform's front end and back end as a single product, in which case each additional data flow is declared in the data-flow manifest and requires the participant's election. Either participant may unlink at any time, revoking the match token and pending leases without affecting a session in progress except to remove the platform's safety partner from the help pathways. The same layers apply to any introduction service, event, or venue through which two people without a trusted channel arrange to meet. One advantage of integration with a dating app is more opportunity to set up and populate a complete profile and be recognized for Safer Yes readiness and willingness as a supportive member of a community pursuing Mission Zero, as described above.

[0318] First-date terms, kept separate from who pays. Where no campus, employer, venue, or community program has already made the system a shared norm, the platform may offer a short set of first-date terms that both people accept, each on his or her own device,

before they meet. The terms are mutual and identical for every gender and orientation: either person may leave at any time; who pays creates no expectation of any kind; and, if the date ever moves toward intimacy, both will use the system. They may also carry each person's advance consent, revocable at any time, to the sealed conflict recording of section 8C, which in a jurisdiction requiring the consent of all parties to a recording can make that function lawful for the encounter. Alongside the terms, either person may offer to split the cost or to cover a part of it, even only the tip; in the common case of a woman meeting a man, picking up a share is a way for her to set the terms of the evening on an equal footing. The platform shows the cost-sharing offer with an express statement that it creates no expectation, and records it as an instrument separate from the terms, with no reference between them, so that a payment can never be read as consideration for any intimate term. Each person's acceptance is signed and time-stamped on his or her own device. Where a person later reports that the other broke the terms, the platform may act under its own terms of service, for example by suspending the account, relying on the signed terms and on an attestation from the reporting person's device of whether the system was used, without receiving any audio, transcript, or consent state of the encounter.

8E. Device and peripheral configurations

[0319] The controller is device-agnostic, and the configurations below are enumerated to make clear that any of them, alone or in combination, is within the disclosure. Each device contributes one or more of: an input channel (voice, touch, gesture, biometric, physiological, neural), an output channel (audio, haptic, light, display), a sensing role (sensor, chaperone, or witness, section 1), a playback target, or a help-stage pathway, and each is bound to a participant under section 4E and classified under section 7 before it may contribute to any role.

[0320] Configurations common today. (a) A single phone on a nightstand, with or without a Bluetooth speaker. (b) One or both phones affixed in a common location (a headboard mount, a nightstand stand, or a wall cradle) facing the bed, so that a single ambient display is visible to both participants and reachable by either for touch affirmation, with the phone's own screen, haptic, and biometric sensor serving as the shared surface described below. (c) A phone plus a smartwatch, fitness band, or smart ring per participant, supplying haptic renewal reminders, touch affirmation, heart-rate and skin-conductance signals, and on-body binding. (d) A voice-assistant smart speaker in any of the three topologies of section 8A. (e) Wireless earbuds or a bone-conduction headset as a private audio channel for check-ins and reminders inaudible to the partner. (f) A tablet or laptop as controller and display. (g) Smart lighting, a smart plug, or a smart-home hub as alarm and ambient outputs. (h) A smart lock, intercom, or doorbell as the access device and knock sensor of section 4. (i) A consumer presence sensor (millimeter-wave radar, passive infrared, or ultrasonic) as the non-imaging room sensor of section 7. (j) A television, streaming stick, or game console as a playback target. (k) A vehicle head unit as described above. (l) A keychain or phone-accessory breath-alcohol sensor (section 6A). (m) A dedicated bedside device as described above. (n) A compatible remote control (a Bluetooth or Bluetooth Low Energy media remote, a camera-shutter remote, or a presentation clicker) kept bedside, whose play, pause, and stop buttons are mapped to the conserved signals of the system: play as affirmation at the enrolled tier, pause as the brake of section 3B, and stop as a stop word, each with the attribution level of the participant bound to the remote under section 4E, or UNATTRIBUTED where the remote is shared. The remote is an input of deliberate simplicity: a button press is firm, unambiguous, and requires no speech, and the remote itself is a physical metaphor grounded in an object whose meaning is universally understood, so that a participant who cannot find words, or whose partner would not hear them, can nonetheless press the one button that everyone already knows means stop. A remote is inexpensive, requires no pairing beyond standard Bluetooth, and because it presents to the operating system as a standard media-key device, the mapping works even where the platform permits no other input.

[0321] Emerging and specialized configurations. (o) A head-worn display or extended-reality headset contributing hand-, eye-, and head-tracking as gesture and interaction inputs, with any camera confined under section 7. (p) A home robot in the chaperone role. (q) A smart bed or mattress sensor reporting presence, movement, and respiration. (r) A connected intimate device acting as a touch-affirmation surface, a haptic output, or an actuator that the controller stops on lapse exactly as it stops media. (s) A smart textile or garment with woven electrodes or pressure sensors. (t) Ultra-wideband anchors for sub-meter presence and separation sensing. (u) A non-invasive neural interface (an electroencephalographic headcap, headband, or in-ear electrode set, or a near-infrared spectroscopy

headband) contributing an attention, arousal, distress, or motor-imagery classifier output as a corroborating signal, and optionally an enrolled imagined cue as a private affirmation, brake, or stop channel for a participant who cannot or prefers not to speak or move, under the neural-data rules of the next paragraph. (v) An implanted or surgically placed neural interface, where a participant already has one for medical reasons, contributing the same classifier outputs under the same rules. (w) A satellite-messaging-capable phone, wearable, or bedside device as the help pathway where no terrestrial network exists. (x) A projector, such as a compact bedside or ceiling-mounted projector, that projects onto a wall or the ceiling during intimacy, as the participants elect, any of: guidance or tips; the session status; a reminder of the words that trigger a pause, a brake, or a stop; and a reminder of the words by which participants may say how happy they are; rendered dimly enough not to disturb the room and, like any shared display, never showing a participant's private settings. (y) A neck-worn choker, necklace, or collar with a pressure or strain sensor, as described in section 8C.

[0322] Neural data in bedroom mode. Neural data is treated as a class more sensitive than imaging. A neural interface may contribute only where the participant wearing it has expressly enabled it at session start, with the session announcement naming it; it may never be enabled by the other participant. Raw electrophysiological or hemodynamic signals are processed on-device, in a buffer that is overwritten, and only a low-dimensional classifier output (a state label, a cue-match result, or a confidence) reaches the controller; no raw signal, spectrogram, embedding, or derived feature is persisted, transmitted, or exposed to any application, and the controller refuses the device unless its firmware or the host operating system attests that confinement. A neural signal is never sufficient on its own to establish affirmation; it corroborates an expressed affirmation, raises the vigilance level, or triggers a check-in, and a neural stop or brake cue is honored with the same priority as a spoken one, because the system fails closed. The event log records that a neural device was present and the state labels it produced, never the signals, and the log-rights provisions of section 7 apply. A neural interface never holds the witness role.

[0323] Shared ambient display and intimacy coordination. The coupled cues of this paragraph serve, for two people alone, the function an intimacy coordinator serves on a film or theater set: a neutral presence that keeps both parties continuously aligned on what has been agreed, signals quietly when agreement needs renewing, and lets either pause without the pause becoming a confrontation, except that here the coordinator is a display both can see and neither has to address. Where a phone or dedicated device is mounted for mutual view, the controller renders a coupled ambient display: a low-luminance field whose color, breathing rate, or glow reflects the joint consent state (steady and warm while both participants are affirmed, slowing and cooling as an affirmation window nears lapse, and shifting to the check-in treatment of FIG. 3 on lapse) so that both participants share one quiet picture of where they stand without either having to ask. The display may be deliberately coarse: in one configuration it shows the joint state only (never which participant's affirmation is nearing lapse, never a score, and never text) so that a glance at it conveys reassurance or a gentle prompt rather than an accusation. In other configurations, selected by the participants at enrollment, the display shows each participant's own state side by side, a countdown of the current affirmation window, a numeric or graded consent level, or short text such as the pending check-in question, and a participant may elect to have his or her own state shown to the partner, that election being one-way and revocable. The coarse configuration is the default under the High-vigilance and Discreet presets and for a first session between participants. Supportive cues (a brief brightening on a mutual affirmation, a pulse acknowledging a renewal, the receipt cue of section 4) are rendered on the same surface. The mounted phone remains the private channel of the participant who owns it: renewal reminders and secret concern cues are delivered to that participant's wearable or earbud when and where available, not to the shared display, and where no private channel is available they are delivered on the shared display only in a form indistinguishable from the display's ordinary activity (a renewal reminder as the same slow cooling that precedes any lapse, a secret concern cue as the response it triggers and never as the cue itself) so that the display reveals nothing about which participant prompted a change. The display is readable only within the room, at a brightness bounded by the ambient light sensor, with no notification, lock-screen preview, or screenshot path exposed, and the controller dims or blanks it when the phone's own sensors report that it has been picked up or unmounted, so that the shared surface cannot be carried out of the room while showing session state.

9. Example method

[0324] In one embodiment a method comprises: (a) receiving, at a user device, enrollment data identifying at least a go word, a stop word, and a close phrase for each of one or more participants and identifying media content or a silent carrier; (b) capturing audio at the user device while the media plays and cancelling from it the playback reference signal; (c) attributing voiced segments to participants (by voiceprint where consented and feasible, otherwise by diarization or proximity) and assigning an attribution level; (d) selecting an active policy by protectively merging each participant's policy; (e) detecting on-device, without a wake word, enrolled go words and stop words, and classifying other utterances by meaning into go, greyzone, or stop with a risk weight; (f) optionally, for a participant who elected it, estimating affect for that participant's own voice; (g) assigning each participant a consent state at the tier of the detected go word, GREYZONE for a greyzone meaning, and WITHDRAWN for a stop word or distress indicator, and computing the joint state as the minimum; (h) maintaining an affirmation timer refreshed by the required-tier state from every participant within a mutual window, and optionally issuing a private renewal reminder and receipt cue; (i) causing the media or silent carrier to play only while no lapse has been signalled and no stop word detected; (j) upon a lapse, fading the media and, after the fade, upon absence of interaction for the check-in, alarm, and help periods in turn, issuing a check-in alert, sounding an alarm, and calling for help, with verified responder entry where designated; (k) upon a greyzone, conducting a clarifying exchange and, where selected, announcing and tracking a reaction script; (l) upon a brake phrase, executing the brake script; (m) upon a stop word, loss of a link, or failure to verify media control, pausing immediately; (n) verifying each media directive by player state and escalating to a fallback adapter on failure; and (o) recording consent-state transitions, directives, acknowledgements, exchanges, scripts, escalation stages, and pathways in a hash-chained log without recording audio, except where a conflict recording is permitted by the participants' settings and preferences, community policy and practice, and law.

10. Hardware and computing environment (FIG. 9)

[0325] Referring to FIG. 9, each of user devices 112 and 114, wearables 117, smart speakers 706 and 714, a dedicated bedside device, and server 102 is a computing device 900 comprising at least one processor 902; memory 904 comprising volatile memory for working state (including the audio ring buffer, consent states, timers, and policy) and non-volatile memory holding the program modules 122 through 138 and the event log; and a system bus or interconnect 930 coupling the processor and memory to the peripherals present on that device. Processor 902 may include a general-purpose CPU together with a neural processing unit or digital signal processor on which the on-device speech recognition, speaker-embedding, affect, and semantic-classification (language) models execute. A secure element or trusted execution environment 906 holds voiceprint templates, the keys under which templates and the event log are encrypted, and the keys used to lease templates over local link 118.

[0326] Input hardware comprises one or more microphones 910 (a single microphone on a phone or wearable, or a far-field array on a smart speaker or dedicated device) with an acoustic echo cancellation front end that receives the playback reference 146 tapped from audio playback path 924; sensors 920 including an inertial measurement unit for motion and interaction detection, photoplethysmographic and skin-conductance sensors on wearables for heart-rate variability and distress indication, and presence or ranging sensors (ultra-wideband, Bluetooth ranging, passive infrared) for the presence gates of the escalation sequence; a physical button 922 usable as a stop or affirmation input, or the play, pause, and stop keys of a Bluetooth media remote mapped to affirmation, brake, and stop as described in section 8E; a biometric sensor 932, being a fingerprint sensor on a phone, tablet, computer, wearable, or button accessory, a wearable's photoplethysmographic, vein-pattern, or bioimpedance sensor, or the device's secure biometric unlock hardware, used for the momentary touch affirmation of section 4D, with matching performed in the sensor's or the operating system's secure hardware and only a match result or signed token delivered to the controller; and touch input on display 916. A home robot present as a chaperone, where used, is a further computing device 900 with its own sensors 920, microphone 910, wireless interface 908, and optionally speaker 912, display 916, and a mobility actuator, reporting participant actions and presence to the controller over local link 118 and rendering alerts under the alert output subsystem and, where enabled and the interruption of the affirmation envelope is severe enough, helping to keep the participants calm and a safe distance apart. A smart bed or bedside sensor, a camera or presence sensor, and an intimate device are each likewise a computing device 900 or a peripheral of one, contributing to

sensors 920 over wireless interface 908; camera frames, where a camera is enabled, are processed on-device for gesture and presence only and are never stored or transmitted. A wearable additionally includes an on-wrist or on-body detector (optical or capacitive) whose state the controller uses to maintain the device-to-participant binding of section 4E. A head-worn display contributes inertial, hand-tracking, and, where present, eye-tracking input as sensors 920, and its camera, where present, is subject to the on-device, non-storage limitation stated for cameras. Sensors 920 preferably include a non-imaging room sensor, most preferably a millimeter-wave radar module, and the sensor classification of section 7 governs which sensors may contribute to which roles. A breath-alcohol sensor 934, in a phone accessory, keychain or wearable device, dedicated or attending device, vehicle, or venue station, supplies the readings of section 6A over wireless interface 908 or a wired accessory port, with each reading time-stamped by clock 928 and held in the participant's own encrypted record. The device and peripheral configurations enumerated in section 8E, including the headboard-mounted shared ambient display, the bedside media remote, and the non-invasive neural interface, are each a computing device 900 or a peripheral of one and contribute to sensors 920, button 922, or display 916 as there described.

[0327] Output hardware comprises a speaker or audio output 912 through which the media, spoken check-ins, alarm tones, and help-stage audio are rendered; a haptic actuator 914 through which check-in taps and continuous alarm vibration are rendered on phones and wearables; a display 916 on which the check-in question and touch targets are shown and on which AI-facilitated intimacy coordination and pre-intimacy and pre-high-voltage checklists may be run; and a status light 918 on a wearable, bedside device, or smart speaker. Lighting device 115 is a separate computing device 900 in its own right (a color-capable bulb, strip, or lamp with its own processor, wireless interface, and light-emitting elements) receiving color, intensity, and pattern commands from alert output subsystem 134 over wireless interface 908. The same haptic actuator 914, lighting device 115, and speaker 912 render the silent ambient signal of section 4C in place of media content.

[0328] Wireless interfaces 908 comprise one or more of Wi-Fi, Bluetooth and Bluetooth Low Energy, ultra-wideband, cellular, and a smart-home radio or bridge interface (Zigbee, Thread, Matter, or a vendor bridge), by which the device communicates with other user devices and wearables over local link 118, with playback targets 140, 142, 144, with lighting device 115 and button 116, with smart speaker platforms as described with respect to FIG. 7, and with server 102, a dating platform 170 where linked, and emergency services 160 over network 106. A dedicated bedside device may include a cellular or satellite interface as a fallback for the help stage when the home network is unavailable, and a phone or wearable with an emergency satellite-messaging capability serves the same purpose. The controller, or a bridge it controls, may act as the controller of a smart-home fabric so that attending devices, lighting devices, and buttons from different vendors are paired and commanded uniformly. Power 926 and a clock 928 complete the device; the clock supplies the time stamps recorded in the event log and drives the affirmation timer and escalation periods. An access device (a smart lock, building intercom or access-control system, gate or garage controller, or hotel door system) is a further networked device, reached over wireless interface 908, a smart-home fabric, or the building's network, that the controller commands to grant entry at the help stage as described in section 4, and that reports its lock state and entry events back to the controller. A knock sensor (a microphone on the inside of the door or in the room, an accelerometer or contact sensor mounted on the door, or the microphone of a doorbell or intercom) supplies the signal for the audibly verified knock described in section 4.

[0329] Every functional component described in this disclosure is implemented by processor 902 executing instructions stored in memory 904 that receive signals from, and send commands to, the specific hardware identified above: the expressed-channel and implied-channel analyzers operate on frames from microphone 910; the consent state fusion component and affirmation timer operate on those analyzers' outputs, on button 922, and on sensors 920; the media control adapter layer issues commands over wireless interfaces 908 and audio playback path 924 and receives player-state events over the same interfaces; the alert output subsystem drives speaker 912, haptic actuator 914, display 916, status light 918, and lighting device 115; the voiceprint attribution component operates on secure element 906; and the event logger writes to non-volatile memory 904 under keys in secure element 906. The functions may be performed by one device or distributed among several, and may be implemented in software, firmware, hardware, or any combination, including in a system on a chip. The stages of the described processes may be reordered, combined, or performed concurrently without departing from the disclosed subject matter. Terms such as "module," "component," "analyzer," "adapter,"

"subsystem," and "engine" refer to any combination of hardware and software that performs the described function on the hardware described.

11. Robotic physical contact and companion robots

[0330] Scope of this section. This section concerns robots that touch people. Home and care robots, including humanoid robots, will increasingly make physical contact with people, in caregiving, bathing and dressing assistance, massage, companionship, and, for adults who choose it, intimacy. Each such contact raises the question this disclosure addresses for contact between people: whether the contact is wanted, continuously, by the person touched. The mechanisms of this disclosure apply to that question unchanged, with the robot in the role of the party whose actions are gated.

[0331] Consent-gated robotic contact. A robot having one or more actuated limbs or end effectors, force or torque sensing, and a controller of this disclosure may make contact with a person only while that person's affirmation for that contact is current. The person's boundary map, described above, identifies body regions and acts that are excluded, permitted, or permitted only after a specific affirmation, and the robot's motion planner treats an excluded region as a keep-out zone. Contact force and pressure are held within limits set for each region, consistent with the power- and force-limiting principles of collaborative-robot safety practice, and lower limits apply wherever the person has set them. Upon a lapse, the robot ramps its force down over the fade period and releases; upon a stop, a brake, or a concern cue, it releases at once and withdraws to a set distance; and upon a detected freeze response, distress indicator, or interruption of breathing, it releases, withdraws, and advances the escalation sequence to the help stage. The robot treats its own contact as the gated output, so that its continuation, its slowing, and its release are perceptible to the person in the same way as the media. FIG. 12 illustrates these states: no contact 1202; contact permitted 1204; lapse 1206; stop, brake, or concern 1208; and distress 1210; together with the invariants 1212 that hold in every state and every variant.

[0332] Companion robots for adult intimacy. Where an adult uses a companion robot for intimacy, the rules above apply with the following additions, which no variant may relax. The robot never applies pressure to the neck, whatever is requested, in keeping with the treatment of strangulation in this disclosure. The robot offers no mode that simulates the overriding of a refusal or that rewards persisting past a stop. The robot, its appearance, its voice, and its persona are never childlike, and the system refuses to operate with any robot or persona that is. Use requires the age assurance described in this disclosure, and the consent-session functions are unavailable to any person not established to be an adult.

[0333] Consent-practice mode for adults. A companion robot, or a virtual partner rendered in a head-worn display, may offer adults a consent-practice mode for learning to manage limits in a warmer state than a classroom allows. The robot models the conversation and signals of this disclosure: it asks before acting; it answers with affirmations, scoped requests, brakes, and stops; it sometimes declines; and it responds positively when the user asks, waits, and honors what he or she is told. The mode tracks, privately for the user, how reliably the user checks in, honors scoped stops and brakes, and recovers from a lapse, and it may be structured in graduated levels that remain below high voltage and that do not require the user to undress. It is available only to adults established by age assurance and, within that limit, only where and to whom the law and the rules of the user's community permit, so that a jurisdiction or community may restrict the mode further, for example by a higher minimum age, a training prerequisite, or a prohibition, but never extend it to a person who is not an adult; and it is offered as practice that supports, and does not replace, the education and training described in this disclosure.

[0334] Although the subject matter has been described in language specific to structural features and methodological acts, the subject matter defined in the claims is not necessarily limited to the specific features or acts described above, which are disclosed as example forms of implementing the claims.

[0335] Combinations expressly contemplated. The features described in the several sections of this disclosure are described together because they are intended to be used together, and each feature may be combined with any other, except where this disclosure states that two features are alternatives to one another or incompatible. Without limiting that statement, each of the following is expressly contemplated in combination with each of the others and with every embodiment described in this disclosure: (a) on-device detection,

continuous and without a wake word, of enrolled go words and stop words and of universal safe words, in microphone audio from which the media content or other output being rendered is cancelled using that content as a reference; (b) a joint consent state computed as the minimum of the consent states of all participants, such that a stop by any one participant withdraws it and only a renewed affirmation by every participant restores it; (c) the distinct responses to a lapse and to a stop, namely a gradual fade, softening, or slowing over a fade period upon a lapse, and an abrupt cut, release, or stop within a stop latency shorter than the fade period upon a stop; (d) the distinction between an interaction, which halts the advancement of escalation, and an affirmation, which alone resumes the gated output; and (e) the escalation sequence of check-in, alarm, and call for help. In particular, each of these features may be combined with the operating-system renewal gate, the smart-speaker and voice-assistant embodiments, the renewal-token message format, the registry of universal safe words, the software development kit and its privacy kernel, the fade at a musical boundary, the classification of stop utterances, the hummed help motif, the wearable detection of a freeze response, the duress affirmation, the access-control, smart-home, smart-bed, intimate-device, recording, and venue embodiments, and the robot embodiments of section 11, in each case in either of the Solo mode and the Duo mode and at any rung of the listening ladder. The examples given for any element are illustrative and not exhaustive.

[0336] Trademarks. Names of products, services, and companies used in this disclosure, including the names of voice assistants and their wake words, are trademarks or service marks of their respective owners and are used only to identify those products and services and to describe the state of the art; their use implies no affiliation with, or endorsement by, their owners.

[0337] Purpose and scope of this disclosure. This provisional application is intended as a roadmap of the many ways in which technology can help reduce, dramatically and durably, the crisis of sexual assault through primary prevention, secondary prevention (including sexual violence interruption), and mitigation, while making intimacy safer, closer, and more rewarding for the people who use it, and it describes more embodiments, features, and combinations than the inventor expects to claim. The inventor may elect not to pursue, in any non-provisional application claiming the benefit of this application, some or many of the claims presented here or of the embodiments described. It is the inventor's intention that subject matter described in this application that is not claimed in any later application, and that has not already been validly claimed by others, be available, upon publication of this disclosure or of any application that claims its benefit and incorporates it, as prior art and as a public resource on which future innovators may build freely. Nothing in this paragraph is a dedication of any subject matter that is claimed in this application or in any application claiming its benefit, a waiver of any right in such subject matter, or a limitation on the scope of any claim.

EXEMPLARY AND ILLUSTRATIVE CLAIMS

The following claims are exemplary and illustrative. They are presented to illustrate the subject matter disclosed and do not limit the scope of this disclosure or of any claims that may later be presented in an application claiming the benefit of this provisional application.

1. A consent-gated media playback system comprising a user device having a microphone, a processor, a memory, and an interface to a media playback target, the memory storing instructions which, when executed by the processor, cause the processor to: detect, by on-device speech recognition on audio from the microphone, an enrolled go word and an enrolled stop word of at least one participant, the detection being performed continuously and without a wake word while media content that the processor caused to be played is playing on the media playback target, using a playback reference signal corresponding to the media content for acoustic echo cancellation of the media content from the audio; maintain an affirmation timer that is reset by detection of the go word and that signals a lapse when the go word is not detected within an affirmation window; cause the media content to play on the media playback target only while no lapse has been signalled and no stop word has been detected; upon a lapse, command the media playback target through the interface to fade the media content over a fade period to a reduced volume, to silence, or to a stop, and confirm from a player state returned through the interface that the command took effect; upon detection of the stop word, command the media playback target to stop the media content abruptly, within a stop latency shorter than the fade period; and resume the media content only upon a subsequent detection of the go word; wherein the media content is rendered audibly in a room in which the at least one participant is present, such that its continuation, its fade, and its abrupt stop are each perceptible to every participant present without reference to a display; and wherein, after the media content has been faded upon a lapse, the instructions further cause the processor, upon absence of any interaction by a participant for a check-in period, to issue a check-in alert and, upon continued absence of interaction, to sound an alarm and then to call for help through a pre-configured pathway, any interaction halting advancement of the check-in alert, alarm, and call for help without resuming the media content.

2. A consent-gated media playback system comprising at least one user device having a microphone and a processor executing a consent-gated media controller, the controller comprising: an expressed-channel analyzer configured to perform on-device speech recognition on audio captured by the microphone and to detect predefined go words and stop words; a consent state fusion component configured to assign a consent state to each of a plurality of participants and to compute a joint consent state as the minimum of the participants' consent states, wherein a stop word from any participant sets the joint consent state to withdrawn; a lapse detector comprising an affirmation timer that is reset by an affirmation from every participant and that signals a lapse when no such affirmation is received within an affirmation window; a media control adapter layer configured to cause media content to play only while the joint consent state is affirmed and no lapse has been signalled, and, upon a lapse, to fade the media content to silence over a fade period while issuing a cue for re-affirmation and to stop the media content if no affirmation is received, and, upon a stop word from any participant, to stop the media content abruptly, within a stop latency shorter than the fade period, rather than fading it; and an escalation sequence configured, after the media content has been softened, muted, or stopped in response to the lapse, to detect the absence of any interaction by the participants and, upon absence of interaction for a check-in period, to issue a check-in alert through an alert output subsystem to at least one of: a speaker, a haptic actuator or display of a wearable device, phone, or other electronic accessory of a participant reached over a wireless interface, and a connected lighting device whose color, intensity, or pattern is changed; upon absence of interaction for an alarm period, to sound an alarm; and upon absence of interaction for a help period, to call for help through at least one pre-configured pathway; wherein interaction of any kind halts advancement of the escalation sequence without resuming the media content, and only an affirmation from every participant resumes the media content.

3. The system of claim 2, wherein go words are enrolled by each participant at two or more tiers including an affirmed tier and an enthusiastic tier, the consent state fusion component assigns each participant a consent state at the tier of the go word detected, and an active policy specifies the tier required to reset the affirmation timer, whereby enthusiasm is determined by the words the participant chose and spoke and not by an assessment of the manner in which they were spoken.

4. The system of claim 2, wherein the expressed-channel analyzer further comprises a semantic classifier executing on the user device and comprising a language model or intent model, configured to assign to each transcribed utterance, and to consent-relevant non-lexical vocalizations, a meaning category selected from at least a go category, a greyzone category, and a stop category, together with a risk weight and a confidence; wherein the thresholds at which the categories are acted upon are asymmetric such that a stop category is acted upon at a lower confidence and risk weight than a go category; wherein a greyzone category assigns the participant a greyzone consent state that does not reset the affirmation timer and governs the joint consent state under the minimum rule; wherein enrolled stop words and go words are detected independently of the classifier and are honored regardless of its output, such that the classifier cannot suppress a stop or override an enrolled go word; wherein the classifier distinguishes an utterance of a stop word spoken alone, an utterance of a stop word with an expressed reason of pain, fear, or distress, and an utterance of a stop word with an explicit scope limited to a particular act, substance, place, or manner, honoring the first as a stop pending clarification, the second as a withdrawal with the reason relayed to a partner of the speaking participant, and the third as a scoped request in the greyzone category that ducks rather than pauses the media and is announced to the partner in the speaker's words, the enrolled stop word pausing the media in every case; wherein the classifier places a partly voiced or non-lexical request in a greyzone sub-category for resolution by a clarifying exchange; wherein an active policy maps meaning categories and risk-weight bands to response patterns; and wherein the classifier is adapted on-device to example phrases supplied by each participant and no transcript is transmitted from the user device.

5. The system of claim 4, wherein, upon a greyzone consent state, the controller conducts a time-limited clarifying exchange delivered, as the speaker's policy elects, audibly through an attending device or speaker, audiovisually through a display, privately through the speaker's own phone, wearable, or earpiece, or by a direction to one or both participants to check a phone or other device on which the exchange is then presented, the exchange consisting of one or more short questions constrained to the speaker's intended scope, duration, intensity, and desired reaction from the partner, phrased to be answerable by a single word, an enrolled word, a touch affirmation, or a button, and presenting stopping and continuing as options of equal prominence; wherein the desired reaction may be selected from reaction scripts pre-agreed at enrollment including at least a neutral or resting-pose script, a pause-and-wait script, a change script, and a stop-and-separate script; wherein upon selection of a script the controller announces the script to the partner and tracks compliance by interaction and presence sensing and by the speaker's subsequent utterances, escalating as a withdrawal if the script is not followed; wherein no answer within the time limit is treated as at least a lapse; and wherein the exchange, the script, and compliance are recorded in an event log.

6. The system of claim 2, further comprising a cooldown command layer in which, upon an enrolled brake phrase or touch from any participant, upon selection in a clarifying exchange, or upon classification of an utterance as a request for a break, the controller enters a brake state distinct from the withdrawn state and executes a brake script agreed by the participants at enrollment, the script comprising: fading the media content and changing a connected lighting device to a brake color; instructing the participants to separate and, where ranging between their devices or a bed or room sensor resolves their positions, confirming separation and prompting again privately if they remain in contact; instructing each participant to adopt a pre-agreed calming posture and delivering, on each participant's own wearable, phone, or accessory, a breathing-paced haptic rhythm for a settling period; at the end of the settling period, asking each participant privately on his or her own device whether he or she is calm and de-aroused and extending the settling period until each has confirmed, optionally requiring in addition that a wearable report the participant's heart rate within a band of an enrolled resting rate for a set interval; upon both confirmations, announcing through a speaker that both participants have confirmed they are calm; then opening a talk step with the media off; and resuming the media content only upon a go word from each participant given after the talk step; wherein a stop word, safe word, button press, or distress indicator takes precedence over the brake script at any point, the escalation sequence remains armed throughout, the brake is recorded as a cooperative pause and not as a withdrawal, and sensed values are shown only to their owner and logged only as pass or fail.

7. The system of claim 2, further comprising an interface to a breath-alcohol sensor, wherein, upon the participant's departure from a drinking occasion detected by check-out, a venue exit station, geofence departure, or a scheduled prompt, the controller prompts the participant for a breath sample, associates the time-stamped reading with the participant's account in the participant's own

encrypted record without transmission to any partner or platform, and, when the reading exceeds a threshold set by the participant or by an institutional policy the participant adopted, applies a consent lockout to that participant for a period computed by a clearance model and a margin policy, the clearance model using a population elimination rate or a per-participant rate fitted from the participant's own calibrated readings, and the margin policy comprising at least one of a modeled-concentration threshold, a minimum elapsed time from the last drink combined with the threshold, fixed padding hours, and early release only by a fresh reading below the threshold; wherein during the lockout the controller refuses to open a session in which the participant is a consenting party and refuses to credit the participant's affirmations, and announces only that a session is not available without disclosing the reading; and wherein the lockout is a rule adopted by the participant in advance and is not an adjudication of capacity.

8. The system of claim 2, further comprising an implied-channel analyzer configured to estimate an affective quality of a participant's speech and non-speech vocalizations, wherein each participant elects, for his or her own voice, one of at least three tone modes: an off mode in which no affect estimate is applied; a private mode in which a distress indicator in that participant's voice causes a private prompt to that participant and is not applied to, displayed to, or logged against any other participant; and a graded mode in which a go word from that participant resets the affirmation timer only when the affect estimate accompanying it is concordant with the go word, and a discordant go word yields an ambiguous consent state that does not reset the affirmation timer and causes a prompt for re-affirmation; and wherein the tone mode in force is recorded with each affirmation in an event log.

9. The system of claim 8, wherein the distress indicator comprises at least one of crying, strained phonation, or a biometric distress signal from a wearable device worn by that participant, wherein the analyzer is calibrated per participant from a neutral speech baseline and a go-word baseline recorded at enrollment, and wherein a distress indicator meeting a withdrawal threshold sets that participant's consent state to withdrawn and invokes an emergency response mechanism.

10. The system of claim 2, wherein the alert output subsystem is further configured, for a participant who has enabled a renewal reminder, to issue, at a configurable lead time before the affirmation timer expires, a private renewal reminder as a haptic pattern to a wearable device, phone, or electronic accessory of each participant whose affirmation has not been received in the current affirmation window, over a wireless interface, without audible change in a room in which the participants are present and without delivery to any other participant; and to issue a private receipt cue to that participant's wearable device, phone, or accessory when the participant's affirmation is credited; wherein the lead time is configurable per participant and is optionally adapted from that participant's observed renewal timing within enrolled bounds.

11. The system of claim 2, wherein, for participants who elect not to play media content during a session, an active policy selects in place of the media content a silent ambient signal comprising at least one of a periodic haptic pulse rendered on each participant's wearable device, phone, or electronic accessory over a wireless interface, a lighting state on a connected lighting device, and a low ambient tone, the silent ambient signal continuing only while no lapse has been signalled and no stop word has been detected, changing character and then ceasing upon a lapse in place of the fade, ceasing abruptly with a distinct haptic pattern upon a stop word, and being followed by the check-in, alarm, and help stages of the escalation sequence unchanged; and wherein the controller selects the silent ambient signal automatically, and announces the selection, when no media playback target is reachable or control of the media content cannot be verified.

12. The system of claim 2, wherein interaction comprises any detected action of a participant—speech of any content, laughter, motion, gesture, touch, a change in a biometric signal, or a button press—detected at any user device, wearable, attending device, or other sensing device present and paired to the controller.

13. The system of claim 12, wherein an attending device in a chaperone role also renders the check-in alert, the alarm, or help-stage output through its speaker, display, light, or motion, a mobile attending device positioning itself to be heard or travelling toward a door; and wherein an attending device in a witness role signs, with a key provisioned at pairing, the interaction and presence events it observed, the signed events being appended to a hash-chained event log as corroboration from a device not controlled by either participant after pairing.

14. The system of claim 12, wherein interaction and presence of the participants are detected by at least one non-imaging sensor whose raw output cannot be rendered into an image from which a person is recognizable, the non-imaging sensor being at least one of a millimeter-wave radar, a low-resolution thermal array, a coarse-zone time-of-flight sensor, an optically blurred sensor, an ultrasonic or acoustic ranging sensor, and a passive infrared sensor; wherein any imaging sensor, including a near-infrared or full-resolution depth camera, participates only where expressly enabled by the participants at session start, with its frames processed on-device for inference only and never persisted, transmitted, or exposed to an application; and wherein an attending device holds a witness role only if its sensors are non-imaging or its imaging sensor is optically or physically incapable of forming a recognizable image.

15. The system of claim 2, wherein the check-in alert is repeated at a repeat interval with increasing intensity and through additional channels including a connected lighting device, a haptic actuator of a wearable device, and a display.

16. The system of claim 2, wherein the alarm comprises an audible tone intended to be heard outside a room in which the participants are present, flashing of a connected lighting device, continuous haptic output on a wearable device, and a spoken warning stating the time remaining before help will be called and the phrase that cancels it.

17. The system of claim 2, wherein the at least one pre-configured pathway comprises an ordered sequence of pathways including two or more of: a message and automated voice call to a trusted contact; a request to a peer emergency-response network for a nearby responder to check on a location of the participants; a loud spoken help stage from the user device addressed to persons nearby; and invocation of a communications link to an emergency dispatch center, the call for help reporting an unattended and unclosed consent session together with the last consent state, the time of last interaction, and the location.

18. The system of claim 17, further comprising an interface to an access device comprising at least one of a smart lock, a building intercom or access-control system, a gate or garage controller, and a venue door system, wherein an active policy designates, in advance and per participant, one or more authorized responders selected from a security professional, a monitoring service, a neighbor, a roommate, a friend, a family member, a building doorman, concierge, or receptionist, a peer responder, an emergency medical technician, a paramedic, a firefighter, a police officer, and an emergency service, and designates for each whether the responder may be admitted; wherein, only at the help stage, upon acceptance of the alert by a designated responder permitted to be admitted, and upon verification at a door that the person present is that responder by at least one of a one-time code, a credential presented by the responder's device, the building's access-control system, a monitoring operator, and an audibly verified knock in which a knock pattern communicated to the responder and not spoken inside a room in which the participants are present is detected by a microphone or door-mounted sensor and matched by pattern timing, impulse signature consistent with that door, and presence at the door, and, where the responder has shared a device location for the response, upon the location being within a configured distance of a dwelling unit containing the door at the time of verification, the controller commands the access device, or a sequence of exterior and interior access devices along a path, to grant entry for a time-limited window while announcing inside the room who is being admitted; wherein a participant may cancel the grant by an OK phrase before it takes effect; wherein the active policy may require a second signal corroborating the absence of interaction before entry is granted; and wherein the grant, the verification method, the responder's identifier, the time of entry, and any cancellation are recorded in an event log.

19. The system of claim 18, wherein the active policy requires the audibly verified knock, alone or in addition to another verification, before entry is granted; the knock pattern is generated for the accepted call for help and expires with it, is delivered to the responder in the alert acceptance message, and is not rendered by any device inside the room; verification compares the pattern of impacts to the expected pattern within a timing tolerance, compares the impulse signature of the impacts to a signature learned for that door, requires agreement between a microphone and a door-mounted accelerometer or contact sensor where one is present, and requires that a non-imaging presence sensor covering the door report a person present at the time of the knock; and a knock failing verification is logged and, after a configured number of failures, the controller falls back to another verification method or declines entry.

20. The system of claim 18, wherein, for a door having no access device the controller can command, the grant instead delivers to the verified responder entry instructions comprising at least one of a lockbox location and a one-time code the controller sets or

reveals, a disclosed spare-key location, a building entry code, and a front-desk contact; and wherein, where a key to the door is held by a trusted neighbor designated in the active policy, the controller alerts the neighbor as a keyholder to admit the responder or to respond in the neighbor's own right, and logs the keyholder's presence at the door as corroboration of the entry; and wherein responder alerting, acceptance, and location sharing are optionally provided by a peer emergency-response platform that returns the accepting responder's identity, acceptance, and location to the controller.

21. The system of claim 17, wherein the call for help to the emergency dispatch center is classified as a no-interaction incident distinguishable from a stop-word incident, and includes room or floor disambiguation data collected at session start.

22. The system of claim 2, wherein a session is ended only by an enrolled close phrase or an explicit confirmation on the user device, and a session that has not been ended is presumed ongoing for purposes of the escalation sequence.

23. The system of claim 2, wherein the controller detects a soft close—the participants' devices leaving the room, the user device idle on a charger for a configured period, or an explicit sleep mode—and ends the session with a single notification instead of advancing the escalation sequence.

24. The system of claim 2, wherein the controller confirms continued presence of the participants by short-range ranging between their devices before advancing the escalation sequence beyond the check-in.

25. The system of claim 2, wherein an enrolled OK phrase spoken during the alarm or help stage cancels pending pathways, returns the escalation sequence to the check-in stage, and causes any pathway already invoked to be logged and notified of the cancellation.

26. The system of claim 2, wherein an enrolled stop word or safe word, an utterance classified by the expressed-channel analyzer as a withdrawal, a button press, or a distress indicator from any participant bypasses the escalation sequence and invokes an emergency response mechanism immediately.

27. The system of claim 2, wherein the affirmation window, fade period, check-in period, alarm period, and help period, and a repeat interval at which the check-in alert is repeated, are each configurable at enrollment and the affirmation window is lengthened within a bounded range while a dynamic consent score remains above a threshold and shortened when it falls.

28. The system of claim 2, wherein the media control adapter layer comprises a plurality of adapters including at least two of: an application-remote adapter that controls a third-party streaming application installed on the user device; a remote-playback adapter that controls a playback device registered to a user's streaming account; a first-party music framework adapter; an operating-system audio-session interruption adapter that pauses whatever audio source is active on the user device; a networked playback device adapter; and a local media library adapter; and a closed-loop verifier configured to confirm from player-state notifications that each fade, stop, or play directive took effect within a latency bound and to escalate to a next adapter when it did not.

29. The system of claim 2, wherein a playback reference signal corresponding to the media content that the controller caused to be played is supplied to the expressed-channel analyzer for acoustic echo cancellation, whereby lyrics of the media content are excluded from go word and stop word detection.

30. The system of claim 2, wherein the plurality of participants share a single user device and the consent state fusion component attributes voiced segments to individual participants by speaker diarization using speaker embeddings enrolled at the start of a session.

31. The system of claim 2, wherein each participant has a user device executing an instance of the controller, the user devices exchange consent-state heartbeats over a local link, and loss of the local link causes the media content to fade and the escalation sequence to begin.

32. The system of claim 2, further comprising an event logger configured to record consent-state transitions, escalation-sequence stages, media directives, verification acknowledgements, pathways invoked, cancellations, and latencies as a hash-chained, append-only log, wherein captured audio is not recorded.

33. A method for gating media playback on continuously affirmed consent, comprising: receiving enrollment data identifying at least a go word, a stop word, and a close phrase for each of a plurality of participants and identifying media content; capturing audio at a user device; attributing voiced segments of the captured audio to individual participants; detecting, on the user device, go words and stop words in the attributed segments; assigning to each participant a consent state and computing a joint consent state as the minimum of the participants' consent states; maintaining an affirmation timer reset upon an affirmation from every participant within a mutual window; causing the media content to play only while the joint consent state is affirmed and the affirmation timer has not expired; upon expiry of the affirmation timer, fading the media content to silence over a fade period while issuing a cue for re-affirmation, and stopping the media content if no affirmation is received; after the media content has been softened, muted, or stopped in response to the expiry, upon absence of any interaction for a check-in period, issuing a spoken check-in and repeating it at increasing intensity; upon absence of any interaction for an alarm period, sounding an alarm; upon absence of any interaction for a help period, calling for help through a pre-configured sequence of pathways and reporting an unattended, unclosed session; halting advancement of the check-in, alarm, and call for help upon any interaction without resuming the media content; resuming the media content only upon an affirmation from every participant; ending the session only upon the close phrase or an explicit confirmation; and recording consent-state transitions, escalation stages, directives, and pathways invoked in a hash-chained event log without recording audio.

34. A non-transitory computer-readable medium storing instructions which, when executed by a processor of a user device having a microphone, cause the processor to perform the method of claim 33.

35. The system of claim 2, further comprising a policy engine configured to receive, from each participant independently, a non-affirmation policy specifying at least: what inputs count as affirmation and from which participants; the affirmation window, fade period, check-in period, alarm period, and help period; a media response on lapse selected from at least fade to silence, fade to a floor volume, duck and hold, crossfade to a designated prompt track, step down a playlist tier, stop, change a lighting device only, haptic cue only, and no change; a prompting form and addressee; which escalation stages are enabled and any sensor gate on advancement; help pathways, their order, and the data shared by each; cancel and resume authority; and logging scope; and to derive an active policy from the participants' policies and a mode preset.

36. The system of claim 35, wherein the policy engine derives the active policy protectively, such that the shortest timing value among the participants governs, a stop from any participant is honored, resumption requires the conjunction of every participant's resume conditions, the help pathways are the union of the participants' pathways with each pathway carrying only the data its owner authorized, and each participant's cancel authority setting is honored without regard to the other participant's settings.

37. The system of claim 35, wherein the media response on lapse is assigned separately for each stage of the escalation sequence.

38. The system of claim 35, wherein the active policy includes a discreet mode in which no audible or visible change is made to the media or a room in which the participants are present upon lapse, prompts are delivered only by haptic output, and the call for help is made silently.

39. The system of claim 2, further comprising a voiceprint attribution component configured, for a participant who has given separate, revocable consent to biometric processing and on a platform determined to support on-device speaker-embedding extraction and protected template storage, to enroll a voiceprint template stored in protected storage of the user device, to extract on-device a speaker embedding from each detected go word and stop word, to compare the embedding with enrolled templates, to apply a liveness check, and to assign an attribution level selected from verified, probable, unattributed, and challenge; wherein an affirmation is credited to a participant at the verified level only when the comparison exceeds a verification threshold and the liveness check passes, and wherein a stop word is honored at any attribution level.

40. The system of claim 39, wherein the voiceprint attribution component determines platform feasibility by an enrollment self-test measuring the separation between the participant's embeddings and reference embeddings on the user device and refuses to enable voiceprint attribution when a measured error rate exceeds a threshold, and wherein, when biometric consent is withheld or the

platform is not feasible, the component attributes utterances by diarization or device proximity at most at the probable level and records the attribution mode in force.

41. The system of claim 39, wherein the voiceprint template is leased to a second device in encrypted form, over a local link or over a transient or sustained encrypted server link, usable only for verification, bound to a session identifier, and destroyed at session close, on loss of the local link, or on revocation by the participant, whichever occurs first.

42. The system of claim 39, wherein the liveness check comprises at least one of: exclusion of audio matching a playback reference signal of the media content; spectral or channel analysis distinguishing speech radiated in a room in which the participants are present from speech emitted by a loudspeaker; correlation of the utterance with a contemporaneous laryngeal-vibration or breathing signal from a wearable device worn by the participant; and a challenge in which the participant is asked to repeat a randomly selected word not previously enrolled and the response is verified.

43. The system of claim 39, wherein an active policy specifies that cancellation of an escalation or resumption of the media content is honored only when the cancelling or affirming utterance is attributed at the verified level to a participant named in the policy, and wherein an attempted cancellation not so attributed is logged and does not halt the escalation sequence.

44. The system of claim 2, wherein an affirmation may be given by a momentary biometric touch on a device enrolled by the participant, selected from a fingerprint sensor of a phone, tablet, computer, wearable device, or button accessory; a physiological sensor of a wearable device matching a pulse-waveform, vein-pattern, or bioimpedance template; and a secure biometric unlock of the device performed in its secure hardware and returning a signature over a session-scoped nonce; wherein no camera image is made available to the controller, the biometric reading is discarded upon matching, the affirmation is attributed to the participant at a physically-verified level, and only the attribution level, the source, a device identifier, and a time stamp are recorded; and wherein an active policy may require the physically-verified level for a participant's renewals, for cancel authority, or as a second factor with a go word.

45. The system of claim 2, wherein the controller maintains a binding between each participant and each wearable device, phone, or electronic accessory of that participant, the binding being established at enrollment by pairing the device from within that participant's own account, refused for a device already bound to another participant, and optionally confirmed at session start by at least one of the device's on-body detection with continuity since its last unlock, a match of a pulse-waveform or motion signature read by the device to the participant's enrolled template or to the participant's phone, a momentary biometric touch on the device, and a secure unlock of the device by the participant; wherein an affirmation, cancellation, or interaction received from a bound device is attributed to the participant to whom it is bound and to no other participant, at an attribution level determined by the strength of the confirmation; wherein a device that reports removal, loses its link, or is re-worn is unbound and its inputs are not credited to any participant until confirmation is repeated; wherein renewal reminders, receipt cues, and private prompts are delivered only to devices bound to the participant they concern; and wherein a stop input from any device is honored regardless of binding.

46. The system of claim 39, wherein each consent-relevant utterance is recorded in a hash-chained event log as a tuple comprising a pseudonymous participant identifier, an utterance class, the attribution level, a score band, the liveness result, the attribution mode in force, and a time stamp, and wherein neither audio nor the speaker embedding is recorded.

47. The system of claim 2, wherein the media content is played through a voice-assistant smart speaker, the controller executes on a user device separate from the smart speaker and performs the on-device speech recognition on audio captured by the user device's microphone, and the media control adapter layer drives the smart speaker through at least one of a third-party voice application interface, a casting interface, a wireless multi-room audio interface, a streaming service's remote-playback interface targeting the smart speaker, and a smart-home device-control interface, the check-in alert, alarm, and call for help being emitted through the smart speaker.

48. The system of claim 47, wherein media-status events returned by the smart speaker's interface are consumed by a closed-loop verifier to confirm that fade and stop directives took effect, a track identifier and playback position reported by the smart speaker

are used as a playback reference signal for acoustic echo cancellation at the user device's microphone, and, where the interface reports no state, the verifier confirms cessation of playback acoustically at the user device's microphone.

49. The system of claim 2, wherein an affirmation is received through a voice-assistant platform's third-party voice application invoked by the platform's wake word and an enrolled invocation phrase, the voice application's backend receives a recognized intent and, where available and enabled by the participant, a platform-generated speaker identifier produced by the platform's native voice recognition, and the consent state fusion component assigns the affirmation a platform-attested attribution level that an active policy may treat as equivalent to verified, equivalent to probable, or insufficient.

50. The system of claim 49, wherein the controller treats a playback-stopped event resulting from the platform's native stop command as a withdrawal signal, and, unless the platform natively recognizes a universal safe word without its wake word, requires, before a session may begin in this configuration, that at least one user device continue to listen for stop words continuously or that a physical stop button be present, and otherwise either refuses to begin the session or begins it only in a reduced-protection mode expressly accepted by each participant whose acceptance the active policy requires.

51. The system of claim 2, wherein the controller executes natively on a smart speaker having a far-field microphone array, performs acoustic echo cancellation using the smart speaker's own playback as a playback reference signal, holds voiceprint templates in a secure element of the smart speaker, and emits the check-in alert, alarm, and call for help through the smart speaker, one or more user devices being optional companions for haptic prompts, presence ranging, and biometric signals.

52. The system of claim 2, wherein the controller, at session setup, enumerates smart speakers present, determines for each which of a speaker-as-endpoint, speaker-as-affirmation-channel, and controller-native-on-speaker configuration it supports, measures interface latency, composes the supported configurations into a single session in which affirmation events from every channel are tagged with their channel and attribution level and processed by the same fusion rules, optionally announces the composed configuration to the participants in plain language, and records it in an event log.

53. The system of claim 2, wherein a session is armed in response to a spoken invocation phrase recognized by a voice-assistant platform following its wake word and routed to the controller by at least one of launching a third-party voice application, triggering a routine configured on the platform, and invoking an application intent on a companion device; wherein, upon arming, the controller optionally announces through a smart speaker a composed configuration and protection level, prompts each participant by voice to speak a go word, and begins the media content only upon receipt within a mutual window of a go word attributed to every participant; and wherein the invocation phrase is not treated as a go word for any participant.

54. The system of claim 53, wherein a speaker identifier produced by the platform's native voice recognition and delivered with the invocation is recorded as a platform-attested attribution of the invoking participant, and the announcement is recorded in an event log together with the invoker's attribution level, channel, topology, and mode.

55. The system of claim 53, wherein, when the composed configuration provides no continuously listening user device and no physical stop button, and the smart speaker does not natively recognize a universal safe word without its wake word, the controller either begins only in a mode that each participant has expressly accepted as reduced protection or, under a policy so configured, refuses to begin.

56. The system of claim 53, wherein a spoken end phrase recognized by the platform following its wake word is treated as a close phrase subject to cancel-and-resume authority of an active policy, and the platform's native stop command is treated as a withdrawal signal.

57. The system of claim 53, wherein the controller conducts mode selection and a pre-interaction checklist by voice through the smart speaker, attributing each spoken answer to a participant, and permits any question to be deferred to a companion device for privacy.

58. The system of claim 53, wherein a companion application on the companion device creates, on a platform supporting user-configured routines with spoken starters, routines whose starters are participant-chosen invocation, close, and OK phrases and whose

actions arm, close, or cancel escalation of a session and set a connected lighting device and speaker volume accordingly, whereby the session may be started by voice on a platform offering no third-party voice application and by phrases that do not name the system.

59. A consent-gated media playback apparatus comprising: at least one user device having a processor, a memory, a microphone, a wireless interface, and at least one of a speaker, a haptic actuator, and a display; at least one wearable device or electronic accessory worn or held by a participant, having a haptic actuator and a wireless interface paired to the user device; a connected lighting device having a wireless interface and controllable color; a media playback target comprising at least one of a streaming application on the user device, an audio output path of the user device, and a networked speaker; and instructions stored in the memory which, when executed by the processor, cause the processor to: detect go words and stop words in audio from the microphone by on-device speech recognition; maintain an affirmation timer reset by an affirmation from every participant and signal a lapse when no such affirmation is received within an affirmation window; cause media content to play on the media playback target only while no lapse has been signalled and no stop word has been detected; upon a lapse, command the media playback target over the wireless interface of the user device or the audio output path to soften, mute, or stop the media content and confirm from a returned player state that the command took effect; thereafter, upon absence of any interaction detected from the microphone, from a motion or presence sensor, from a touch input, or from a physical button for a check-in period, command the haptic actuator of the wearable device or electronic accessory over its wireless interface, command the connected lighting device over its wireless interface to change color, and render a spoken check-in through the speaker; upon absence of interaction for an alarm period, render an alarm tone through the speaker and command the connected lighting device to flash; upon absence of interaction for a help period, transmit a call for help over the wireless interface of the user device to at least one pre-configured pathway; and record each lapse, alert, and command with a time stamp in a hash-chained log in the memory.

60. The apparatus of claim 59, further comprising a secure element in which voiceprint templates and log encryption keys are held, wherein the instructions further cause the processor to extract a speaker embedding from each detected go word and stop word, compare it with the voiceprint templates in the secure element, and record an attribution level with each affirmation in the hash-chained log.

61. The apparatus of claim 59, wherein the user device is a smart speaker having a far-field microphone array and an audio playback path from which a playback reference signal is tapped and supplied to the on-device speech recognition for acoustic echo cancellation.

62. The system of claim 2, wherein the participants comprise an enrolled participant and an unenrolled partner, an active policy designates that affirmation is awaited only from the enrolled participant, the joint consent state is computed over enrolled participants only, a stop word from the unenrolled partner is nonetheless honored, the configuration is announced at session start where the active policy and applicable law so require, and the unenrolled partner may become a session-scoped guest participant by a spoken enrollment whose data is deleted at session close.

63. The system of claim 2, further comprising an interface to a dating platform, wherein, upon mutual opt-in by two matched users within the platform, the platform passes a pseudonymous match token to each user's device; each user completes enrollment of go words, stop word, close phrase, policy, trusted contacts, and optionally a voiceprint template on his or her own device without transmission to the platform; and the platform receives from the system at most an outcome signal selected from a normal session close and an invocation of the help stage, and only where the user's policy permits.

64. The system of claim 63, wherein, upon the two users meeting, their devices perform a handshake over a local link by short-range ranging or by scanning a displayed code, the handshake binding the match token to each enrolled participant and to the device present, releasing any pending session-scoped voiceprint template lease, and arming a session, whereby an affirmation attributed by voiceprint is traceable to the matched account.

65. The system of claim 63, wherein, before the two users meet, the controller runs a check-in timer whose missed check-in escalates through the escalation sequence to a trusted contact and, where provided, to a safety partner designated by the dating platform as a help pathway.

66. The system of claim 63, wherein a dating policy preset is applied by default upon linking, comprising a shortened affirmation window, the escalation sequence enabled through the help stage, help pathways beginning with a trusted contact, and cancel authority requiring the verified voice of at least whichever user proposed the session.

67. The system of claim 63, wherein either user may unlink the system from the dating platform at any time, unlinking revoking the match token and pending leases without ending a session in progress, and wherein the platform is prevented from receiving audio, enrolled words, consent states, escalation stages, locations, or voiceprint data.

68. The system of claim 35, wherein the policy engine computes on the user device, from frequency ranges entered privately by each participant, a target cadence comprising the overlap of the ranges, and records the time of each session; wherein, at session start, the policy engine shortens the affirmation window from the value otherwise in force by an amount that increases with the extent to which the interval since the participants' preceding session exceeds the target cadence, the shortened window being bounded below by a first-session default and the shortening being limited to a fraction of the difference between the value otherwise in force and the first-session default; wherein the policy engine restores the window toward the value otherwise in force as sessions are held within the target cadence; wherein the shortening is announced at session start without attribution to either participant and may be declined for the session by either participant; and wherein neither participant's frequency range is disclosed to the other participant.

69. The system of claim 17, wherein, upon acceptance of the call for help by a responder, the controller renders through a speaker in a room in which the participants are present a pre-enrolled melodic motif, distinct from the media content, indicating that help has been acknowledged and is on the way, and renders a changed form of the motif upon verification of the responder's arrival at a door; and wherein the expressed-channel analyzer recognizes the same motif when hummed, whistled, or sung by a participant, by matching a key- and tempo-independent representation of its pitch intervals and inter-onset intervals against an enrolled template after acoustic echo cancellation of a playback reference signal, treats the recognized motif as a request for help that advances the escalation sequence to the help stage, and renders the motif back through the speaker as an acknowledgement that the request was received.

70. The system of claim 39, further comprising a solo mode in which one enrolled participant operates the controller without enrollment of another party present, wherein a session announcement is delivered privately to the enrolled participant's own wearable device, earpiece, or phone; wherein the expressed-channel analyzer processes only speech attributed by the voiceprint attribution component to the enrolled participant at the verified or probable level and discards other speech from a ring buffer without transcription, classification, or logging, except that a stop word in any voice is honored and logged as an unattributed stop without retaining audio or any speaker feature; and wherein the controller determines from a jurisdiction table, according to a region associated with the user device, whether the solo mode may operate without an audible announcement and whether a conflict recording may be armed, and, where the jurisdiction table indicates that all-party consent is required, either requires an audible announcement or restricts the session to non-audio inputs of the enrolled participant.

71. The system of claim 2, wherein a participant enrolls, privately and without disclosure to any other participant, a secret concern cue selected from a word, a touch pattern, a gesture, a button pattern, and a vocal cue recognized by acoustic signature, pitch contour, or rhythm, and binds the cue to one of a plurality of modes of concern including at least a greyzone mode, a brake mode, and an emergency mode; and wherein, upon recognition of the cue, the controller executes the response defined for the bound mode while announcing only that response, and represents the response, and not the cue or the participant who gave it, in any output perceptible to another participant.

72. The system of claim 2, wherein the media control adapter layer obtains the times of upcoming boundaries of the media content from at least one of section or audio-analysis metadata of a streaming service, time-synchronized lyrics, a queue and track-duration state of a media player, and on-device beat tracking or structural segmentation of a playback reference signal, the boundaries comprising at least one of an internal boundary within a track, selected from the end of a bar, a phrase, a chorus, a verse, a bridge, and a lyric line, and an external boundary between tracks; wherein the adapter layer schedules the start of the fade upon a lapse to coincide with a boundary at or before expiry of the affirmation timer, or with a boundary after expiry delayed by no more than a maximum deferral specified by an active policy, an affirmation received during the deferral cancelling the fade and the deferral not being

counted toward any period of the escalation sequence; and wherein the stop of the media content upon a stop word is executed without regard to any boundary.

73. The system of claim 72, wherein the adapter layer further aligns to a boundary at least one of a crossfade to a designated prompt track, the start of a next track, a resumption of the media content after renewed affirmation, and the issuance of a renewal reminder, a crossfade being matched in tempo and beat phase between the outgoing and incoming media; and wherein the deferral applied and the type and source of the boundary are recorded with a corresponding media directive in an event log.

74. The system of claim 72, wherein a response mapper of the media control adapter layer selects and sequences tracks of the media content according to their durations or the positions of their internal boundaries so that a track, or a section within a track, ends within a tolerance of the expiry of the current affirmation window, and re-plans the selection each time the affirmation timer is reset; wherein the selection follows changes in the length of the affirmation window during a session and, for a period of the session designated by the participants, is constrained to durations, tempi, and boundary spacings associated with that period; and wherein the selection neither lengthens the affirmation window nor delays a lapse response beyond the maximum deferral.

75. The system of claim 4, wherein each participant enrolls a boundary map comprising at least one of words to be used and words not to be used for the participant's body and its parts, areas that may not be touched or may be touched only after a specific affirmation, and acts designated welcome, conditional, or excluded; wherein the boundary map is held in the participant's own encrypted record and presented to another participant only upon the participant's election; and wherein the semantic classifier resolves a scoped stop or greyzone utterance that refers to an item in the boundary map against that item, and the clarifying exchange presents the item in the participant's own words.

76. The system of claim 68, wherein each participant privately enters a role preference selected from fixed roles, reciprocation within each session, and alternation at a selected interval; wherein, only where the preferences of both participants permit alternation, the controller suggests, by random selection or by rotation, an assignment of roles for a session and its duration, and tracks privately for each participant the balance of roles against that participant's own stated preference; and wherein either participant may decline a suggestion without stating a reason and no record of a decline is presented to the other participant.

77. The system of claim 76, wherein the suggestion is presented on a display as a randomizer that the participants actuate together, the randomizer being weighted toward the assignment that moves the balance of roles toward the participants' stated preferences; wherein, in an aftercare period following a session, each participant is privately asked whether the balance of roles feels right, has shifted too far toward one participant, or has shifted too far toward the other, and the answers are combined with the recorded role history into a balance meter; and wherein the balance meter is displayed to each participant, and to both participants only where both have elected sharing, and is optionally compared with a reference range selected by the participants from a published community guideline, the comparison being presented as information and not as a requirement.

78. The system of claim 76, wherein the controller maintains privately for each participant a receptive load comprising the number of sessions in a rolling period in which that participant took a receptive role for a designated act, compares the receptive load with a reference frequency selected by the participant from a published dose-response guideline or set by the participant or the participant's clinician, and, where the receptive load of one participant approaches or exceeds the reference frequency while the other participant's receptive load is below it, weights the role suggestion toward exchange of roles and presents a private notice to the participant whose load is high; wherein the controller applies, where elected, a soft balance bound under which the weighting increases as the ratio of the participants' receptive loads departs from equality beyond a configured ratio; wherein aftercare responses reporting pain, bleeding, or other symptoms increase the weighting and present to the reporting participant information on seeking clinical care; and wherein the weighting and notices never prevent a session or a role chosen by the participants, and a participant may exempt himself or herself from the weighting for a stated medical or anatomical reason without disclosing the reason to the other participant.

79. The system of claim 71, wherein a participant enrolls, privately, a duress variant of an affirmation, selected from a go word accompanied by an enrolled additional word or cadence, a touch pattern, a button sequence, and a vocal cue, the duress variant being

indistinguishable to another participant from an ordinary affirmation; and wherein, upon recognizing the duress variant, the controller does not credit an affirmation, leaves the media content and any output perceptible to another participant unchanged, silently invokes at least one help pathway designated by the participant for duress, and records the duress only in the participant's own sealed record.

80. The system of claim 2, wherein the controller prevents any participant's device from receiving another participant's location, event log, settings, or cadence data except upon that other participant's own election made on that participant's own device and revocable without notice; and wherein, for a participant who has excluded pressure on the neck in an enrolled boundary map and has enabled a respiration or blood-oxygen sensor of a wearable device or a non-imaging room sensor, an interruption of breathing or a fall in oxygen saturation beyond an enrolled threshold is treated as a distress indicator that bypasses the escalation sequence.

81. The system of claim 32, wherein, upon election by a participant, the controller contributes at a reporting interval, for each of a plurality of event classes including at least a greyzone state, a brake, a withdrawal, and an invocation of the help stage, a count of events perturbed by a local differential-privacy mechanism or submitted through a secure-aggregation protocol, without any participant identifier, without any time stamp finer than the reporting interval, and without any location finer than a community.

82. The system of claim 2, wherein the controller maintains a current stage of a session selected from an ordered plurality of stages including at least a lower stage comprising sexual contact and a high stage comprising sexual acts, the boundary between the lower stage and the high stage being set by default, from a jurisdiction table, at the line drawn by applicable law between sexual contact and a sexual act; wherein ascent to a higher stage requires a stage-entry affirmation, distinct from an affirmation renewing the current stage, from every participant whose affirmation is required by the mode of the session; wherein each stage has its own affirmation window, the window being shorter at higher stages; and wherein any participant may descend to a lower stage by a downshift input without any affirmation, the descent not being treated as a withdrawal and not advancing the escalation sequence.

83. The system of claim 82, wherein ascent to the high stage requires, for each participant whose affirmation is required, an affirmation attributed at a verified or physically verified attribution level where supported, and otherwise the controller announces the lower attribution level and records it; wherein, in a duo mode in which every party present is enrolled, stage-entry affirmations are required from every participant, and, in a solo mode, from the enrolled participant only, the controller recording that only that participant's affirmation was obtained; and wherein each stage transition, downshift, and attribution level is recorded in an event log.

84. The system of claim 82, wherein the participants may agree at session start a stage plan identifying the stages they expect to reach, after which a transition within the plan requires only an abbreviated confirmation; wherein the participants may elect a simplified configuration in which adjacent stages are merged while a gate at the legal line is preserved; and wherein relaxation of the gate at the legal line is available only upon express joint election, reconfirmed at each session start, unavailable in a first session between the participants, and revocable by either participant.

85. The system of claim 82, wherein, in a configuration selected for a session conducted through a voice-assistant smart speaker without a display, the stages are grouped into a first envelope comprising one or more lower stages of the ordered plurality and a second envelope comprising one or more higher stages including the high stage; wherein the session begins in the first envelope; wherein entry to the second envelope requires an affirmation at an enthusiastic tier registered by every participant whose affirmation is required by the mode of the session, the controller announcing the transition; wherein registration of that affirmation grants a lease of the second envelope for the remainder of the session, subject to renewal within an affirmation window whose duration is the same as or different from that of the first envelope; and wherein the lease ends upon any stop, withdrawal, or descent into the first envelope, after which re-entry to the second envelope requires a new registration.

86. The system of claim 2, wherein, after a session has reached a high stage of intimacy, upon detection of wind-down cues comprising at least one of a declared finish, relaxed conversation or laughter, a declining heart rate reported by a wearable device, and stillness of the participants reported by a non-imaging sensor, attributed to every participant whose affirmation is required by the mode of the session, the controller enters a wind-down state in which the affirmation window is suspended and a single prompt to close the session is offered; wherein, upon departure from the room of an enrolled participant after a high stage in the absence of any risk indicator, the controller grants an afterglow grace period during which the escalation sequence advances no further than a private

check-in delivered to that participant's own devices and no audible output is made in the room, and, upon expiry of the grace period without response and without any risk indicator, closes the session with a notification permitting the session to be reopened, recording the reason for closure; and wherein any risk indicator, including a prior stop or brake, a distress indicator, an impairment flag, a secret concern cue, or a stop from any participant, ends the wind-down state or grace period and restores the escalation sequence.

87. The system of claim 2, wherein whether the alarm stage produces audible output in a room in which the participants are present is elected by an enrolled participant from a plurality of options, presented together with information on the outcomes associated with each, the options including at least an audible alarm in the room, a silent alarm with help pathways only, an alarm audible only outside the room, and a help-adjacent alarm whose audible output in the room is deferred until a responder has accepted the call for help and is reported to be within a configured distance of the room.

88. The system of claim 2, further comprising a wireless handheld input device selected from a panic button, a media remote control, a camera-shutter remote, a presentation clicker, and a game controller, wherein distinct press patterns, including at least single, double, and multiple presses and holds of distinct durations, are mapped by enrollment to distinct signals selected from an affirmation, a brake, a stop, a session close, a concern cue, a duress affirmation, and a silent call for help; and wherein recognition of each signal is acknowledged by a distinct haptic pattern on the input device or on a wearable device of the participant, without audible output.

89. The system of claim 2, wherein the controller provides a guard state, distinct from a consent session, in which no affirmation is awaited and no media is gated, but in which stop words, brake inputs, concern cues, duress affirmations, button patterns, and distress indicators are honored and the help pathways are armed; wherein the guard state is entered upon an enrolled cryptic arming input selected from a distinctive turn of phrase addressed to a voice assistant in the form of an ordinary request, a button or remote pattern, a touch or gesture on a wearable device, a phone placed in a particular orientation, and an ordinary command to a connected device, the arming input not naming the system and producing only the ordinary response to the request and a private haptic acknowledgement; and wherein a consent session may be begun from the guard state only by the affirmations the session requires, the guard state otherwise continuing until closed by the participant who armed it.

90. The system of claim 32, wherein each entry of the event log records an identifier and a cryptographic hash of each recognition and classification model whose output produced the entry; wherein each released version of each such model is archived together with its published error rates measured across accents, speech conditions including distressed and whispered speech, and acoustic environments; and wherein, upon a claim that an enrolled stop word, safe word, or concern cue was spoken but not recognized, the archived model identified in the log is made available for a reproduction test in which the claimant's renditions of the cue, or a conflict recording where one exists, are processed by that model under the recorded conditions, the result being reportable as evidence that the cue was or was not likely to have been missed.

91. The system of claim 39, wherein the controller permits a participant to designate in advance a voice-relay property for an emergency condition selected from a stop word, a brake input, a concern cue, a duress affirmation, and a button pattern; wherein, upon the emergency condition being triggered by that participant, the controller, in addition to stopping the media content, opens a relay in which speech attributed by the voiceprint attribution component to the triggering participant at the verified or probable level, or attributed to that participant by the enrolled device or button that triggered the condition, is recognized on the user device and rendered through a loudspeaker in a room in which the participants are present at an elevated volume, in a form elected in advance by the participant from verbatim repetition, paraphrase, and summary, while speech attributed to any other voice is not relayed, whether such speech is transcribed or retained being governed by conflict-recording settings independent of the relay; wherein the relay remains open, relaying only the triggering participant, until the participants separate or the triggering participant deactivates it; wherein the relay is rendered in a synthesized voice and is preceded by a distinctive tone; and wherein, when a recognition confidence is below a threshold, the controller asks the triggering participant a confirming question and relays only a confirmed utterance.

92. The system of claim 39, wherein the voiceprint template is leased to a second device in encrypted form, over a local link or over a transient or sustained encrypted server link, usable only for verification and is bound, with the consent of the participant whose

template it is, to a relationship identifier shared by that participant and another participant, the lease persisting on the second device across sessions while that consent continues, carrying an expiry that is renewed only while that consent continues, and being destroyed upon termination of the relationship by either participant, upon revocation by the participant whose template it is, upon deletion of the template, or upon expiry, whichever occurs first.

93. The system of claim 35, further comprising a variant interface through which a separately signed software variant, executing in a sandbox without access to the microphone, sensors, or an event log except through the interface, supplies additional vocabulary, prompts, media, stages, or rules to the policy engine; wherein the controller accepts from the variant only settings that are equal to or more protective than a set of core invariants enforced by the controller, the core invariants including that a stop word is always honored, that absence of affirmation causes a lapse rather than an extension of the affirmation window, and that the escalation to the call for help remains available to every participant; and wherein, where participants in a session are associated with different variants, the controller applies, for each core invariant, the more protective of their settings.

94. The system of claim 2, further comprising a neck-worn wearable device in the form of a choker, necklace, or collar, comprising a pressure or strain sensor and a wireless interface, wherein the wearable device, upon its enabling by the participant wearing it, reports to the controller a compressive force on the neck exceeding an enrolled threshold, and wherein the controller treats the report as a distress indicator that advances the escalation sequence to the help stage regardless of any affirmation then in force.

95. The system of claim 35, wherein the policy engine, upon election by both participants, maintains a relationship profile bound to a relationship identifier; lengthens at least one of the affirmation window and the check-in period, up to a ceiling set by the participants, upon certification by both participants of a milestone defined by them; and shortens it toward a first-session default upon a stop word, a concern cue, a lapse that reaches the check-in stage, a negative aftercare report, or a request by either participant; wherein lengthening requires the agreement of both participants and shortening requires the request of only one; and wherein no lengthening disables the stop word, the abrupt stop, or the escalation to the call for help.

96. The system of claim 2, wherein, for participants who elect not to play media content during a session, the controller renders no continuous signal while more than a lead time remains in the affirmation window, and renders, through a wearable device, phone, or accessory of each participant, a haptic pattern, or through a connected lighting device a change of lighting, only upon the affirmation window coming within the lead time of expiry or expiring, the haptic pattern increasing in insistence as expiry approaches and passes; and wherein, absent an affirmation, the controller proceeds through the check-in, alarm, and help stages of the escalation sequence.

97. The system of claim 1, wherein, for a session of a plurality of participants, the instructions designate the participants in rotation, one for each successive affirmation window, and reset the affirmation timer in each window only upon detection of an enrolled enthusiastic go word of the participant designated for that window, each participant being designated at least once in every run of consecutive windows equal in number to the participants; and wherein a stop word of any participant is honored in every window.

98. The system of claim 17, wherein the controller stores, upon registration of the session location, a dispatchable bedroom location comprising a civic address, a building and unit identifier, a floor, a room identifier, and wayfinding instructions from the building entrance to the room, confirmed by a scan, by the user device, of a room tag affixed in the room selected from a near-field communication tag, an optically readable code, and an ultra-wideband or Bluetooth beacon; wherein the controller estimates the vertical position of the user device from a barometric pressure sensor and the horizontal position from a fusion of satellite positioning and nearby wireless signals, detects a movement of the device beyond a threshold from the registered location, and requests re-confirmation of the location upon such a movement; and wherein the call for help includes the dispatchable bedroom location and wayfinding instructions.

99. A method of activating a consent safety system, comprising: enrolling, at a user device, a media selection selected from a playlist, a track, an artist, and a station as an activation cue for the consent safety system; detecting, by a controller, that the enrolled media selection has been chosen for playback, by voice, by touch, or through a media-control interface, without a command that

names the consent safety system; in response to the detecting, arming the consent safety system, including listening for a stop word and enabling an escalation to a call for help, without an audible announcement of the arming where a policy and applicable law permit; and playing the enrolled media selection.

100. The method of claim 99, further comprising, in response to the arming, opening an affirmative consent envelope in which playback of the media selection continues only while an affirmation from each participant is received within an affirmation window, and, upon a lapse of the affirmation window, softening or stopping the playback and issuing a check-in.

101. The system of claim 2, wherein each participant selects a listening level from an ordered plurality of levels comprising at least: no microphone listening; wake-word-only listening, in which a safe word is recognized only following a wake word; continuous on-device keyword spotting for the participant's enrolled safe words without transcription; recognition, without enrollment, of universal safe words; and on-device semantic analysis of the participant's utterances while a session is in an intimacy-active state; wherein analysis of each participant's speech does not exceed the level that participant selected, a reduction of level takes effect immediately, and an increase of level requires an affirmation of each participant whose speech would newly be analyzed.

102. The system of claim 4, further comprising a private agent executing on the user device or under end-to-end encryption that, to the extent enabled by each participant, conducts clarifying exchanges in affirmation, greyzone, and withdrawal states, supports a participant's disclosure of a sensitive fact bearing on comfort or safety at a time and in words the participant chooses, and, before an ascent to a higher stage of intimacy, flags an item of the participants' stated ground rules, stored as a checklist in an encrypted record of the participants, for which no confirmation entry exists in an event log of a session; wherein the agent may add a check-in, a greyzone, or a withdrawal but cannot supply an affirmation on behalf of any participant.

103. The system of claim 35, wherein, for a pair of participants without a history of sessions together, the active policy sets a short affirmation window and lengthens the affirmation window available to that pair only after sessions completed without a lapse, a withdrawal, or a flagged concern.

104. The system of claim 45, wherein the controller classifies each binding as presumed, where the device is owned by or registered to the participant without a contemporaneous confirmation, or as proven, where a biometric signature obtained by the device as it is put on or picked up, or while it is worn or held, identifies the participant; wherein an input through a presumed-bound device is assigned at most a probable attribution level and an input through a proven-bound device may be assigned a physically verified attribution level; and wherein a proven binding lapses upon a break in on-body or in-hand continuity.

105. The system of claim 17, wherein the controller maintains no persistent connection to any server, opens an outbound connection only upon a hazardous or emergency condition or a greyzone requiring review beyond a room in which the participants are present, computes the consent state on the user device, and transmits a minimized payload, encrypted to its recipient and signed by the user device where the recipient supports it, through at least one transport selected from an HTTPS request, an on-demand session closed after delivery, a push notification carrying an opaque token, an end-to-end encrypted message, a request through an oblivious relay, an SMS, MMS, or RCS message including a text to an emergency number, a satellite message, and a peer-to-peer or mesh relay through neighboring devices.

106. The system of claim 105, wherein, where an operating system of the user device requires user intervention to send an SMS message, the controller prepares the message, including a recipient, text, and location, in a native messaging composer of the user device for sending by a single confirmation.

107. The system of claim 32, wherein, upon an emergency condition, the controller freezes, against destruction at session close, data that the participants consented to use during a session, including voiceprint templates, biometric matches, attribution records, and the event log, and seals the frozen data in encrypted form in an evidence locker on the user device, or on a server where law, contract, and the participants' preferences allow, accessible upon lawful process or to a participant entitled to review it, each access being logged.

108. The system of claim 2, wherein a participant enrolls a media selection as belonging to a safe-word category, and selection of that media selection for playback during a session is treated as a stop or a call for help, the resulting escalation proceeding, according to an active policy, either perceptibly in a room in which the participants are present or without any change in the room.

109. The system of claim 2, wherein the controller mixes the media content in real time with transitions or generated splices that mark the opening of an affirmation window, an ascent to a higher stage, or an approach of expiry, and, upon an approaching expiry of a participant's affirmation, mixes in a track, artist, or motif that the participant has pre-registered as a signal of a need for reaffirmation.

110. The system of claim 2, further comprising a lighting feedback device positioned to be visible to the participants and driven by the controller to a hue, saturation, or color representing the consent state, wherein an affirmation shifts the color immediately toward a color representing affirmed consent.

111. The system of claim 110, wherein the lighting feedback device is a smartphone mounted in a room in which the participants are present and driven remotely by another device of the system, each camera of the smartphone being covered, closed, or disabled while the smartphone serves as the lighting feedback device.

112. The system of claim 18, wherein, upon verification of a responder at the door, the controller delays unlocking for a courtesy interval specified by the active policy and announced in the room and to the responder, the interval being shortened or waived by a participant or by the active policy when an emergency requires.

113. The system of claim 29, wherein the controller compares lyric or transcript data of each track selected for the media content against the go words, stop words, safe words, and universal safe words in force for a session, and excludes the track, skips or mutes a segment containing a match, or substitutes an edited version of the track.

114. The system of claim 68, wherein either participant may record an initiation manually after the fact, individually or in a periodic review paced by a calendar period, a menstrual cycle, or a virtual cycle, the record identifying the initiating participant and an outcome of the initiation and being optionally confirmed by the other participant, and wherein manually recorded initiations are counted in determining an initiation balance between the participants.

115. The system of claim 114, further comprising a button, wherein a short press of the button records an initiation and a long press of the same button signals for help.

116. The system of claim 88, wherein the input device is a commercially available media remote, camera-shutter remote, presentation clicker, or game controller whose buttons are mapped according to their natural analogs, including play as an affirmation, pause as a brake, stop as a stop, directional buttons as directional guidance, and volume buttons as harder and softer.

117. The system of claim 2, wherein each signal received or rendered by the controller is available through an alternative modality selected for a participant with a disability, including captions, light, and haptic patterns in place of audible output, and gestures, tap patterns, typed text, sign-language signs recognized on the user device, switch inputs, and augmentative and alternative communication devices in place of speech, each alternative modality carrying the same authority, timing, attribution, and logging as the modality it replaces.

118. A method of ambient safety signaling, performed by a computing device, comprising: rendering a perceptible signal selected from playback of media content, a haptic pattern, and a lighting state; continuing the perceptible signal only while an affirmation is received from each of one or more persons within an affirmation window; upon expiry of the affirmation window without such an affirmation, changing the perceptible signal gradually over a fade period; upon detection of a stop input from any of the persons, changing the perceptible signal abruptly in a manner distinguishable from the gradual change; after the gradual change, issuing a check-in and distinguishing an interaction of a person from an affirmation, such that an interaction halts escalation without resuming the perceptible signal; and, upon continued absence of any interaction through a help period, transmitting a call for help.

119. The method of claim 118, wherein the one or more persons comprise at least one of a lone worker, a person living alone, a person exercising alone, a person on a date, a passenger in a vehicle, and a person recovering from a medical procedure.

120. The method of claim 118, wherein the affirmation window, the check-in, and the call for help are computed and initiated on the computing device; an event log of the method is encrypted under a key held only by devices of the persons; and no server operated by a provider of the media content receives audio, a transcript, or a history of affirmations.

121. A smart speaker comprising a far-field microphone array, a loudspeaker, a processor, and a memory storing an operating system that causes the processor to: play media content through the loudspeaker during a consent session in which playback continues only while an affirmation is renewed within an affirmation window by each participant whose affirmation the session requires; while the media content plays, perform on-device keyword spotting, without first detecting a wake word, for a set of universal safe words adopted by a maker of the smart speaker, by an industry, or under a policy applicable to a locale of the smart speaker, using the media content being played as a reference for echo cancellation; upon detecting a universal safe word, stop the playback of the media content and signal a safety pathway configured by a user of the smart speaker; and retain no audio, and generate no transcript, of speech that does not match a universal safe word.

122. A method performed by a media streaming server, comprising: receiving, from a client device, a request to begin a check-in-gated playback session for one or more participants; streaming media content to a playback target associated with the session; receiving, from one or more client devices, renewal tokens each attributed to a participant; upon failing to receive, within an affirmation window, a renewal token attributed to each participant, transmitting to the playback target an instruction to fade the media content and transmitting to the client devices an instruction to issue a check-in; and, upon a failure to receive any renewal token or interaction token through a further period, transmitting a help notification to a recipient designated for the session; wherein the renewal tokens carry no audio and no transcript.

123. A method performed by a mobile device, comprising: maintaining a safety session having an affirmation window renewed by an input of a user of the mobile device; rendering no warning while more than a lead time remains in the affirmation window, and rendering a haptic warning of increasing insistence as the affirmation window approaches and passes expiry; upon continued absence of an input through a help period, preparing a message comprising a recipient selected from an emergency number and a designated contact, a text describing the help request, and a location of the mobile device; and sending the message automatically where an operating system of the mobile device permits, or presenting the message in a native messaging composer for sending by a single confirmation where the operating system requires user intervention; wherein the mobile device maintains no persistent connection to a server during the safety session.

124. A method performed by a media playback system, comprising: playing media content in a room during a session in which playback continues only while an affirmation is received from each participant within an affirmation window; determining, from metadata of the media content or from analysis of its audio, a time of a musical boundary of the media content selected from the end of a bar, a phrase, a section, a lyric line, and a track; upon a lapse of the affirmation window, beginning a fade of the media content at a musical boundary at or before the lapse, or at a musical boundary after the lapse that is delayed from the lapse by no more than a maximum deferral; cancelling the fade upon an affirmation received from each participant before the fade completes; and, upon a stop input from any participant, stopping the media content without regard to any musical boundary.

125. The method of claim 124, further comprising selecting and sequencing tracks of the media content so that a track, or a section within a track, ends within a tolerance of the expiry of the affirmation window, and re-planning the selection upon each renewal of the affirmation window.

126. A method performed by an operating system of a user device, comprising: receiving, from an application through an application programming interface, a request to place audio output of the user device under a renewal gate having a renewal window and identifying one or more participants; thereafter, continuing audio output of whichever audio source is active on the user device only while renewal events attributed to each participant are received within the renewal window; upon expiry of the renewal window without a renewal event from each participant, fading the audio output at an audio mixer of the operating system, regardless of which application produces the audio; upon a stop event from any participant, muting the audio output at the audio mixer within a stop

latency shorter than the fade; and reporting the state of the renewal gate to the application, whereby the renewal gate is enforced even where the application producing the audio provides no control interface.

127. A method performed by an access-control system of a dwelling, comprising: receiving, from a safety controller associated with a room of the dwelling, a grant request identifying a responder who has accepted a call for help issued by the safety controller upon an unanswered escalation of a session in the room; verifying, at a door of the dwelling, that a person present is the responder by at least one of a one-time code, a credential presented by a device of the responder, and a knock pattern that was delivered to the responder with the call for help and is not rendered by any device inside the room; confirming that a location reported by the device of the responder is within a configured distance of the dwelling; unlocking the door for a time-limited window only upon both the verifying and the confirming, unless a cancellation is received from the room before the window opens; and recording the grant, the verification method, and any cancellation.

128. A method performed by a user device, comprising: during a session in which media content plays only while an affirmation is renewed, recognizing on the user device a stop utterance of a participant; classifying the stop utterance, by a language model or intent model executing on the user device, as one of a bare stop, a stop accompanied by an expressed reason of pain, fear, or distress, and a stop of explicitly limited scope; pausing the media content upon an enrolled stop word in every case; upon a stop of explicitly limited scope that is not an enrolled stop word, reducing the volume of the media content without pausing it and announcing the request to another participant in the words of the participant; upon a stop accompanied by an expressed reason, stopping the media content and relaying the reason to another participant; and acting upon a stop classification at a lower confidence than is required to act upon a continuation classification.

129. A method performed by a computing device in a room, comprising: enrolling a melodic motif as a call for help; while media content plays through a loudspeaker in the room, cancelling the media content from microphone audio using a reference of the media content; recognizing the motif when hummed, whistled, or sung by a participant by matching a key-independent and tempo-independent representation of its pitch intervals and inter-onset intervals against the enrolled motif; upon recognizing the motif, transmitting a call for help and rendering an acknowledgement through the loudspeaker; and, upon acceptance of the call for help by a responder, rendering through the loudspeaker a motif, distinct from the media content, indicating that help is on the way.

130. A method performed by a receiving device, comprising: receiving, during a consent session, messages each comprising a protocol version, a session identifier generated randomly for the session, a participant pseudonym scoped to the session, an event type selected from at least a renewal, a stop, a brake, an interaction, a close, and a call for help, an attribution level, a monotonic counter, and a time stamp, each message containing no audio and no transcript and being signed by a key of an originating device; verifying the signature of each message and rejecting any message whose counter or time stamp is not fresh; upon a verified stop message from any participant, ceasing media playback controlled by the receiving device; and, upon failing to receive a verified renewal message for each participant of the session within an affirmation window, entering a lapse state in which media playback controlled by the receiving device is faded.

131. The method of claim 130, wherein each message transmitted beyond a local link is encrypted to the receiving device under hybrid public-key encryption, and wherein a stop message identifies the entry of a versioned registry of universal safe words that produced it.

132. A non-transitory computer-readable medium storing a software development kit which, when executed on a user device as part of a third-party application, causes the user device to: provide to the application an interface comprising operations to open a consent session identifying participants and a policy, to register renewal, stop, brake, concern, and close inputs, and to subscribe to lapse, check-in, alarm, help, and state-change events; execute speech and sensor processing, an affirmation timer, and an escalation sequence within a privacy kernel of the kit on the user device and not within the application; provide to the application only session states and events, and never raw audio, sensor frames, or transcripts; control media playback through the application or the operating system of the user device so that playback continues only while affirmation is current; and record each data flow it performs in a data-flow manifest of the application.

133. A method performed by a device, comprising: storing a versioned registry of universal safe words, each entry comprising a phrase, a language tag, phonetic variants, and a category selected from at least stop, brake, and help, and the registry specifying a maximum response latency for each category; advertising the version of the registry supported by the device to other devices of a consent session; while a consent session or guard state is active, recognizing on the device, without enrollment and without a wake word, any entry of the registry for the locale of the device; and, upon recognizing an entry, executing the response of its category within the maximum response latency, the response to a stop entry comprising ceasing media playback.

134. A connected intimate device comprising an actuator, a wireless interface, and a controller configured to: receive, during a consent session, an affirmation state from a consent controller; continue actuation only while the affirmation of each participant whose affirmation the session requires, including the person on whom the device acts, is current; upon a lapse, reduce the intensity of actuation gradually over a fade period; upon a stop input from any participant, cease actuation within a stop latency shorter than the fade period; resume actuation only upon a fresh affirmation; and accept actuation commands from a remote controller operated by another person only while the affirmation of the person on whom the device acts is current.

135. A method performed by a venue system for a room of a venue, comprising: receiving an opt-in of a guest for a stay; creating a stay-scoped profile holding inputs enrolled by the guest and a policy of the guest; operating venue-owned devices in the room, comprising at least one of a room-control panel, a smart speaker, a lighting device, and a bedside button, to conduct a consent session in which an output continues only while affirmation is renewed within an affirmation window and an escalation sequence follows a lapse; upon a help stage of the escalation sequence, routing an alert, where the policy of the guest permits, to security staff of the venue together with a dispatchable location of the room, without disclosing content of the session to the venue; and erasing the stay-scoped profile at checkout unless the guest elects to transfer it to a device of the guest.

136. A method performed by a voice assistant, comprising: upon request of participants, before an ascent to a higher stage of intimacy, conducting a checklist of ground rules stored in an encrypted record of the participants by presenting each item and recording the confirmation of each participant in an event log on a device of the participants; flagging an item for which a confirmation of each participant is not recorded, and declining to register a stage-entry affirmation until each flagged item is confirmed or waived by each participant; upon an utterance classified as a greyzone, conducting a clarifying exchange with the participant who made it; and processing the utterances of the participants on the device without transmitting a transcript; wherein the voice assistant may add a check-in, a greyzone state, or a withdrawal but cannot register an affirmation on behalf of any participant.

137. A method performed by a user device, comprising: generating, for a reporting period, an attestation signed by a key held in secure hardware of the user device, the attestation stating that a release of a consent safety system having an attested build hash was installed on the user device, that self-tests of the release passed, and that the release was armed for at least a stated share of the reporting period, the share being computed without times of day; and transmitting the attestation to a recipient selected from an insurer, an institution, and a regulator; wherein the attestation discloses no information on whether, when, or with whom any session occurred.

138. The method of claim 137, wherein the attestation is provided as a zero-knowledge proof or an anonymous credential establishing the stated facts without identifying the user device.

139. A method performed by a server of a media streaming service, comprising: hosting a shared playback session that two or more accounts of the service have joined and in which a single playback target in a room plays media content; receiving renewal tokens each attributed, by the account through which it was sent, to a participant using that account; continuing playback on the single playback target only while a renewal token attributed to each account whose affirmation the session requires is received within an affirmation window; upon failing to receive such a renewal token for any such account, instructing the playback target to fade the media content; and, upon a stop token from any joined account, instructing the playback target to stop the media content within a stop latency shorter than the fade.

140. A method performed by an earbud comprising an inertial sensor, comprising: during a consent session, detecting from the inertial sensor a deliberate head gesture of the wearer; classifying the head gesture as a nod or a shake; transmitting to a consent

controller a renewal attributed to the wearer upon a nod, and a brake upon a shake; and rendering, through the earbud and to the wearer alone, a renewal reminder before expiry of an affirmation window and a receipt cue upon crediting of a renewal.

141. A method performed by a first wearable device bound to a first participant, comprising: detecting a deliberate gesture of the first participant; determining, by ultra-wideband or other short-range ranging with a second wearable device bound to a second participant, that the second wearable device is within a threshold distance; only upon so determining, signing a renewal message attributed to the first participant and transmitting it to the second wearable device; receiving from the second wearable device a signed renewal message attributed to the second participant; and continuing a gated output of a consent session only while a verified renewal from each participant has been received within an affirmation window.

142. The method of claim 126, wherein the request to place the audio output under the renewal gate is made upon activation of a focus mode of the operating system designated for intimacy, and wherein, while the focus mode is active, the operating system suppresses notifications and calls other than prompts of the consent session, emergency calls, and calls from contacts designated by the user, and ends the focus mode and the renewal gate upon a close input, a soft close, or a time set by the user.

143. A method performed by a wearable device worn by a participant, comprising: during an active consent session, monitoring movement and at least one physiological signal selected from heart rate, heart-rate variability, respiration, and skin conductance; detecting a pattern, relative to an enrolled baseline of the participant, comprising an abrupt cessation of movement together with a change in the at least one physiological signal, persisting beyond a set interval; upon detecting the pattern, delivering to the participant alone a private check-in answerable by a single tap or squeeze; continuing the session upon such an answer; and, absent an answer within a response period, signaling a consent controller to pause a gated output of the session and to advance an escalation sequence; wherein detection of the pattern is not presented to any other participant and is never credited as an affirmation.

144. A method performed by a recording device, comprising: recording intimate content of two or more adults only while each person being recorded has a current affirmation within an affirmation window; pausing the recording upon a lapse of any such affirmation, and stopping it upon a stop input from any such person; signing each recorded segment with a content credential asserting that each person recorded had affirmed consent to recording and, where elected, to publication, and identifying each person by a revocable pseudonymous credential; and, upon receiving a signed revocation of consent to publication referencing the content credential, transmitting the revocation to each platform to which the content was published for removal or restriction of the content.

145. A method performed by a smart-home hub acting as a controller of a smart-home fabric, comprising: receiving, from a consent controller of a room, stage events of a consent session selected from at least a lapse, a check-in, an alarm, a help stage, and a close; executing, for each stage event, a scene defined for that stage across devices of the fabric, the scene comprising at least a change of a lighting device and a change of a speaker; and, upon a help-stage event accompanied by a verified-responder grant, commanding a lock of the fabric to grant entry for a time-limited window.

146. A smart bed comprising a frame, occupancy and movement sensors, a respiration sensor, a panic button mounted in a standardized location of the bed, a wireless interface, and a controller configured to: report occupancy, interaction, and respiration to a consent controller during a consent session; report an interruption of respiration of an occupant beyond an enrolled threshold as a distress indicator; and, upon a press of the panic button, transmit a call for help regardless of the state of any consent session.

147. A robot comprising: one or more actuated limbs or end effectors; force or torque sensing; and a controller configured to: make physical contact with a person only while an affirmation of that person for the contact is current within an affirmation window; plan motion so as to exclude body regions designated as excluded in a boundary map of the person, and to enter a region designated as conditional only after a region-specific affirmation; hold contact force within a limit set for each region; upon a lapse of the affirmation, reduce contact force gradually over a fade period and release the person; upon a stop, brake, or concern input from the person, release the person within a stop latency shorter than the fade period and withdraw to a set distance; and, upon a distress indicator, a detected freeze response, or an interruption of breathing of the person, release and withdraw from the person and transmit a call for help.

148. The robot of claim 147, wherein the controller refuses any command to apply pressure to the neck of the person, whether or not the person requests it.

149. The robot of claim 147, being a companion robot for adult intimacy, wherein the controller provides no mode that simulates the overriding of a refusal, refuses to operate with a childlike appearance, voice, or persona, and permits a consent session only with a person established by age assurance to be an adult.

150. The robot of claim 149, further providing a consent-practice mode, available only to a person established by age assurance to be an adult and, within that limit, only where and to whom applicable law and the rules of the person's community permit, in which the robot requests consent before acting, renders affirmations, scoped requests, brakes, stops, and declinations, and records privately for the person measures of how reliably the person requests consent, honors scoped stops and brakes, and recovers from a lapse, the mode comprising graduated levels below a high stage of intimacy that do not require the person to undress.

151. A method performed by an evaluation server, comprising: receiving, from each of a plurality of devices of a consent safety system, each device having been assigned at random and with the consent of its participants to one of two or more variants of a feature of the system, counts of event classes comprising at least two of a greyzone state, a brake, a lapse, a withdrawal, and a help-stage invocation, each count being labelled only with the assigned variant and a reporting interval and being perturbed by local differential privacy or submitted through secure aggregation; estimating, from the counts aggregated across the devices, a difference between the variants in the rate of at least one of the event classes, together with an uncertainty of the difference; and designating a variant as a default of the feature according to the estimated differences; wherein no count identifies a participant, a session, a time of day, or a location finer than a community.

152. The method of claim 151, further comprising stopping the assignment of a variant before the end of the evaluation upon the estimated rate of help-stage invocations or withdrawals for that variant exceeding the corresponding rate for another variant by a preset margin.

153. A method performed by a computing device, comprising: rendering a perceptible output during a session only while an affirmation input from each participant of the session is received within an affirmation window; upon expiry of the affirmation window without such an affirmation input, changing the perceptible output gradually over a fade period; upon a stop input from any participant, changing the perceptible output abruptly, within a stop latency shorter than the fade period, in a manner perceptibly distinguishable from the gradual change; and resuming the perceptible output after either change only upon a fresh affirmation input from each participant.

154. A method performed by a device that plays media content, comprising: playing media content through a loudspeaker during a session; capturing audio by a microphone of the device while the media content plays; cancelling the media content from the captured audio using, as a reference, the media content that the device is causing to be played; detecting in the cancelled audio, on the device and without first detecting a wake word, an enrolled or universal safe word and an enrolled affirmation word of each participant; upon detecting a safe word, stopping playback of the media content; and continuing playback only while an affirmation word of each participant is detected within an affirmation window.

155. The method of claim 118, wherein the perceptible signal is playback of media content, and the affirmation is an enrolled go word detected by on-device speech recognition performed continuously and without a wake word on audio from which the media content has been cancelled using a playback reference signal.

156. The method of claim 118, wherein the perceptible signal is playback of media content on a media playback target, and changing the perceptible signal gradually comprises commanding the media playback target to fade the media content and confirming, from a player state returned by the media playback target, that the command took effect.

157. The method of claim 118, wherein the perceptible signal is rendered audibly in a room in which the one or more persons are present, such that its continuation, its gradual change, and its abrupt change are each perceptible to every person present without reference to a display.

158. The method of claim 126, further comprising, while the audio output is under the renewal gate: capturing audio by a microphone of the user device; cancelling the audio output from the captured audio using, as a reference, the audio output at the audio mixer; detecting in the cancelled audio, on the user device and without first detecting a wake word, an enrolled go word of each participant and an enrolled or universal stop word; treating detection of the go word of a participant as a renewal event attributed to that participant and detection of a stop word as a stop event; and, after the audio output has been faded upon expiry of the renewal window, issuing a check-in and, upon continued absence of any interaction by a participant, sounding an alarm and then calling for help through a pre-configured pathway, any interaction halting advancement of the check-in, alarm, and call for help without resuming the audio output.

159. The method of claim 130, wherein entering the lapse state comprises fading the media playback controlled by the receiving device over a fade period, and ceasing that media playback upon a verified stop message is performed within a stop latency shorter than the fade period; wherein the receiving device resumes that media playback, after the lapse state or after ceasing, only upon a verified renewal message for each participant of the session received thereafter; and wherein a verified message whose event type is an interaction halts advancement of an escalation sequence of the session, comprising a check-in, an alarm, and a call for help, without resuming that media playback and without being treated as a renewal.

160. The method of claim 133, wherein the device recognizes the entries of the registry while media content that the device causes to be played is playing through a loudspeaker, by cancelling the media content from microphone audio using, as a reference, the media content being played; wherein, during the consent session, the device continues playback of the media content only while an affirmation of each participant is received within an affirmation window and, upon a lapse of the affirmation window, fades the media content over a fade period longer than the maximum response latency for the stop category; and wherein the device retains no audio, and generates no transcript, of speech that does not match an entry of the registry.

161. The non-transitory computer-readable medium of claim 132, wherein the speech and sensor processing executed within the privacy kernel comprises detecting, on the user device and without first detecting a wake word, an enrolled go word of each participant and an enrolled or universal stop word in microphone audio from which media content being played is cancelled using that media content as a reference; wherein the privacy kernel computes a consent state for each participant and a joint consent state of the consent session as the minimum of the participants' consent states; and wherein controlling the media playback comprises, upon a lapse, fading the media playback over a fade period and, upon a stop input from any participant, stopping the media playback within a stop latency shorter than the fade period.

162. The smart speaker of claim 121, wherein the operating system further causes the processor to detect, by the keyword spotting and using the media content being played as the reference for echo cancellation, an enrolled go word of each participant as the affirmation of that participant; upon a lapse of the affirmation window, to fade the media content over a fade period; and, upon detecting a universal safe word, to stop the playback of the media content within a stop latency shorter than the fade period.

163. The method of claim 124, wherein the affirmation of each participant is an enrolled go word of that participant detected by a device of the media playback system, without first detecting a wake word, in microphone audio from which the media content is cancelled using, as a reference, the media content being played; and further comprising, after the fade completes without such an affirmation, issuing a check-in and, upon continued absence of any interaction by a participant, sounding an alarm and then calling for help through a pre-configured pathway, any interaction halting advancement of the check-in, alarm, and call for help without resuming the media content.

164. A method performed by a controller of a consent safety system, comprising: learning, on a device of a participant and over a plurality of sleep periods of the participant, a room in which the participant sleeps, from at least one of a charging cycle of a phone or wearable device of the participant and a wake-sleep cycle of the participant; associating the room with a fixed device having a microphone that is nearest to the phone or wearable device during the sleep periods, as determined by short-range wireless or acoustic ranging; designating the room as a bedroom of the participant upon confirmation by the participant; thereafter, upon detecting that a person is present in a space matching the designated bedroom, causing the fixed device to become safe-word sensitive, in which state

the fixed device detects on the fixed device, without first detecting a wake word, at least the universal safe words of a registry for its locale in audio from the microphone from which any media content the fixed device is playing is cancelled using that media content as a reference, and retains no audio and generates no transcript of speech that does not match a safe word; and, upon detecting a safe word, stopping any media content the fixed device is playing and executing a response configured for the safe word comprising at least one of a check-in, an alarm, and a call for help through a pathway pre-configured by the participant; wherein a fixed device in a room not designated as a bedroom does not become safe-word sensitive by reason of the learning.

165. The method of claim 164, wherein, where a phone of the participant is charged in a room other than the room in which the participant sleeps, the learning gives greater weight to at least one of sleep detected by a wearable device worn by the participant, the room in which an alarm of the participant is dismissed, the room whose fixed device last detects a voice of the participant before a sleep period and first detects it after, and snoring detected in a room; and wherein, where the signals remain in conflict, no room is designated until the participant confirms one.

166. The method of claim 164, wherein the learning and designating are performed separately for each of a first participant and a second participant, from that participant's own devices, so that a bedroom of each is designated, each with its associated fixed device; and wherein the fixed device associated with either designated bedroom, while safe-word sensitive, detects, in addition to the universal safe words, enrolled safe words of either participant that the participant has shared with the devices of the other.

167. The method of claim 164, wherein the fixed device becomes safe-word sensitive by reason of the designation only where the participant has been established, by attestation or by inference from patterns of use, to be at or above a required minimum age, the participant being treated as below the minimum age where such signals are absent, conflict, or suggest otherwise.

168. A method performed by a smart speaker in a room, comprising: detecting, by an on-device acoustic event classifier operating on audio from a microphone of the smart speaker, snoring in the room; determining that snoring has recurred in the room on a plurality of nights during night hours, retaining only a count or flag of such nights and no audio of the snoring; determining from the recurrence that the room is a bedroom environment; determining whether a resident of the room is at or above a required minimum age, by attestation or by inference from patterns of use, the resident being treated as below the minimum age where such a determination is absent or in conflict; and, only upon determining both that the room is a bedroom environment and that the resident is at or above the minimum age, entering a safe-word sensitive state in which, while the room is occupied, the smart speaker detects on the smart speaker, without first detecting a wake word, universal safe words of a registry for its locale and, upon detecting a universal safe word, stops any media content it is playing and signals a safety pathway configured by the resident.

169. The method of claim 164, wherein learning the room comprises detecting a sleep period of the participant from physiological signals sensed by a wearable device worn by the participant, determining the room from short-range ranging between the wearable device and the fixed device during the sleep period, and determining a dwelling in which the room lies from a satellite position of, or a wireless network connected to, the phone or wearable device of the participant, whereby a sleeping site of the participant away from the participant's home may also be designated upon confirmation by the participant.

170. The method of claim 164, wherein learning the room comprises detecting a wake-up alarm set on the phone or wearable device of the participant at night and turned off after an interval, since a last use of that device by the participant, consistent with a sleep period; determining the room in which the alarm sounded and was turned off; and giving that room greater weight as the room in which the participant sleeps where the alarm was turned off within a short interval of sounding or where the fixed device detected the sound of the alarm.

171. The method of claim 164, further comprising: recording, on a phone of the participant during the sleep periods, a fingerprint of the place where the phone rests, comprising at least one of signal strengths of wireless networks detected by the phone, a barometric pressure, and a local magnetic field; and, upon the phone determining that its current surroundings match the fingerprint, causing the phone to become safe-word sensitive, in which state the phone detects on the phone, without first detecting a wake word, at least the universal safe words of the registry and retains no audio and generates no transcript of speech that does not match a safe word, and, upon detecting a safe word, executes the response configured for the safe word.

172. A method performed by a phone of a participant, comprising: inferring a plurality of sleep periods of the participant from at least one of a charging cycle of the phone, a wake-sleep cycle of the participant, and a wake-up alarm set on the phone at night and turned off after an interval consistent with a sleep period; recording, during the sleep periods, a fingerprint of the place where the phone rests, comprising at least one of signal strengths of wireless networks detected by the phone, a barometric pressure, and a local magnetic field; designating that place as a bedroom of the participant upon confirmation by the participant; thereafter, upon determining that the current surroundings of the phone match the fingerprint, becoming safe-word sensitive, in which state the phone detects on the phone, without first detecting a wake word, at least universal safe words of a registry for its locale and retains no audio and generates no transcript of speech that does not match a safe word; and, upon detecting a safe word, executing a response configured for the safe word comprising at least one of a check-in, an alarm, and a call for help through a pathway pre-configured by the participant; wherein the fingerprint and the designation are stored only on the phone.

173. A method performed by a user device having a microphone, comprising: determining that the user device is in a bedroom of a user, from at least one of a designation of the bedroom learned from where the user device rests during sleep periods of the user, a room tag of the bedroom read by the user device, and an assignment of the user device to a bedroom in a smart-home application; entering a bedroom mode upon a bedtime signal selected from connection of the user device to electrical power after a configured time of night, activation of a sleep, bedtime, or do-not-disturb mode of an operating system of the user device, setting of a wake-up alarm for the following morning, execution of a bedtime routine of a voice assistant, and activation of a bedtime scene of a smart-home system, only where the bedtime signal occurs while the user device is determined to be in the bedroom; while in the bedroom mode, detecting on the user device, without first detecting a wake word, at least universal safe words of a registry for its locale in audio from the microphone from which any media content that the user device is playing, including sleep sounds, is cancelled using that media content as a reference, and retaining no audio and generating no transcript of speech that does not match a safe word; upon detecting a safe word while in the bedroom mode, stopping any media content that the user device is playing and advancing an escalation sequence comprising a private check-in to the user on a device of the user, an alarm, and a call for help through a pathway pre-configured by the user, the escalation sequence advancing upon absence of a response to the check-in within a check-in period, halting upon a response of the user, and advancing directly to the call for help upon a safe word of a help category of the registry; and leaving the bedroom mode upon a wake signal selected from a configured morning time, disconnection of the user device from electrical power in the morning, an end of a sleep, bedtime, or do-not-disturb mode, and the turning off of a wake-up alarm.

174. The method of claim 173, wherein the user device enters the bedroom mode only where the user has been established, by attestation or by inference from patterns of use, to be at or above a required minimum age, the user being treated as below the minimum age where such signals are absent, conflict, or suggest otherwise.

175. The method of claim 173, further comprising, while in the bedroom mode, detecting by short-range wireless detection that a device of a second person, other than a device of the user, is present in the bedroom, and thereupon privately offering the user, or, where the user has so set, automatically entering, a guard state in which, in addition to the universal safe words, enrolled safe words, a brake, concern cues, and button patterns of the user are honored and help pathways are armed.

176. A method performed by a user device having a microphone, comprising: entering a bedroom mode of the user device upon any of a bedtime signal, a determination that the user device is in a bedroom of a user, and an explicit command of the user; upon and by reason of entering the bedroom mode, automatically becoming safe-word sensitive, in which state the user device detects on the user device, without first detecting a wake word, at least one safe phrase in audio from the microphone, the at least one safe phrase comprising at least a universal emergency phrase adopted under a policy applicable to a locale of the user device, and retains no audio and generates no transcript of speech that does not match a safe phrase; and, upon detecting a safe phrase while in the bedroom mode, executing a response configured for the safe phrase comprising at least one of a check-in, an alarm, and a call for help through a pathway pre-configured by the user.

177. The method of claim 176, wherein the at least one safe phrase consists of the universal emergency phrase, and wherein the response to its detection comprises raising a greyzone concern by a private check-in to the user and, absent a response of the user within a check-in period, escalating to the call for help.

178. The method of claim 176, wherein the at least one safe phrase further comprises enrolled safe words, safe phrases, and greyzone words of the user, each bound to a response of a category selected from at least a greyzone category, a brake category, a stop category, and a help category.

179. The method of claim 176, wherein the user device is a fixed device, and the determination that the user device is in a bedroom of a user comprises the user device having been assigned, in a registration of the user device or in a smart-home application, to a room or a device group labeled as a bedroom, including a guest bedroom, whereby the user device enters the bedroom mode, and becomes safe-word sensitive, by reason of the assignment without any learned designation of the bedroom.

180. The method of claim 179, wherein, as set by the user, the user device is safe-word sensitive by reason of the assignment only while the room is detected to be occupied.

181. The method of claim 141, wherein the deliberate gesture is a clasp of a hand of the first participant with a hand of the second participant, and the renewal message attributed to the first participant is signed only where the consent session has privately prompted the first participant for an affirmation at a decision point of the session; the motion or grip force sensed by the first wearable device matches an enrolled clasp pattern of the first participant that differs from sustained hand-holding; the second wearable device reports, within a matching interval, a matching clasp sensed at the wrist of the second participant; and the relative orientation of the two wearable devices indicates that the participants face each other; whereby hand contact that does not meet these conditions, including ordinary hand-holding, is never treated as a renewal.

182. The method of claim 181, further comprising transmitting from the first wearable device, through the clasped hands by human body communication or by a vibration pattern of a haptic actuator of the first wearable device sensed by an inertial sensor of the second wearable device, a signed token comprising a session identifier, a pseudonym of the first participant scoped to the session, a counter, and a time stamp, the second wearable device treating the renewal message as verified only upon receiving the token through the clasp; and, only upon an election of both participants, exchanging over an encrypted short-range radio link bound to the token a record selected by both participants from at least an attestation of test results and a confirmation of ground rules.

183. The method of claim 176, wherein the explicit command of the user comprises a touch message, composed on a wearable device of the user as a tap pattern, a drawn sketch, or a transmitted heartbeat, that matches an enrolled arming pattern and is sent by the user from that wearable device to the user device.

184. The system of claim 89, wherein the enrolled cryptic arming input comprises a touch message, composed on a wearable device of a participant as a tap pattern, a drawn sketch, or a transmitted heartbeat, that matches an enrolled arming pattern and is sent by the participant to a device of the participant or to a trusted contact, the guard state being armed upon its receipt while the touch message appears, to anyone who sees it, only as an ordinary message.

185. A method of joining a first participant and a second participant to a consent session where only the first participant wears a wearable device having a body-coupling electrode, comprising: upon an initiating input of the first participant on the wearable device, detecting an onset of skin contact between a hand of the first participant and the second participant from a change in the coupling seen at the body-coupling electrode together with an enrolled clasp pattern sensed by an inertial sensor of the wearable device; upon the onset, advertising from a first phone bound to the wearable device an ephemeral pairing invitation that carries no identity of either participant; selecting as a second phone, by ranging from the wearable device or the first phone, the nearest phone other than the first phone that answers the invitation, the selection being made only where that phone lies within a threshold distance and is nearer by a margin than every other phone that answers; agreeing a session key between the first phone and the second phone by an authenticated key exchange; and joining the first phone and the second phone to the consent session only upon a confirmation entered by the first participant on the first phone and a confirmation entered by the second participant on the second phone, the pairing being cancelled

upon release of the contact, or upon expiry of a pairing interval, before both confirmations; wherein the joining is not credited as an affirmation of either participant.

186. The method of claim 185, wherein the confirmation of the second participant is entered by a touch on a touch screen of the second phone while the contact continues, further comprising injecting from the body-coupling electrode a code that passes through the body of the first participant, the contact, and the body of the second participant to the touch screen, the second phone treating the confirmation as valid only upon its touch controller receiving the code as a modulation of the sensed touch signal.

187. The method of claim 185, wherein each of the first phone and the second phone displays a confirmation code derived from the session key before the confirmation of its participant is entered; and wherein none of the invitation, the key exchange, and the joining is reported to a server, and no record of the pairing is retained after the consent session ends beyond what each participant elects to keep.

188. The system of claim 2, wherein the controller is further configured to compute, on the user device, a reward credit for a participant upon a protective act selected from enrollment of the participant, configuration of a safe word or a help pathway, completion of a check-in or an aftercare step, and an honored stop, the credit for an honored stop being no less than the credit for a session in which the media content continued; wherein the reward credit is independent of whether intimate activity occurred, of its frequency, and of the enrollment or affirmation of any other participant; and wherein the controller discloses to a reward issuer only an anonymous credential attesting that a qualifying protective act occurred, without the time of any session, the identity of any other participant, or any audio, transcript, or consent state of a session.

189. The system of claim 188, wherein the controller further produces, at the request of the participant, an anonymous attestation that a preventive configuration comprising at least a safe word recognized without a wake word and a help pathway is in place on the user device, for presentation to an institution that grants a benefit therefor, the attestation revealing no data of any session.

190. The system of claim 188, wherein the controller further maintains, only on the devices of the participants, a gift offered by one participant to another in recognition of a protective act, the gift being selected from nonsexual goods, services, and acts, being redeemable only upon a confirmation entered by the offering participant at the time of redemption, and being withdrawable by the offering participant at any time without a stated reason; and wherein a gift comprising physical touch is delivered only within a session in which the consent of each participant is current.

191. The method of claim 182, wherein the token is transmitted by human body communication, and wherein the second wearable device further detects the clasp from a variation in the amplitude of the body-coupled signal carrying the token, as received by the second wearable device, caused by changes in the pressure and area of skin contact between the clasped hands, and treats the renewal message as verified only where that variation matches the clasp sensed by an inertial or grip-force sensor of the second wearable device.

192. The method of claim 185, wherein the enrolled clasp pattern comprises an initiating squeeze by the first participant, the method further comprising detecting a responding count of squeezes by the second participant from variation in the coupling seen at the body-coupling electrode or from the inertial sensor, a count of zero, or no response within a response interval, cancelling the pairing, and a count of one or more being recorded as an indicated consent tier of the second participant that increases with the count; and wherein, after the joining, each participant confirms on his or her own phone the indicated consent tier and a self-attestation of sobriety and completes a pre-session checklist before any stage above a first stage of the consent session is unlocked, the self-attestation never overriding an impairment lockout.

ABSTRACT OF THE DISCLOSURE

A privacy-preserving consent system for safer, closer intimacy plays shared media, typically music, or another perceptible signal only while every participant renews affirmation within an affirmation window. On-device recognition detects safe words, gestures, touches, and button presses, and the joint consent state is the minimum across participants. A lapse fades the signal and a stop cuts it at once; continued absence of interaction escalates through a check-in and an alarm to a call for help, with verified responder entry. Media control is verified closed-loop and can follow musical boundaries. Optional features include voiceprint and wearable attribution, a classifier for scoped stops, brake scripts, staged envelopes, a listening ladder, operating-system and smart-speaker integration, a renewal-token standard, a cross-platform universal safe-word registry, safe-word listening in recognized bedrooms, and dating, venue, smart-home, and intimate-device integrations. Processing is preferably on-device, and the system is a signaling aid, not a determination of consent.